

Canonical GHZ configurations - cluster P4

species: electron
polarization cluster: P4
source cluster configurations: 81
canonical configurations collected: 54

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

cluster retained-component count distribution:

- 2 components: 54 configurations
- 4 components: 8 configurations
- 5 components: 3 configurations
- 6 components: 14 configurations
- 7 components: 2 configurations

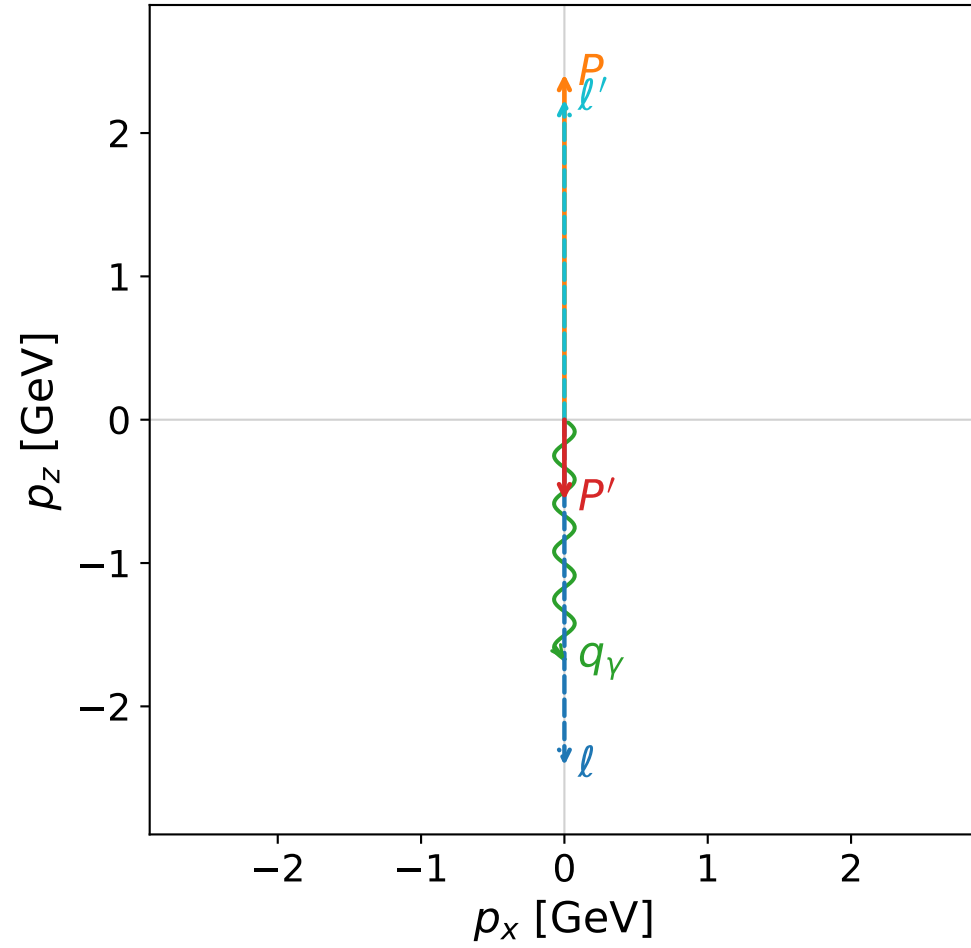
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_04\combined\
canonical_dghz_polarization_cluster_04_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

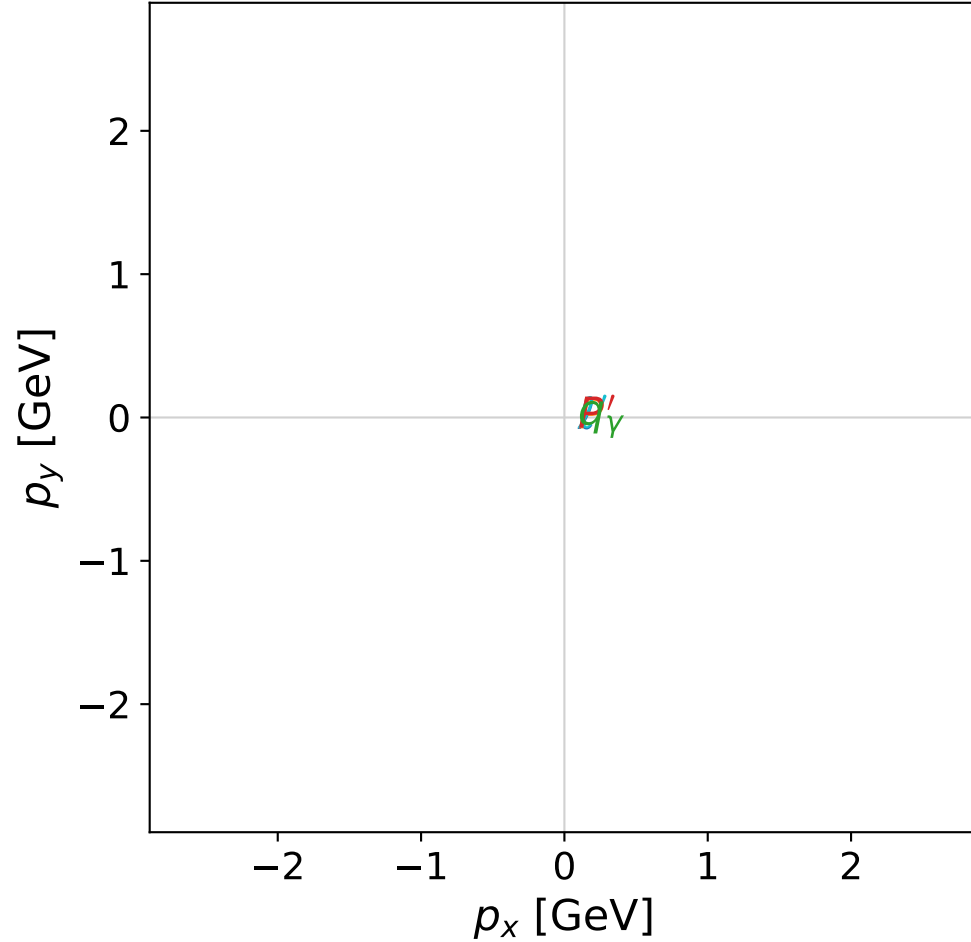


dghz_mixing_angles_polarization_04_local_minimum_8
 $d_{\text{GHZ}}=2.4276\text{e-}08$
 region: polarization_cluster_04_local_minimum_8
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0008
 lepton mixing angle: $\alpha_e=2.332037$ rad
 incoming lepton: $-0.68982|+\rangle +0.723981|-\rangle$
 proton mixing angle: $\alpha_p=2.3320298$ rad
 incoming proton: $-0.689815|+\rangle +0.723986|-\rangle$

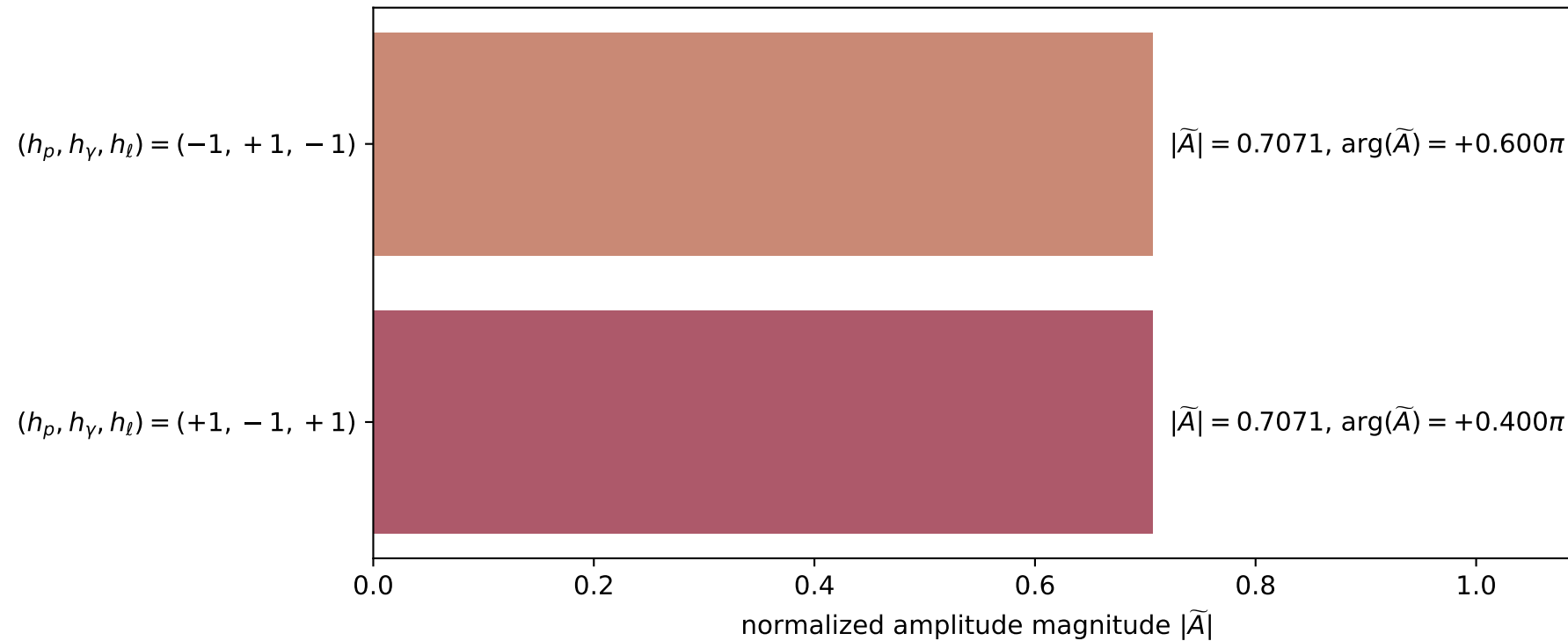
$s=24.9835$, $\sqrt{s}=4.99835$, $|\vec{P}|=2.41116$, $|\vec{P}'|=0.561095$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=4.05473$, $\phi_\gamma=1.2574$
 $E_\gamma=1.67212$, $\theta_{l'}=0$, $\phi_{l'}=4.56305$
 $Q^2=21.5386$, $x_B=0.923711$, $t=-6.60174$, $W^2=2.6587$, $y=0.967382$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4112$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4112$
 P $E= 2.5872$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4112$
 l' $E= 2.2332$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2332$
 P' $E= 1.0930$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.5611$
 q_γ $E= 1.6721$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6721$

x--y plane

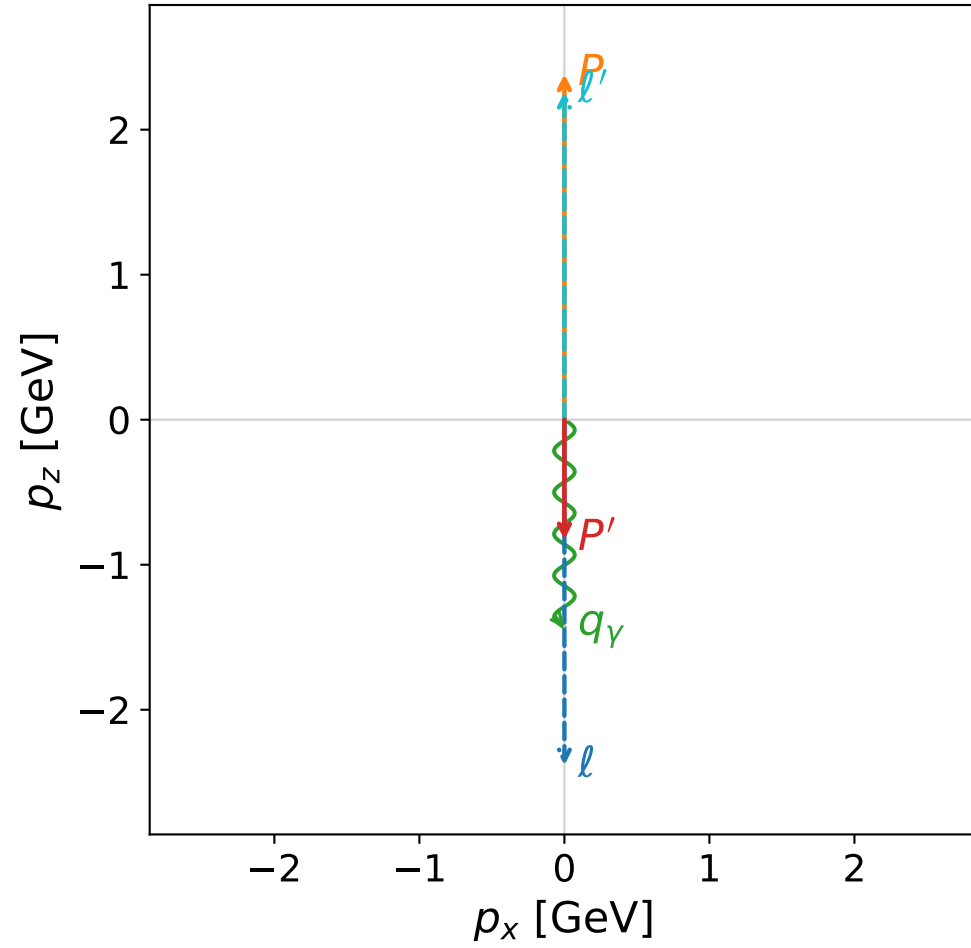


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_9

$d_{\text{GHZ}}=2.43084\text{e-}08$

region: polarization_cluster_04_local_minimum_9

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0009

lepton mixing angle: $\alpha_e=2.380064$ rad

incoming lepton: $-0.723782|+\rangle + 0.690029|-\rangle$

proton mixing angle: $\alpha_p=2.3800831$ rad

incoming proton: $-0.723795|+\rangle + 0.690015|-\rangle$

$s=24.4448$, $\sqrt{s}=4.94417$, $|\vec{P}|=2.38311$, $|\vec{P}'|=0.827879$

$\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14928$, $\phi_\gamma=4.99248$

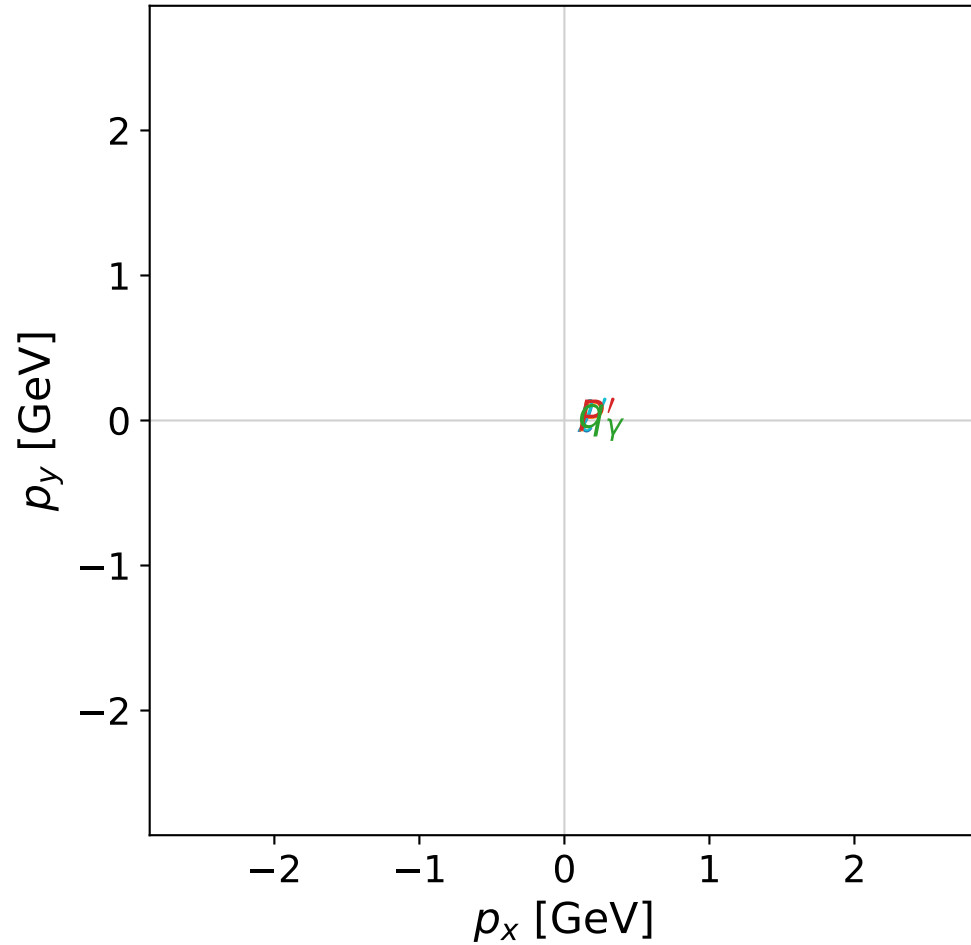
$E_\gamma=1.4326$, $\theta_{l'}=0$, $\phi_{l'}=1.26818$

$Q^2=21.5479$, $x_B=0.946724$, $t=-8.5944$, $W^2=2.09243$, $y=0.965859$

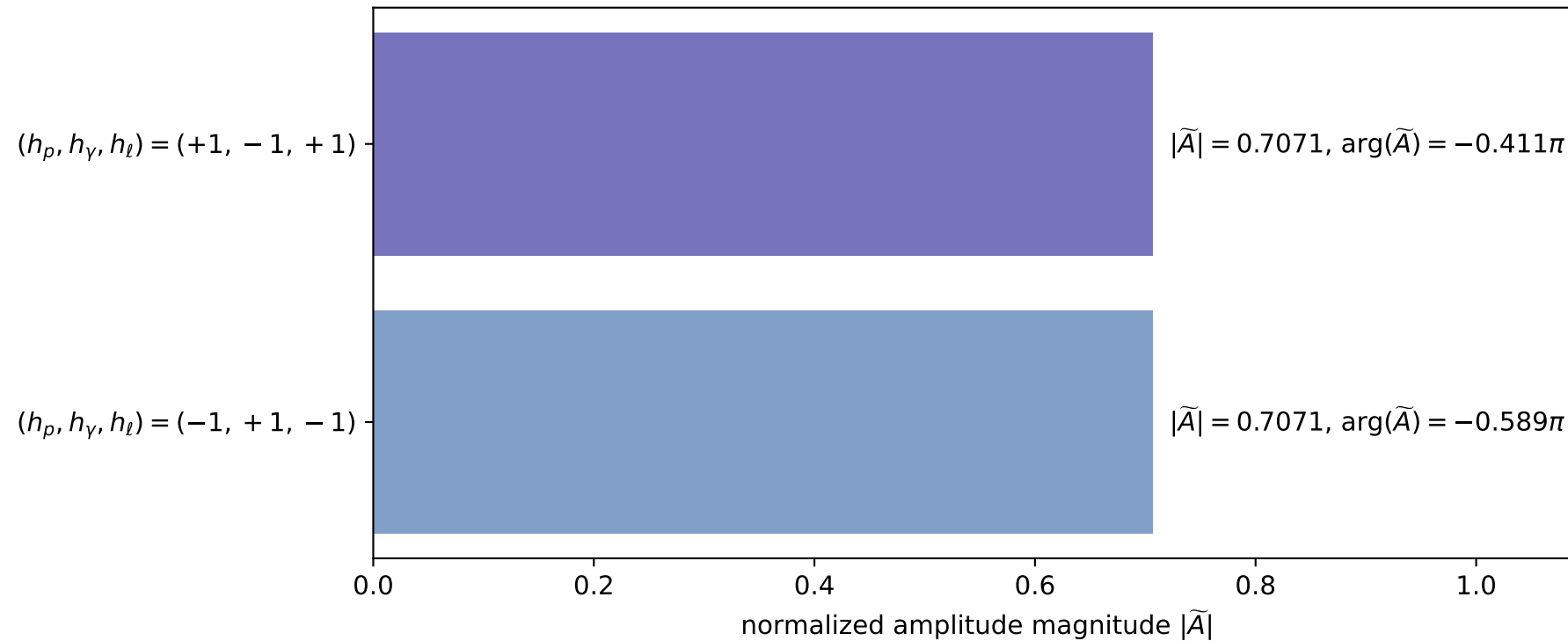
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	2.3831	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.3831
P	$E=$	2.5611	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.3831
l'	$E=$	2.2605	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.2605
P'	$E=$	1.2511	$p_x=$	-0.0000	$p_y=$	-0.0000	$p_z=$	-0.8279
q_γ	$E=$	1.4326	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-1.4326

x--y plane

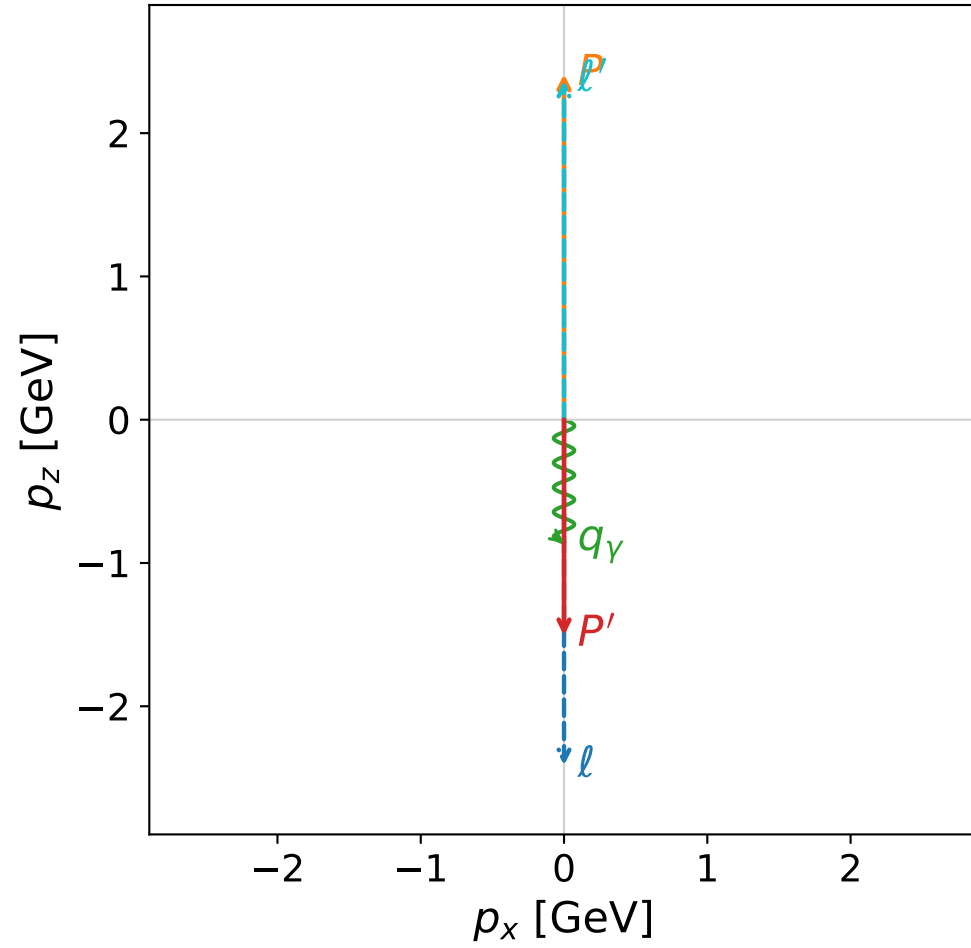


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

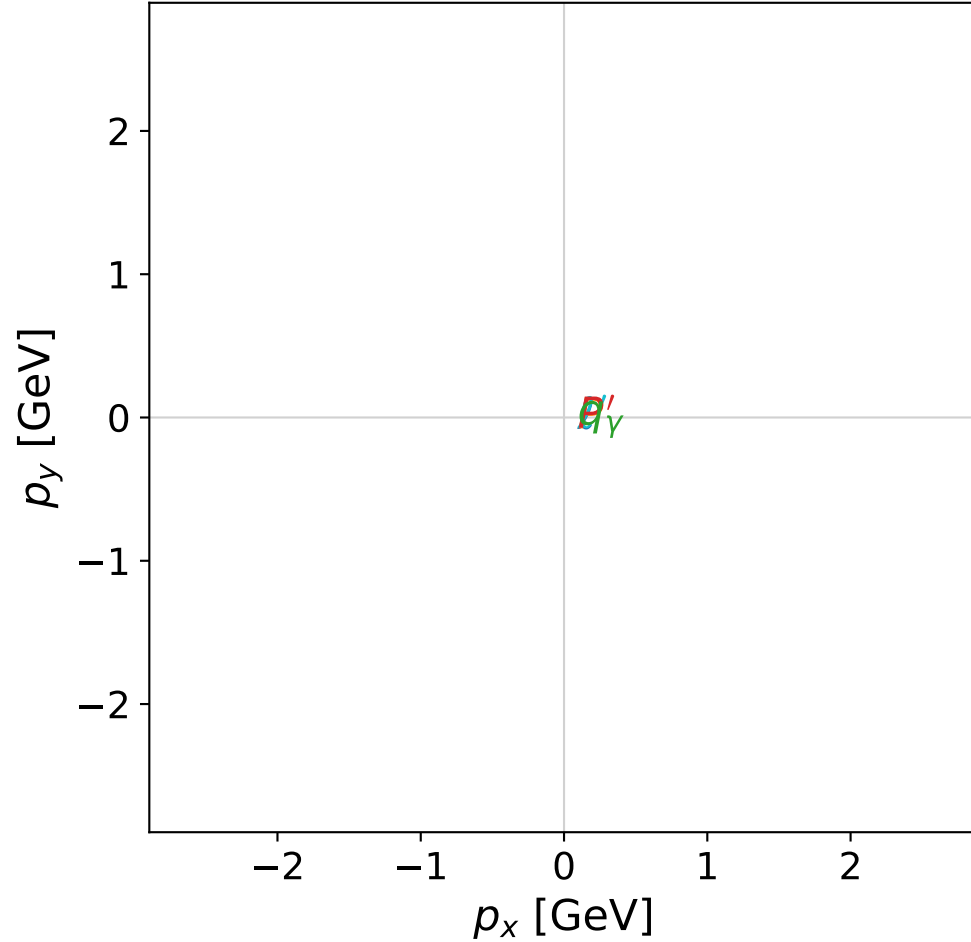


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



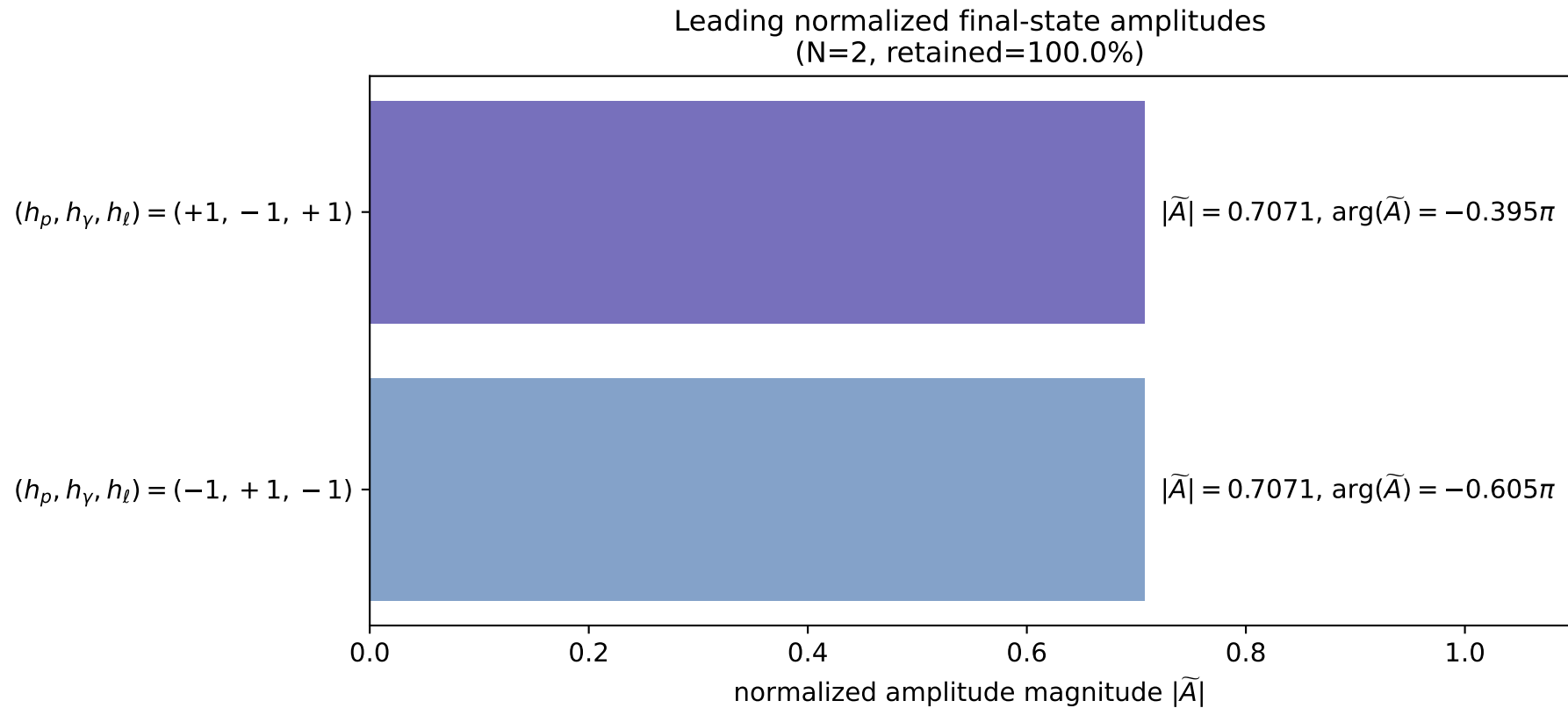
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_18
 $d_{\text{GHZ}}=2.51612\text{e-}08$
 region: polarization_cluster_04_local_minimum_18
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0018
 lepton mixing angle: $\alpha_e=2.5703729$ rad
 incoming lepton: $-0.841242|+\rangle +0.540659|-\rangle$
 proton mixing angle: $\alpha_p=2.5703847$ rad
 incoming proton: $-0.841248|+\rangle +0.540649|-\rangle$

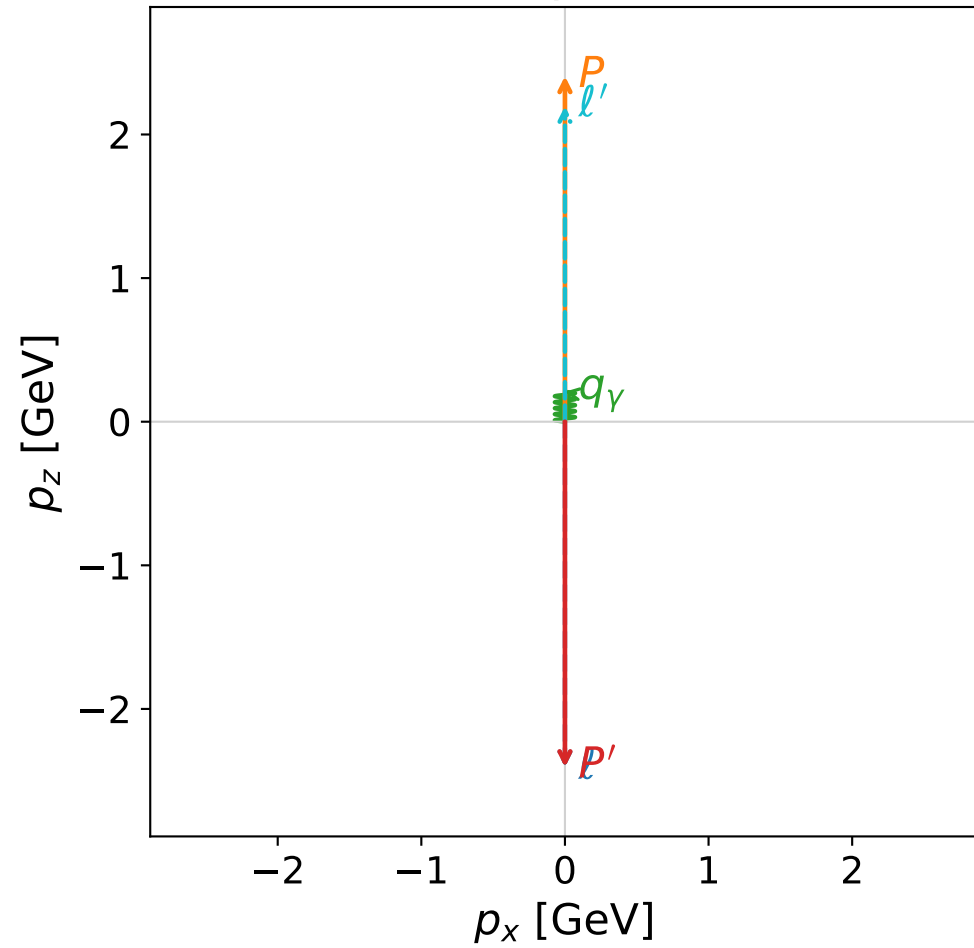
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.50685$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.529804$, $\phi_\gamma=5.04178$
 $E_\gamma=0.859098$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.10903$
 $Q^2=22.8268$, $x_B=0.980219$, $t=-14.6965$, $W^2=1.34049$, $y=0.965478$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3660$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.3660$
 P' $E= 1.7750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5069$
 q_γ $E= 0.8591$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.8591$

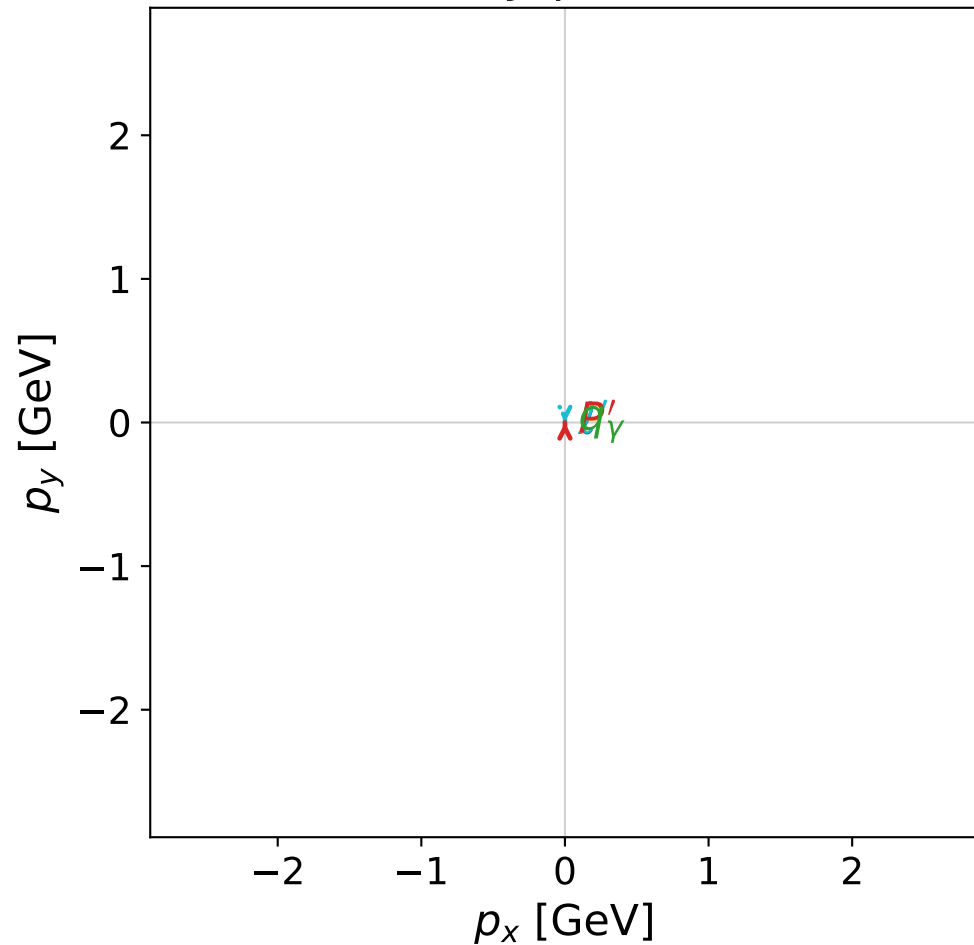


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

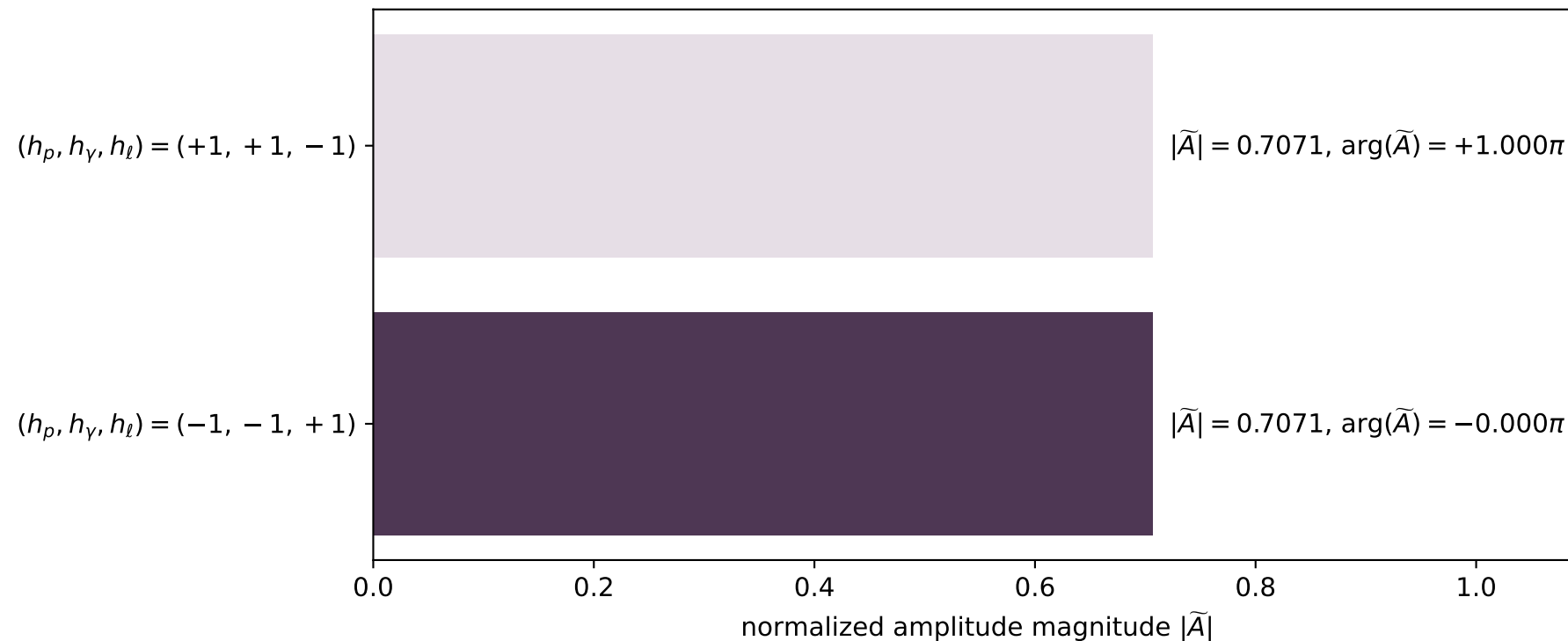


dghz_mixing_angles_polarization_04_local_minimum_22
 $d_{\text{GHZ}}=2.56605\text{e-}08$
 region: polarization_cluster_04_local_minimum_22
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0022
 lepton mixing angle: $\alpha_e=2.4937434$ rad
 incoming lepton: $-0.797384|+\rangle +0.603473|-\rangle$
 proton mixing angle: $\alpha_p=2.218653$ rad
 incoming proton: $-0.603479|+\rangle +0.797379|-\rangle$

$s=24.8915$, $\sqrt{s}=4.98914$, $|\vec{P}|=2.4064$, $|\vec{P}'|=2.4064$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.87557$, $\phi_\gamma=0.0606964$
 $E_\gamma=0.208957$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.01716$
 $Q^2=21.1516$, $x_B=0.91027$, $t=-23.163$, $W^2=2.96488$, $y=0.967722$

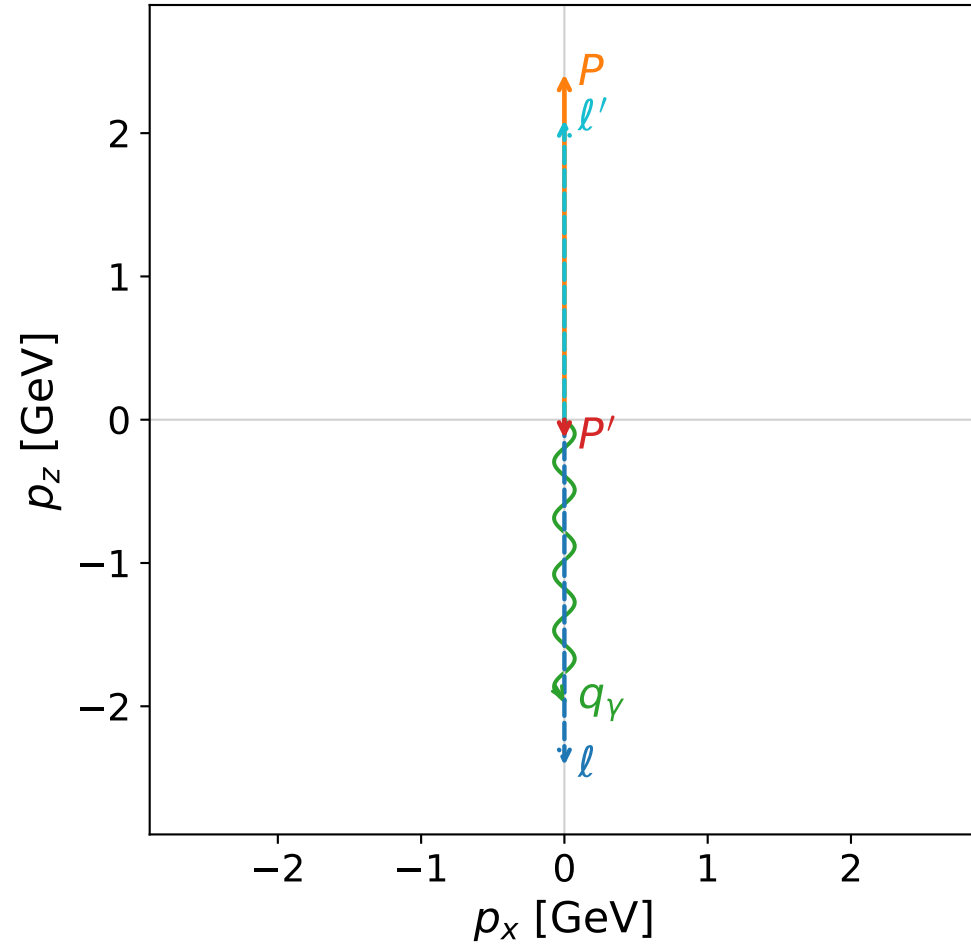
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.4064	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.4064
P	$E=$	2.5827	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.4064
ℓ'	$E=$	2.1974	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	2.1974
P'	$E=$	2.5827	$p_x=$	-0.0000	$p_y=$	0.0000	$p_z=$	-2.4064
q_γ	$E=$	0.2090	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.2090

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

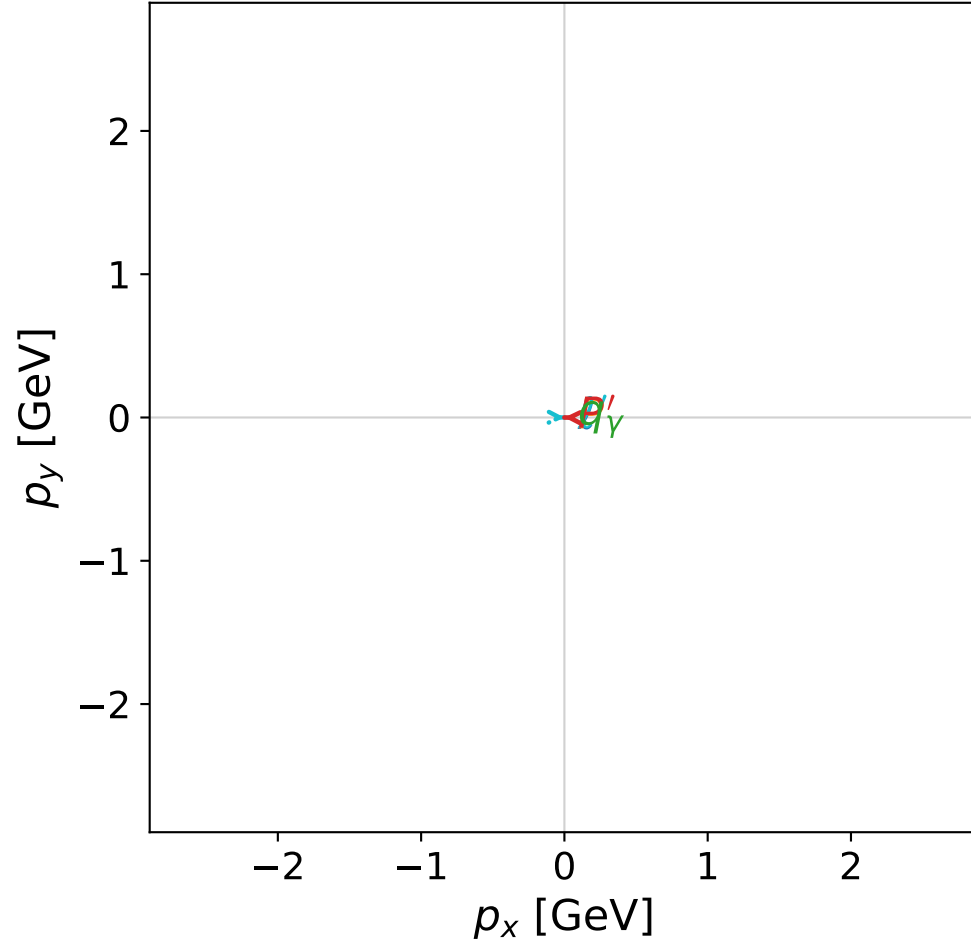


dghz_mixing_angles_polarization_04_local_minimum_29
 $d_{\text{GHZ}}=2.60368\text{e-}08$
 region: polarization_cluster_04_local_minimum_29
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0029
 lepton mixing angle: $\alpha_e=2.3616845$ rad
 incoming lepton: $-0.710978|+\rangle +0.703214|-\rangle$
 proton mixing angle: $\alpha_p=2.3617041$ rad
 incoming proton: $-0.710992|+\rangle +0.7032|-\rangle$

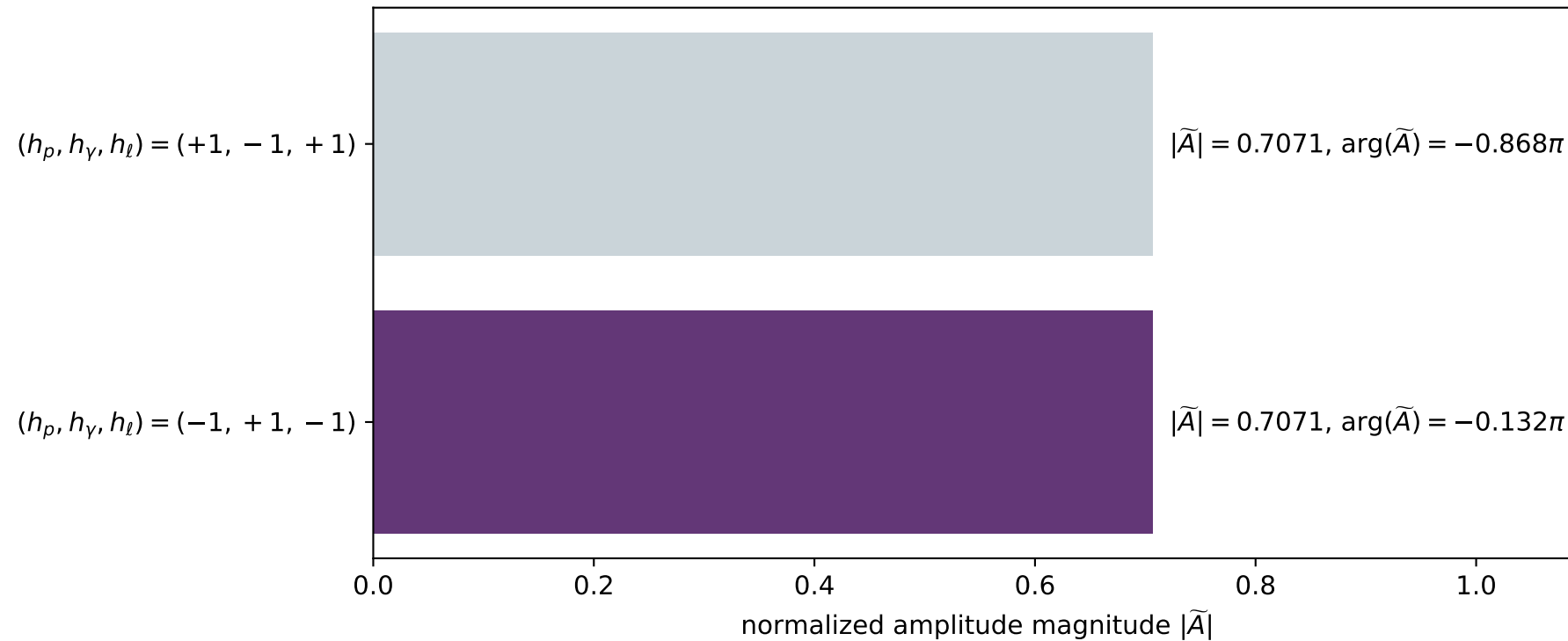
$s=24.9994$, $\sqrt{s}=4.99994$, $|\vec{P}|=2.41198$, $|\vec{P}'|=0.13022$
 $\theta_{P'}=3.14095$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.1261$, $\phi_\gamma=0.397731$
 $E_\gamma=1.96136$, $\theta_{l'}=3.98045\text{e-}05$, $\phi_{l'}=6.2677$
 $Q^2=20.1795$, $x_B=0.86298$, $t=-3.77005$, $W^2=4.08383$, $y=0.969481$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0916$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 2.0916$
 P' $E= 0.9470$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= -0.1302$
 q_γ $E= 1.9614$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9614$

x--y plane

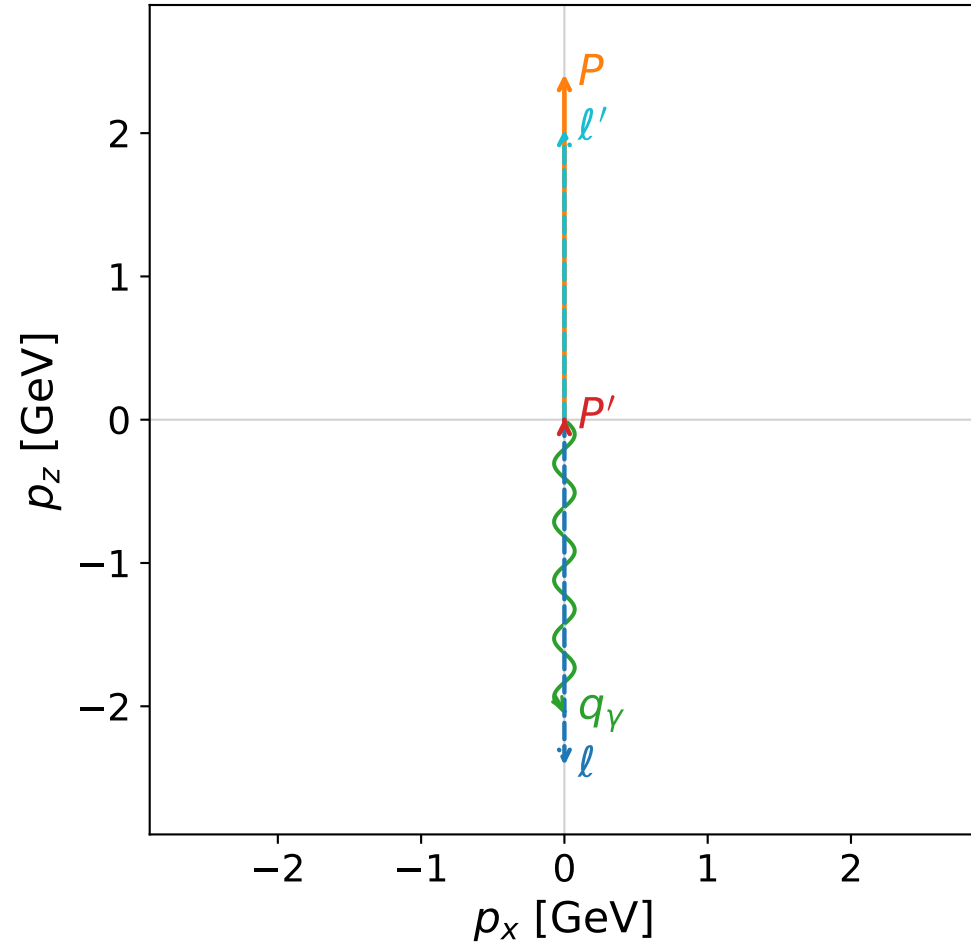


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

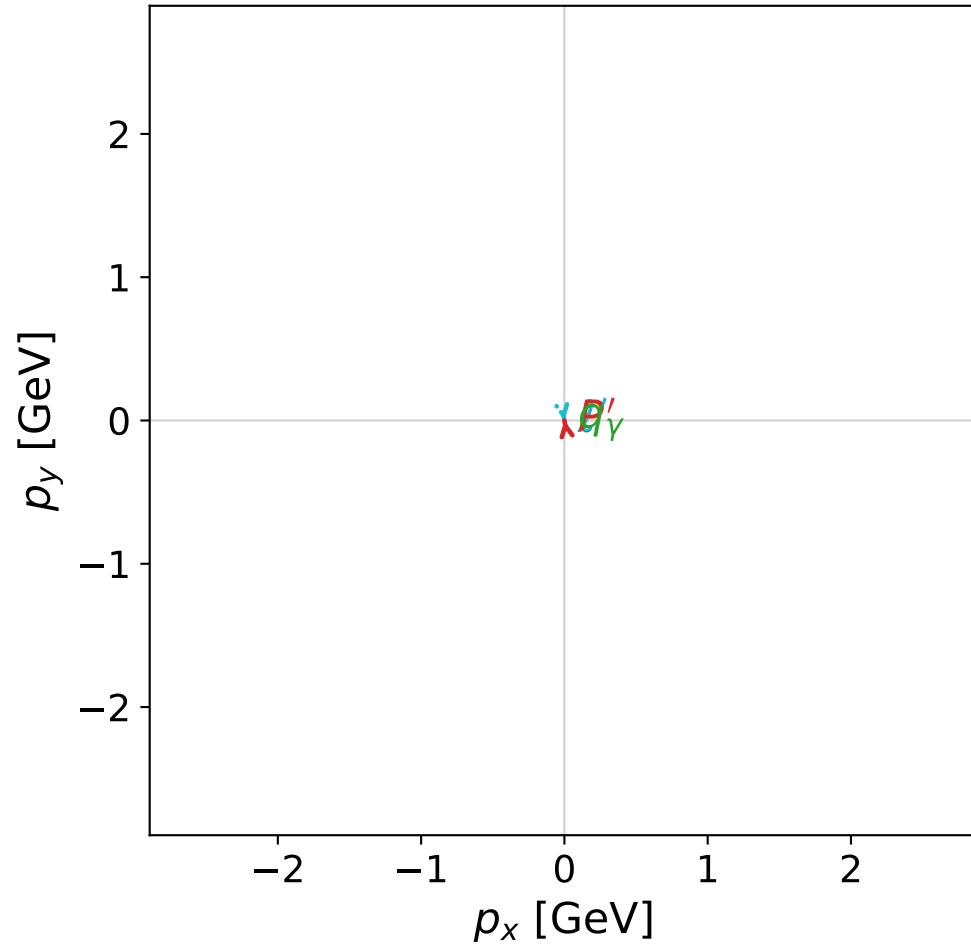


dghz_mixing_angles_polarization_04_local_minimum_30
 $d_{\text{GHZ}}=2.62358\text{e-}08$
 region: polarization_cluster_04_local_minimum_30
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0030
 lepton mixing angle: $\alpha_e=2.3547844$ rad
 incoming lepton: $-0.706109|+\rangle +0.708103|-\rangle$
 proton mixing angle: $\alpha_p=2.3547913$ rad
 incoming proton: $-0.706114|+\rangle +0.708098|-\rangle$

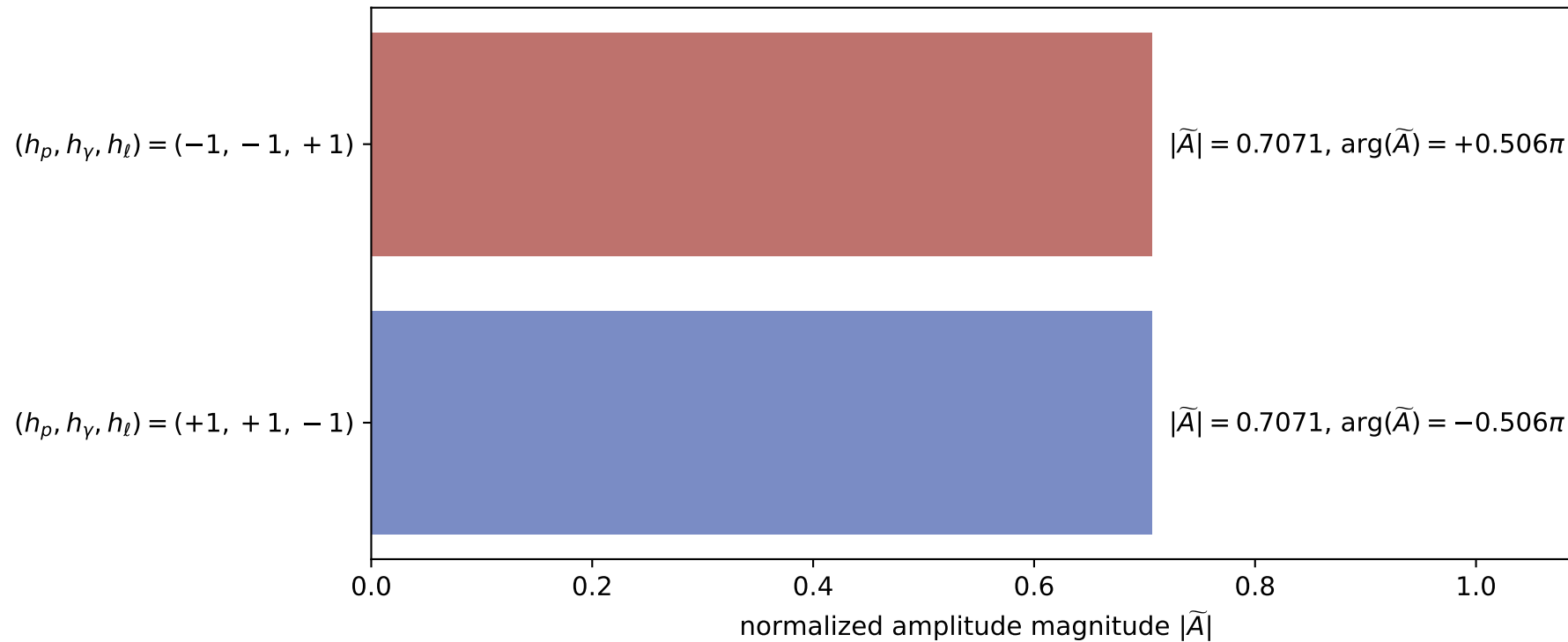
$s=24.9986$, $\sqrt{s}=4.99986$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0112972$
 $\theta_{P'}=0.0085095$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.74098$, $\phi_\gamma=1.5897$
 $E_\gamma=2.03655$, $\theta_{\ell'}=4.7467\text{e-}05$, $\phi_{\ell'}=4.88257$
 $Q^2=19.5391$, $x_B=0.834792$, $t=-3.0411$, $W^2=4.74669$, $y=0.970447$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0252$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= 2.0252$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0001$ $p_z= 0.0113$
 q_γ $E= 2.0365$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0365$

x--y plane

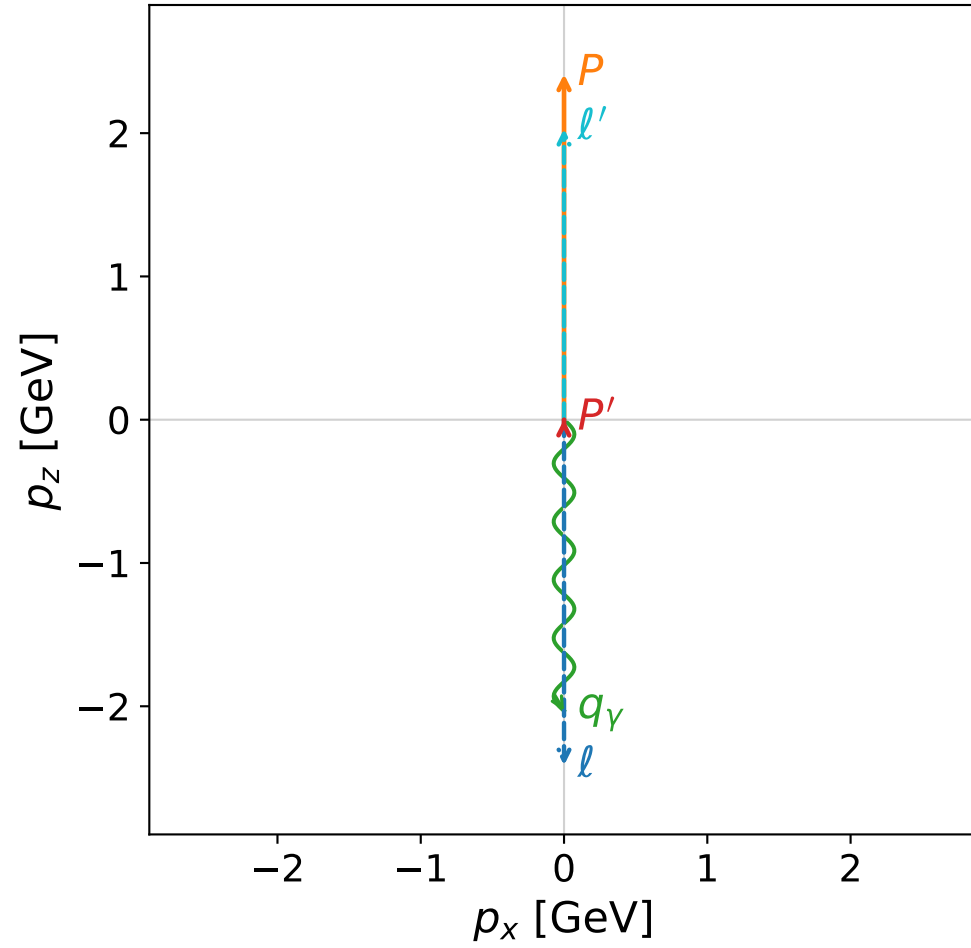


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

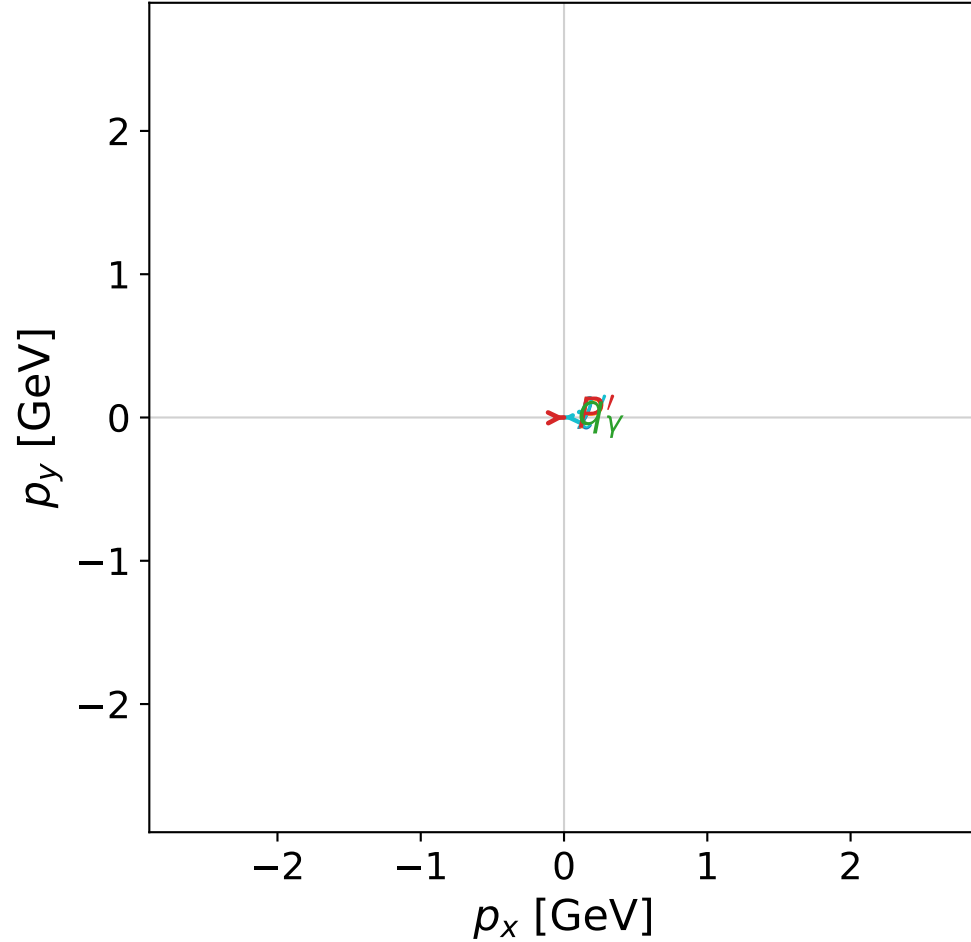


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



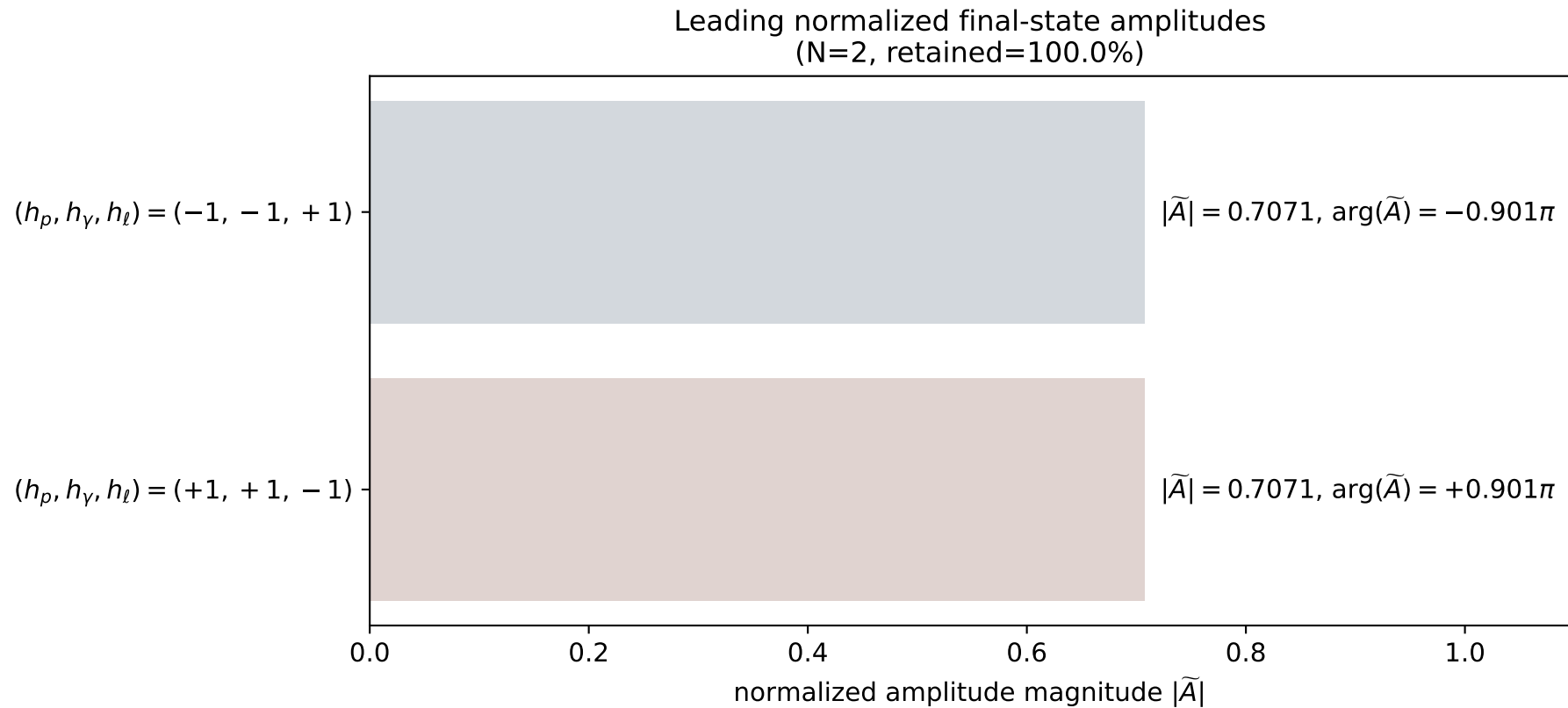
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_39
 $d_{\{\rm GHZ\}}=2.64421\text{e-}08$
 region: polarization_cluster_04_local_minimum_39
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0039
 lepton mixing angle: $\alpha_e=2.4732875$ rad
 incoming lepton: $-0.784873|+\rangle +0.619657|-\rangle$
 proton mixing angle: $\alpha_p=2.4732933$ rad
 incoming proton: $-0.784877|+\rangle +0.619652|-\rangle$

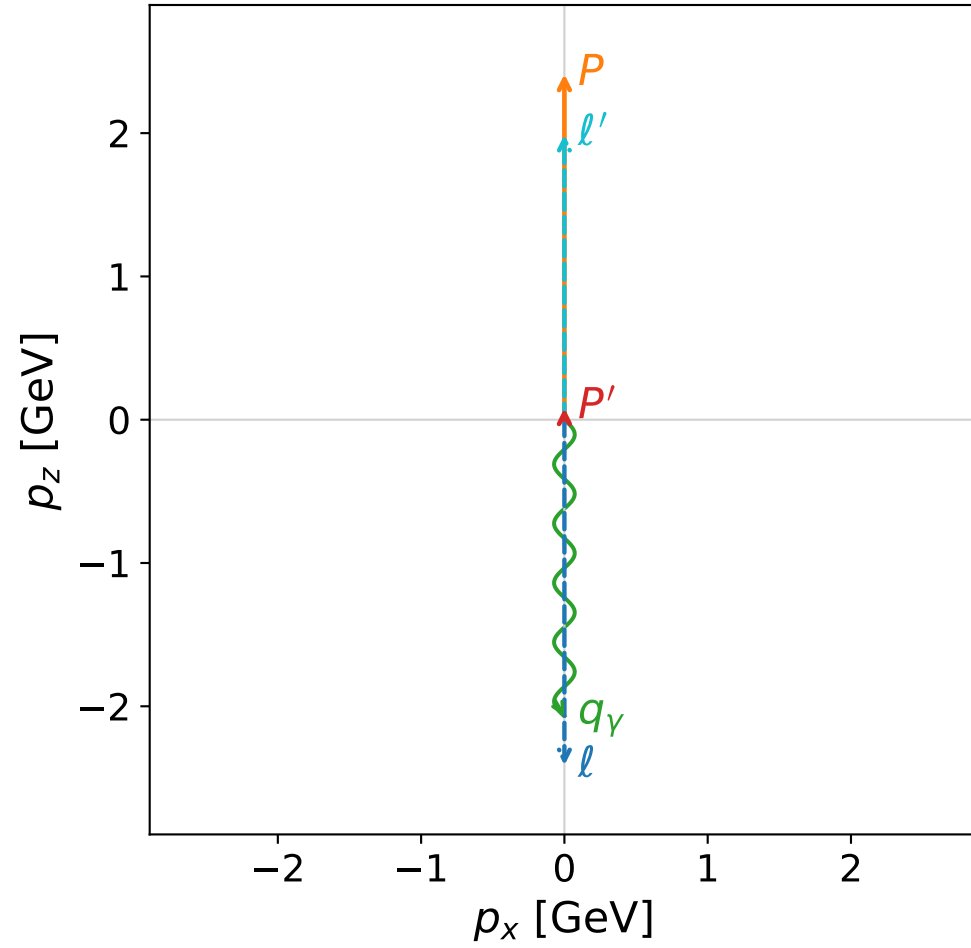
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00214662$
 $\theta_{p'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.019635$, $\phi_\gamma=3.45408$
 $E_\gamma=2.03207$, $\theta_{l'}=1.03817\text{e-}05$, $\phi_{l'}=3.16123$
 $Q^2=19.5848$, $x_B=0.836754$, $t=-3.08503$, $W^2=4.70074$, $y=0.970381$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.0299$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0299$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0021$
 q_γ $E= 2.0321$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0321$

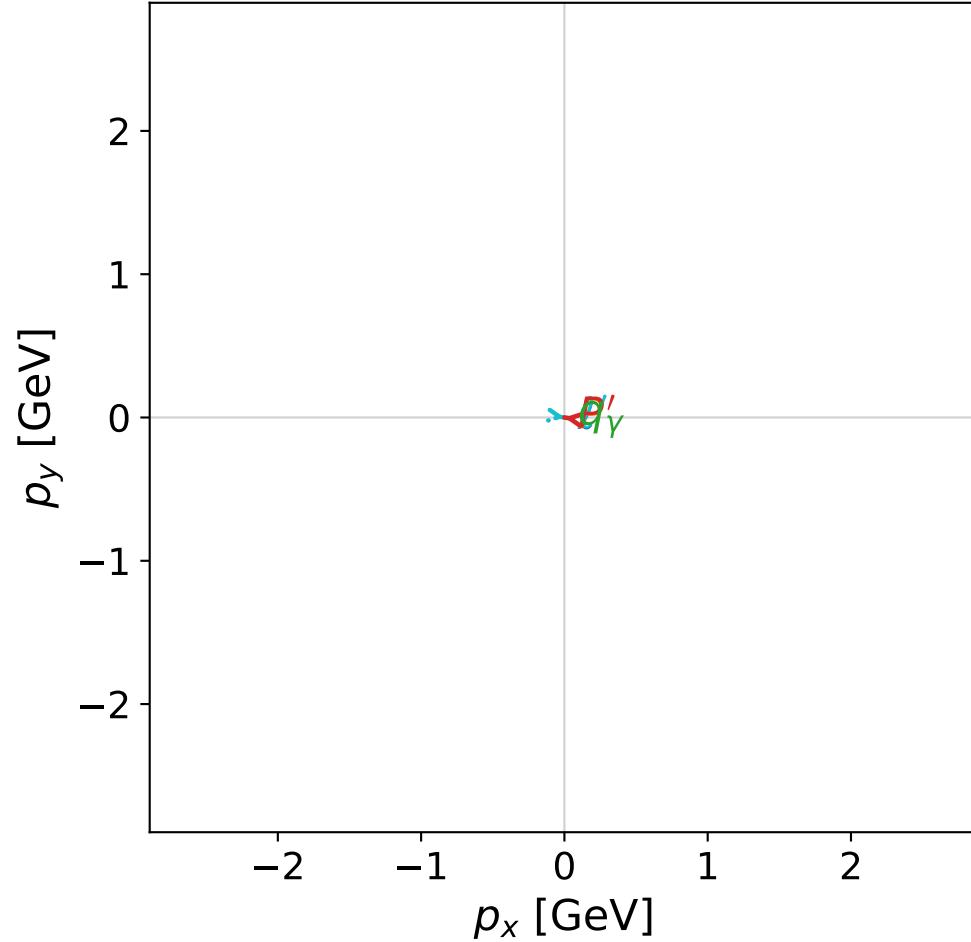


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



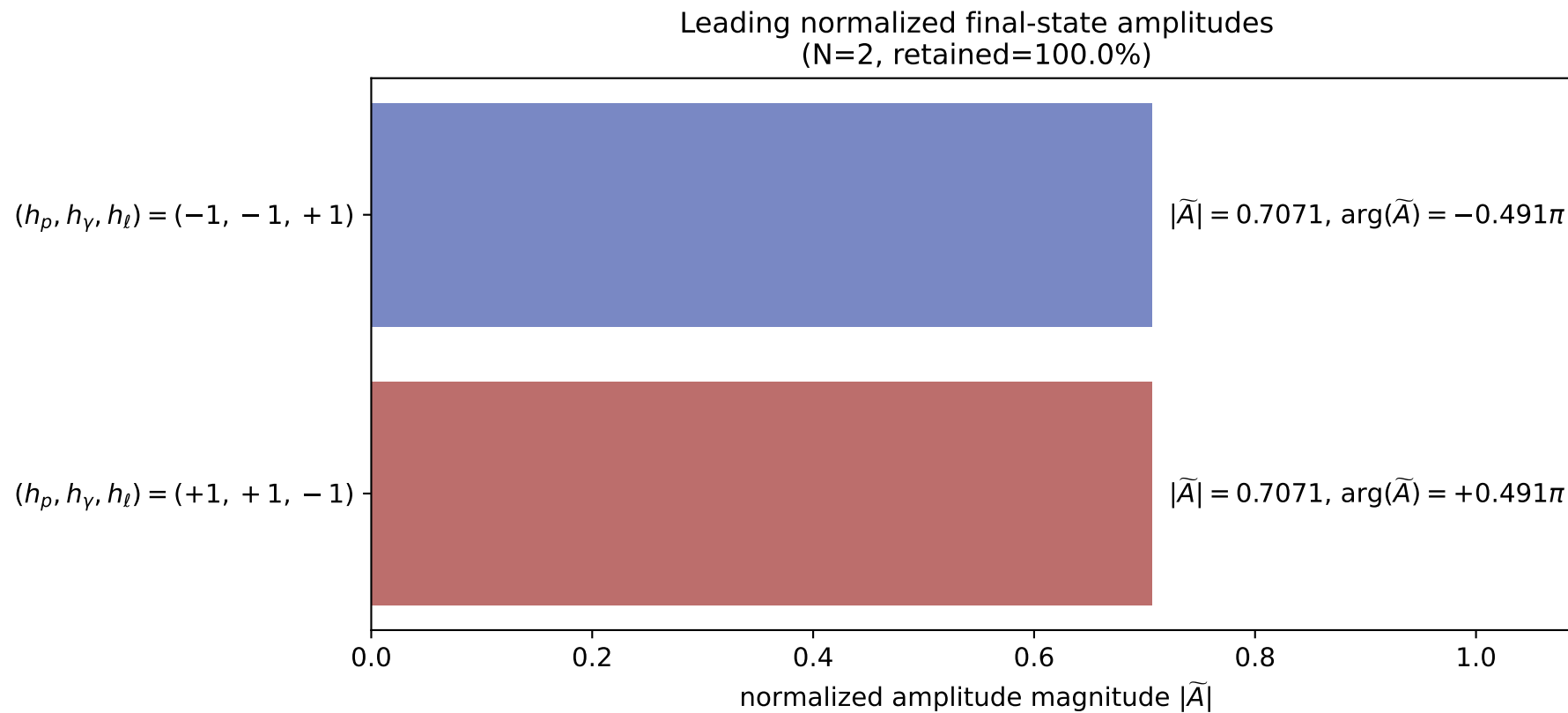
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_50
 $d_{\text{GHZ}}=2.68327\text{e-}08$
 region: polarization_cluster_04_local_minimum_50
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0050
 lepton mixing angle: $\alpha_e=2.2455924$ rad
 incoming lepton: $-0.624738|+\rangle + 0.780834|-\rangle$
 proton mixing angle: $\alpha_p=2.2455942$ rad
 incoming proton: $-0.624739|+\rangle + 0.780833|-\rangle$

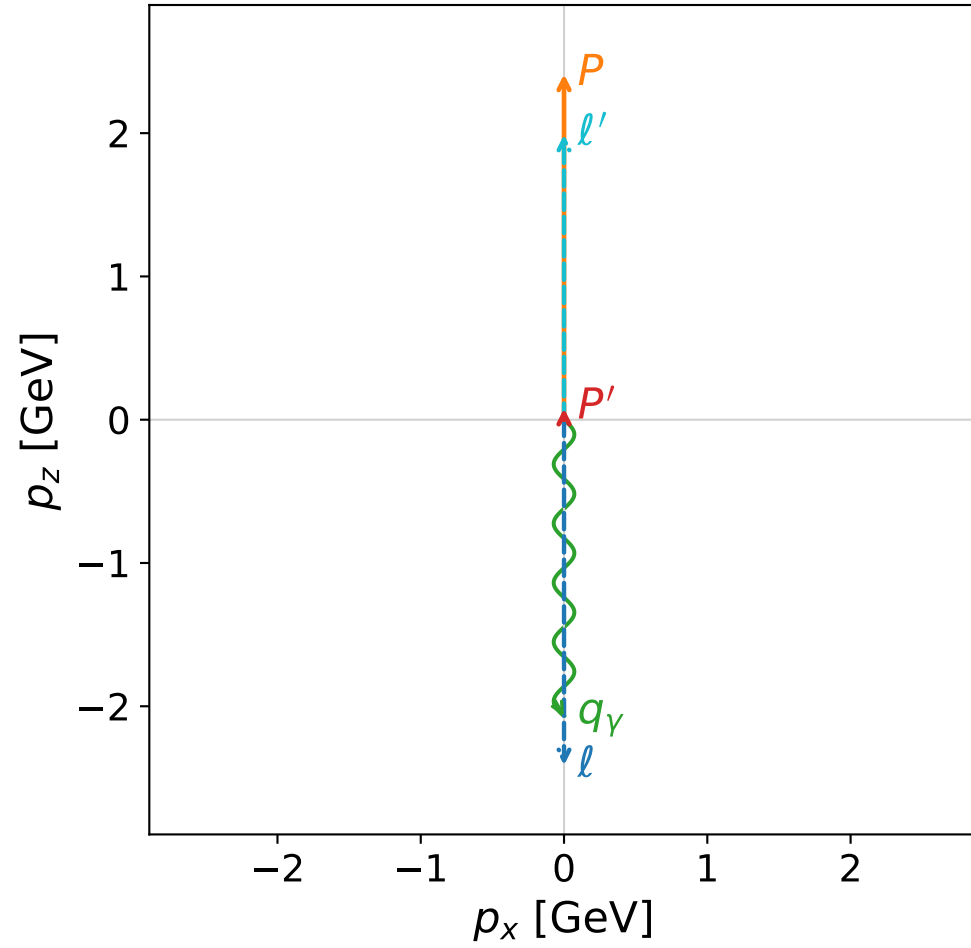
$s=24.9974$, $\sqrt{s}=4.99974$, $|\vec{P}|=2.41188$, $|\vec{P}'|=0.0814227$
 $\theta_{P'}=7.88291\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.99389$, $\phi_\gamma=4.74182$
 $E_\gamma=2.06982$, $\theta_{\ell'}=3.23049\text{e-}07$, $\phi_{\ell'}=6.13548$
 $Q^2=19.183$, $x_B=0.819169$, $t=-2.72062$, $W^2=5.11448$, $y=0.970983$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 1.9884$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9884$
 P' $E= 0.9415$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0814$
 q_γ $E= 2.0698$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0698$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

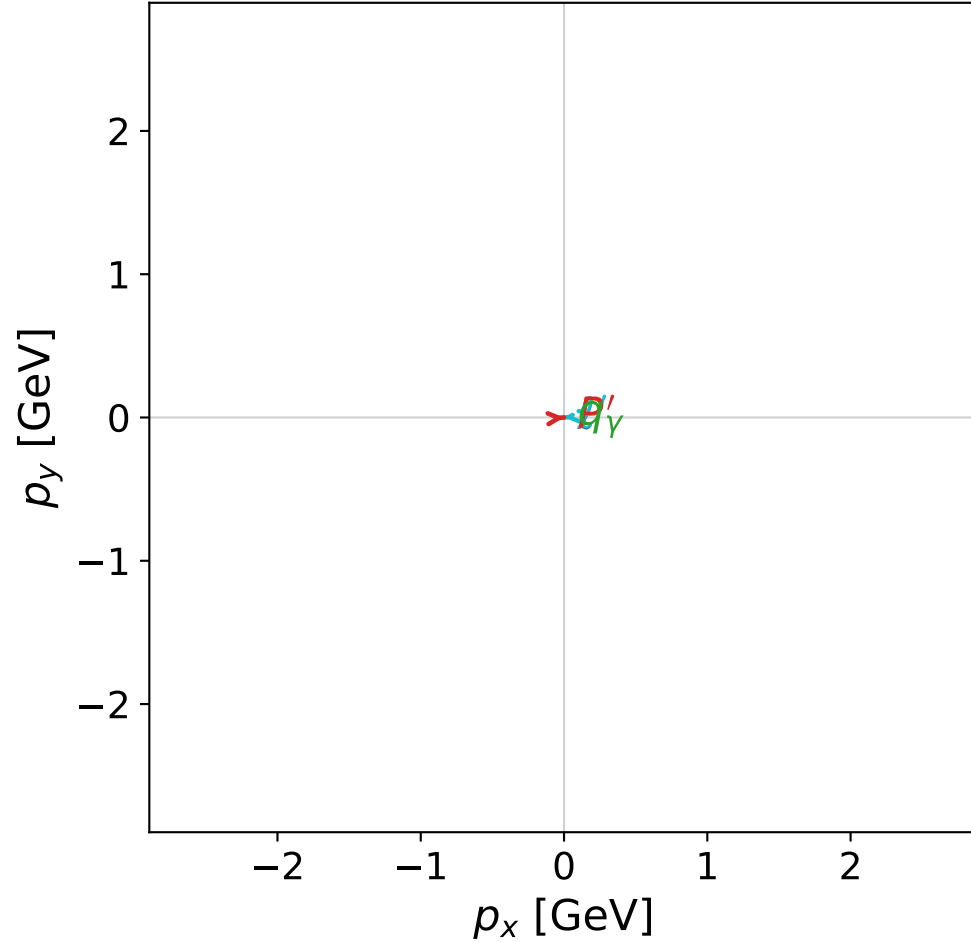


dghz_mixing_angles_polarization_04_local_minimum_59
 $d_{\{\rm GHZ\}}=2.72599\text{e-}08$
 region: polarization_cluster_04_local_minimum_59
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0059
 lepton mixing angle: $\alpha_e=2.4105855$ rad
 incoming lepton: $-0.744502|+\rangle +0.66762|-\rangle$
 proton mixing angle: $\alpha_p=2.4105852$ rad
 incoming proton: $-0.744502|+\rangle +0.66762|-\rangle$

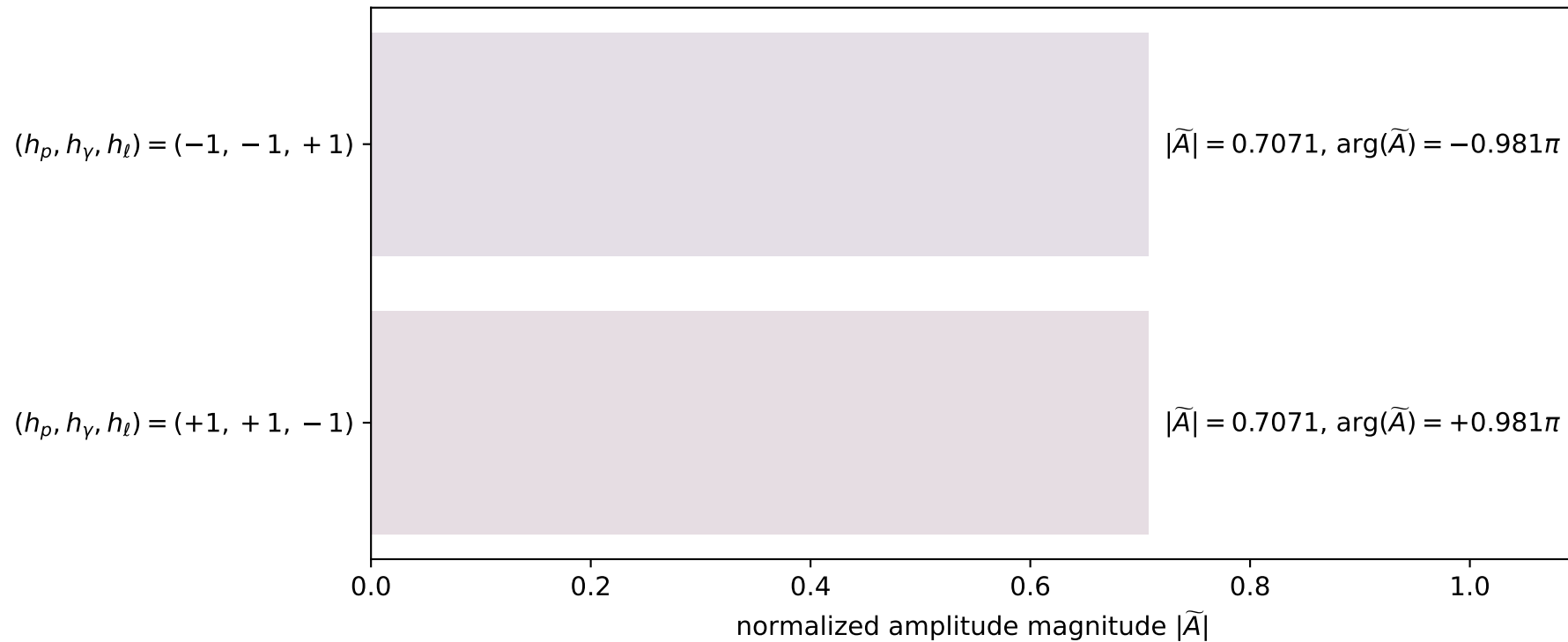
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0792932$
 $\theta_{P'}=0.00083419$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0772728$, $\phi_\gamma=3.20129$
 $E_\gamma=2.06897$, $\theta_{\ell'}=3.32443\text{e-}05$, $\phi_{\ell'}=3.21887$
 $Q^2=19.1966$, $x_B=0.819668$, $t=-2.73017$, $W^2=5.10319$, $y=0.970969$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9897$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 1.9897$
 P' $E= 0.9413$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0793$
 q_γ $E= 2.0690$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0690$

x--y plane

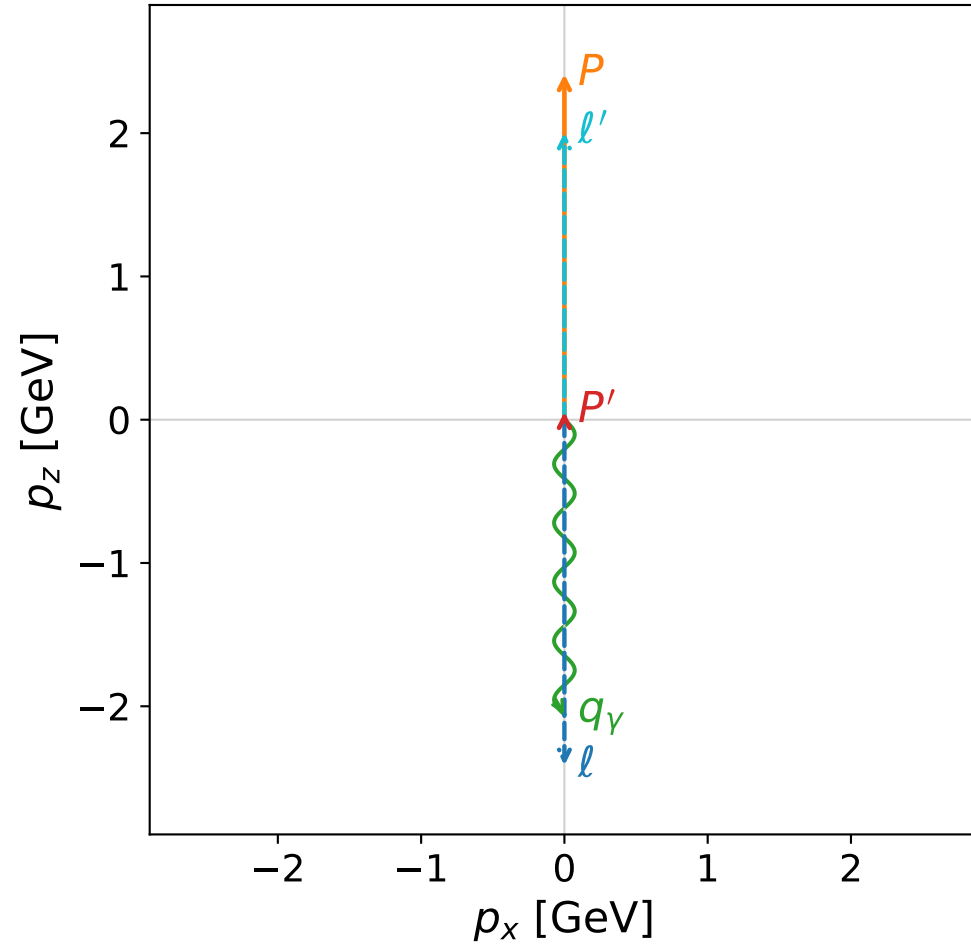


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

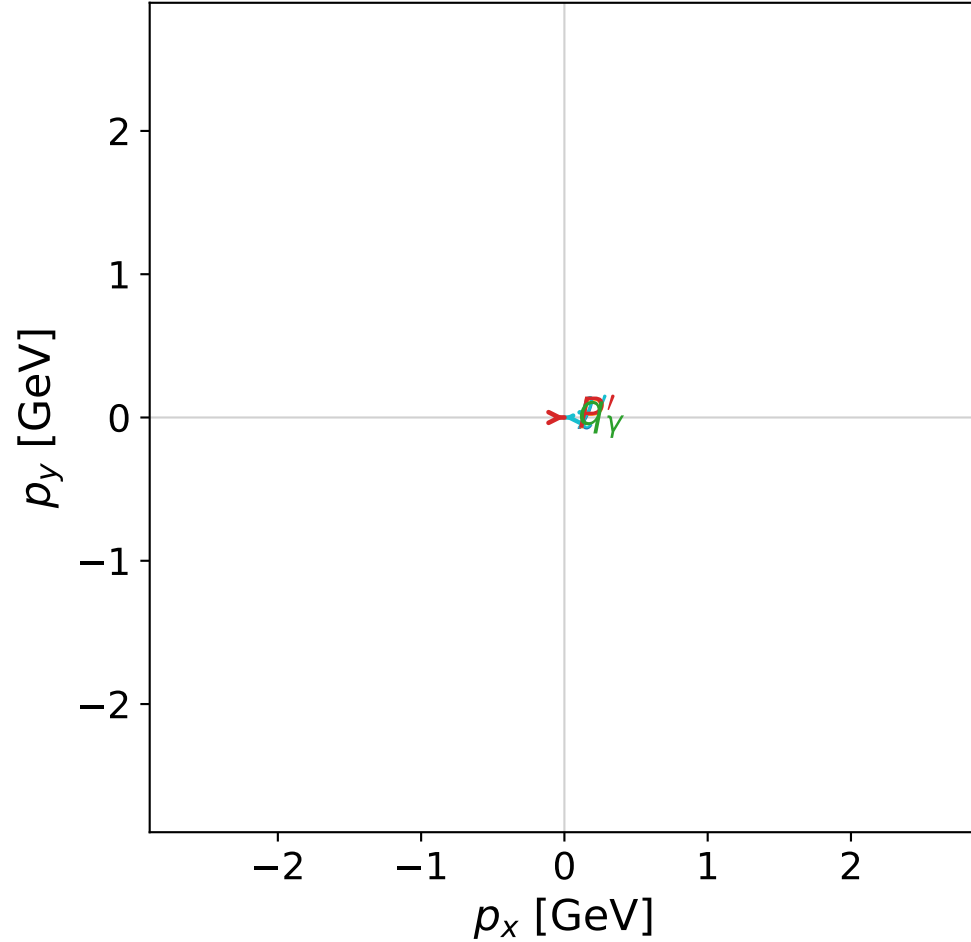


dghz_mixing_angles_polarization_04_local_minimum_63
 $d_{\text{GHZ}}=2.75718\text{e-}08$
 region: polarization_cluster_04_local_minimum_63
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0063
 lepton mixing angle: $\alpha_e=2.4612083$ rad
 incoming lepton: $-0.777331|+\rangle +0.629092|-\rangle$
 proton mixing angle: $\alpha_p=2.461193$ rad
 incoming proton: $-0.777321|+\rangle +0.629104|-\rangle$

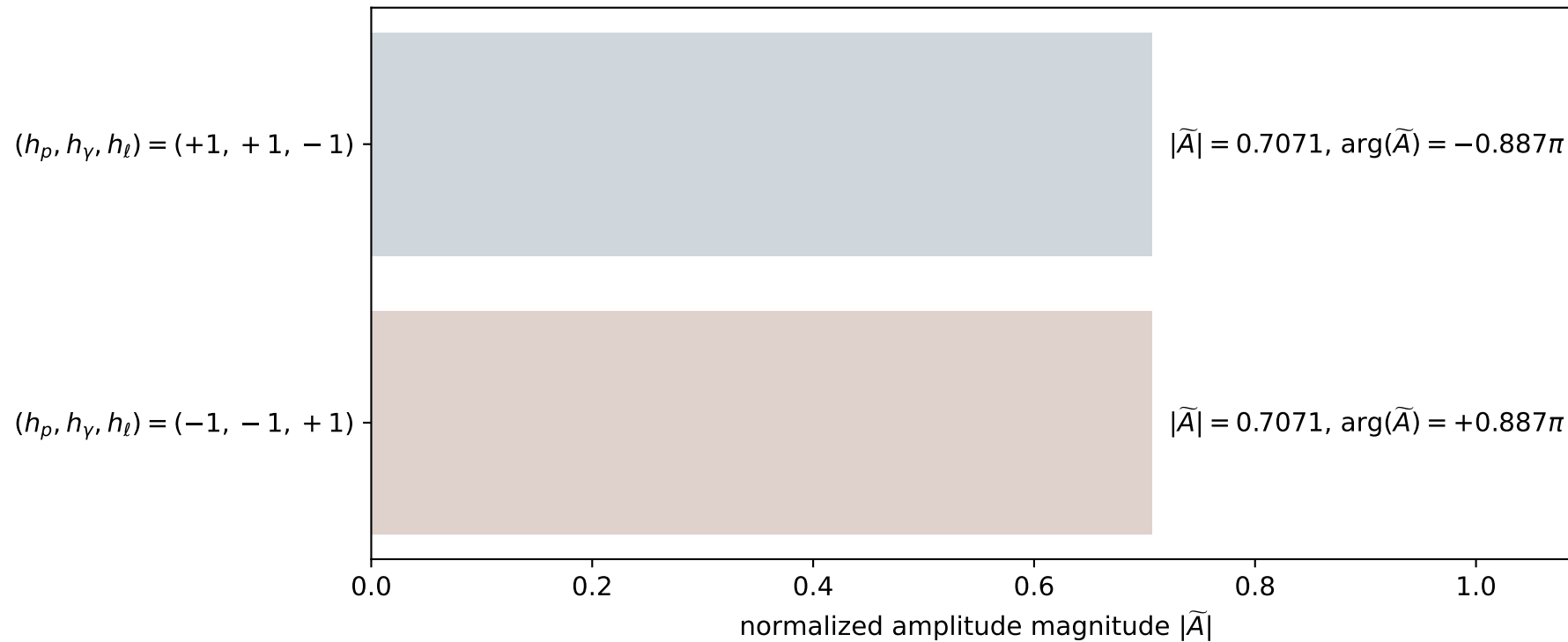
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0542678$
 $\theta_{P'}=0.00132352$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.28313$, $\phi_\gamma=2.78563$
 $E_\gamma=2.05735$, $\theta_{\ell'}=3.58569\text{e-}05$, $\phi_{\ell'}=3.14154$
 $Q^2=19.3259$, $x_B=0.825355$, $t=-2.8417$, $W^2=4.96918$, $y=0.970773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0031$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0031$
 P' $E= 0.9396$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0543$
 q_γ $E= 2.0573$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

x--y plane

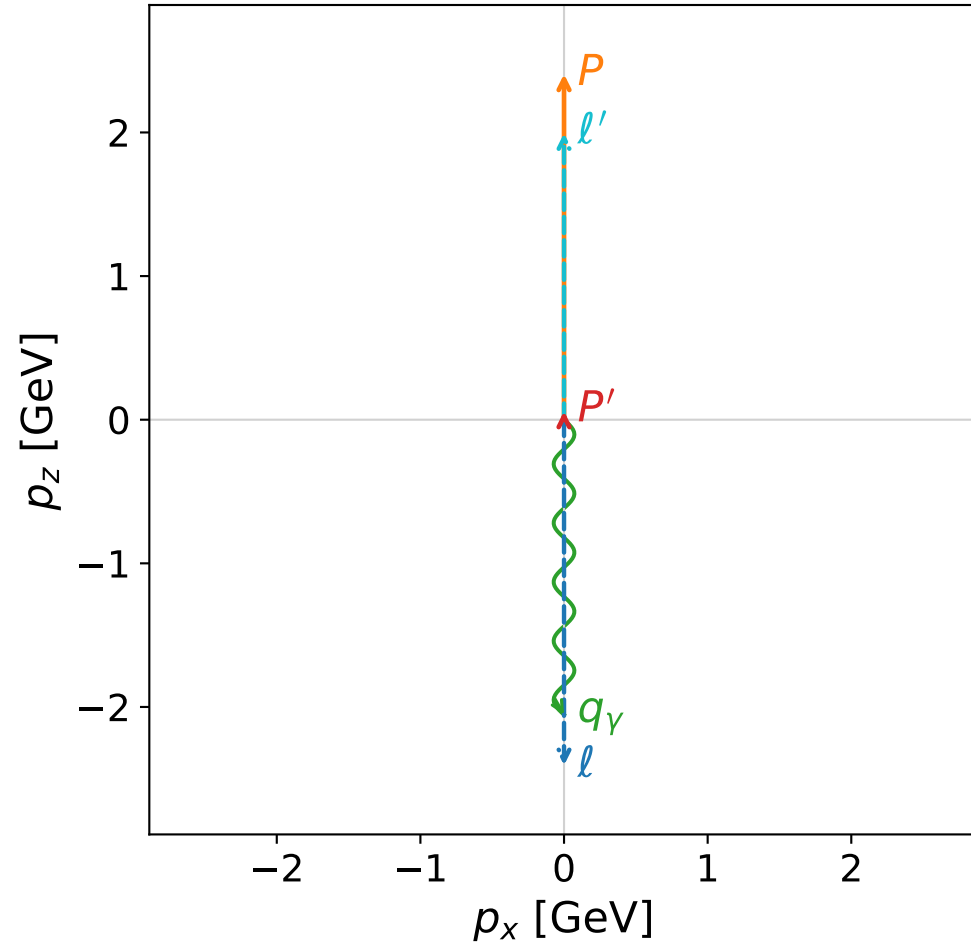


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

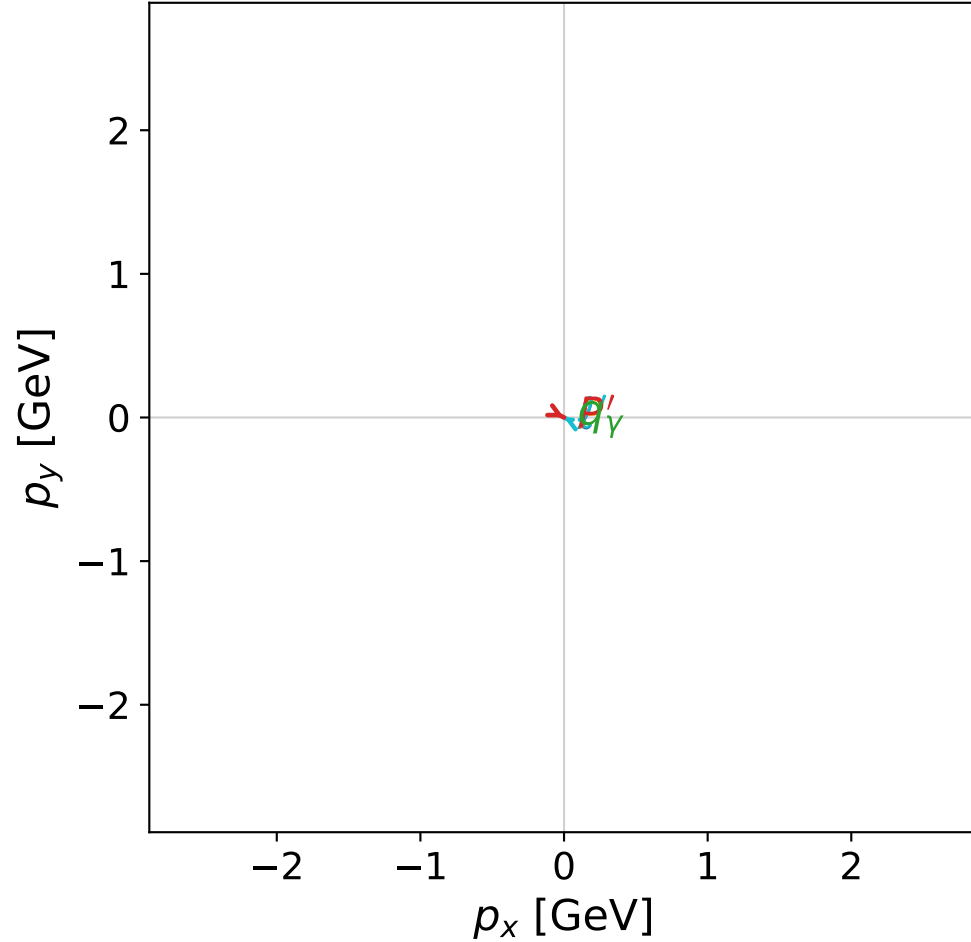


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



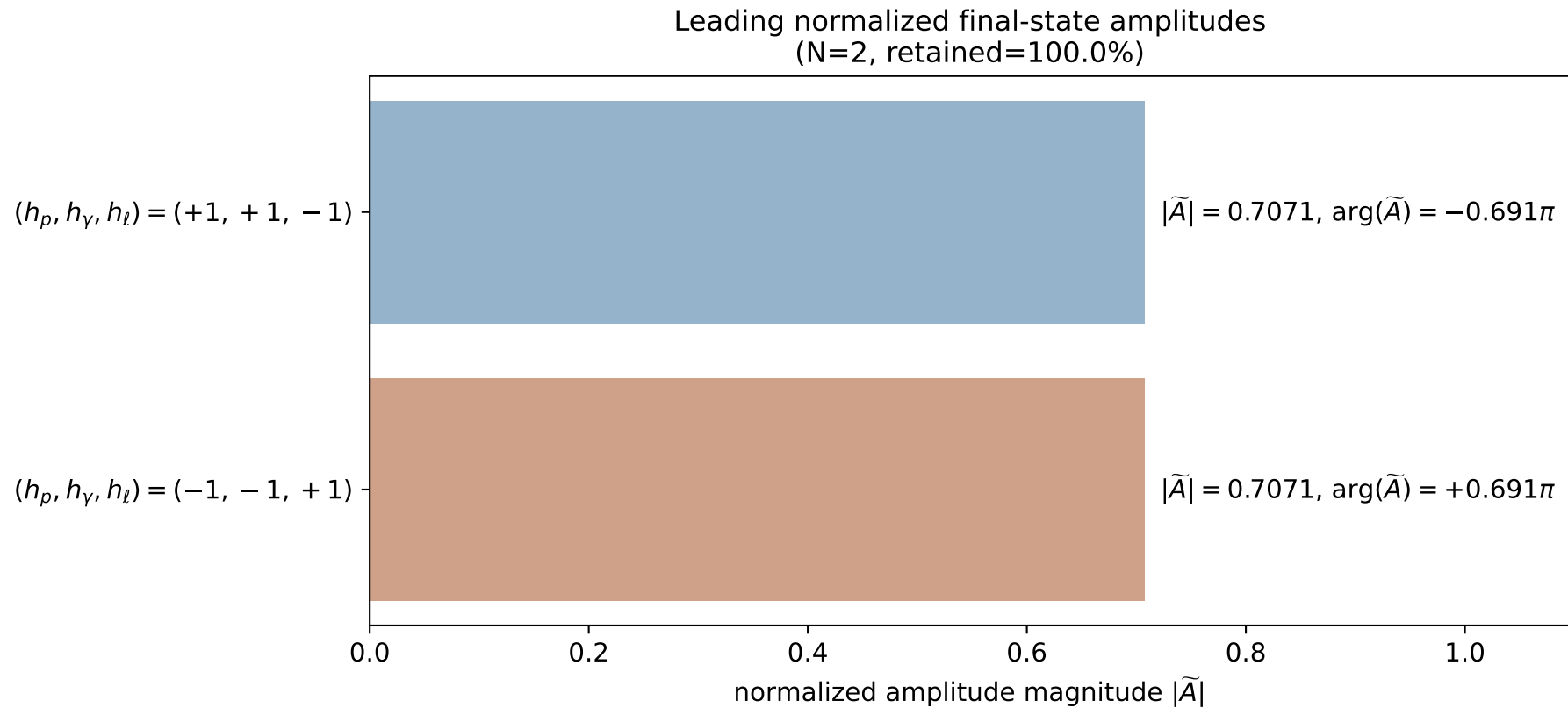
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_64
 $d_{\text{GHZ}}=2.76858\text{e-}08$
 region: polarization_cluster_04_local_minimum_64
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0064
 lepton mixing angle: $\alpha_e=2.2443831$ rad
 incoming lepton: $-0.623793|+\rangle +0.781589|-\rangle$
 proton mixing angle: $\alpha_p=2.2443657$ rad
 incoming proton: $-0.62378|+\rangle +0.7816|-\rangle$

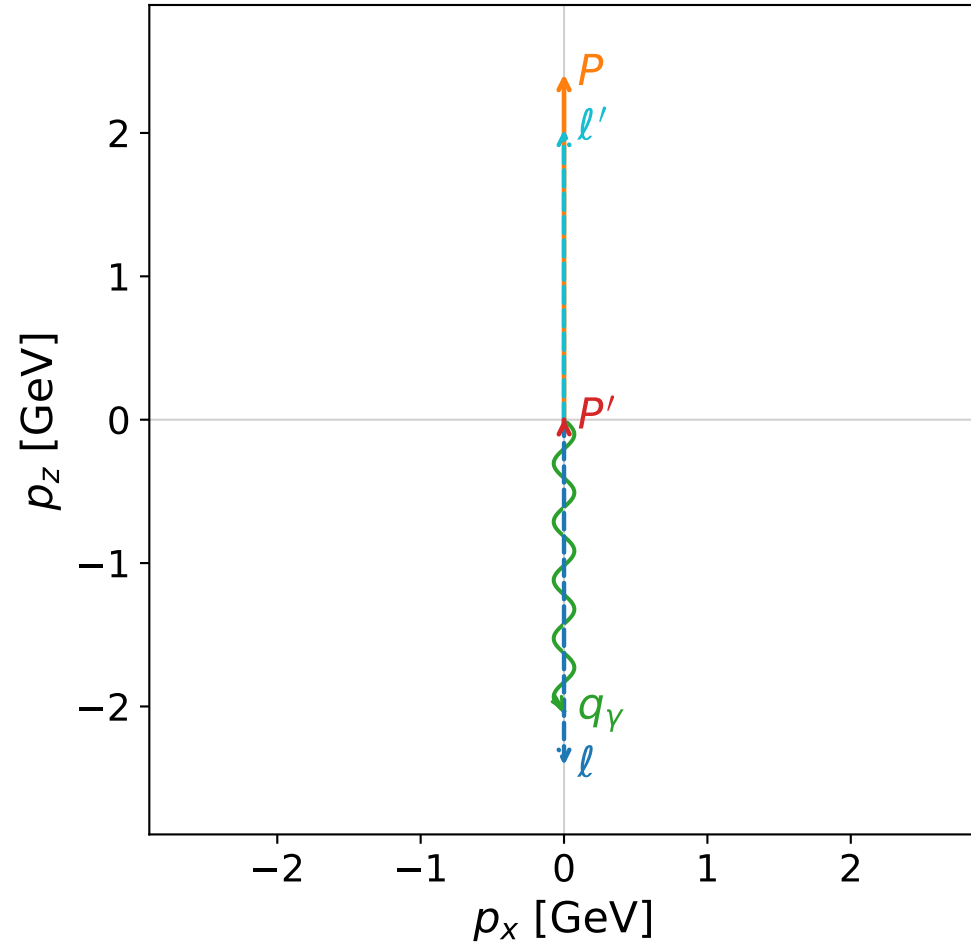
$s=24.887$, $\sqrt{s}=4.98869$, $|\vec{P}|=2.40616$, $|\vec{P}'|=0.0592441$
 $\theta_{P'}=7.76676\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.81389$, $\phi_\gamma=2.17126$
 $E_\gamma=2.05403$, $\theta_\ell=2.30665\text{e-}06$, $\phi_\ell=2.67229$
 $Q^2=19.1991$, $x_B=0.823871$, $t=-2.80969$, $W^2=4.98427$, $y=0.970691$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4062$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4062$
 P $E= 2.5825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4062$
 ℓ' $E= 1.9948$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9948$
 P' $E= 0.9399$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0592$
 q_γ $E= 2.0540$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0540$

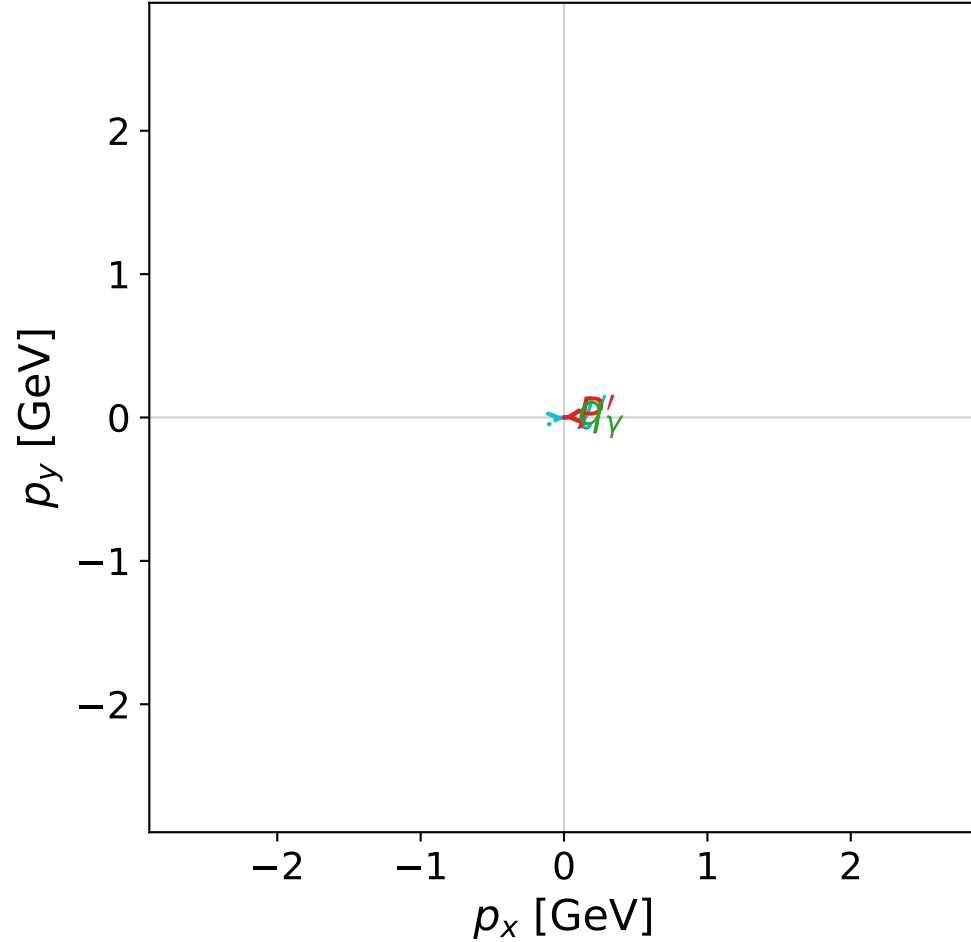


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



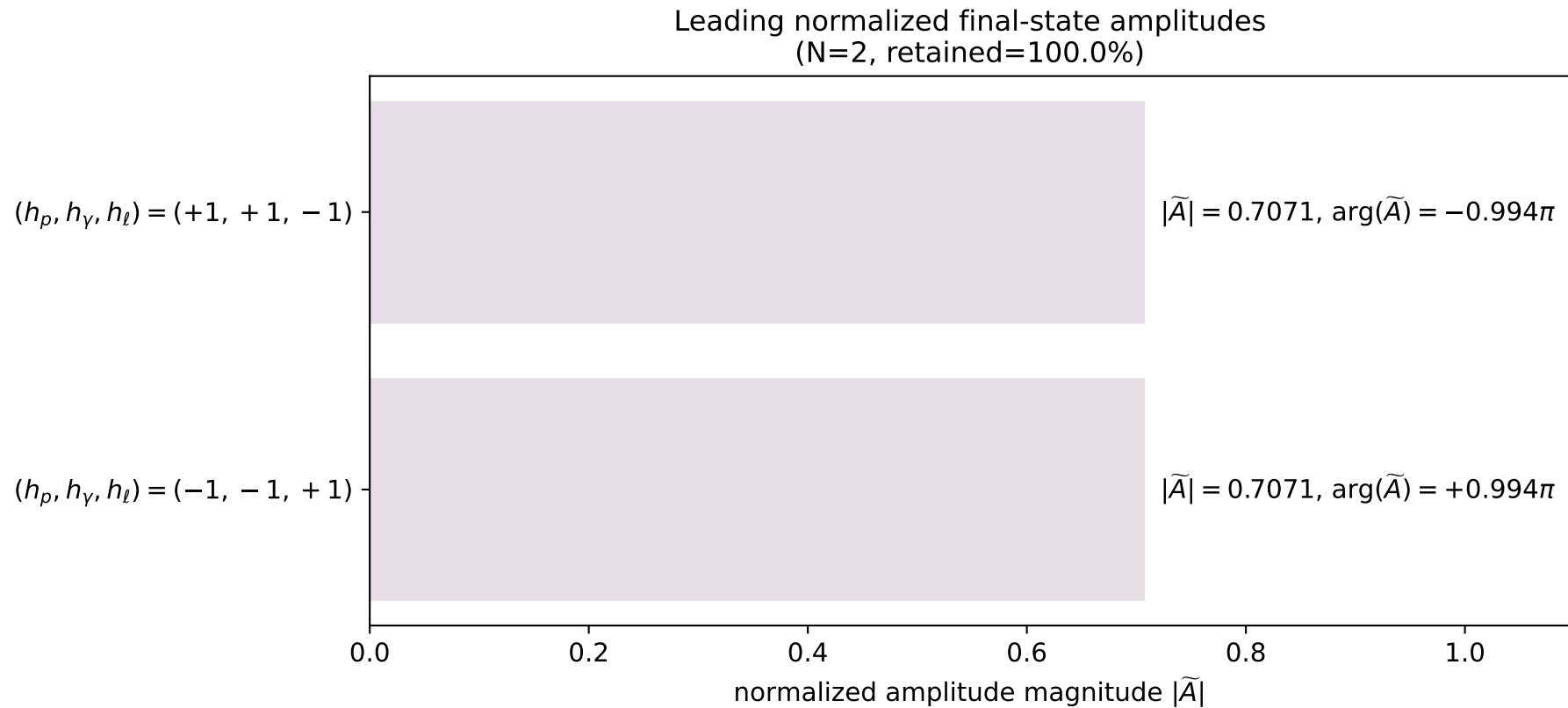
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_74
 $d_{\text{GHZ}}=2.93202\text{e-}08$
 region: polarization_cluster_04_local_minimum_74
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0074
 lepton mixing angle: $\alpha_e=2.1227555$ rad
 incoming lepton: $-0.524356|+\rangle +0.851499|-\rangle$
 proton mixing angle: $\alpha_p=2.12275$ rad
 incoming proton: $-0.524352|+\rangle +0.851502|-\rangle$

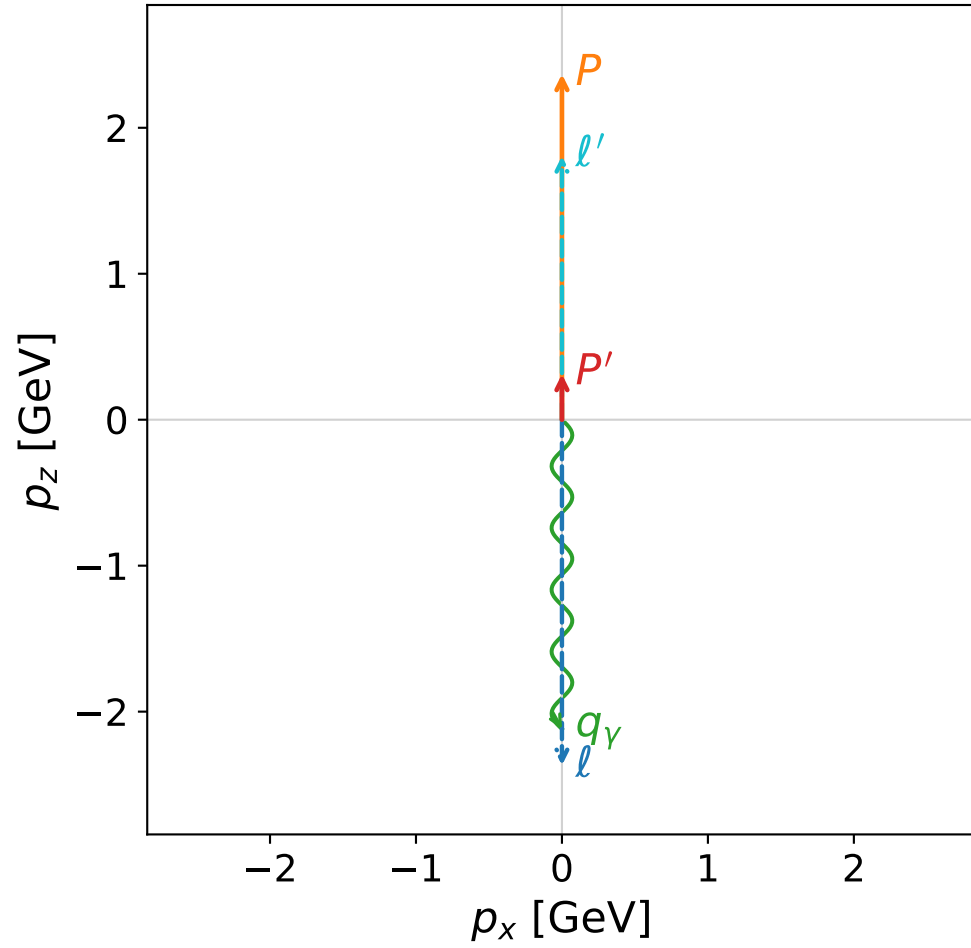
$s=24.9688$, $\sqrt{s}=4.99688$, $|\vec{P}|=2.4104$, $|\vec{P}'|=0.00957823$
 $\theta_{P'}=0.00965936$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.2353$, $\phi_\gamma=3.12295$
 $E_\gamma=2.0342$, $\theta_{l'}=4.56964\text{e-}05$, $\phi_{l'}=0.0937025$
 $Q^2=19.5206$, $x_B=0.835073$, $t=-3.04662$, $W^2=4.73517$, $y=0.970402$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 l' $E= 2.0246$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 2.0246$
 P' $E= 0.9380$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 0.0096$
 q_γ $E= 2.0342$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0342$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

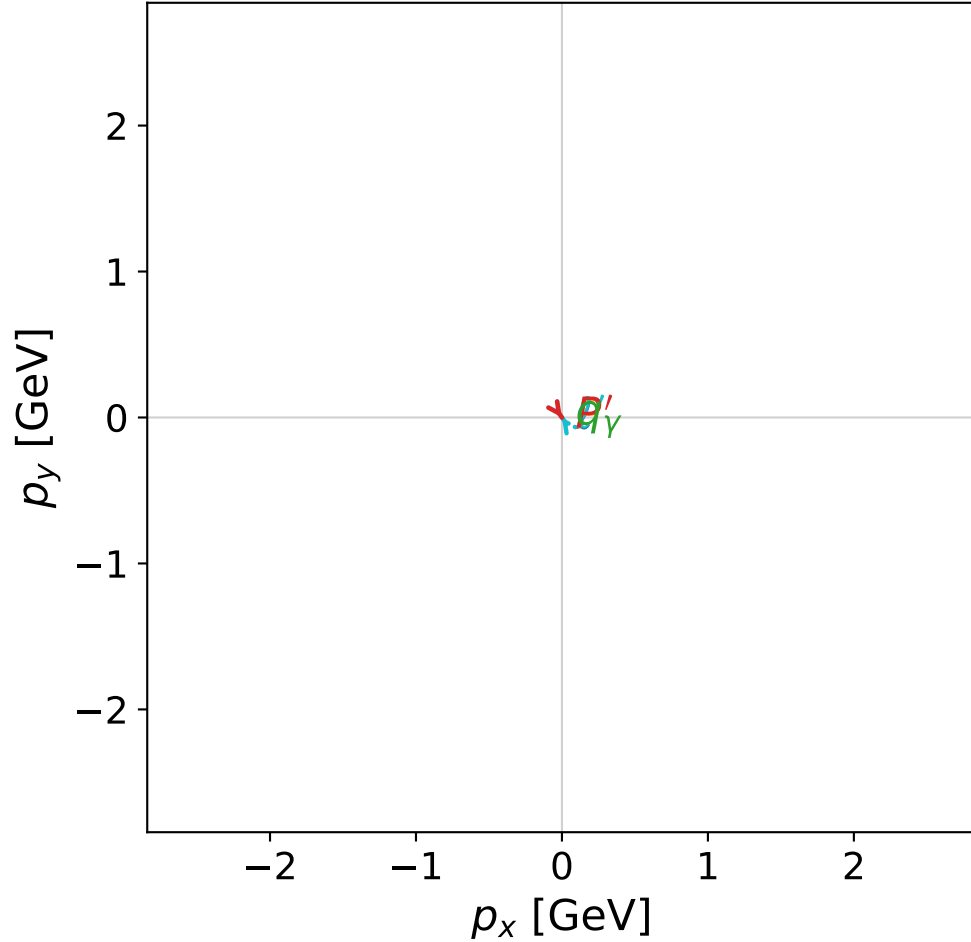


dghz_mixing_angles_polarization_04_local_minimum_80
 $d_{\text{GHZ}}=3.02153\text{e-}08$
 region: polarization_cluster_04_local_minimum_80
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0080
 lepton mixing angle: $\alpha_e=2.3404957$ rad
 incoming lepton: $-0.695919|+\rangle + 0.71812|-\rangle$
 proton mixing angle: $\alpha_p=2.3404627$ rad
 incoming proton: $-0.695896|+\rangle + 0.718143|-\rangle$

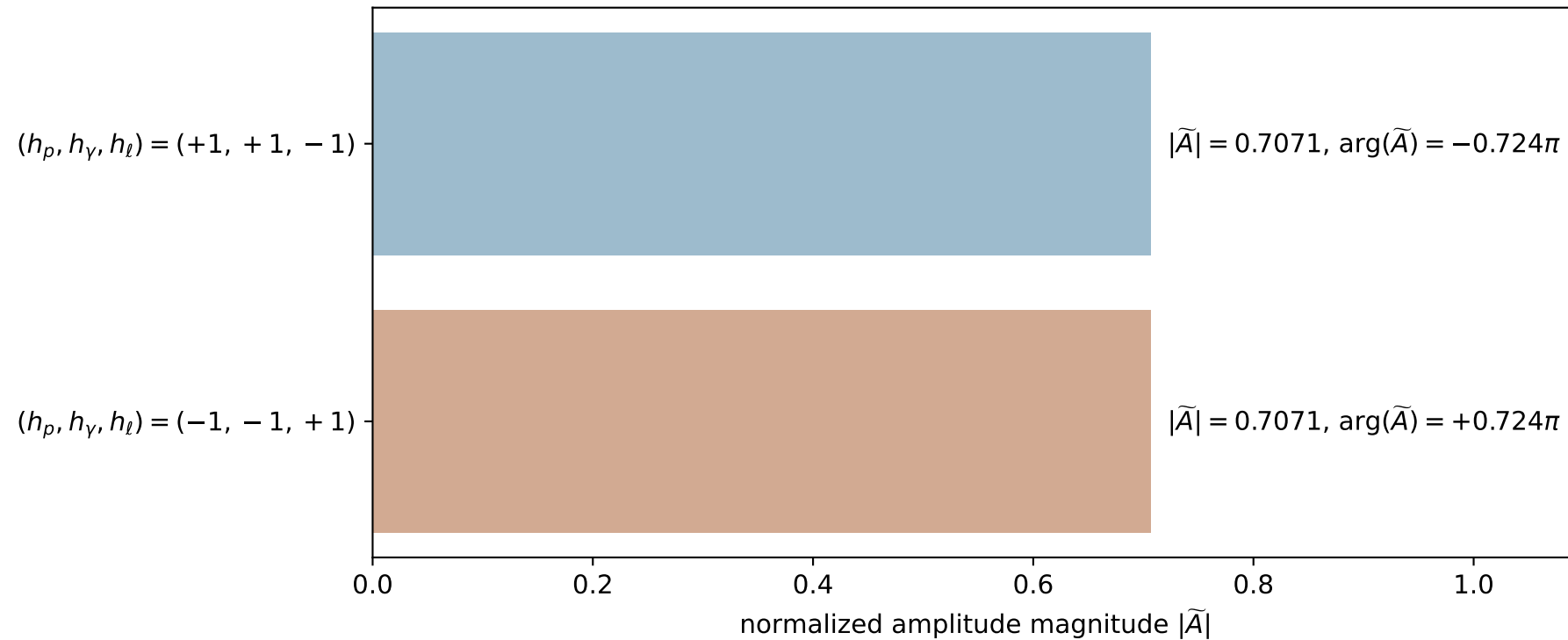
$s=24.1511$, $\sqrt{s}=4.91438$, $|\vec{P}|=2.36767$, $|\vec{P}'|=0.31225$
 $\theta_{P'}=9.71376\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.3268$, $\phi_\gamma=2.27603$
 $E_\gamma=2.11901$, $\theta_{l'}=1.67875\text{e-}06$, $\phi_{l'}=2.18521$
 $Q^2=17.1113$, $x_B=0.756322$, $t=-1.79708$, $W^2=6.39291$, $y=0.9722$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3677$
 P $E= 2.5467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3677$
 l' $E= 1.8068$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8068$
 P' $E= 0.9886$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.3123$
 q_γ $E= 2.1190$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1190$

x--y plane

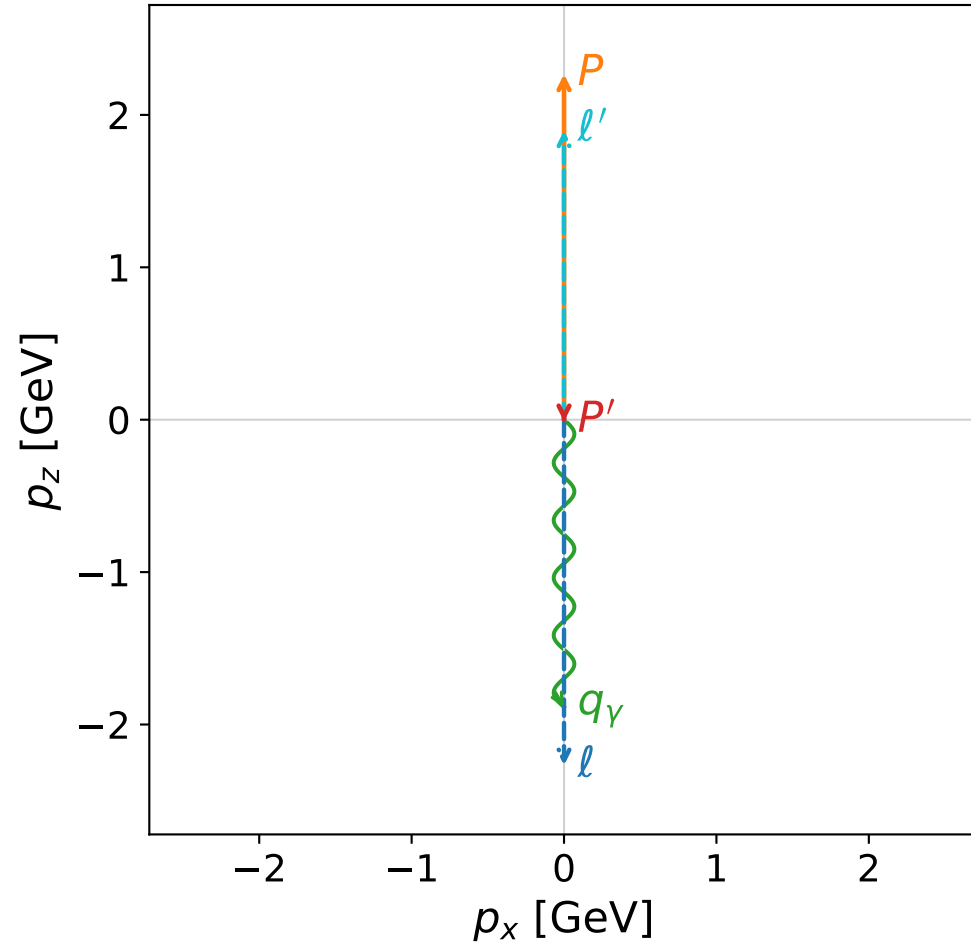


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

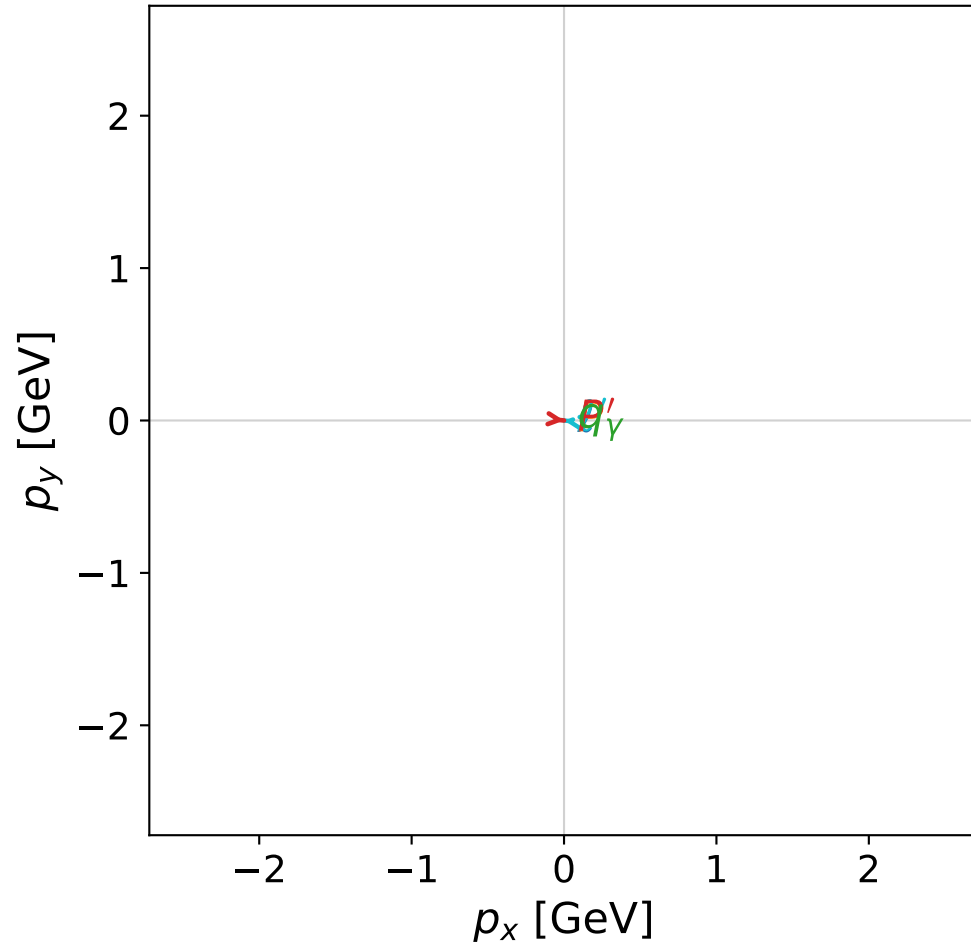


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



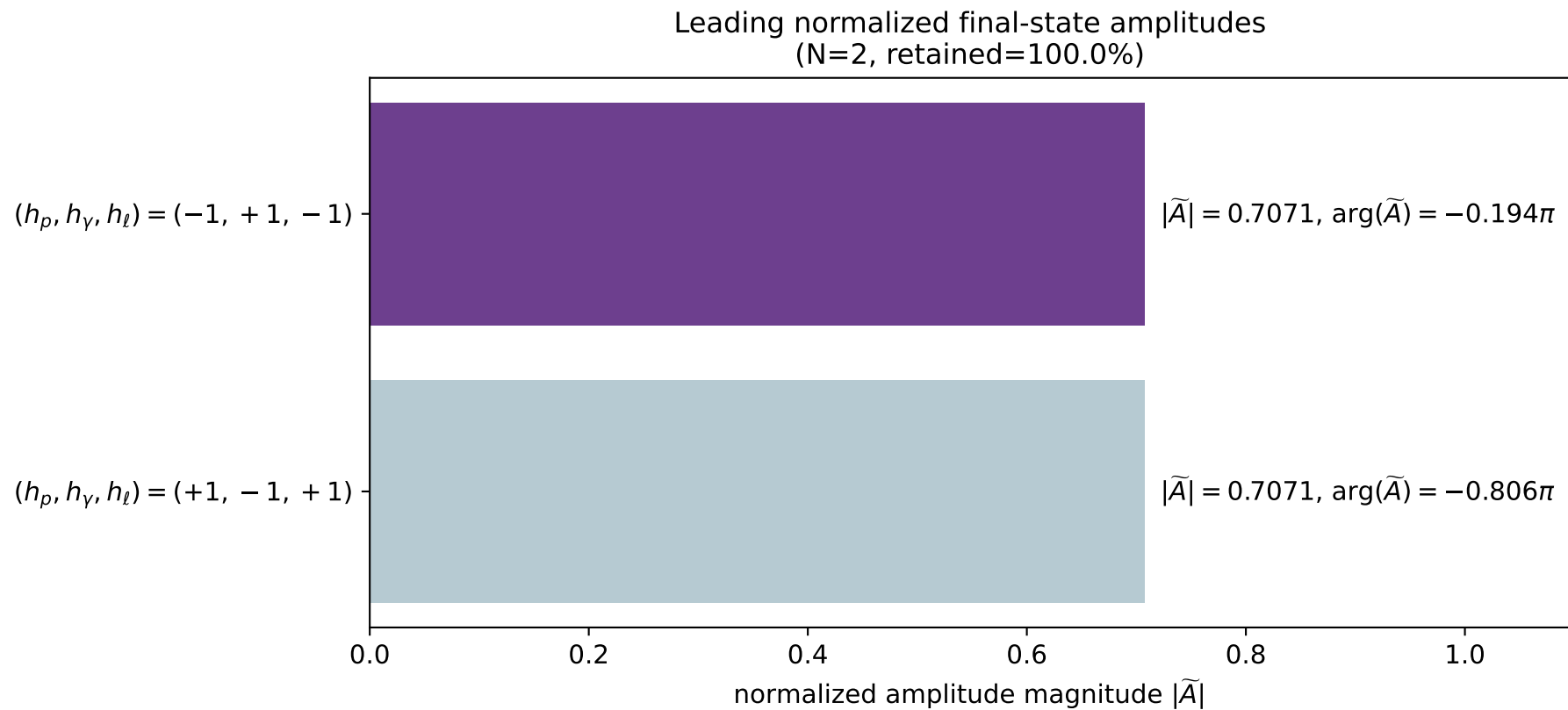
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_81
 $d_{\text{GHZ}}=3.03827\text{e-}08$
 region: polarization_cluster_04_local_minimum_81
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0081
 lepton mixing angle: $\alpha_e=2.3663814$ rad
 incoming lepton: $-0.714273|+\rangle +0.699867|-\rangle$
 proton mixing angle: $\alpha_p=2.3663611$ rad
 incoming proton: $-0.714259|+\rangle +0.699881|-\rangle$

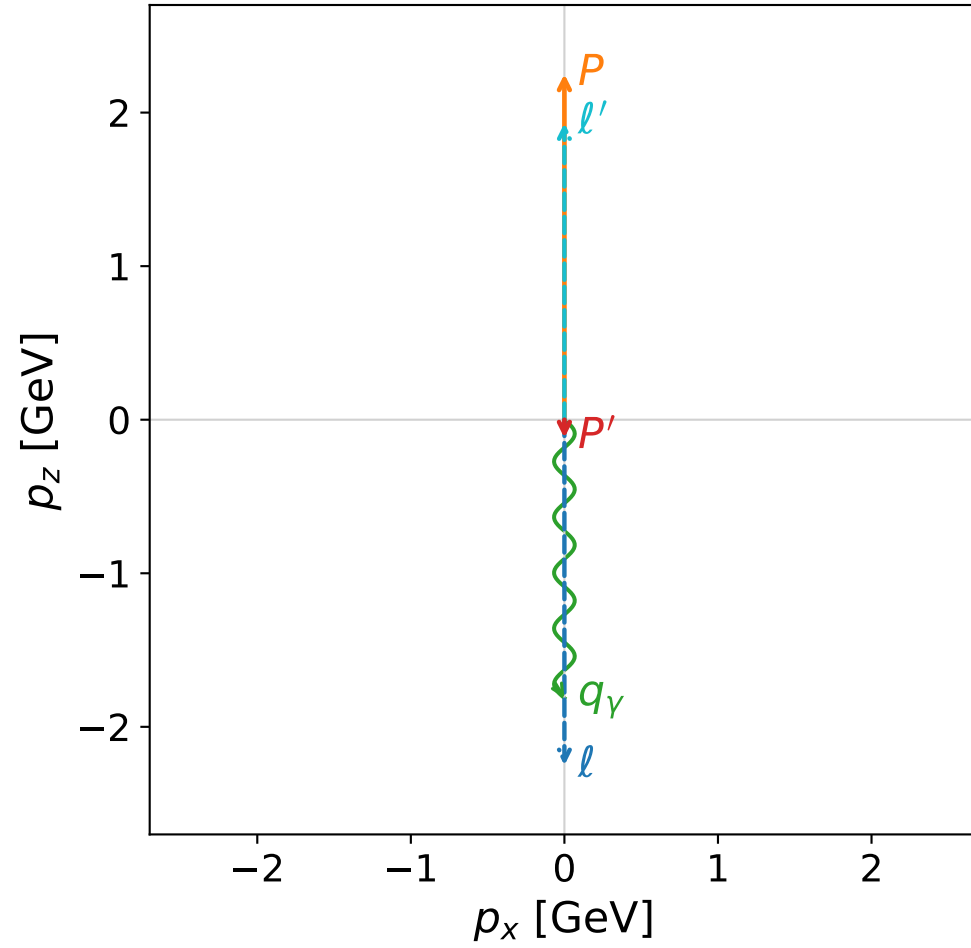
$s=22.294$, $\sqrt{s}=4.72165$, $|\vec{P}|=2.26765$, $|\vec{P}'|=0.0124921$
 $\theta_{P'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.17498$, $\phi_\gamma=3.64321$
 $E_\gamma=1.88554$, $\theta_{\ell'}=7.48922\text{e-}07$, $\phi_{\ell'}=3.03338$
 $Q^2=17.2163$, $x_B=0.831433$, $t=-2.90107$, $W^2=4.37032$, $y=0.966967$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2677$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2677$
 P $E= 2.4540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2677$
 ℓ' $E= 1.8980$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8980$
 P' $E= 0.9381$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0125$
 q_γ $E= 1.8855$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8855$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

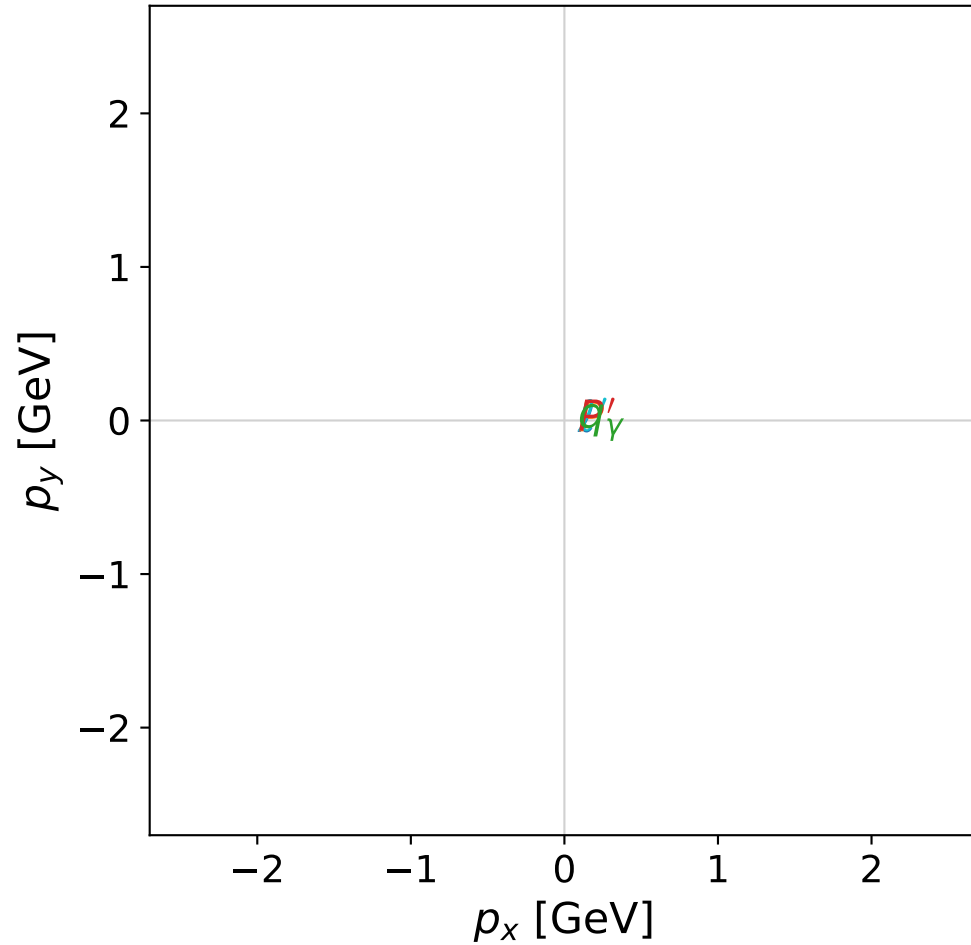


dghz_mixing_angles_polarization_04_local_minimum_82
 $d_{\text{GHZ}}=3.04333\text{e-}08$
 region: polarization_cluster_04_local_minimum_82
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0082
 lepton mixing angle: $\alpha_e=2.2763558$ rad
 incoming lepton: $-0.64846|+\rangle +0.761249|-\rangle$
 proton mixing angle: $\alpha_p=2.2763635$ rad
 incoming proton: $-0.648466|+\rangle +0.761244|-\rangle$

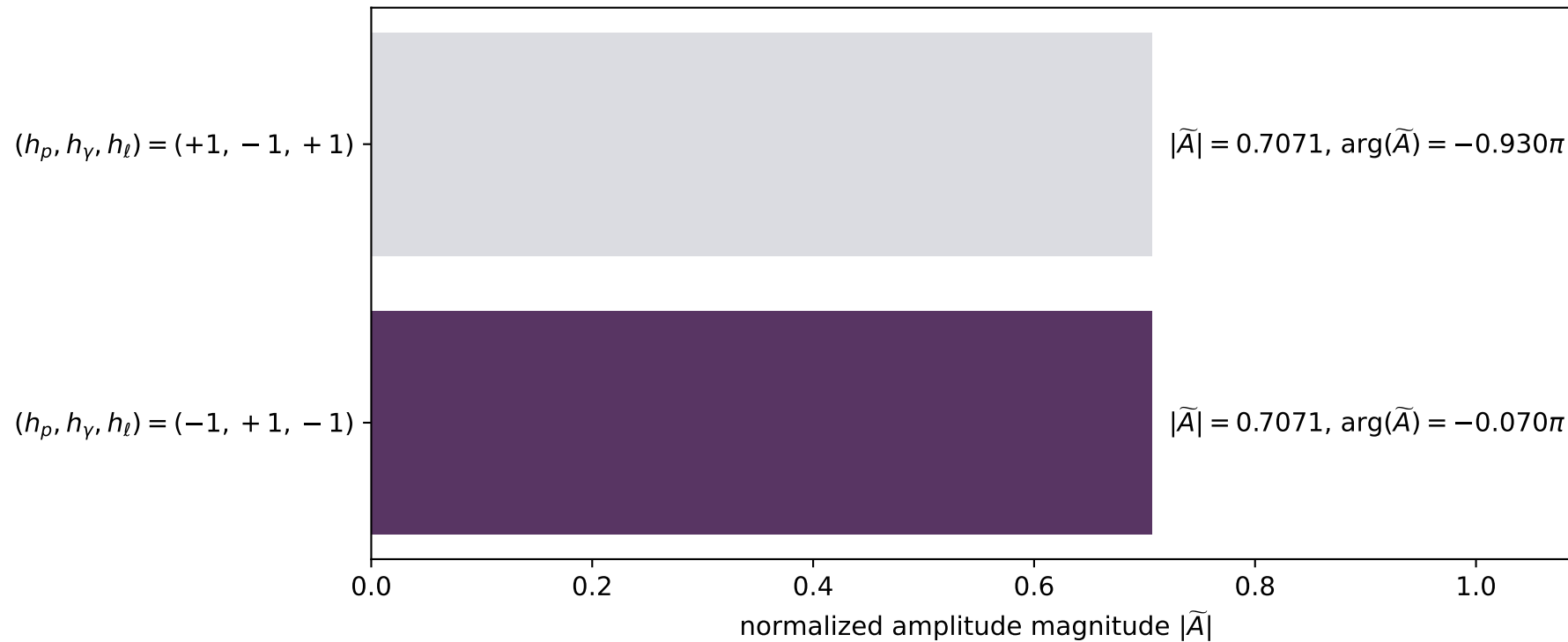
$s=21.9839$, $\sqrt{s}=4.68869$, $|\vec{P}|=2.25052$, $|\vec{P}'|=0.118469$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.11906$, $\phi_\gamma=3.36218$
 $E_\gamma=1.81239$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.205792$
 $Q^2=17.3817$, $x_B=0.852909$, $t=-3.3839$, $W^2=3.87747$, $y=0.965662$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2505$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2505$
 P $E= 2.4382$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2505$
 ℓ' $E= 1.9309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9309$
 P' $E= 0.9455$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1185$
 q_γ $E= 1.8124$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8124$

x--y plane

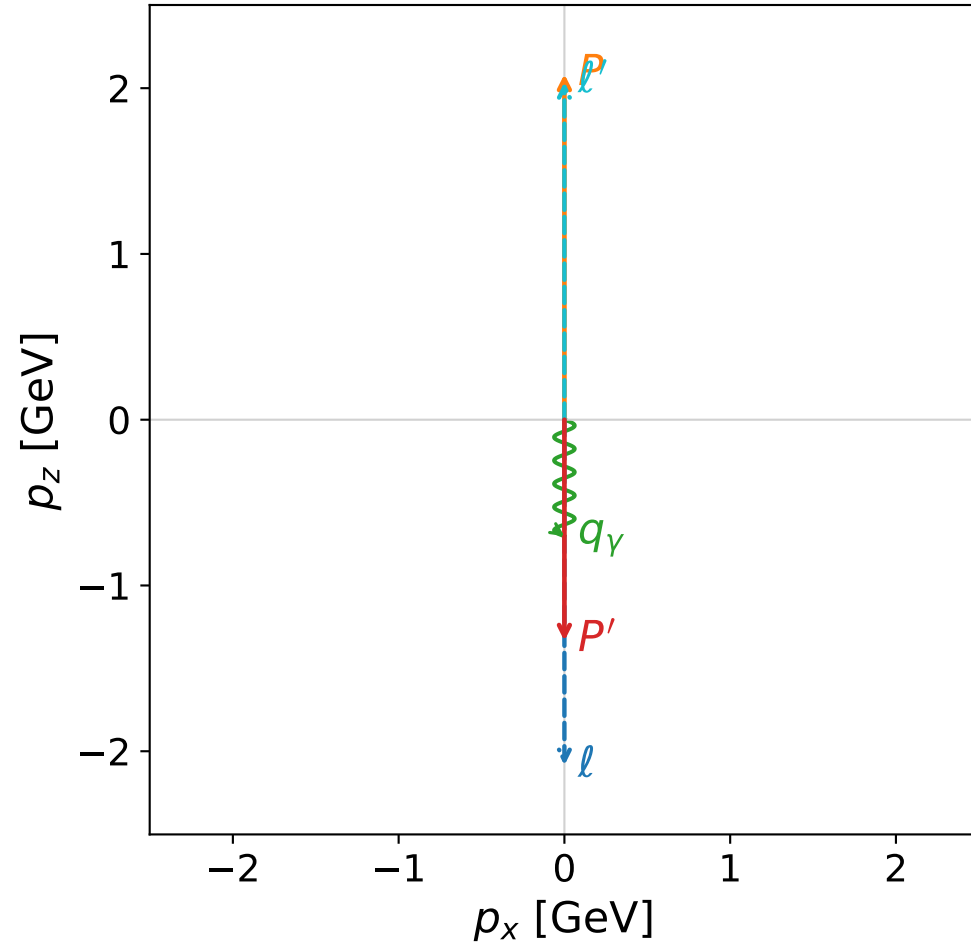


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

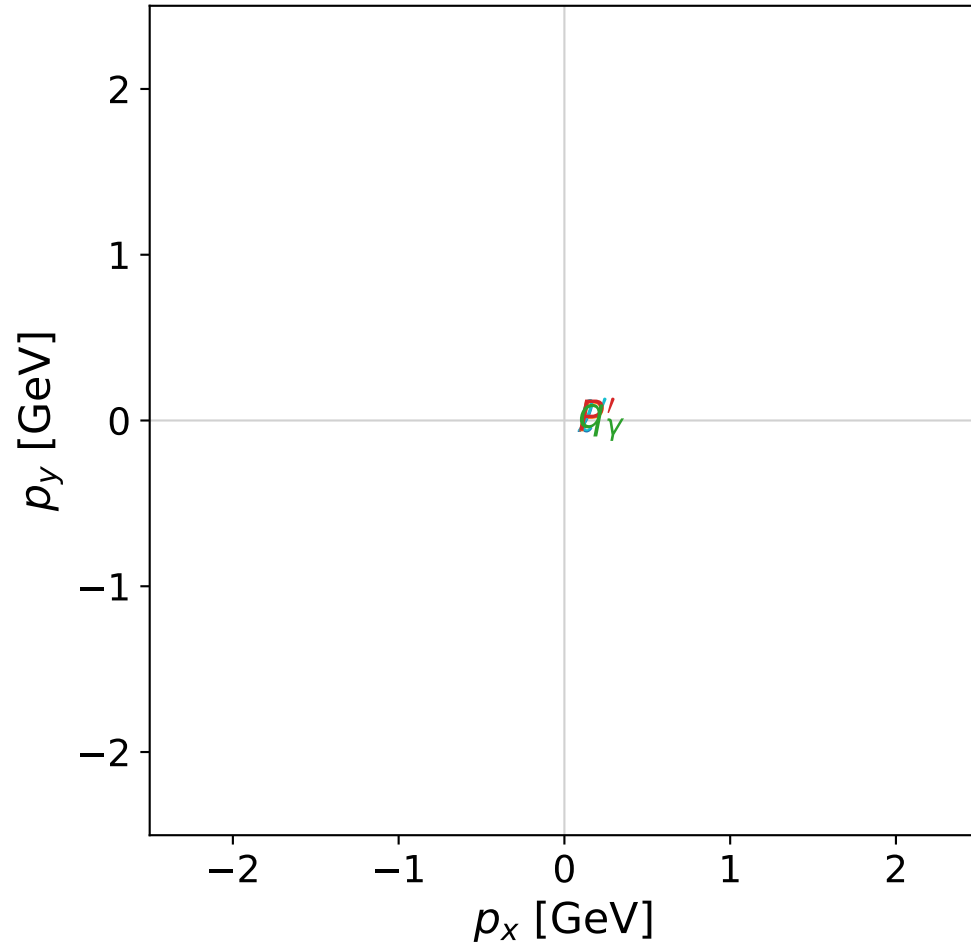


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



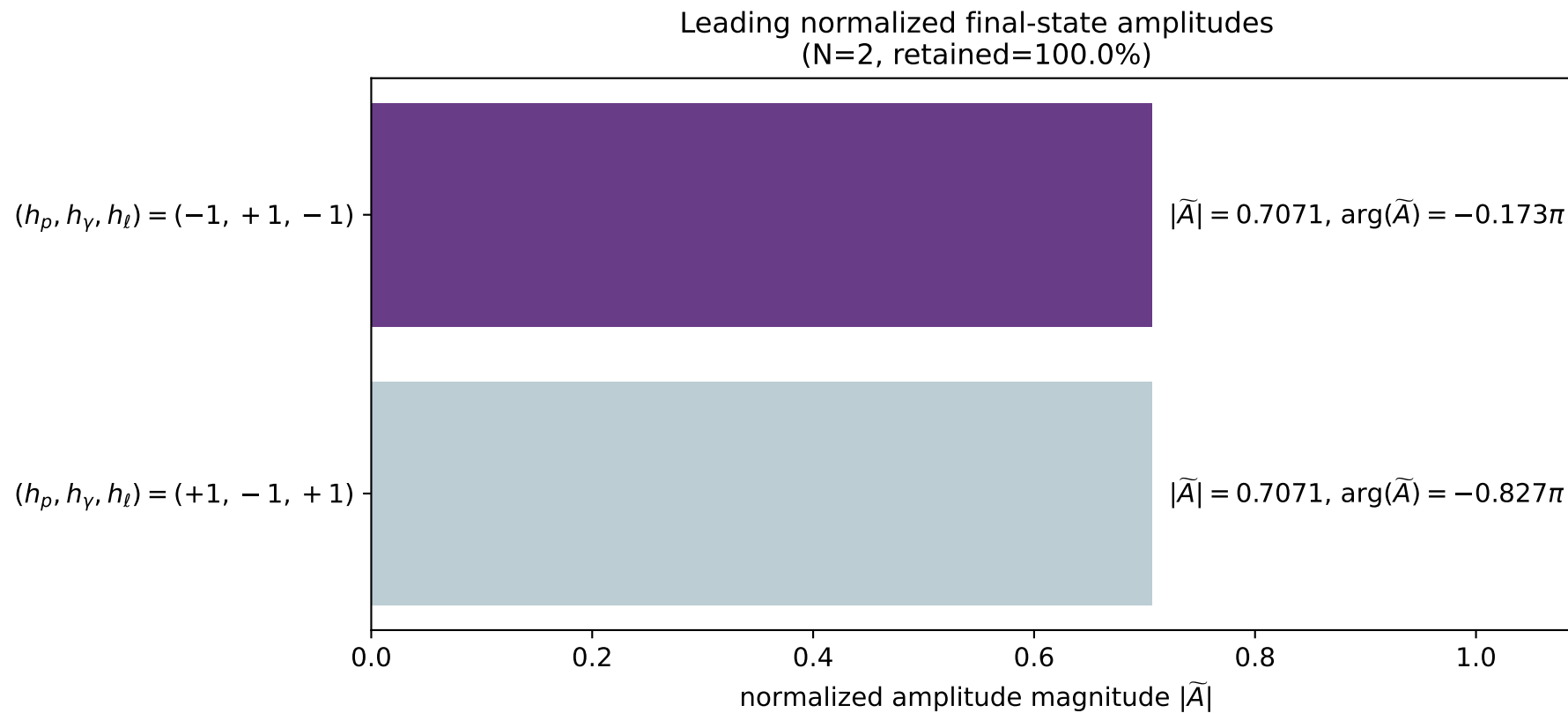
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_84
 $d_{\text{GHZ}}=3.10329\text{e-}08$
 region: polarization_cluster_04_local_minimum_84
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0084
 lepton mixing angle: $\alpha_e=2.2903682$ rad
 incoming lepton: $-0.659063|+\rangle +0.752088|-\rangle$
 proton mixing angle: $\alpha_p=2.2903511$ rad
 incoming proton: $-0.65905|+\rangle +0.752099|-\rangle$

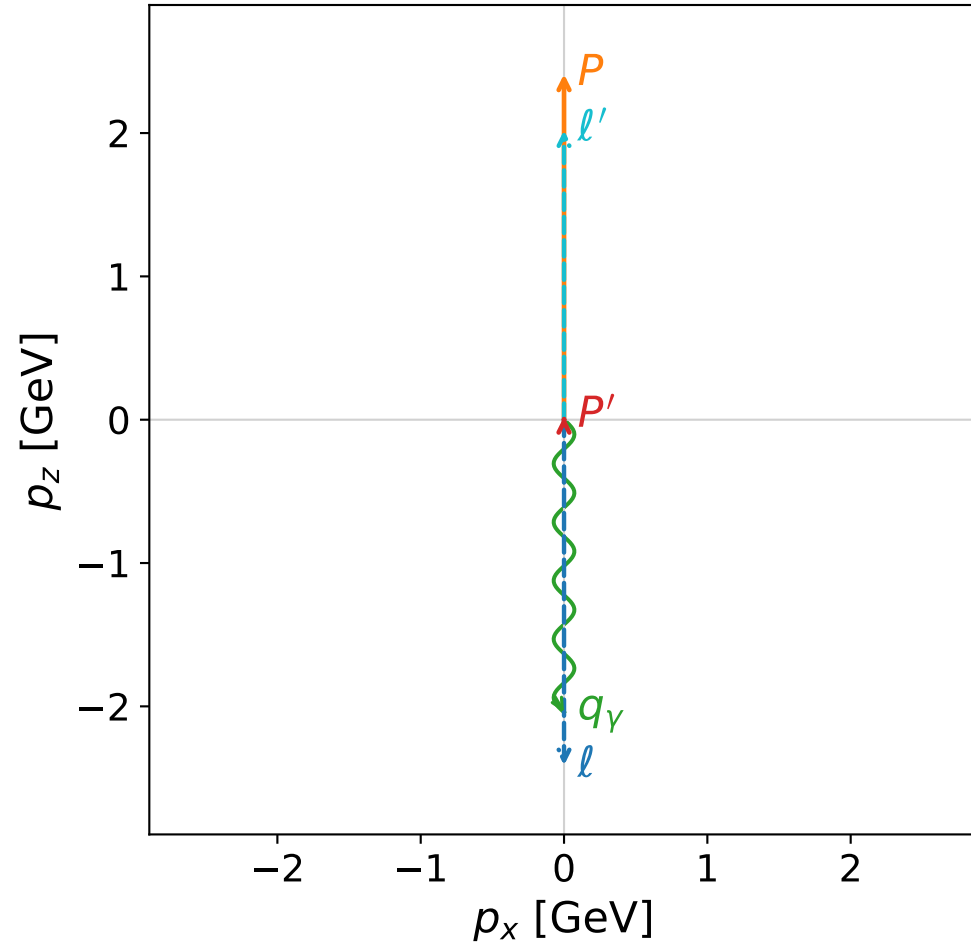
$s=19.1051$, $\sqrt{s}=4.37093$, $|\vec{P}|=2.08482$, $|\vec{P}'|=1.33443$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0100816$, $\phi_\gamma=3.68413$
 $E_\gamma=0.702689$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.69661$
 $Q^2=16.9881$, $x_B=0.976044$, $t=-11.2623$, $W^2=1.2968$, $y=0.955001$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0848$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0848$
 P $E= 2.2861$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0848$
 ℓ' $E= 2.0371$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0371$
 P' $E= 1.6311$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3344$
 q_γ $E= 0.7027$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.7027$

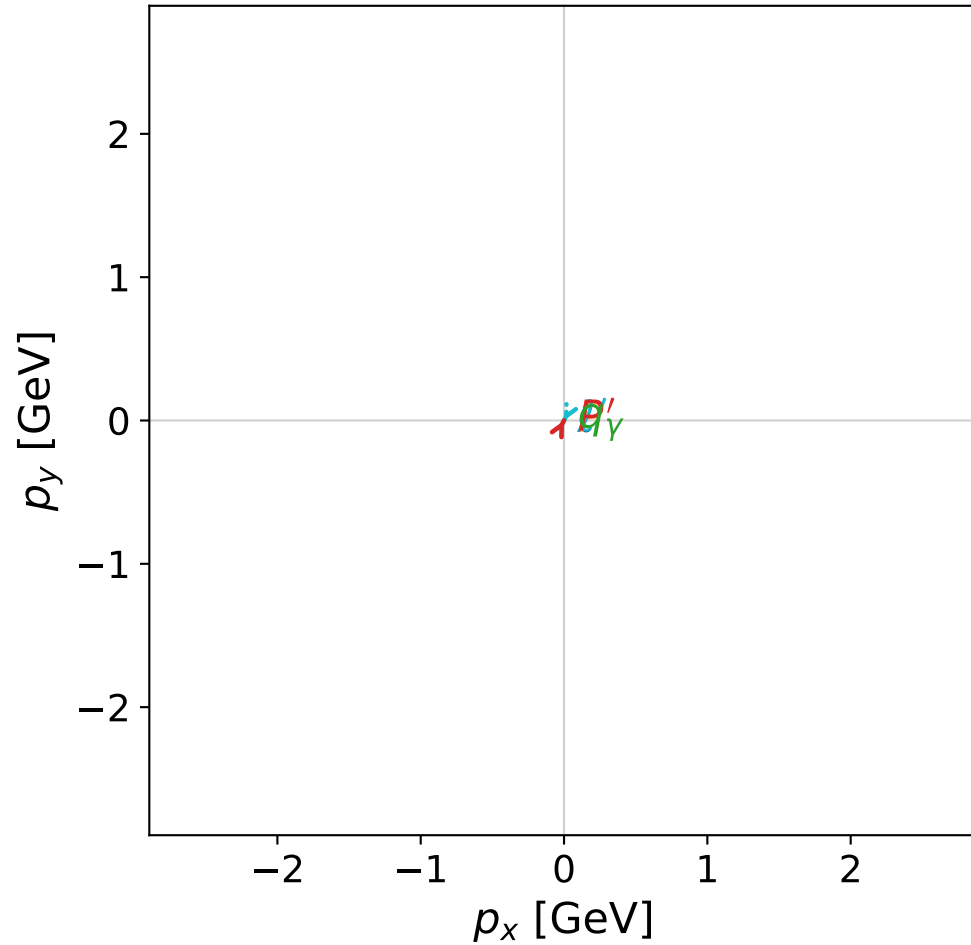


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



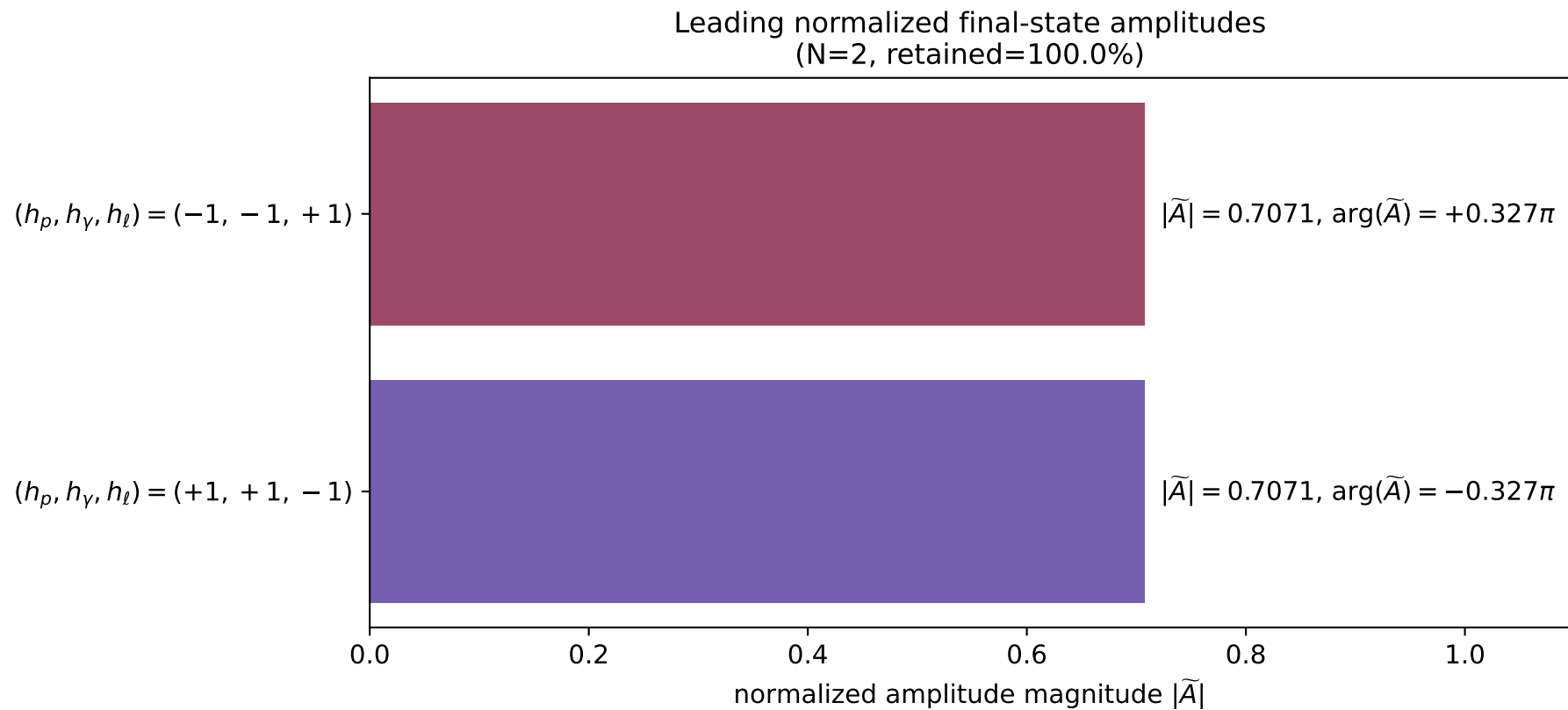
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_85
 $d_{\text{GHZ}}=3.10962\text{e-}08$
 region: polarization_cluster_04_local_minimum_85
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0085
 lepton mixing angle: $\alpha_e=2.1008763$ rad
 incoming lepton: $-0.505602|+\rangle +0.862767|-\rangle$
 proton mixing angle: $\alpha_p=2.1008833$ rad
 incoming proton: $-0.505608|+\rangle +0.862763|-\rangle$

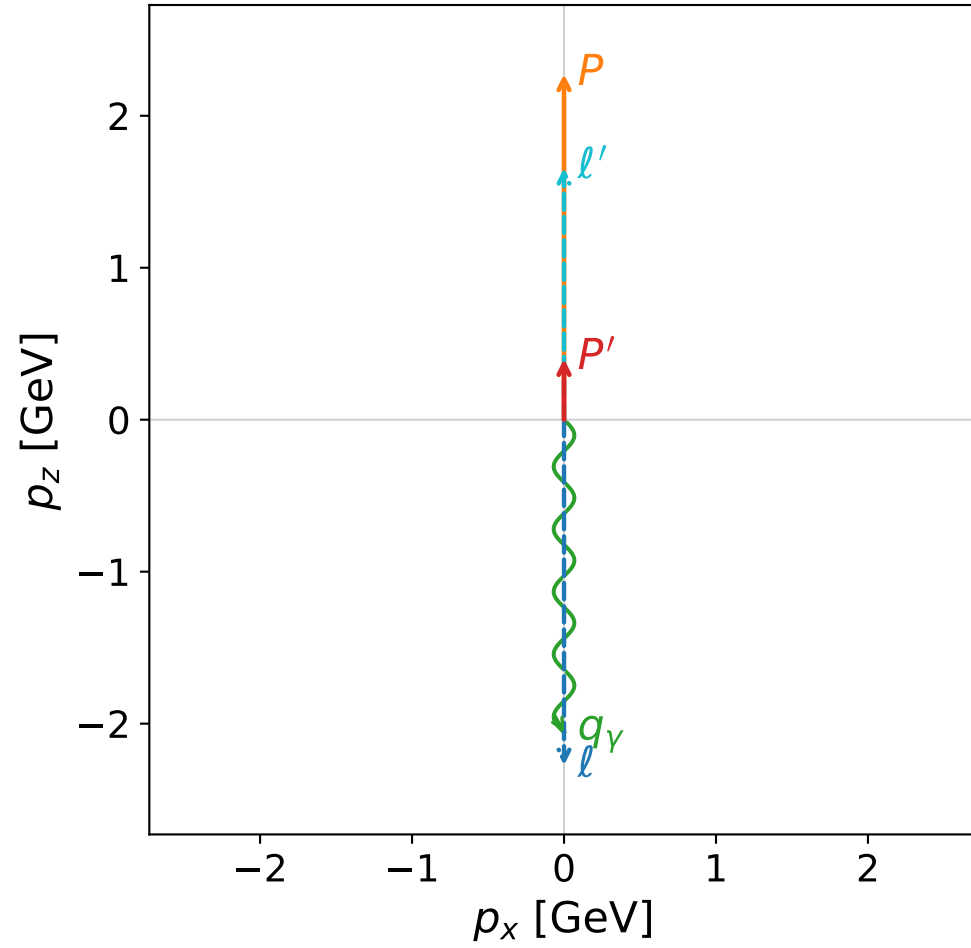
$s=24.9853$, $\sqrt{s}=4.99853$, $|\vec{P}|=2.41125$, $|\vec{P}'|=0.0216116$
 $\theta_{P'}=0.00498868$, $\theta_\gamma=3.14159$, $\phi_{P'}=1.09327$, $\phi_\gamma=1.0281$
 $E_\gamma=2.04095$, $\theta_{\ell'}=5.33904\text{e-}05$, $\phi_{\ell'}=4.23487$
 $Q^2=19.4765$, $x_B=0.832523$, $t=-2.99111$, $W^2=4.79789$, $y=0.970509$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$
 P $E= 2.5873$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$
 ℓ' $E= 2.0193$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0193$
 P' $E= 0.9382$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0216$
 q_γ $E= 2.0409$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0409$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

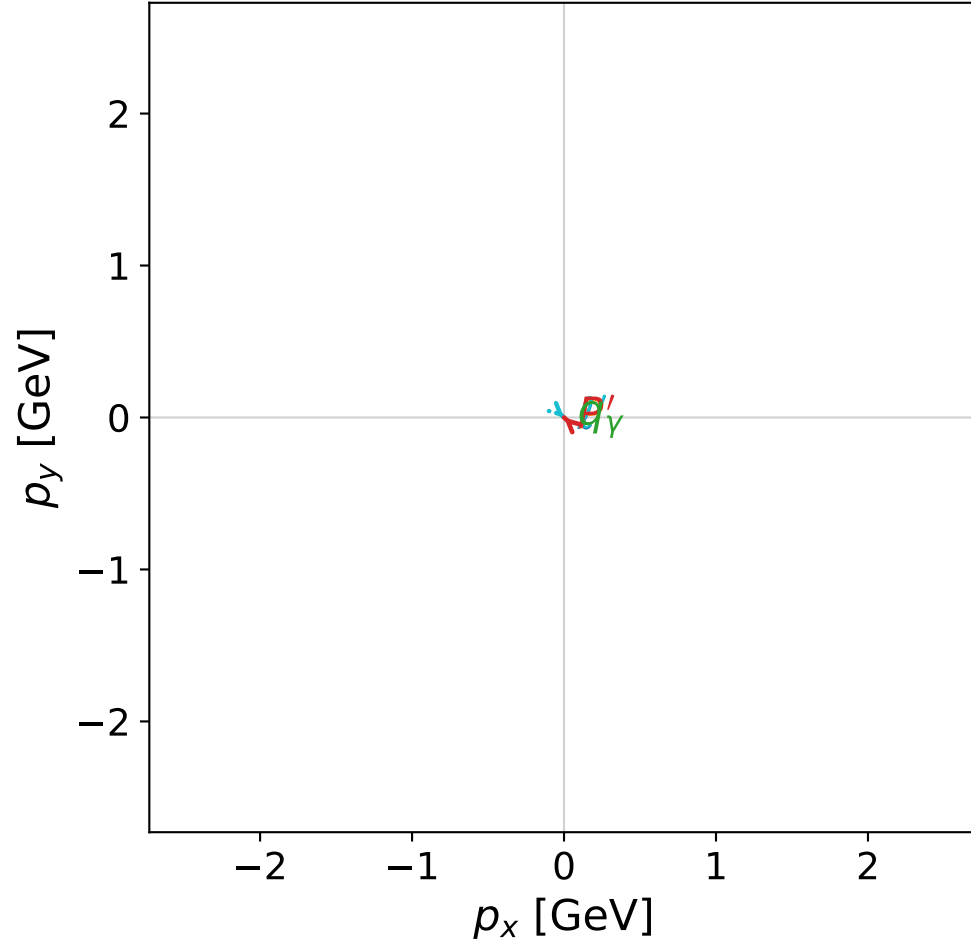


dghz_mixing_angles_polarization_04_local_minimum_88
 $d_{\{\rm GHZ\}}=3.20878\text{e-}08$
 region: polarization_cluster_04_local_minimum_88
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0088
 lepton mixing angle: $\alpha_e=2.2717872$ rad
 incoming lepton: $-0.644975|+\rangle +0.764203|-\rangle$
 proton mixing angle: $\alpha_p=2.2717877$ rad
 incoming proton: $-0.644976|+\rangle +0.764203|-\rangle$

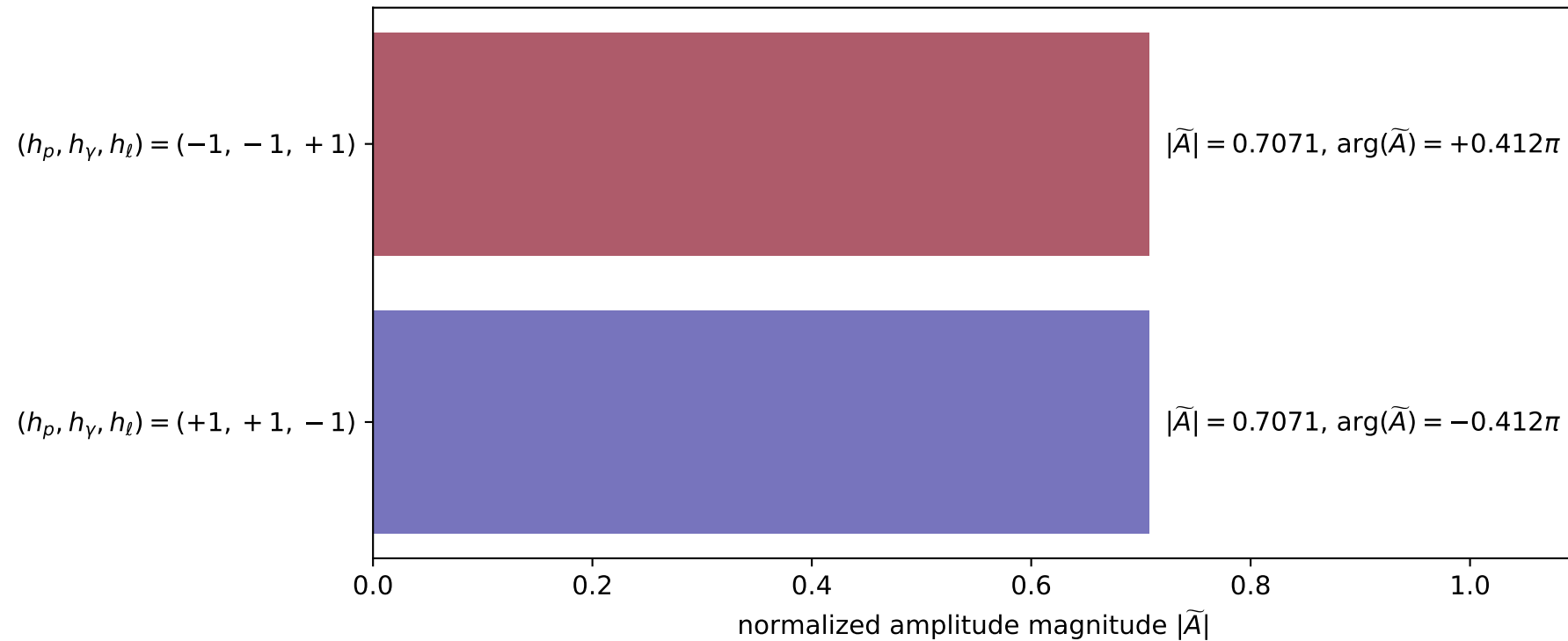
$s=22.4118$, $\sqrt{s}=4.73411$, $|\vec{P}|=2.27413$, $|\vec{P}'|=0.400283$
 $\theta_{p'}=1.00102\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{p'}=2.40749$, $\phi_\gamma=1.29355$
 $E_\gamma=2.05728$, $\theta_{l'}=2.41656\text{e-}07$, $\phi_{l'}=5.54909$
 $Q^2=15.0729$, $x_B=0.720637$, $t=-1.43729$, $W^2=6.72301$, $y=0.971395$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2741$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2741$
 P $E= 2.4600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2741$
 l' $E= 1.6570$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.6570$
 P' $E= 1.0198$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.4003$
 q_γ $E= 2.0573$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0573$

x--y plane

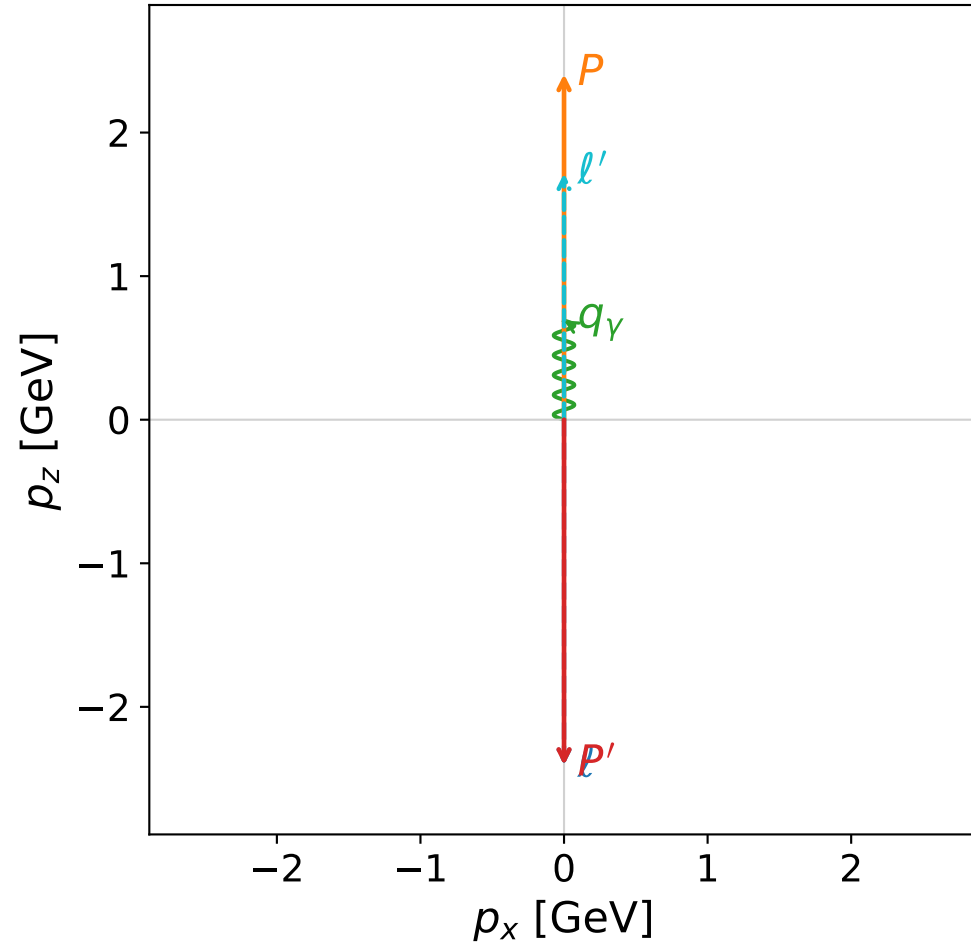


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

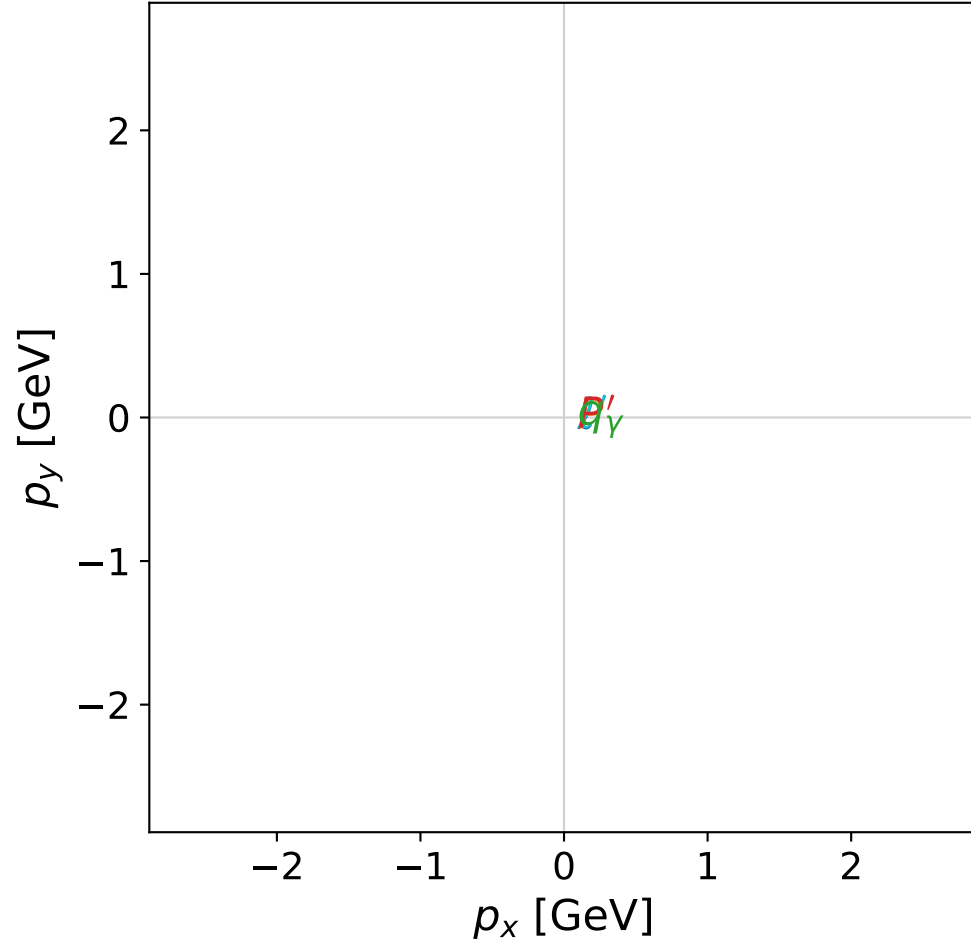


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



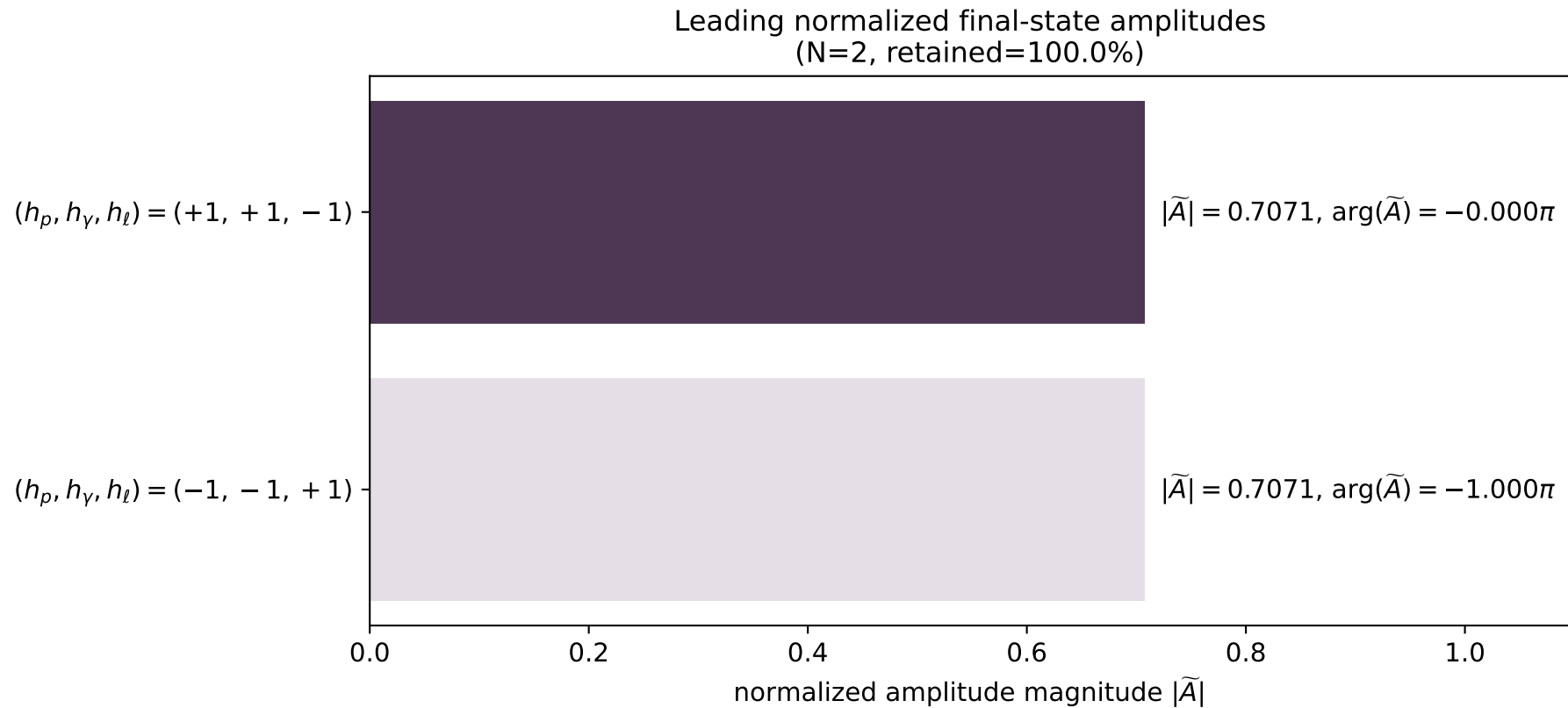
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_91
 $d_{\{\text{rm GHZ}\}}=3.29073\text{e-}08$
 region: polarization_cluster_04_local_minimum_91
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0091
 lepton mixing angle: $\alpha_e=2.4970427$ rad
 incoming lepton: $-0.79937|+\rangle +0.600839|-\rangle$
 proton mixing angle: $\alpha_p=2.2153473$ rad
 incoming proton: $-0.60084|+\rangle +0.79937|-\rangle$

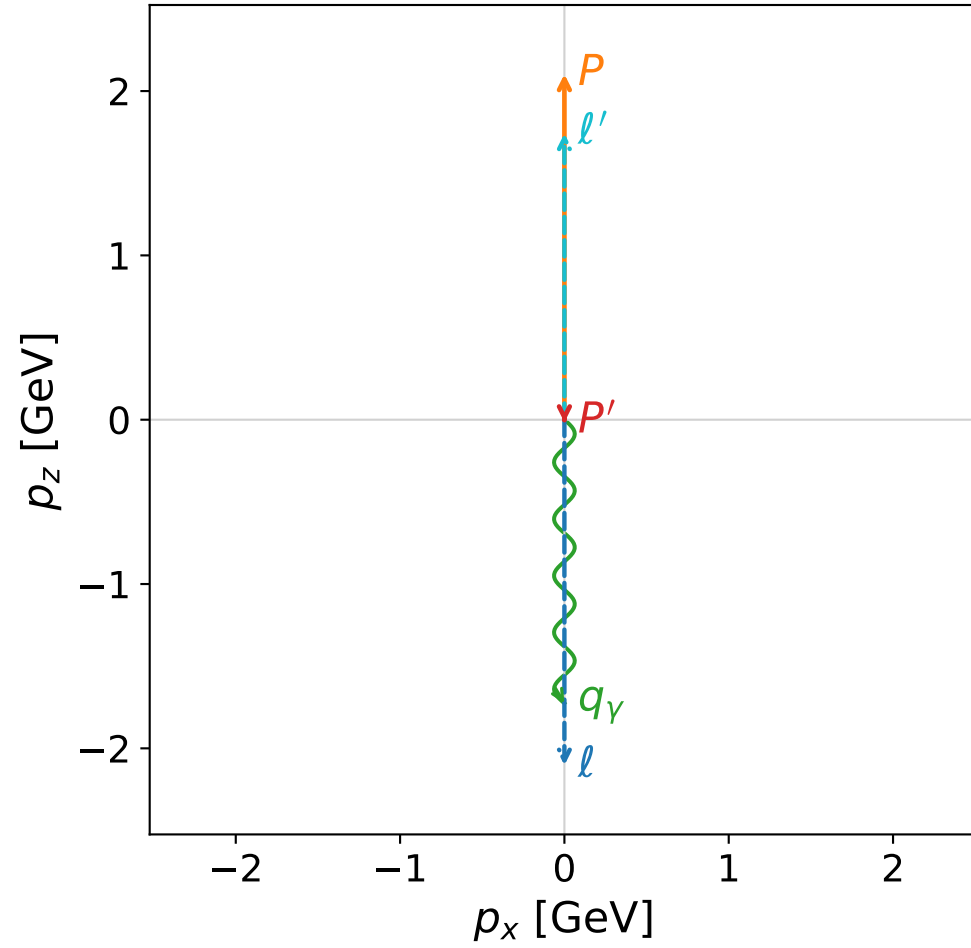
$s=24.9047$, $\sqrt{s}=4.99046$, $|\vec{P}|=2.40708$, $|\vec{P}'|=2.40708$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=4.19237$, $\phi_Y=1.989$
 $E_Y=0.691167$, $\theta_{l'}=0$, $\phi_{l'}=1.05077$
 $Q^2=16.5213$, $x_B=0.705442$, $t=-23.1761$, $W^2=7.77833$, $y=0.974816$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4071$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4071$
 P $E= 2.5834$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4071$
 ℓ' $E= 1.7159$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7159$
 P' $E= 2.5834$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4071$
 q_Y $E= 0.6912$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.6912$

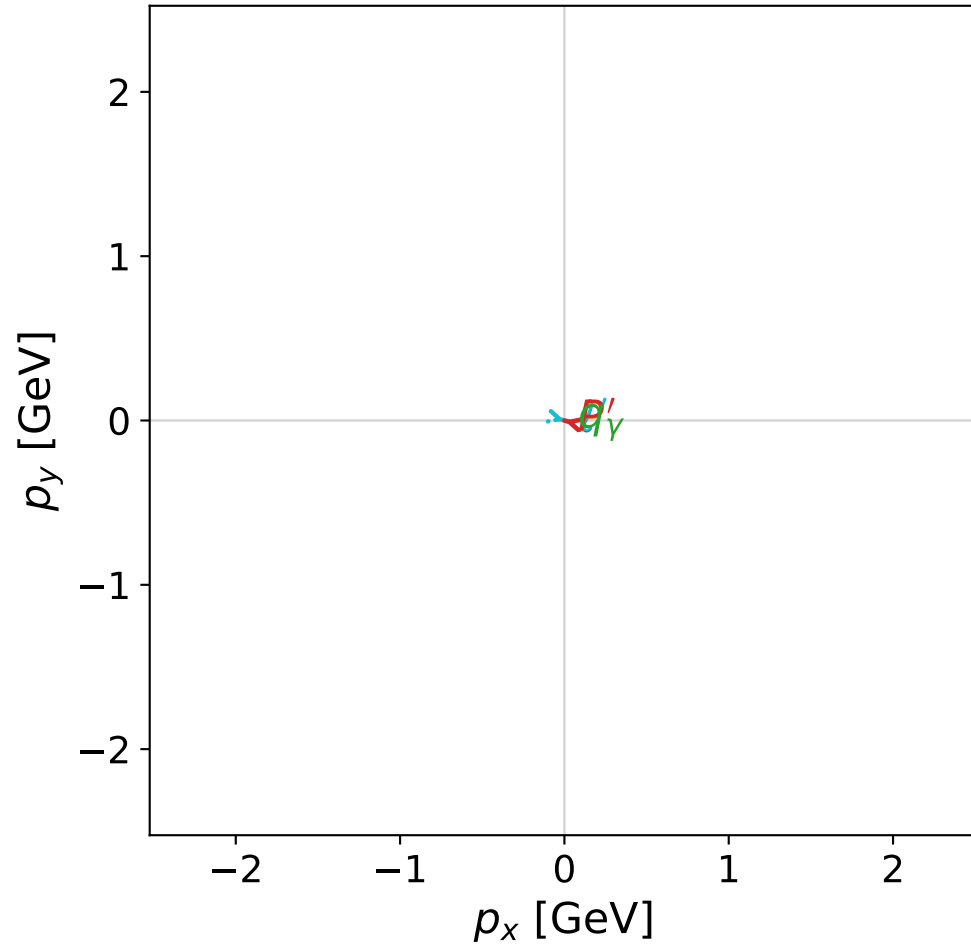


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



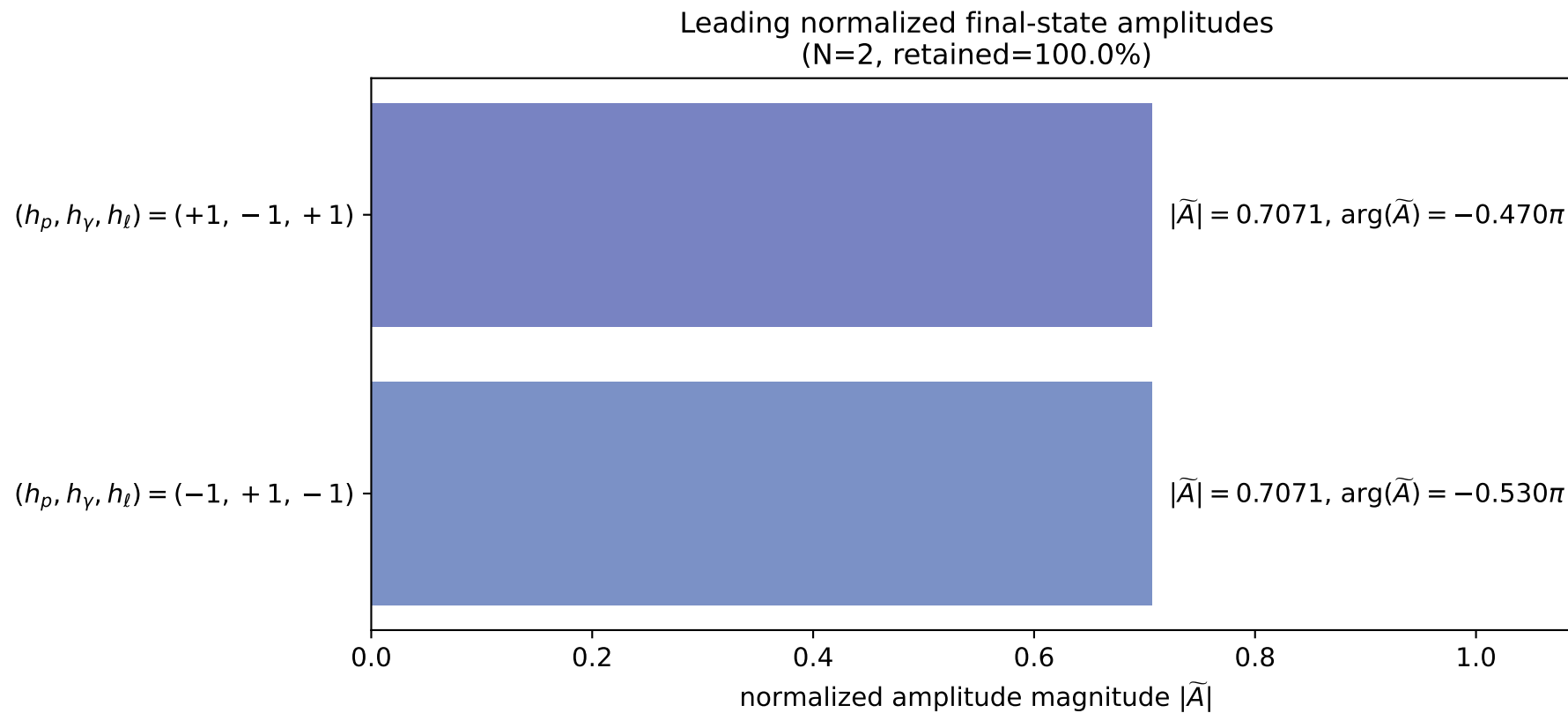
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_98
 $d_{\text{GHZ}}=3.56394\text{e-}08$
 region: polarization_cluster_04_local_minimum_98
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0098
 lepton mixing angle: $\alpha_e=2.3614211$ rad
 incoming lepton: $-0.710793|+\rangle +0.703401|-\rangle$
 proton mixing angle: $\alpha_p=2.3614305$ rad
 incoming proton: $-0.7108|+\rangle +0.703395|-\rangle$

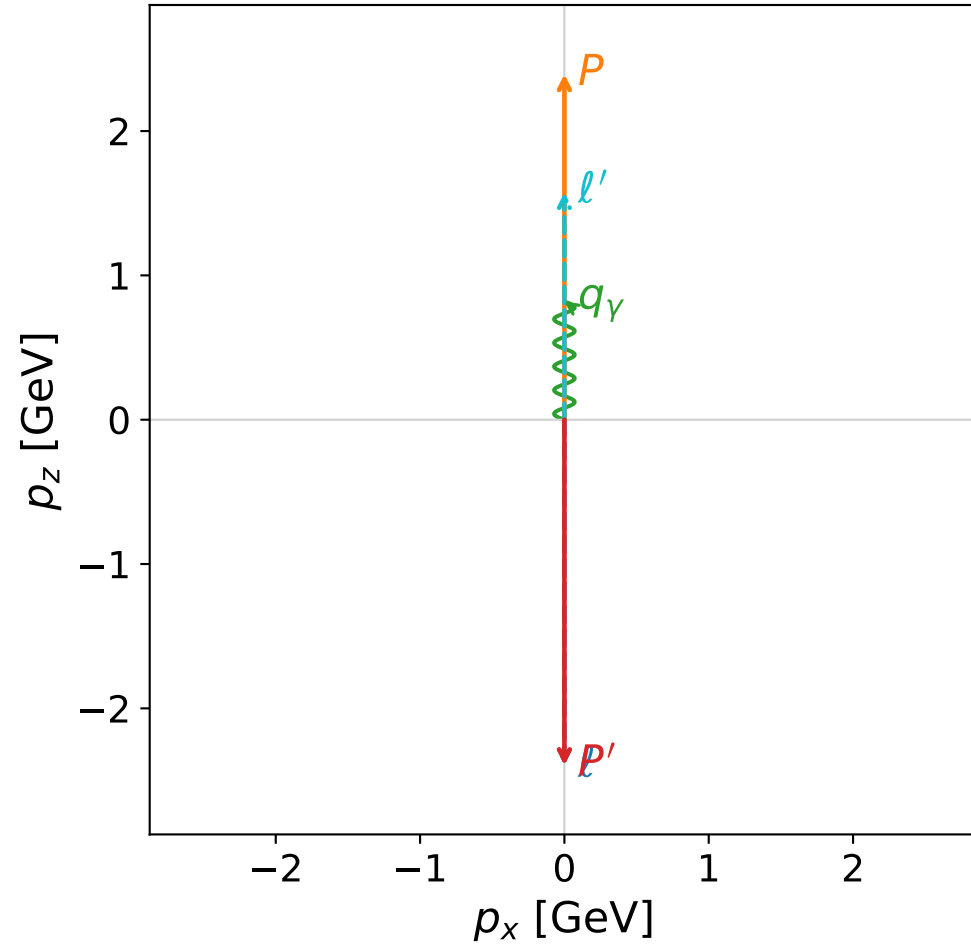
$s=19.4131$, $\sqrt{s}=4.40603$, $|\vec{P}|=2.10317$, $|\vec{P}'|=0.0160914$
 $\theta_{P'}=3.14148$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.86856$, $\phi_\gamma=1.39128$
 $E_\gamma=1.7259$, $\theta_{\ell'}=1.00525\text{e-}06$, $\phi_{\ell'}=6.01016$
 $Q^2=14.6548$, $x_B=0.821572$, $t=-2.6288$, $W^2=4.06257$, $y=0.962461$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1032$
 P $E= 2.3029$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1032$
 ℓ' $E= 1.7420$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.7420$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0161$
 q_γ $E= 1.7259$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7259$

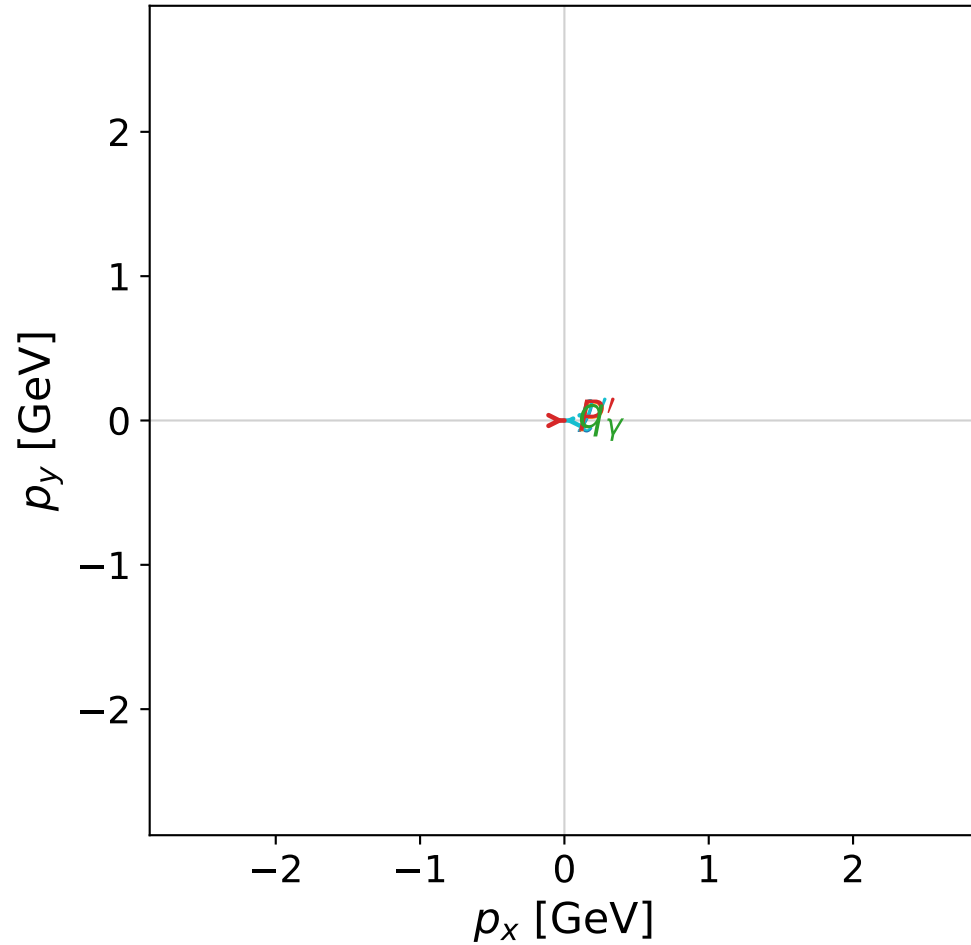


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

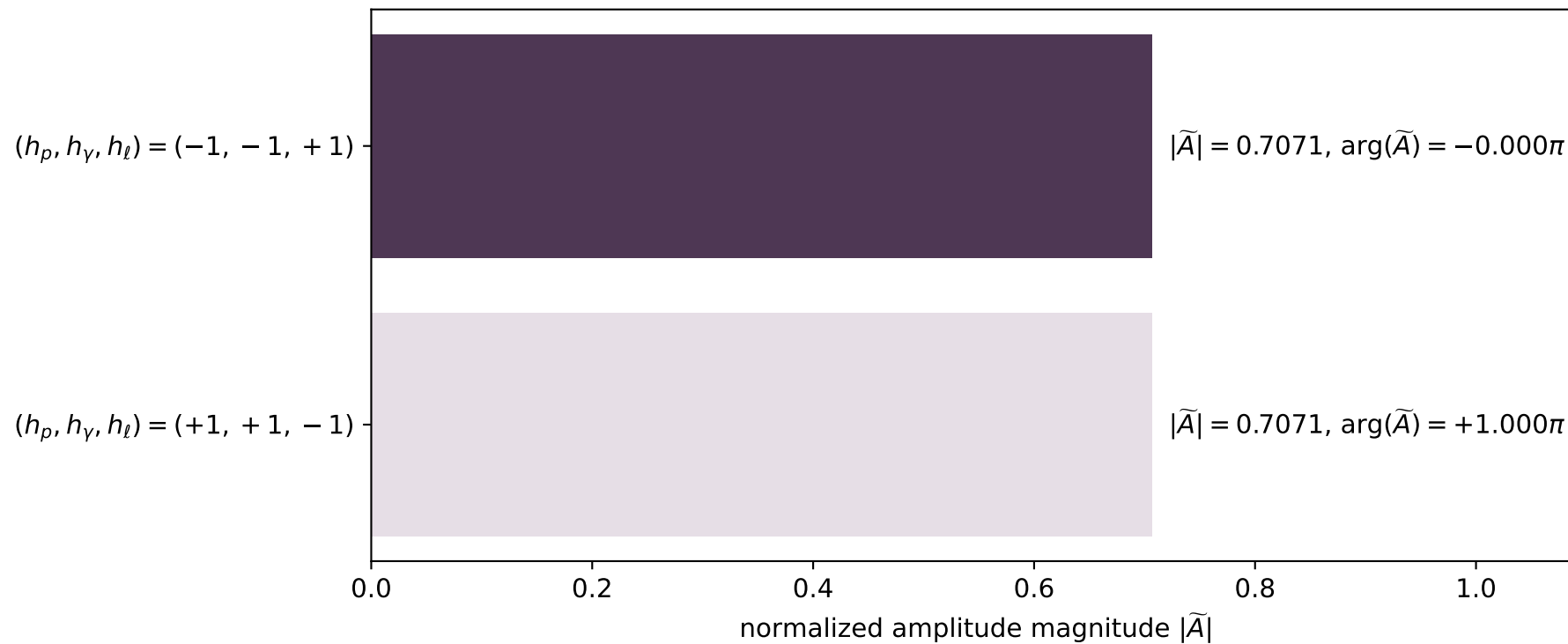


dghz_mixing_angles_polarization_04_local_minimum_101
 $d_{\text{GHZ}}=3.64201\text{e-}08$
 region: polarization_cluster_04_local_minimum_101
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0101
 lepton mixing angle: $\alpha_e=2.19789$ rad
 incoming lepton: $-0.586794|+\rangle +0.809736|-\rangle$
 proton mixing angle: $\alpha_p=2.514492$ rad
 incoming proton: $-0.809732|+\rangle +0.5868|-\rangle$

$s=24.6624$, $\sqrt{s}=4.96613$, $|\vec{P}|=2.39448$, $|\vec{P}'|=2.39448$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=0.290949$, $\phi_Y=6.12241$
 $E_Y=0.819028$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.43254$
 $Q^2=15.0896$, $x_B=0.64973$, $t=-22.9341$, $W^2=9.01465$, $y=0.976527$

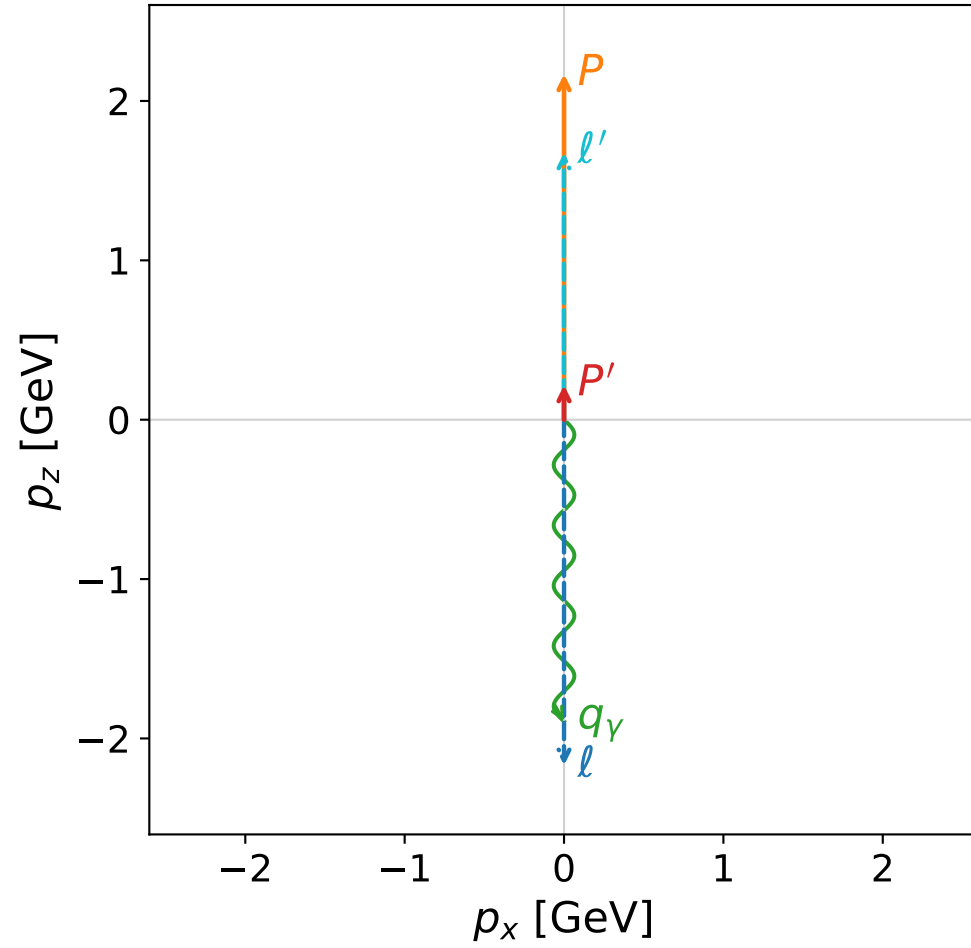
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3945$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3945$
 P $E= 2.5716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3945$
 ℓ' $E= 1.5755$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5755$
 P' $E= 2.5716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3945$
 q_Y $E= 0.8190$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8190$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

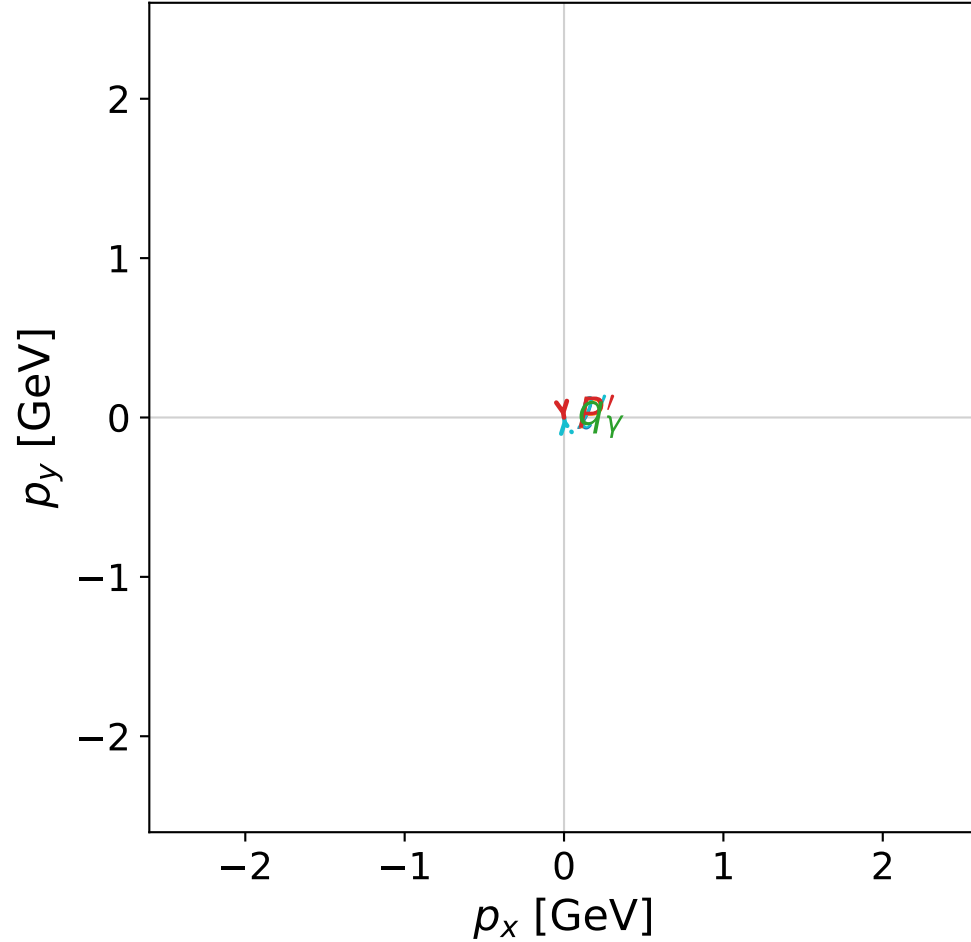


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



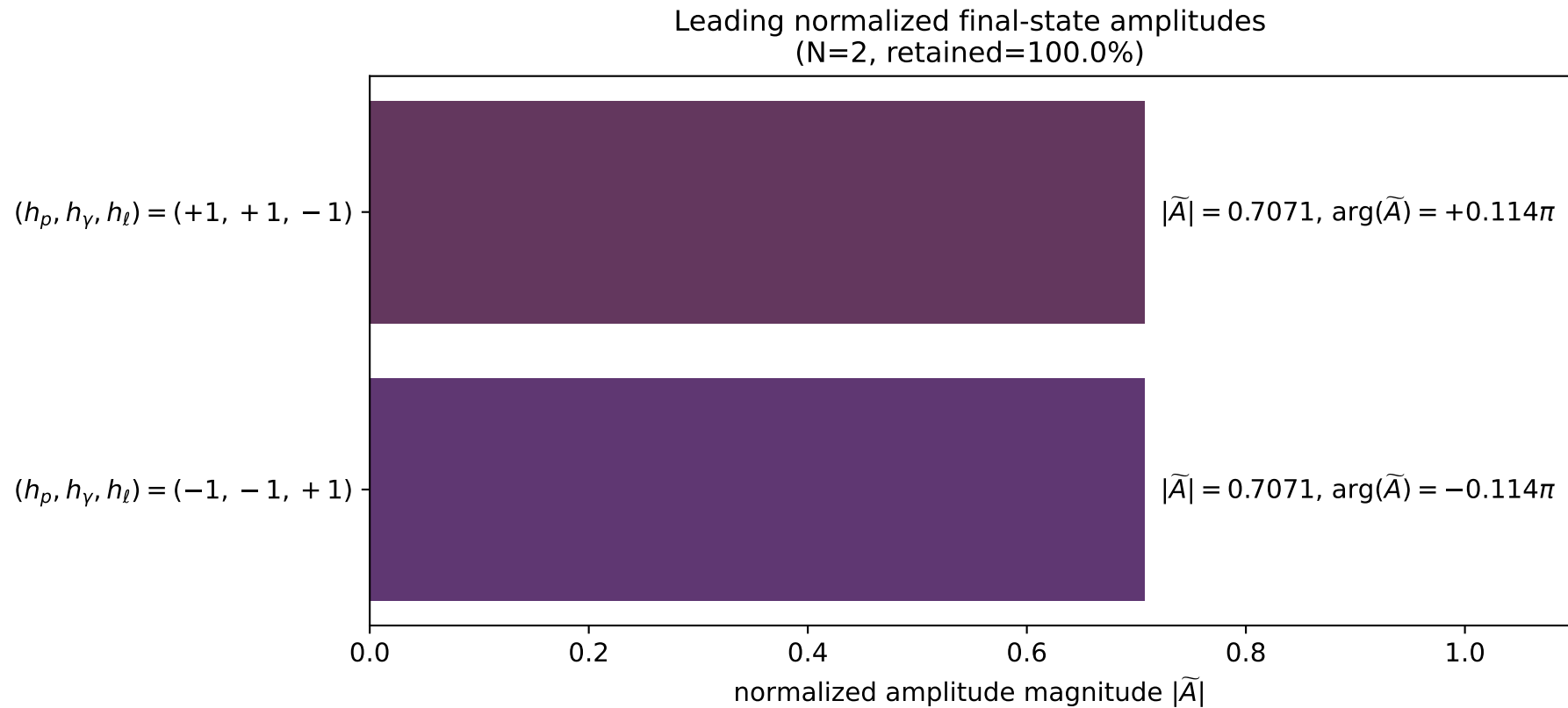
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_105
 $d_{\text{GHZ}}=3.82625\text{e-}08$
 region: polarization_cluster_04_local_minimum_105
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0105
 lepton mixing angle: $\alpha_e=2.5411507$ rad
 incoming lepton: $-0.825086|+\rangle +0.565007|-\rangle$
 proton mixing angle: $\alpha_p=2.5411495$ rad
 incoming proton: $-0.825085|+\rangle +0.565008|-\rangle$

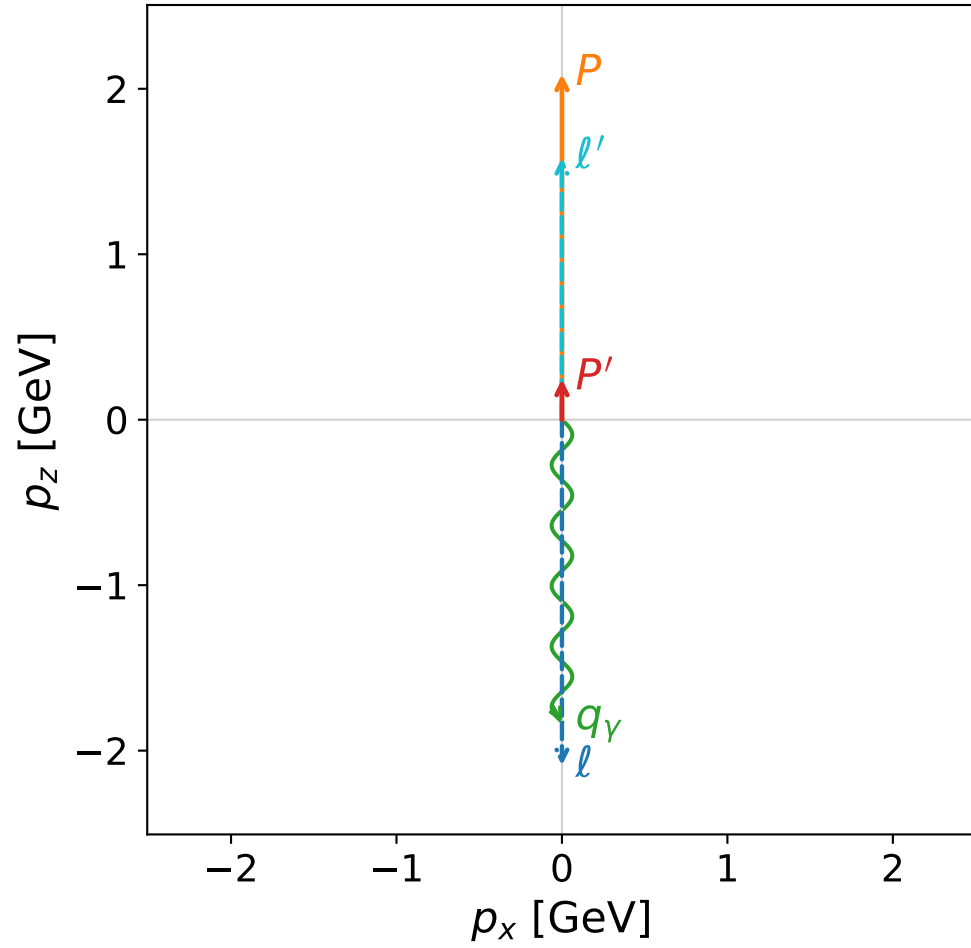
$s=20.5321$, $\sqrt{s}=4.53124$, $|\vec{P}|=2.16853$, $|\vec{P}'|=0.216611$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_Y=3.14159$, $\phi_{P'}=4.86682$, $\phi_Y=5.92443$
 $E_Y=1.89258$, $\theta_{l'}=9.91298\text{e-}06$, $\phi_{l'}=1.72522$
 $Q^2=14.5376$, $x_B=0.765079$, $t=-1.84994$, $W^2=5.34368$, $y=0.966881$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1685$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1685$
 P $E= 2.3627$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1685$
 l' $E= 1.6760$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6760$
 P' $E= 0.9627$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.2166$
 q_Y $E= 1.8926$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.8926$

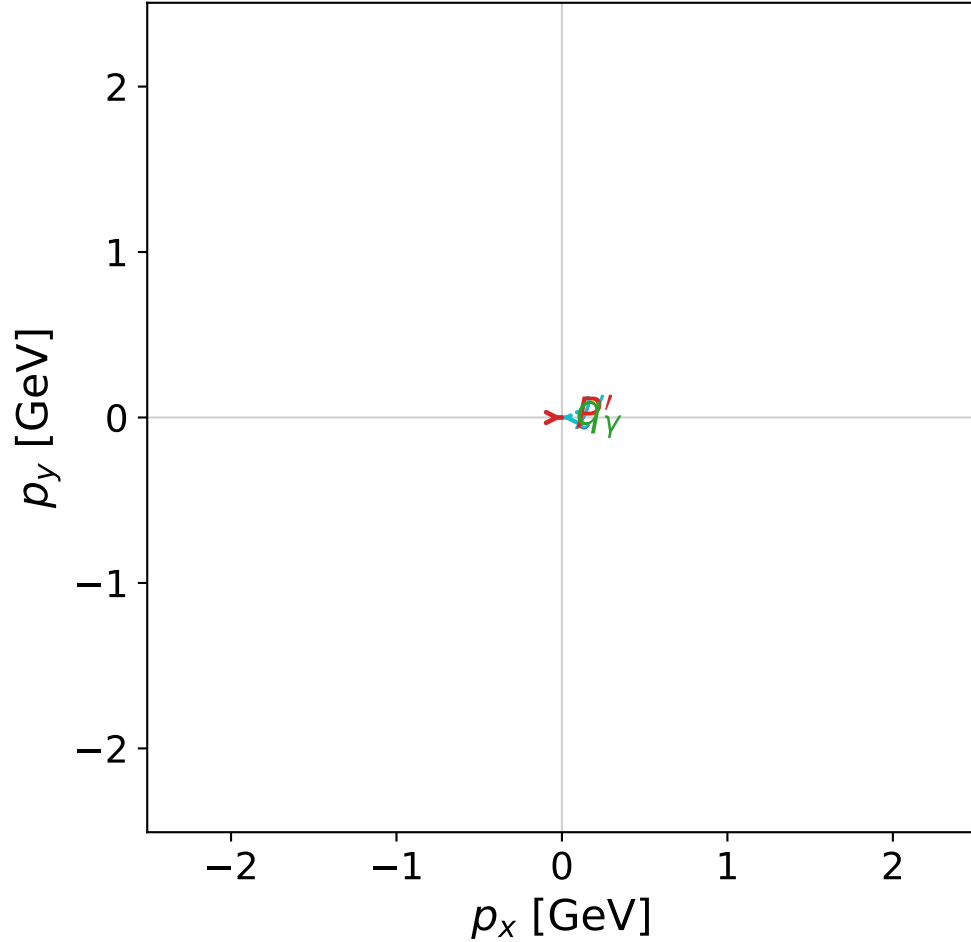


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



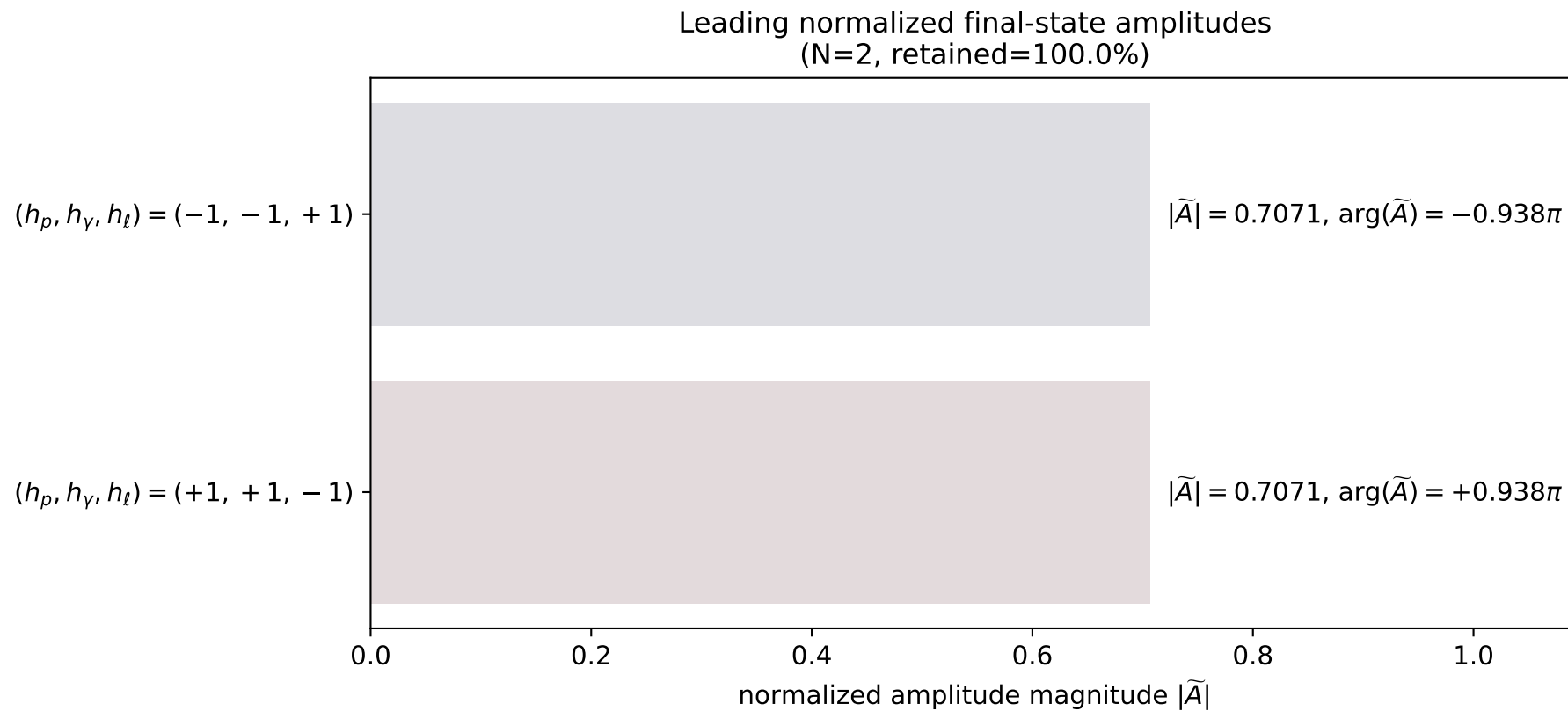
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_108
 $d_{\text{GHZ}}=3.93085\text{e-}08$
 region: polarization_cluster_04_local_minimum_108
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0108
 lepton mixing angle: $\alpha_e=2.3726047$ rad
 incoming lepton: $-0.718615|+\rangle +0.695408|-\rangle$
 proton mixing angle: $\alpha_p=2.3726093$ rad
 incoming proton: $-0.718618|+\rangle +0.695405|-\rangle$

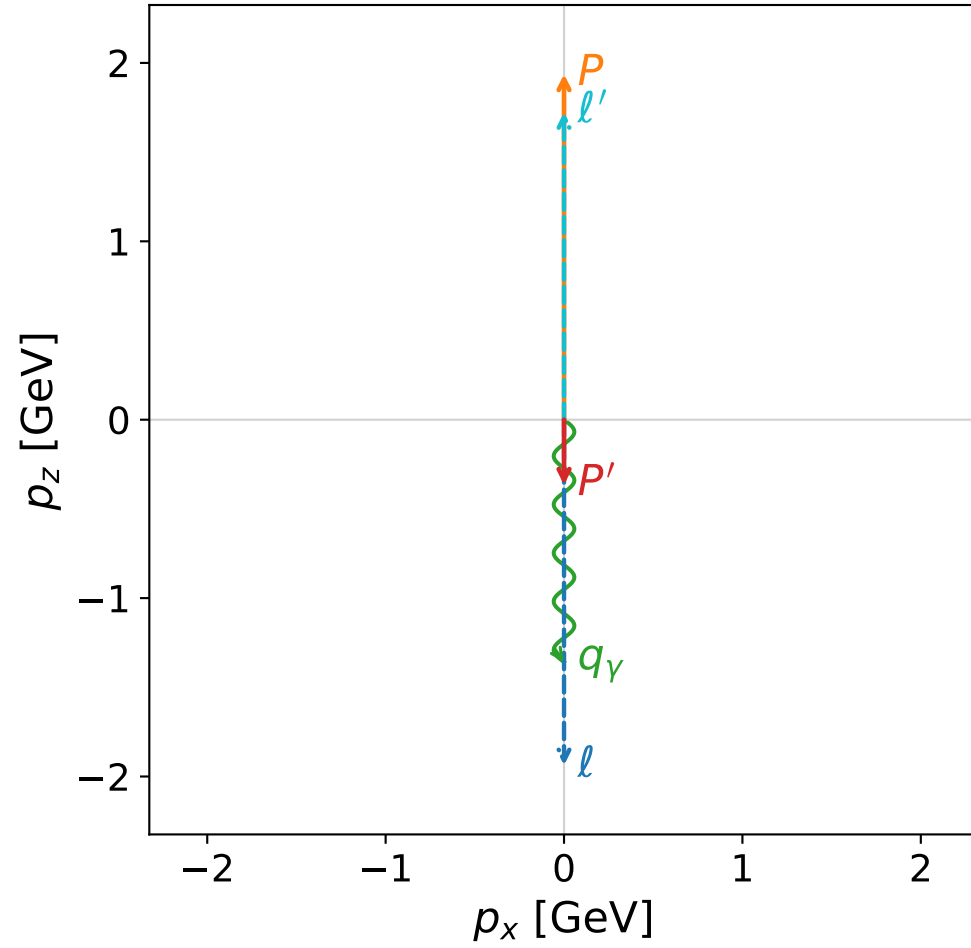
$s=19.1697$, $\sqrt{s}=4.37832$, $|\vec{P}|=2.08868$, $|\vec{P}'|=0.24491$
 $\theta_{P'}=0.000152337$, $\theta_Y=3.14159$, $\phi_{P'}=0.00207501$, $\phi_Y=3.33601$
 $E_Y=1.82689$, $\theta_{\ell'}=2.35835\text{e-}05$, $\phi_{\ell'}=3.14367$
 $Q^2=13.217$, $x_B=0.74867$, $t=-1.65659$, $W^2=5.31684$, $y=0.965237$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.0887$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0887$
 P $E= 2.2896$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0887$
 ℓ' $E= 1.5820$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.5820$
 P' $E= 0.9694$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2449$
 q_Y $E= 1.8269$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.8269$

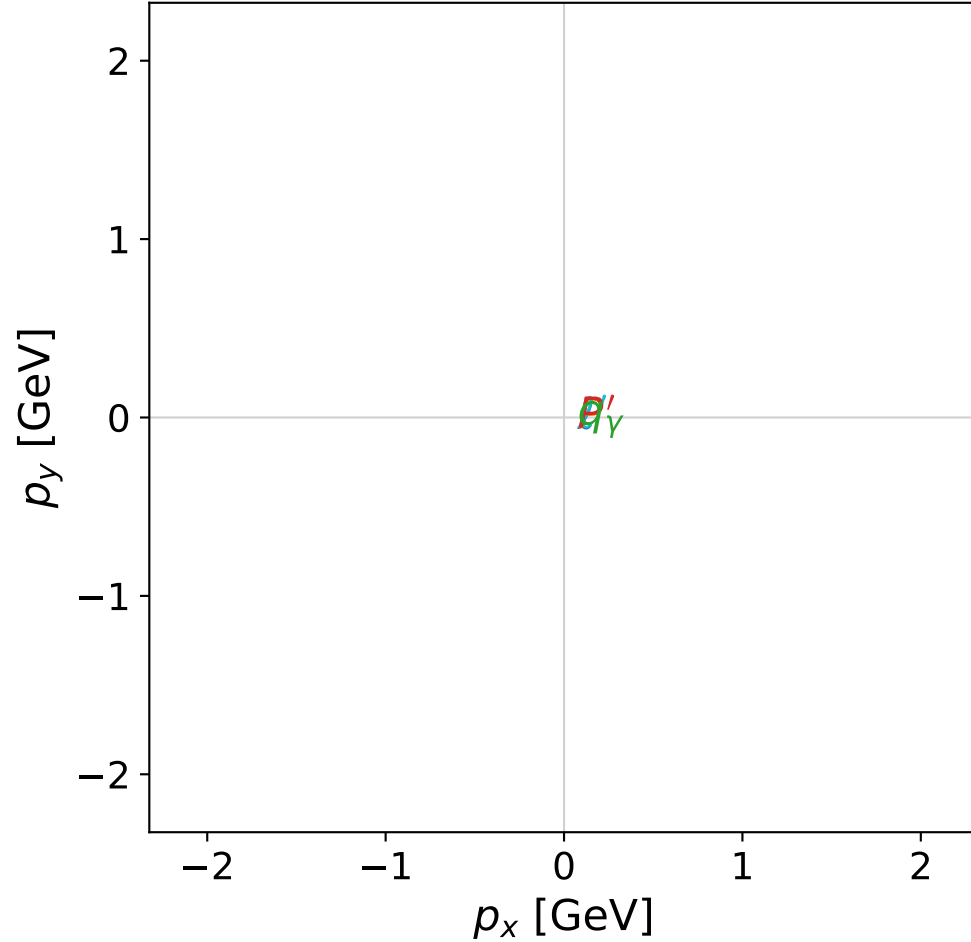


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



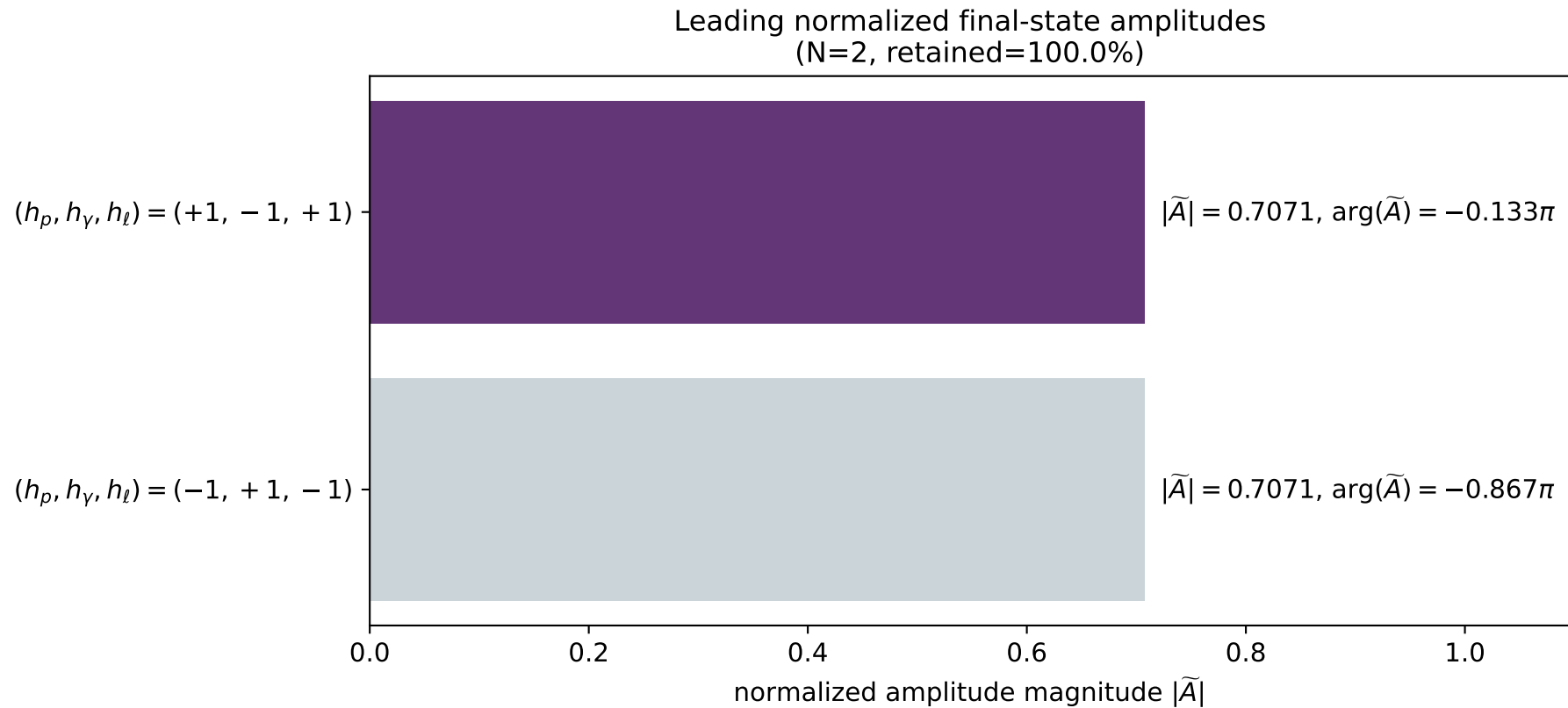
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_109
 $d_{\{\text{rm GHZ}\}}=3.93746\text{e-}08$
 region: polarization_cluster_04_local_minimum_109
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0109
 lepton mixing angle: $\alpha_e=2.4147598$ rad
 incoming lepton: $-0.747283|+\rangle +0.664506|-\rangle$
 proton mixing angle: $\alpha_p=2.4147771$ rad
 incoming proton: $-0.747294|+\rangle +0.664493|-\rangle$

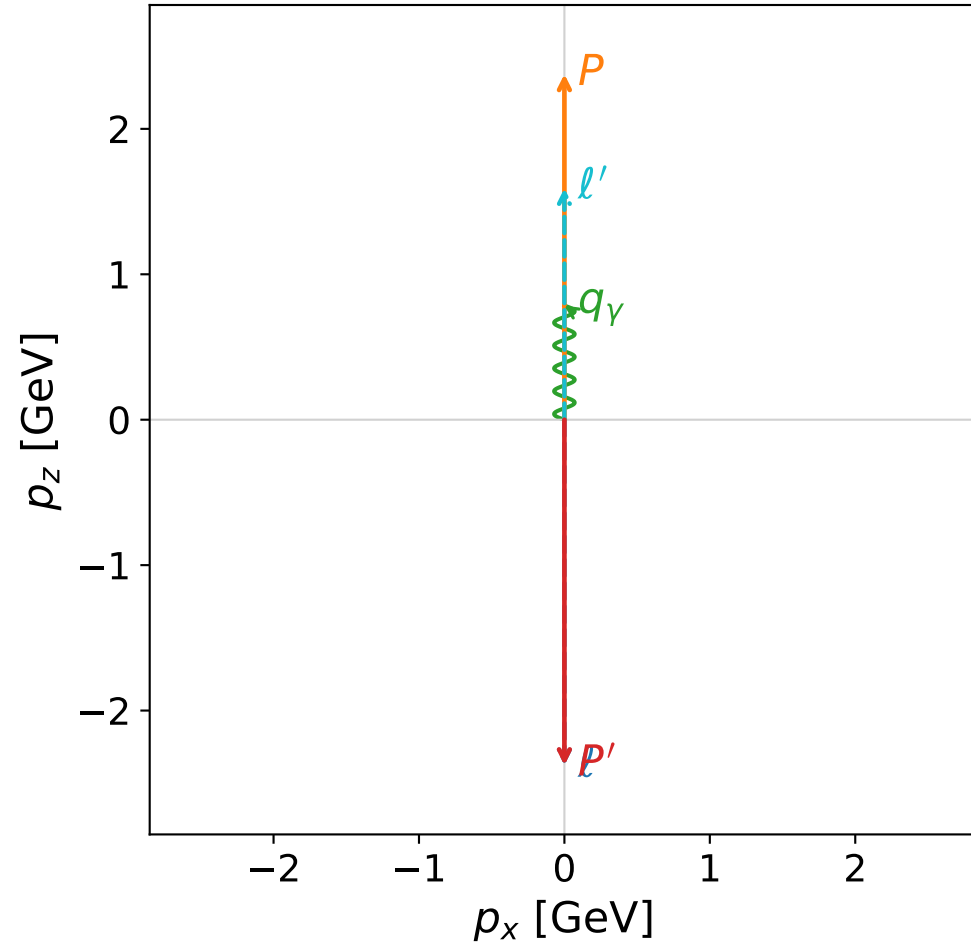
$s=16.7248$, $\sqrt{s}=4.0896$, $|\vec{P}|=1.93723$, $|\vec{P}'|=0.365737$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.18745$, $\phi_\gamma=5.86465$
 $E_\gamma=1.35854$, $\theta_{l'}=0$, $\phi_{l'}=2.56558$
 $Q^2=13.3613$, $x_B=0.884674$, $t=-3.99128$, $W^2=2.62161$, $y=0.953176$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9372$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9372$
 P $E= 2.1524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9372$
 ℓ' $E= 1.7243$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.7243$
 P' $E= 1.0068$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.3657$
 q_γ $E= 1.3585$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3585$

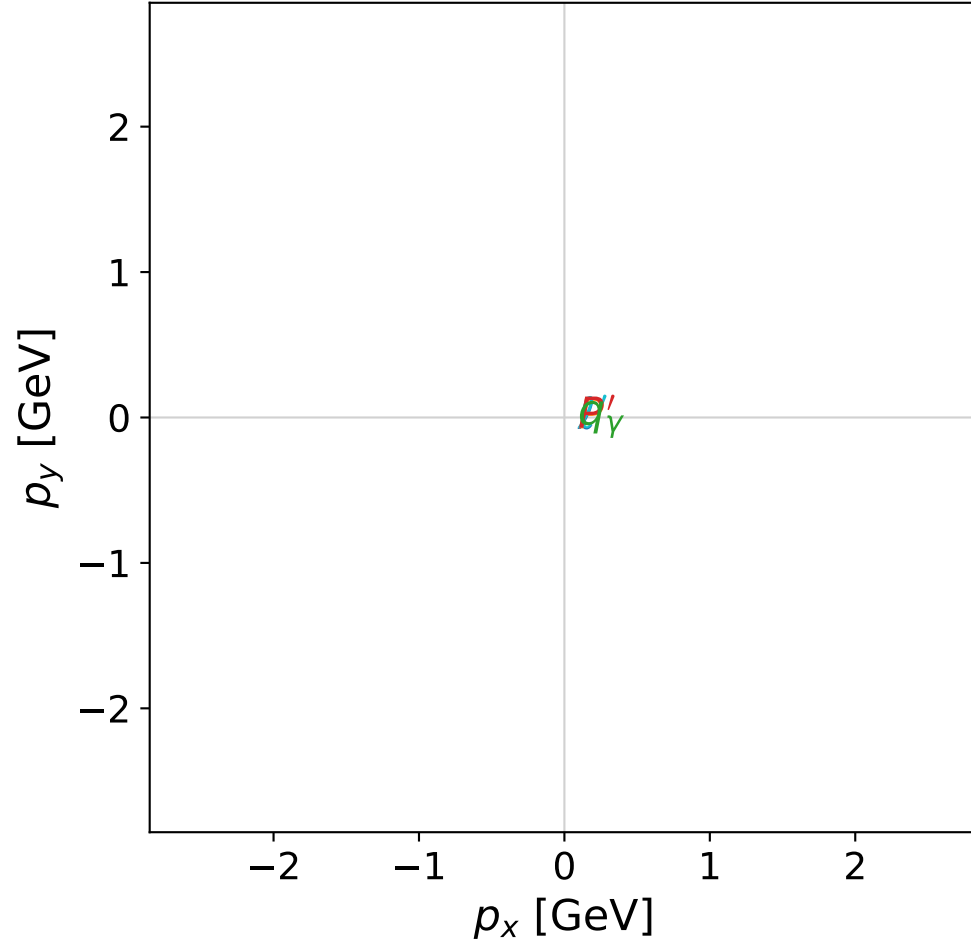


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

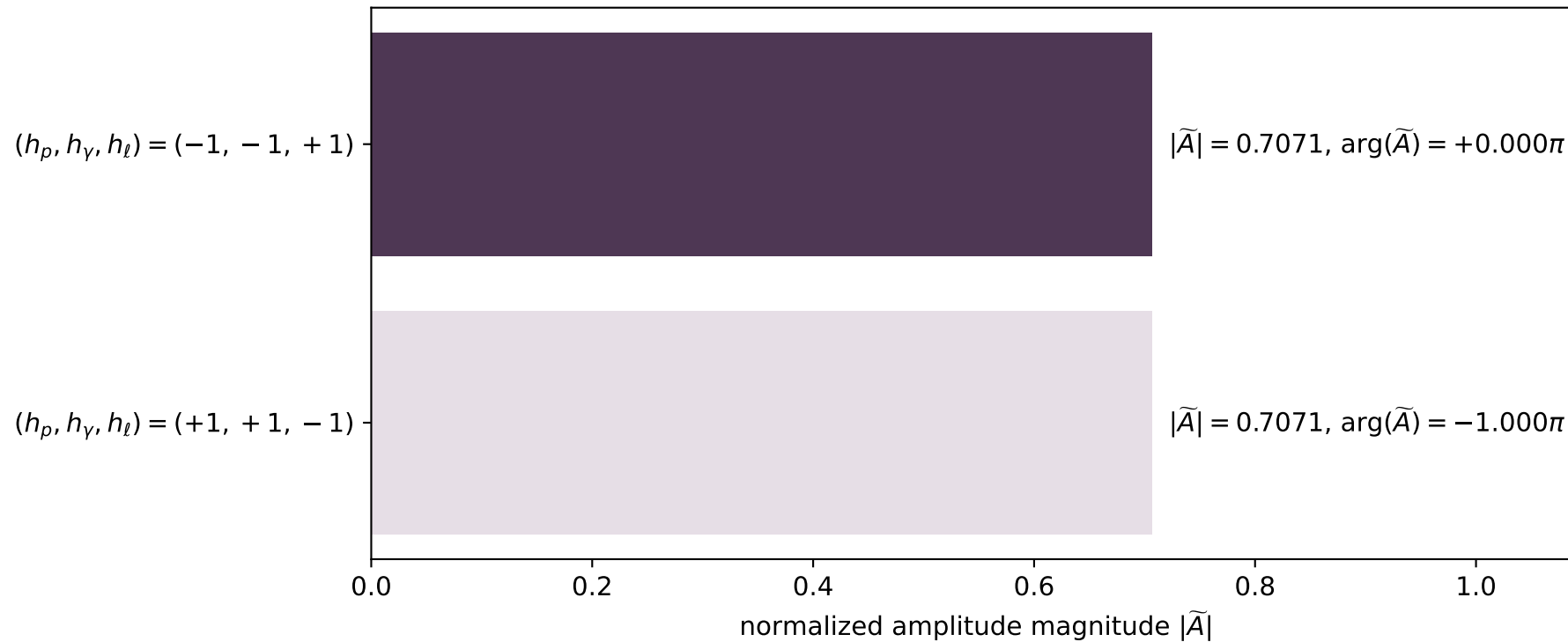


dghz_mixing_angles_polarization_04_local_minimum_117
 $d_{\text{GHZ}}=4.36167\text{e-}08$
 region: polarization_cluster_04_local_minimum_117
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0117
 lepton mixing angle: $\alpha_e=2.0273398$ rad
 incoming lepton: $-0.440848|+\rangle +0.897582|-\rangle$
 proton mixing angle: $\alpha_p=2.6850445$ rad
 incoming proton: $-0.89758|+\rangle +0.440852|-\rangle$

$s=24.3182$, $\sqrt{s}=4.93135$, $|\vec{P}|=2.37646$, $|\vec{P}'|=2.37646$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.60836$, $\phi_\gamma=5.61305$
 $E_\gamma=0.784824$, $\theta_{l'}=0$, $\phi_{l'}=0.466771$
 $Q^2=15.1299$, $x_B=0.66155$, $t=-22.5903$, $W^2=8.62032$, $y=0.975768$

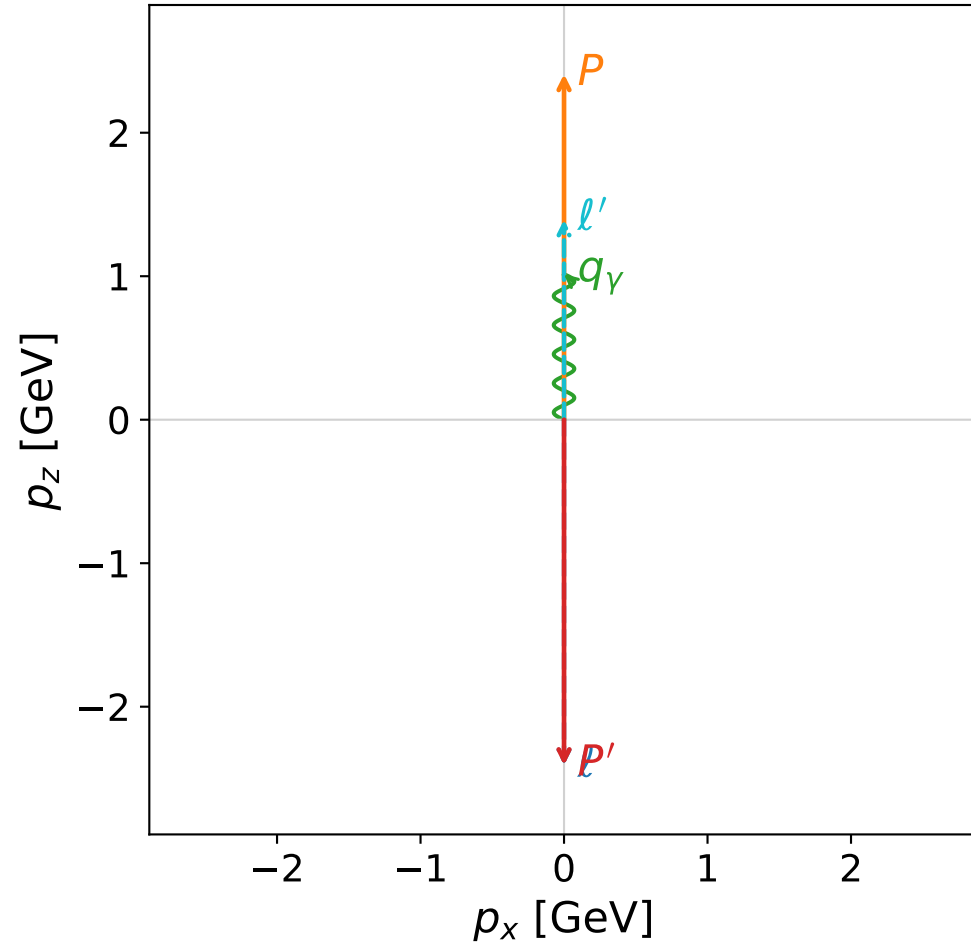
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3765$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3765$
 P $E= 2.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3765$
 ℓ' $E= 1.5916$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5916$
 P' $E= 2.5549$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3765$
 q_γ $E= 0.7848$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7848$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

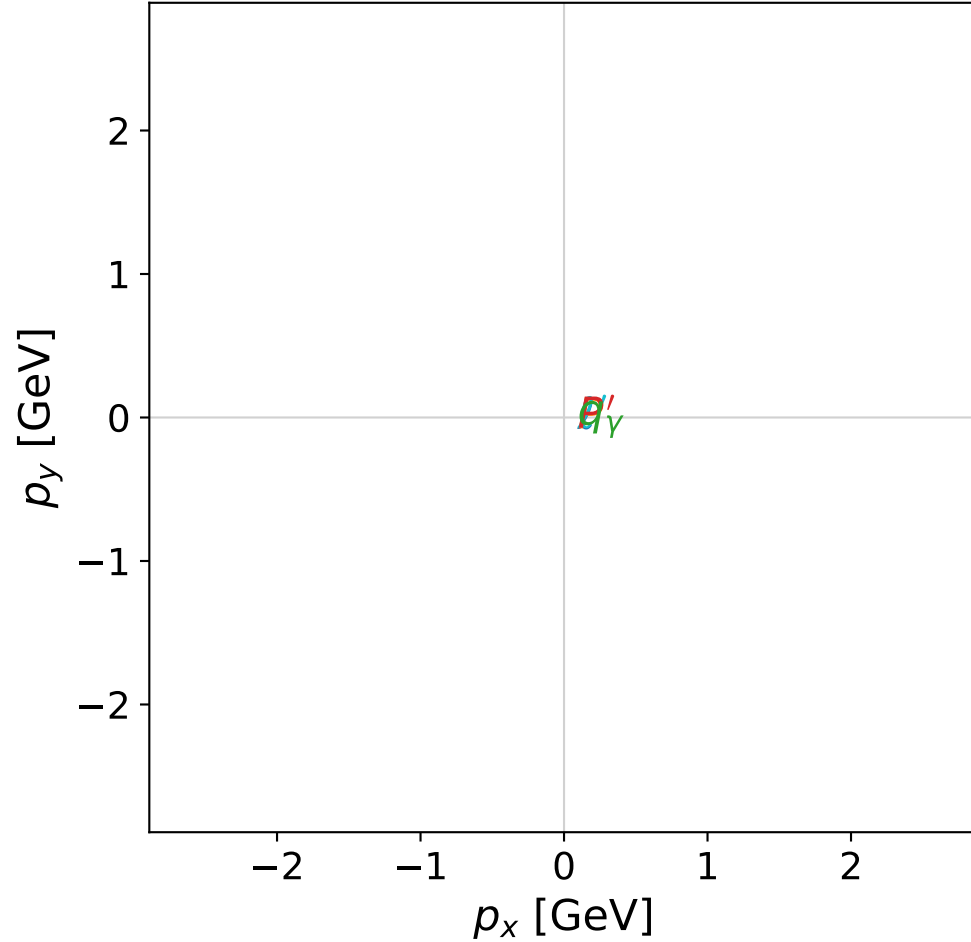


dghz_mixing_angles_polarization_04_local_minimum_119
 $d_{\{\rm GHZ\}}=4.46361\text{e-}08$
 region: polarization_cluster_04_local_minimum_119
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0119
 lepton mixing angle: $\alpha_e=2.099717$ rad
 incoming lepton: $-0.504602|+\rangle +0.863352|-\rangle$
 proton mixing angle: $\alpha_p=2.61268$ rad
 incoming proton: $-0.863356|+\rangle +0.504595|-\rangle$

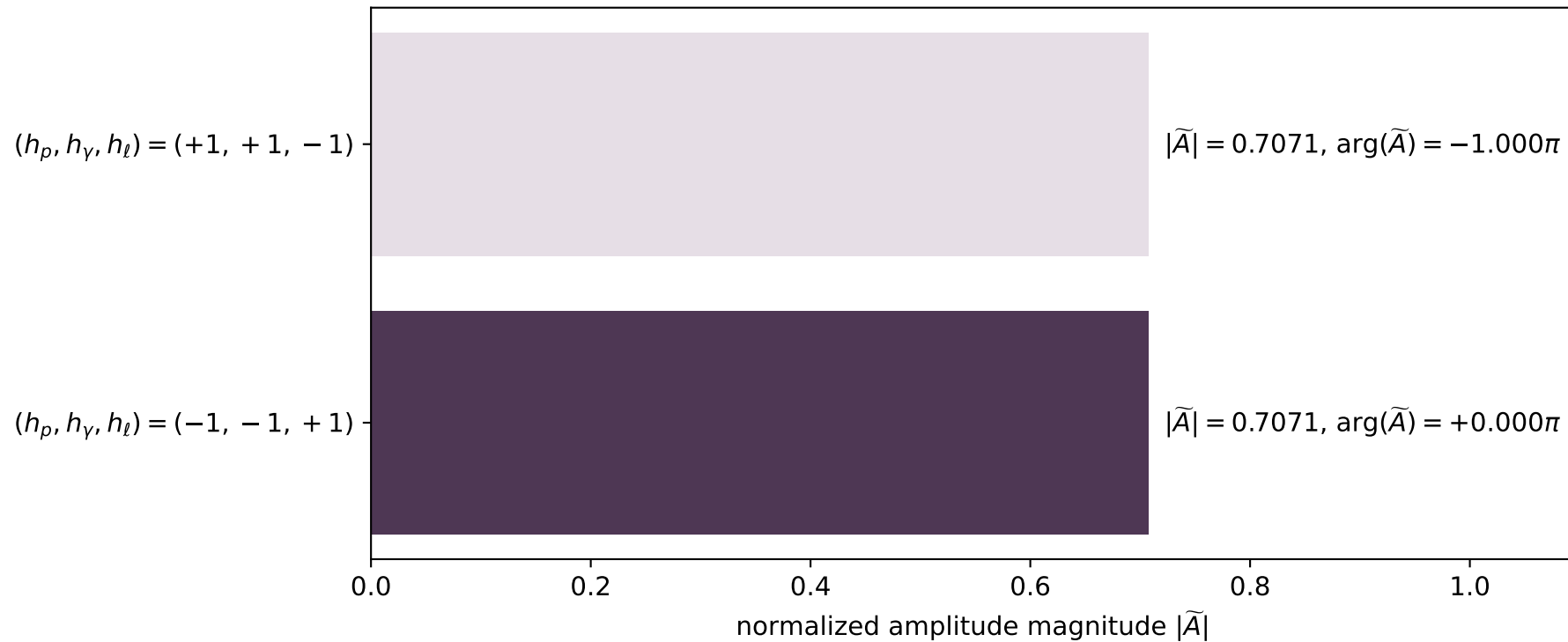
$s=24.93$, $\sqrt{s}=4.993$, $|\vec{P}|=2.40839$, $|\vec{P}'|=2.40839$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.04535$, $\phi_\gamma=0.354409$
 $E_\gamma=1.01446$, $\theta_{l'}=0$, $\phi_{l'}=0.903761$
 $Q^2=13.4285$, $x_B=0.569998$, $t=-23.2014$, $W^2=11.0102$, $y=0.979573$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4084$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4084$
 P $E= 2.5846$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4084$
 l' $E= 1.3939$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3939$
 P' $E= 2.5846$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4084$
 q_γ $E= 1.0145$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0145$

x--y plane

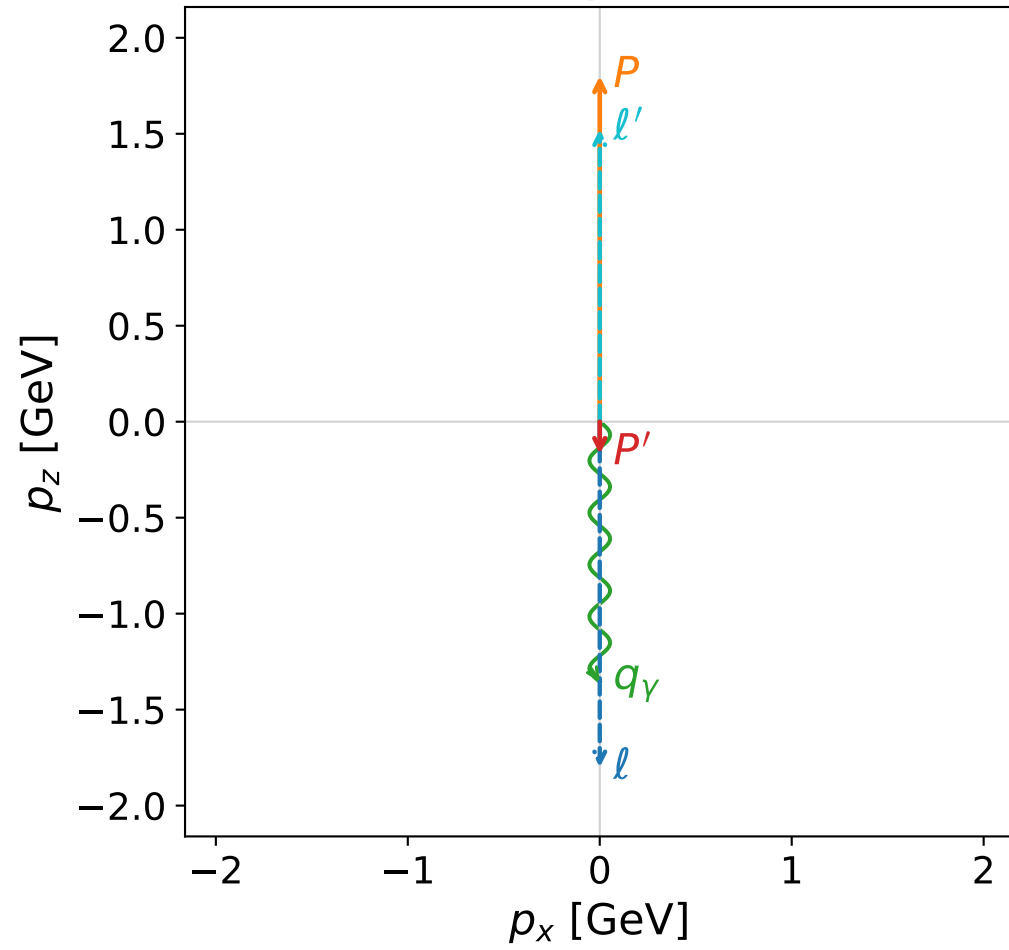


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

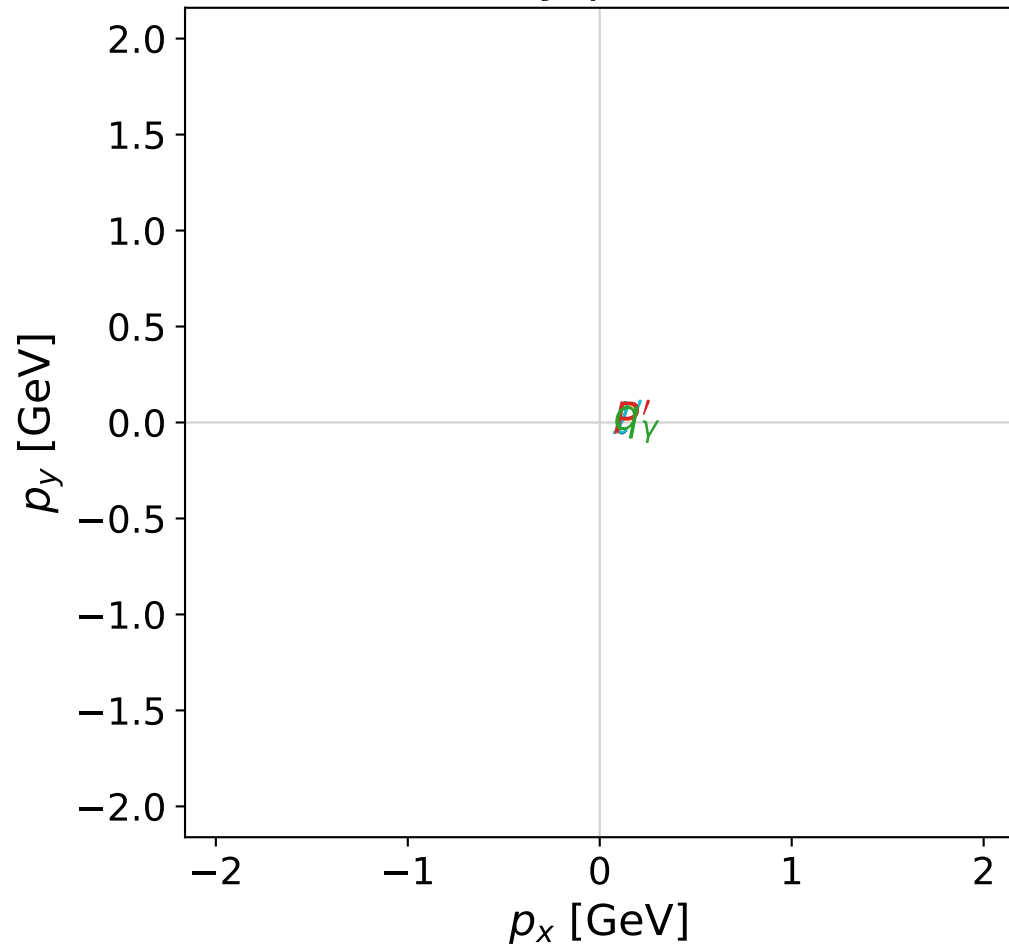


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



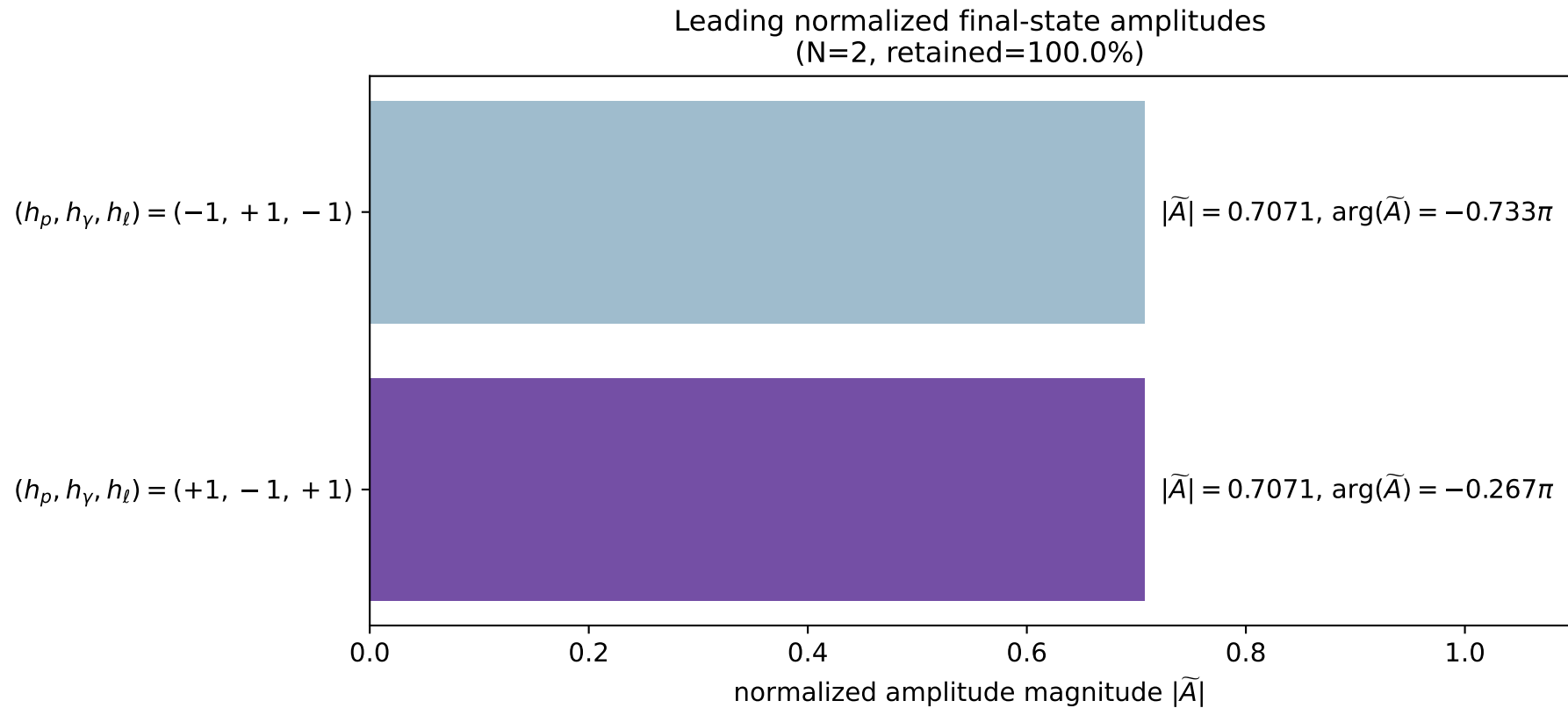
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_122
 $d_{\{\rm GHZ\}}=4.77096\text{e-}08$
 region: polarization_cluster_04_local_minimum_122
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0122
 lepton mixing angle: $\alpha_e=2.3265008$ rad
 incoming lepton: $-0.685802|+\rangle +0.727789|-\rangle$
 proton mixing angle: $\alpha_p=2.3264837$ rad
 incoming proton: $-0.685789|+\rangle +0.7278|-\rangle$

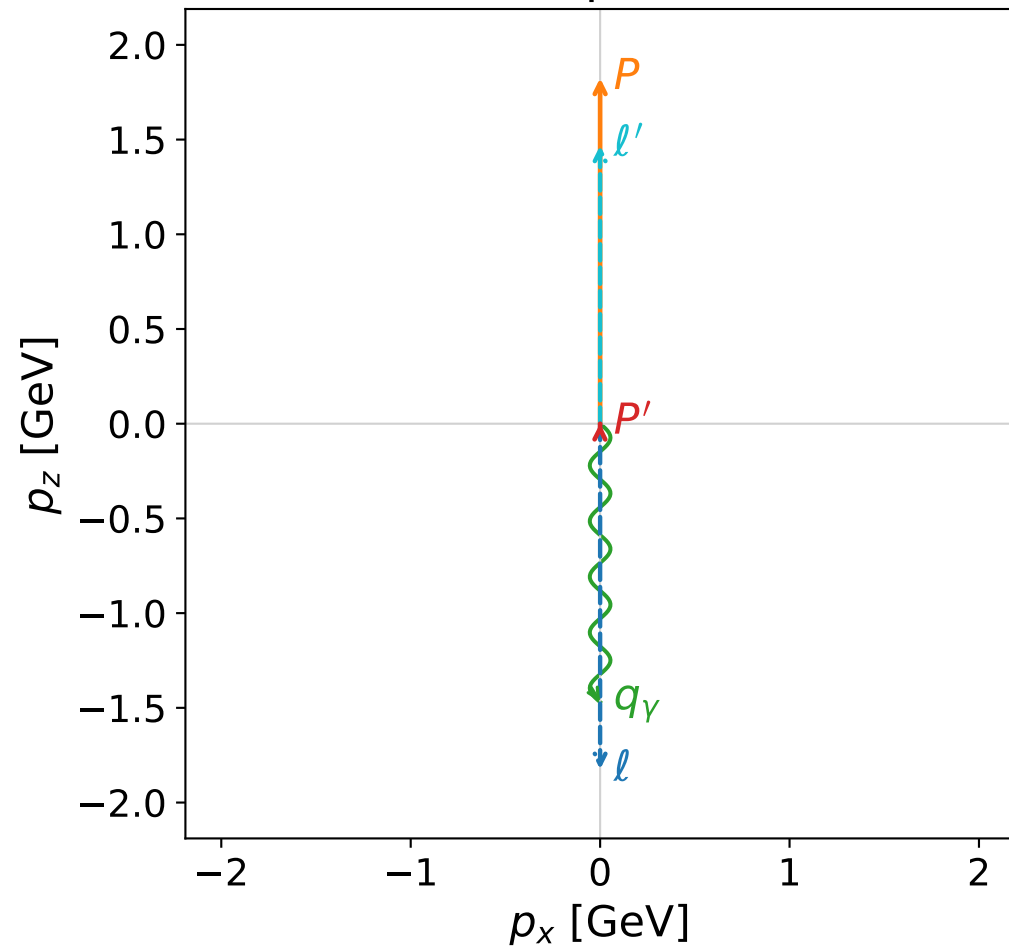
$s=14.673$, $\sqrt{s}=3.83054$, $|\vec{P}|=1.80042$, $|\vec{P}'|=0.16869$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.10573$, $\phi_\gamma=5.44421$
 $E_\gamma=1.3544$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.20483$
 $Q^2=10.9688$, $x_B=0.837731$, $t=-2.71733$, $W^2=3.00452$, $y=0.949273$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8004$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8004$
 P $E= 2.0301$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8004$
 ℓ' $E= 1.5231$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5231$
 P' $E= 0.9530$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1687$
 q_γ $E= 1.3544$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3544$

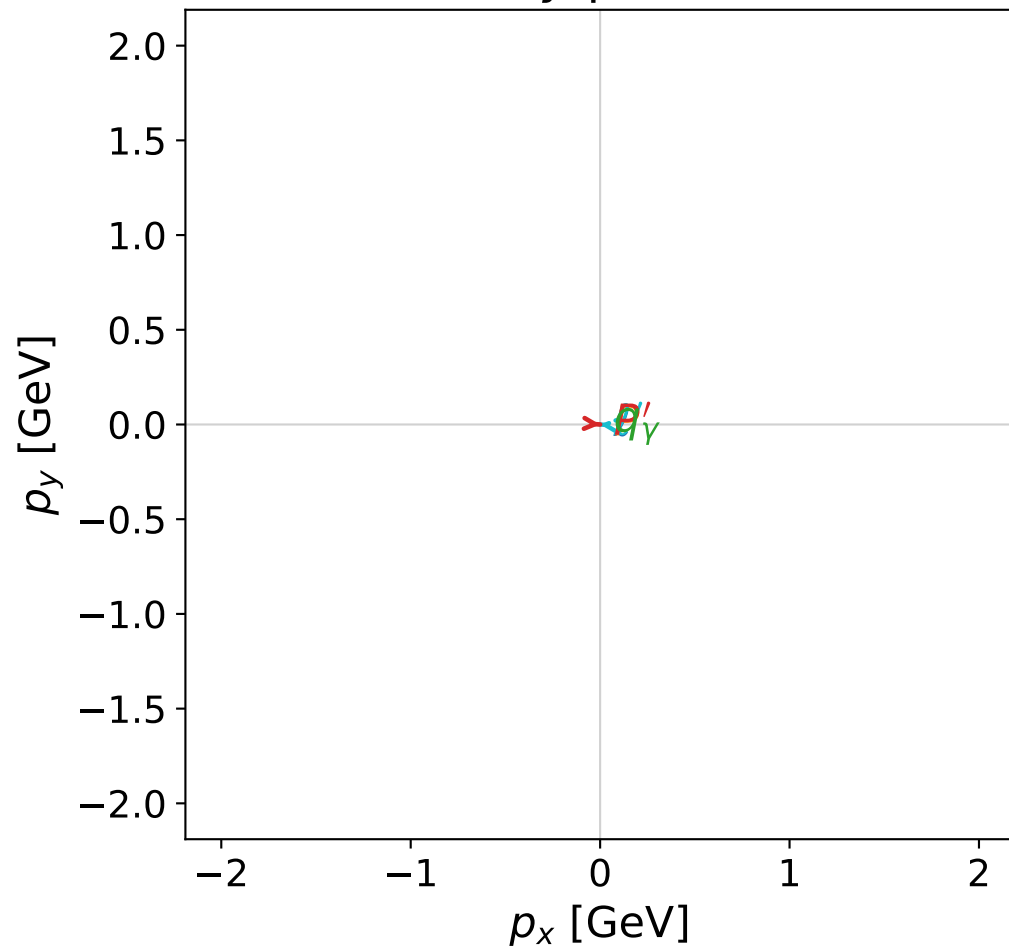


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



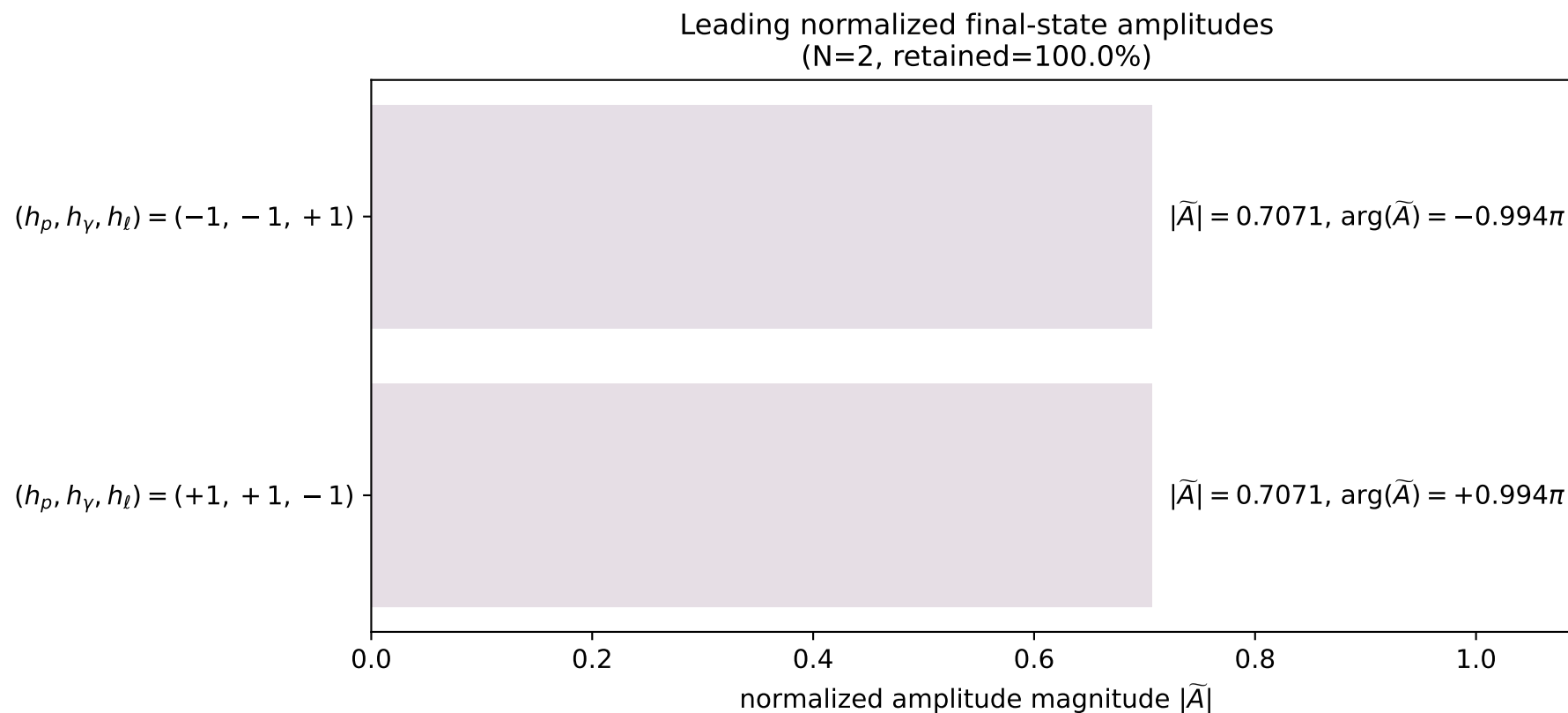
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_123
 $d_{\text{GHZ}}=4.77825\text{e-}08$
 region: polarization_cluster_04_local_minimum_123
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0123
 lepton mixing angle: $\alpha_e=2.3820098$ rad
 incoming lepton: $-0.725123|+\rangle + 0.688619|-\rangle$
 proton mixing angle: $\alpha_p=2.3820131$ rad
 incoming proton: $-0.725126|+\rangle + 0.688617|-\rangle$

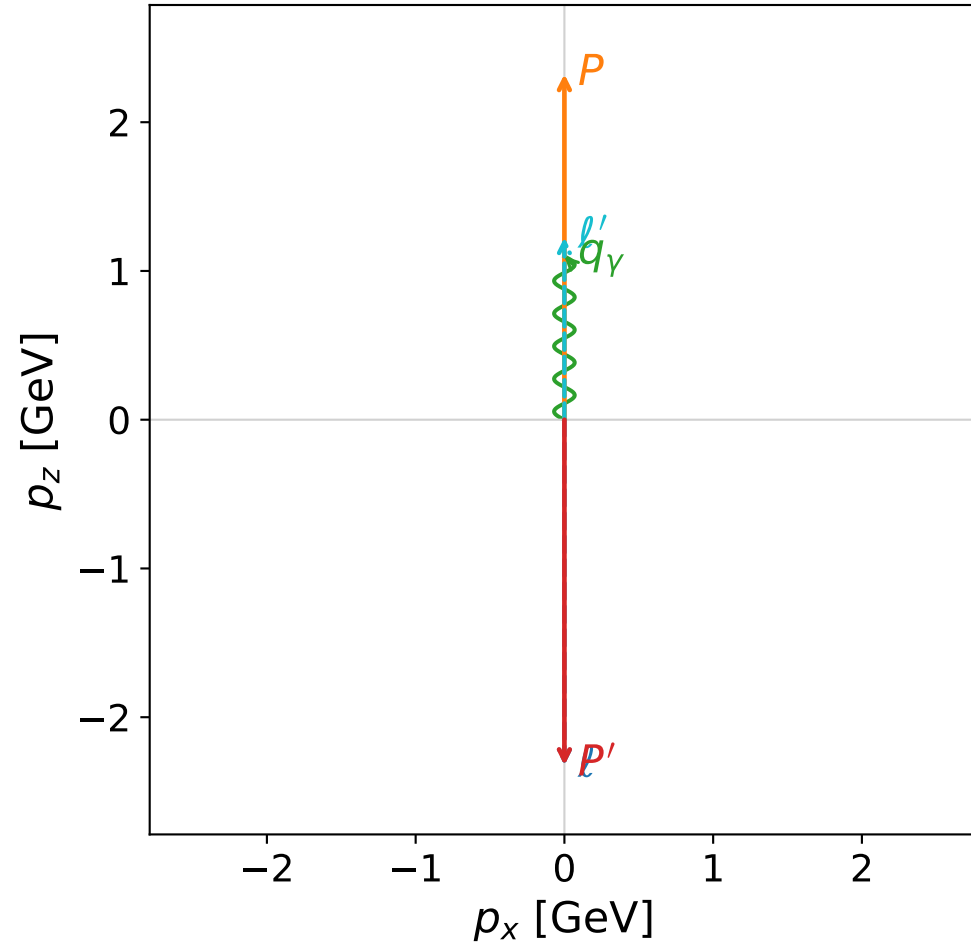
$s=15.0226$, $\sqrt{s}=3.8759$, $|\vec{P}|=1.82445$, $|\vec{P}'|=0.00112548$
 $\theta_{P'}=0.0124896$, $\theta_Y=3.14159$, $\phi_{P'}=6.21424$, $\phi_Y=3.15921$
 $E_Y=1.46951$, $\theta_{\ell'}=9.57262\text{e-}06$, $\phi_{\ell'}=3.07264$
 $Q^2=10.716$, $x_B=0.795185$, $t=-2.08473$, $W^2=3.63996$, $y=0.952862$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8244$
 P $E= 2.0515$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8244$
 ℓ' $E= 1.4684$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4684$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0011$
 q_Y $E= 1.4695$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4695$

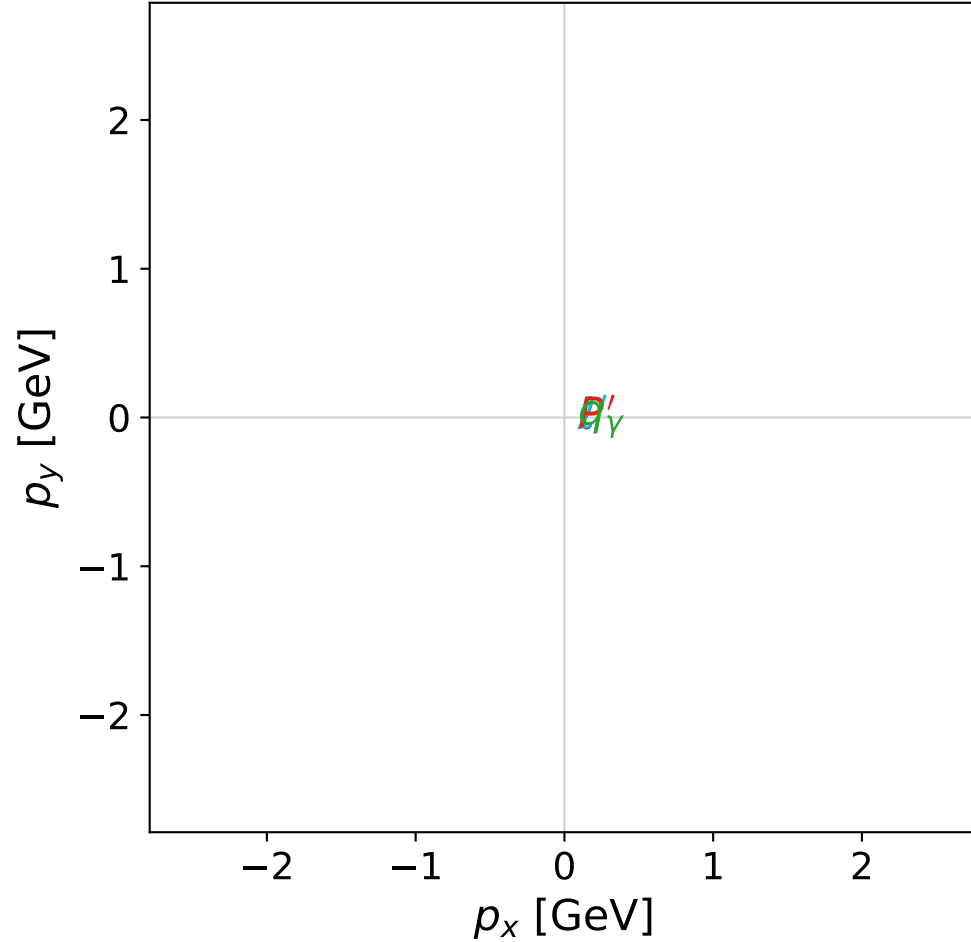


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



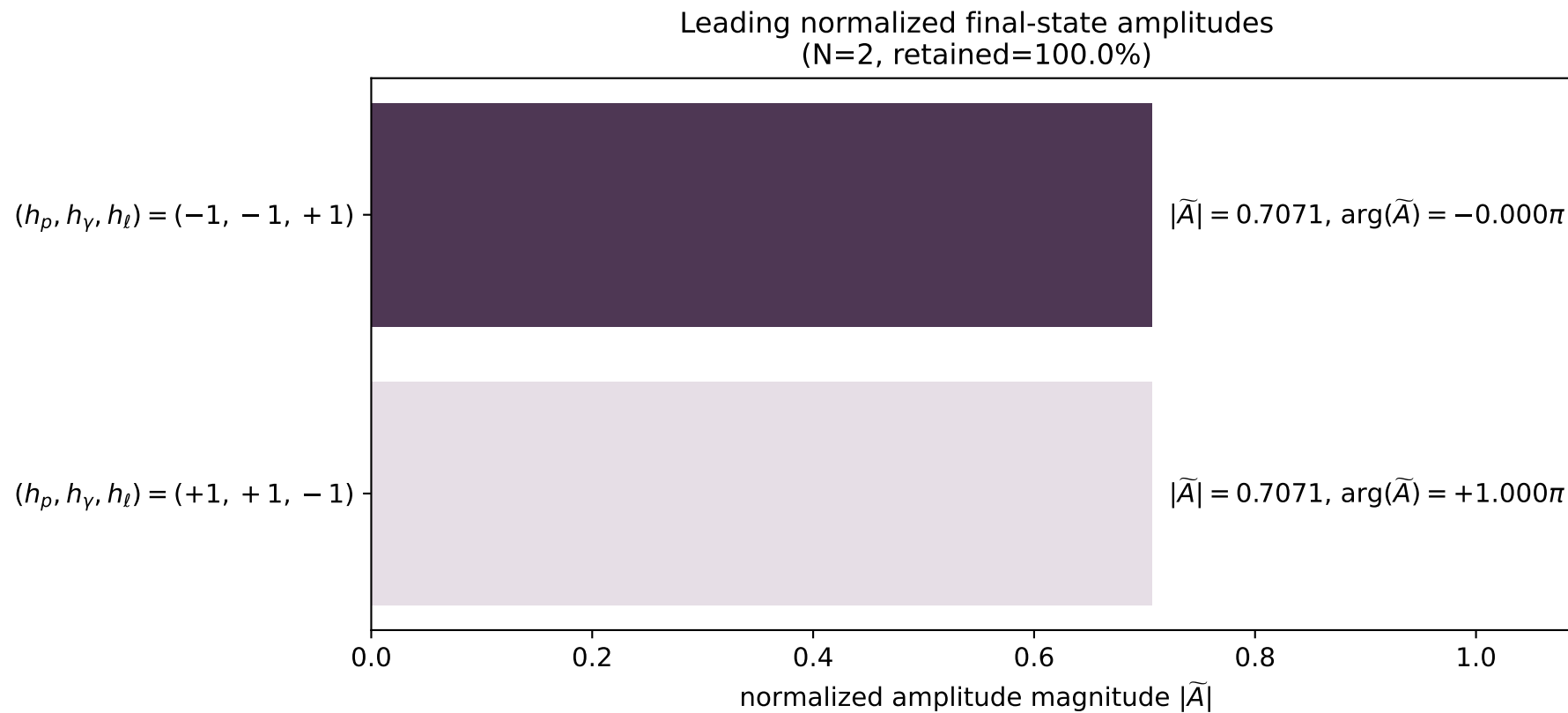
x--y plane



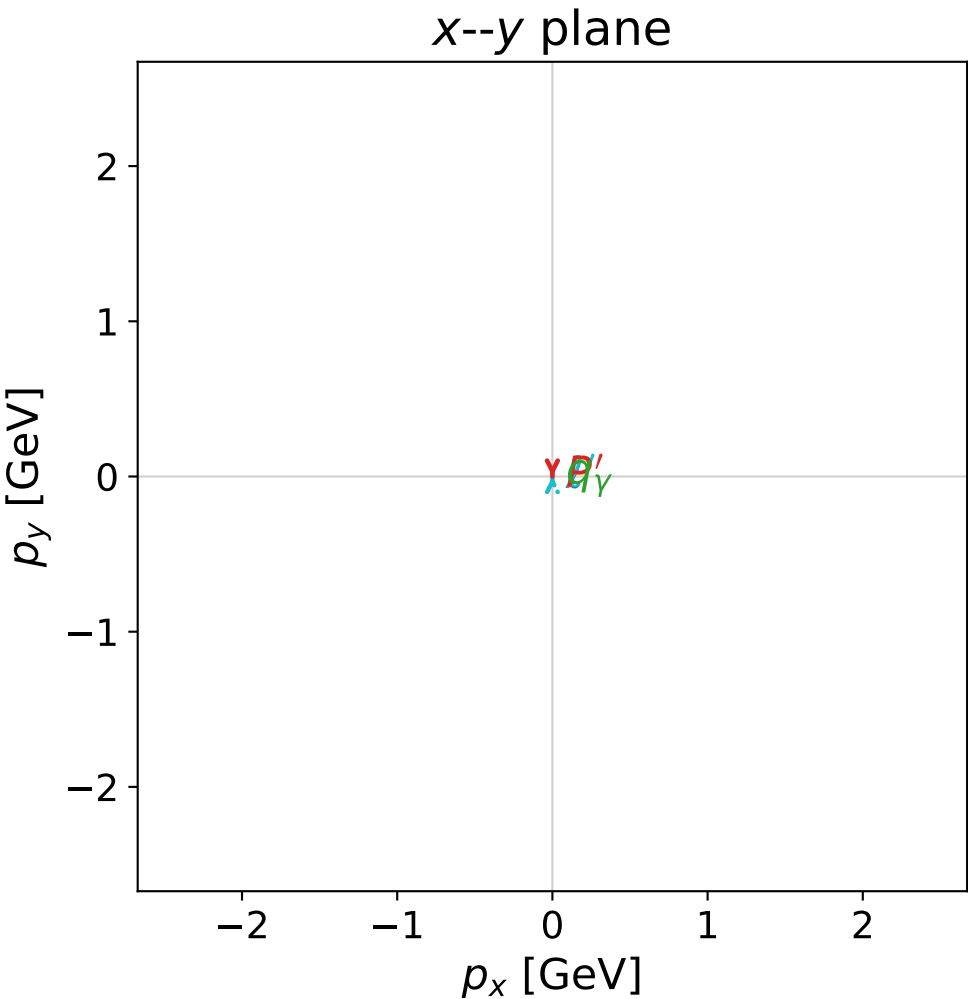
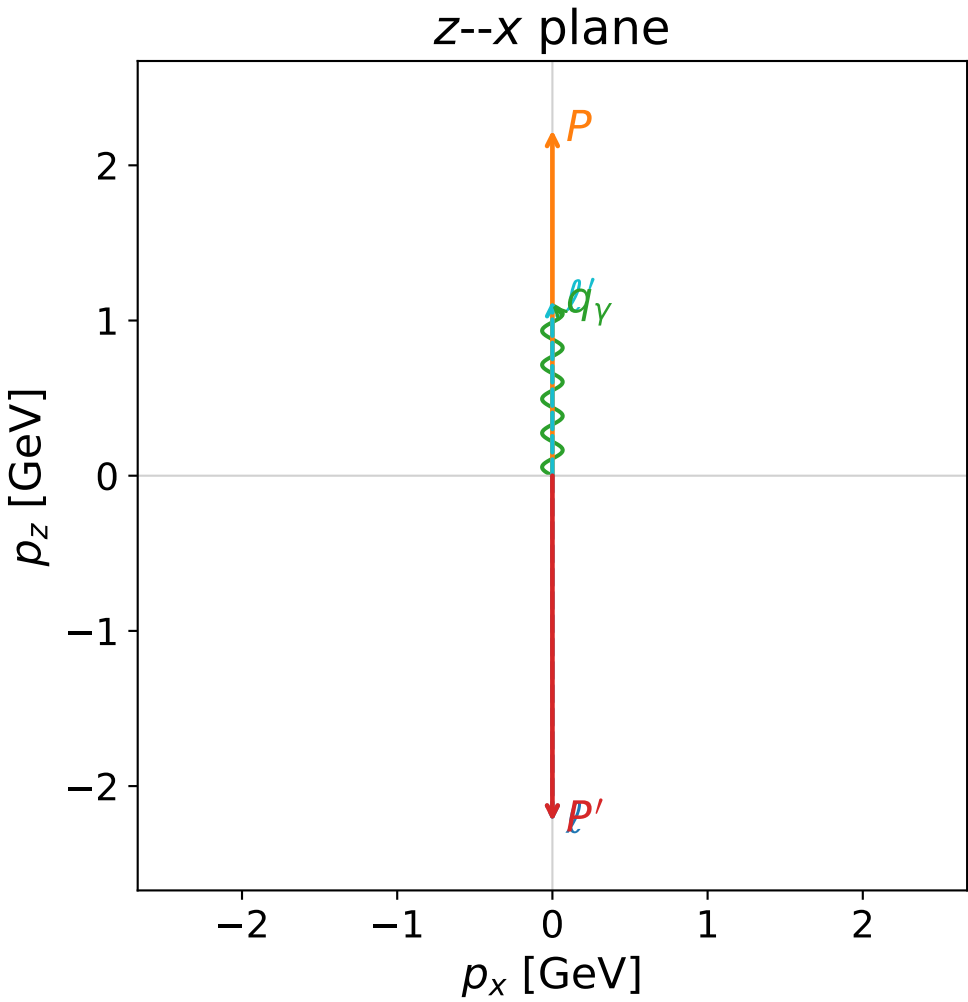
dghz_mixing_angles_polarization_04_local_minimum_125
 $d_{\text{GHZ}}=4.83377\text{e-}08$
 region: polarization_cluster_04_local_minimum_125
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0125
 lepton mixing angle: $\alpha_e=2.1954113$ rad
 incoming lepton: $-0.584785|+\rangle +0.811188|-\rangle$
 proton mixing angle: $\alpha_p=2.5169716$ rad
 incoming proton: $-0.811185|+\rangle +0.58479|-\rangle$

$s=23.3175$, $\sqrt{s}=4.82882$, $|\vec{P}|=2.32331$, $|\vec{P}'|=2.32331$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.30408$, $\phi_\gamma=5.61598$
 $E_\gamma=1.09792$, $\theta_{l'}=0$, $\phi_{l'}=5.44568$
 $Q^2=11.3878$, $x_B=0.517838$, $t=-21.591$, $W^2=11.4831$, $y=0.980098$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3233$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 P $E= 2.5055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3233$
 ℓ' $E= 1.2254$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.2254$
 P' $E= 2.5055$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.3233$
 q_γ $E= 1.0979$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0979$



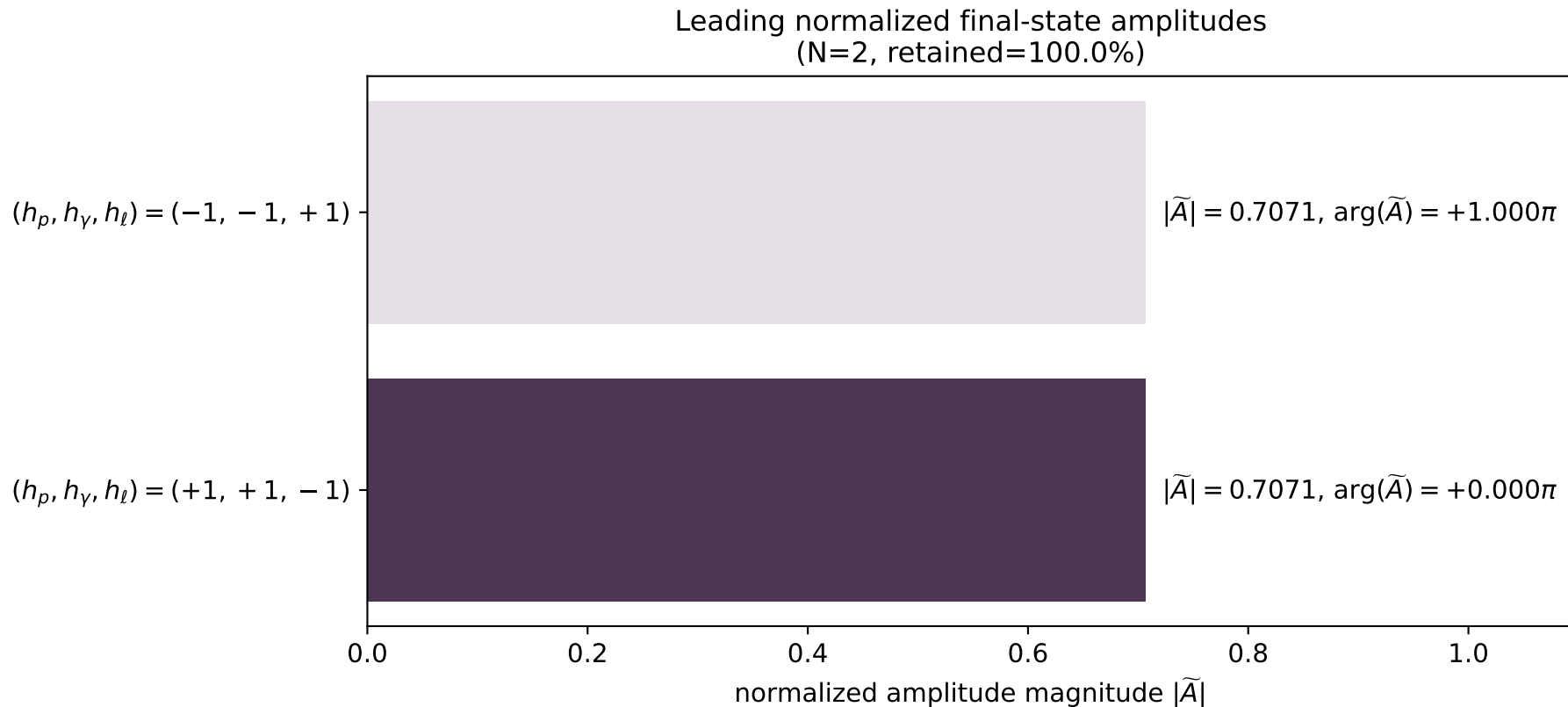
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



```
dghz_mixing_angles_polarization_04_local_minimum_133
d_{\rm GHZ}=5.36187e-08
region: polarization_cluster_04_local_minimum_133
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0133
lepton mixing angle: alpha_e=2.2324916 rad
incoming lepton: -0.614455|+> +0.788952|->
proton mixing angle: alpha_p=2.4798937 rad
incoming proton: -0.788949|+> +0.614458|->
```

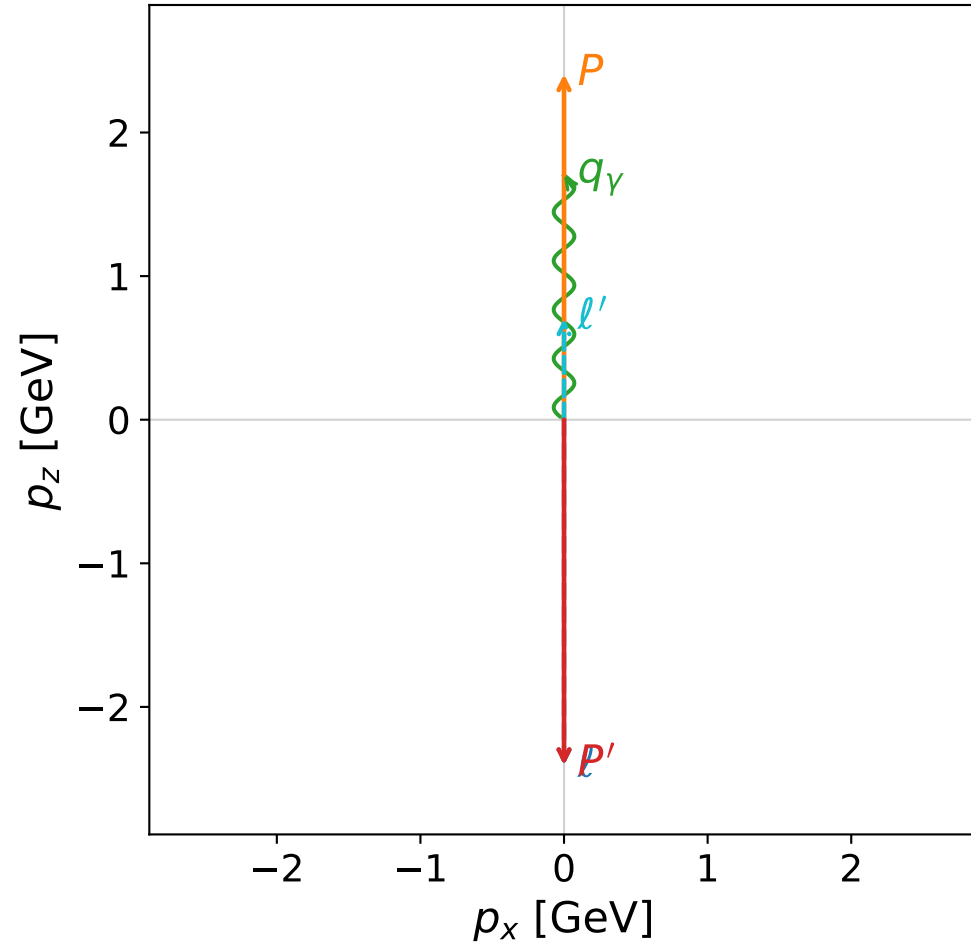
```
s=21.5586, sqrt(s)=4.64312, |P|=2.22681, |P'|=2.22681
theta_{p'}=3.14159, theta_Y=0, phi_{p'}=4.93115, phi_Y=3.73044
E_Y=1.09901, theta_{l'}=0, phi_{l'}=1.78956
Q^2=10.0456, x_B=0.496047, t=-19.8348, W^2=11.0856, y=0.97933
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.2268 p_x=  0.0000 p_y=  0.0000 p_z= -2.2268
P      E=  2.4163 p_x=  0.0000 p_y=  0.0000 p_z=  2.2268
l'     E=  1.1278 p_x= -0.0000 p_y=  0.0000 p_z=  1.1278
P'     E=  2.4163 p_x=  0.0000 p_y= -0.0000 p_z= -2.2268
q_Y    E=  1.0990 p_x= -0.0000 p_y= -0.0000 p_z=  1.0990
```

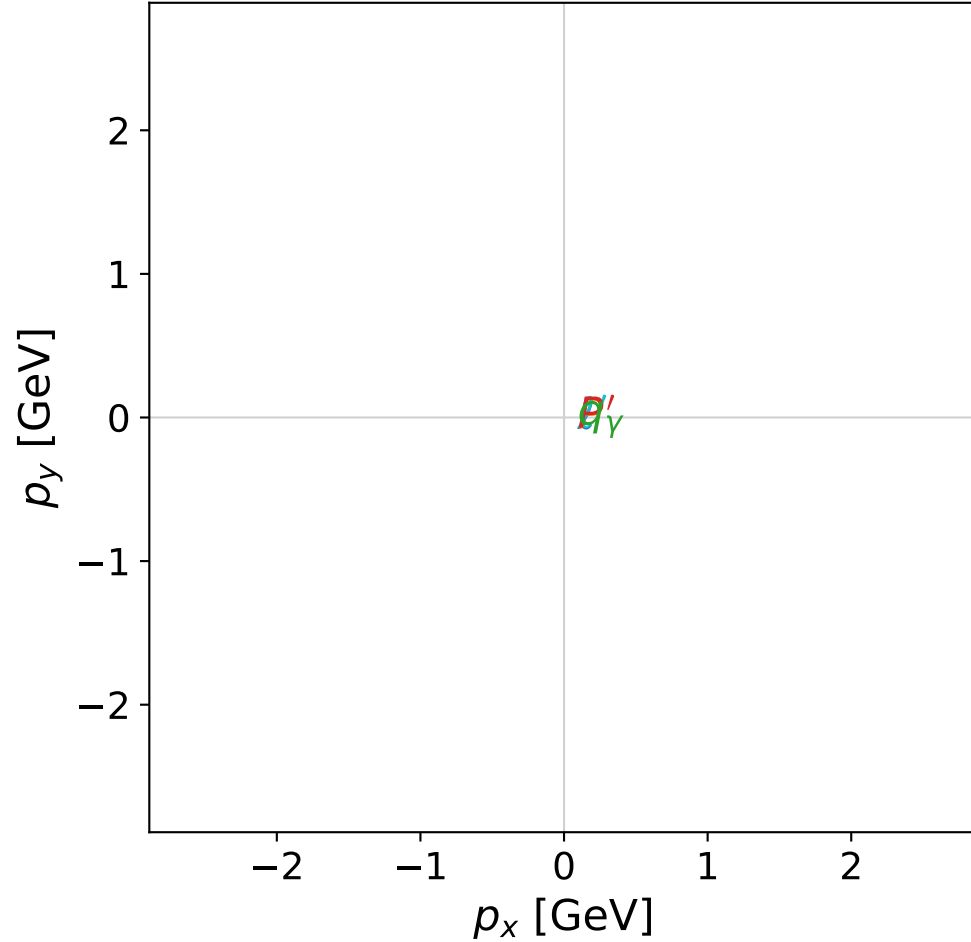


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



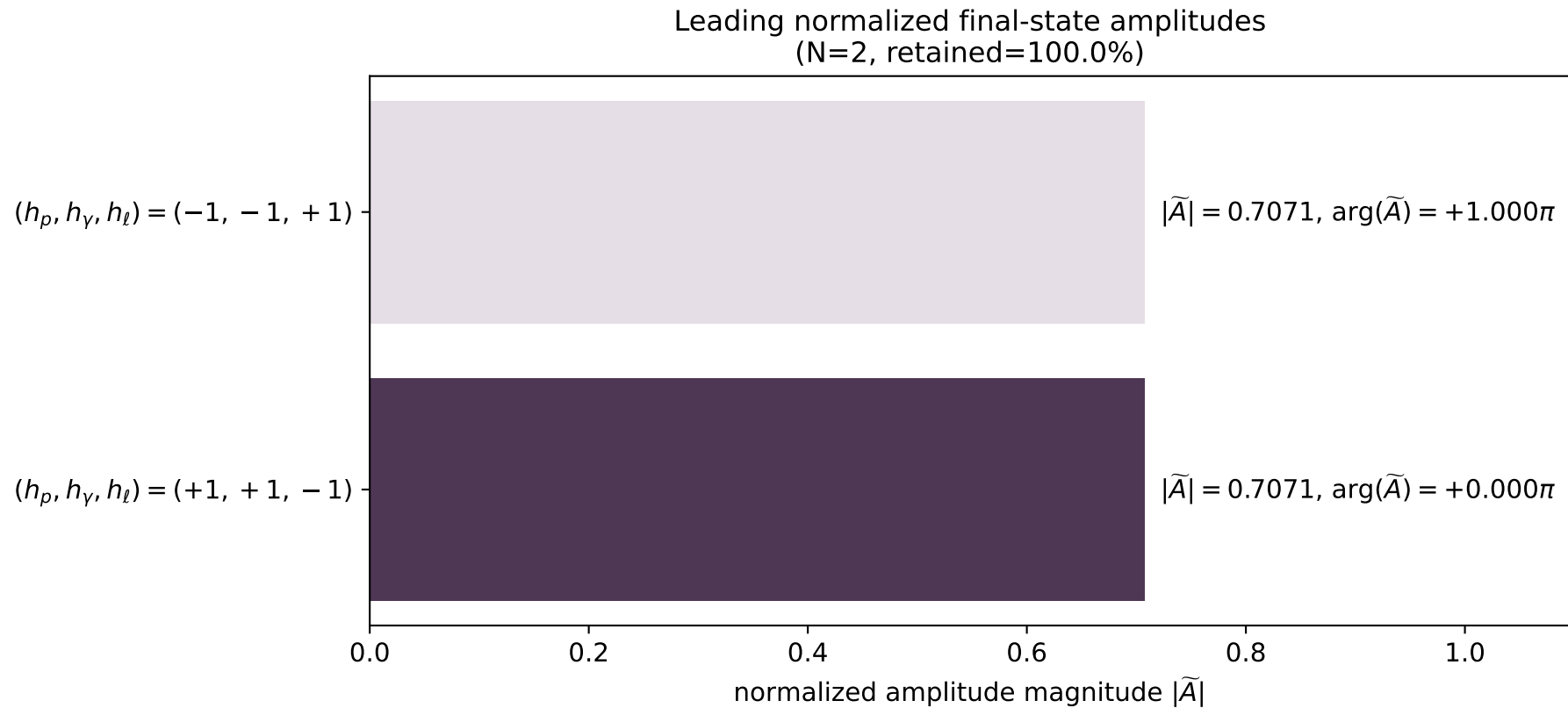
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_145
 $d_{\{\rm GHZ\}}=7.88665\text{e-}08$
 region: polarization_cluster_04_local_minimum_145
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0145
 lepton mixing angle: $\alpha_e=2.4645339$ rad
 incoming lepton: $-0.779419|+\rangle +0.626503|-\rangle$
 proton mixing angle: $\alpha_p=2.2478416$ rad
 incoming proton: $-0.626493|+\rangle +0.779427|-\rangle$

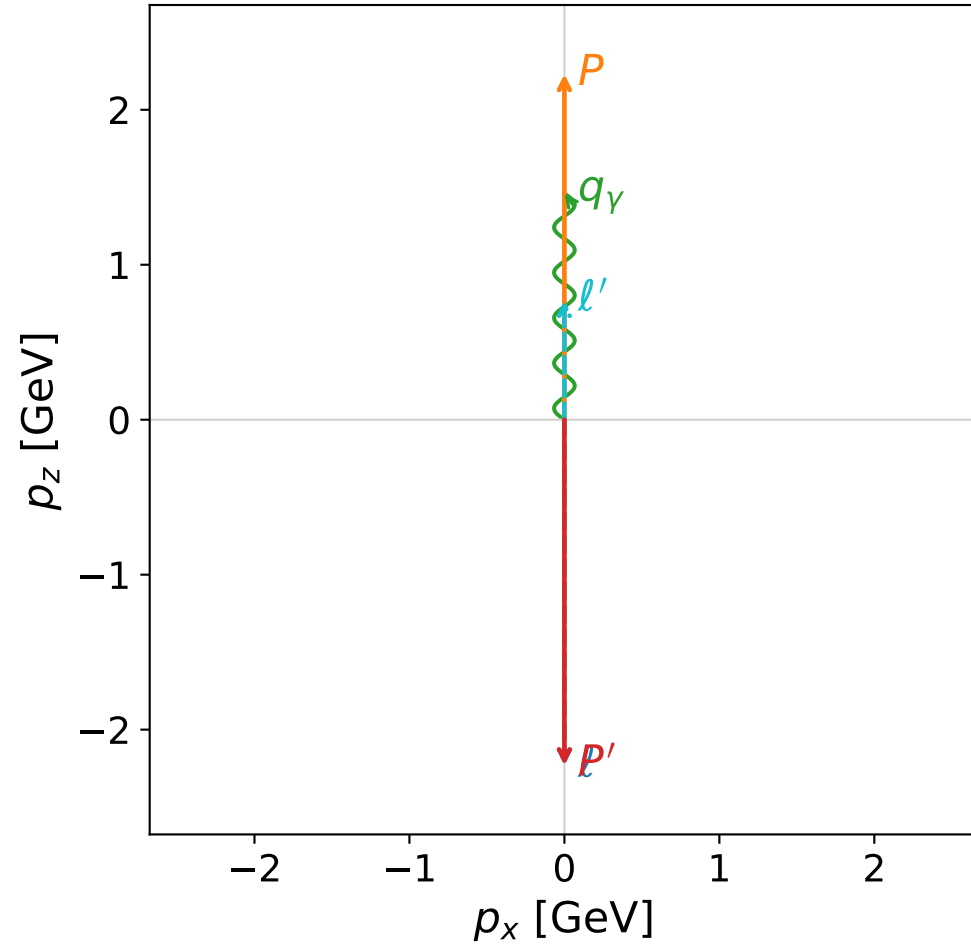
$s=24.8957$, $\sqrt{s}=4.98956$, $|\vec{P}|=2.40661$, $|\vec{P}'|=2.40661$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=2.62423$, $\phi_Y=3.19394$
 $E_Y=1.70223$, $\theta_{l'}=0$, $\phi_{l'}=5.76582$
 $Q^2=6.78069$, $x_B=0.285293$, $t=-23.1671$, $W^2=17.8666$, $y=0.989656$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4066$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 P $E= 2.5829$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4066$
 l' $E= 0.7044$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7044$
 P' $E= 2.5829$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4066$
 q_Y $E= 1.7022$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.7022$

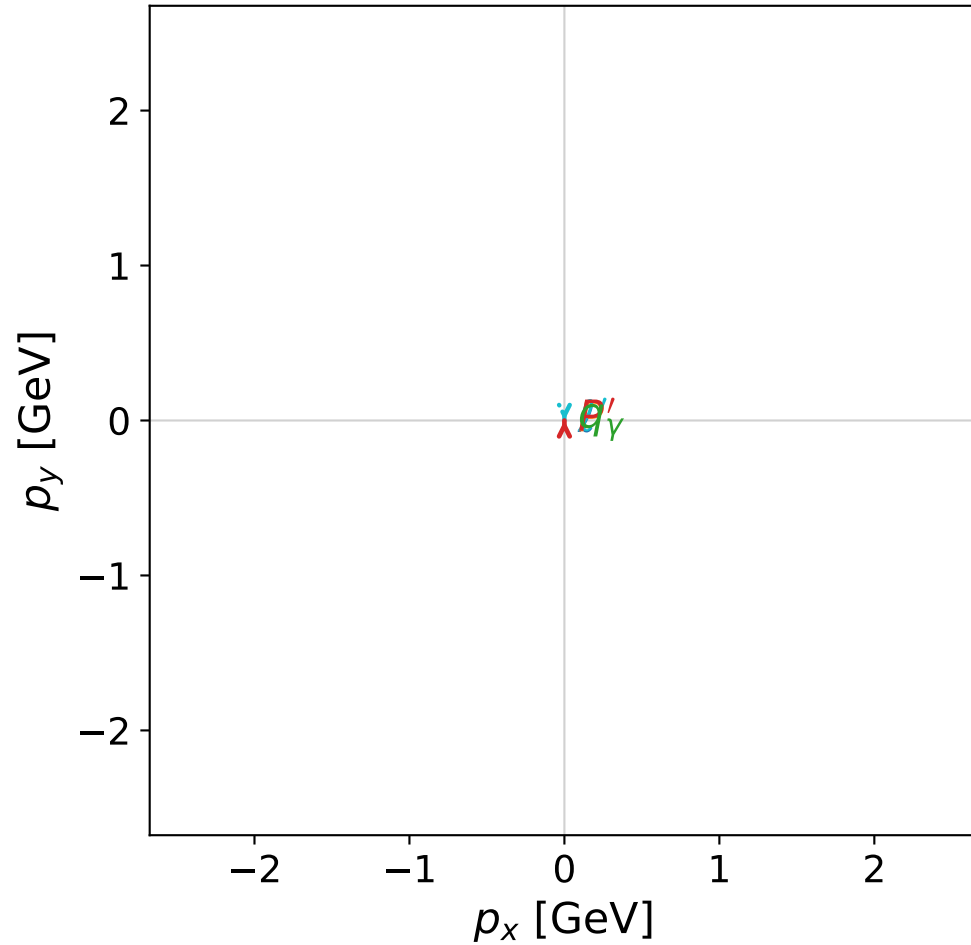


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



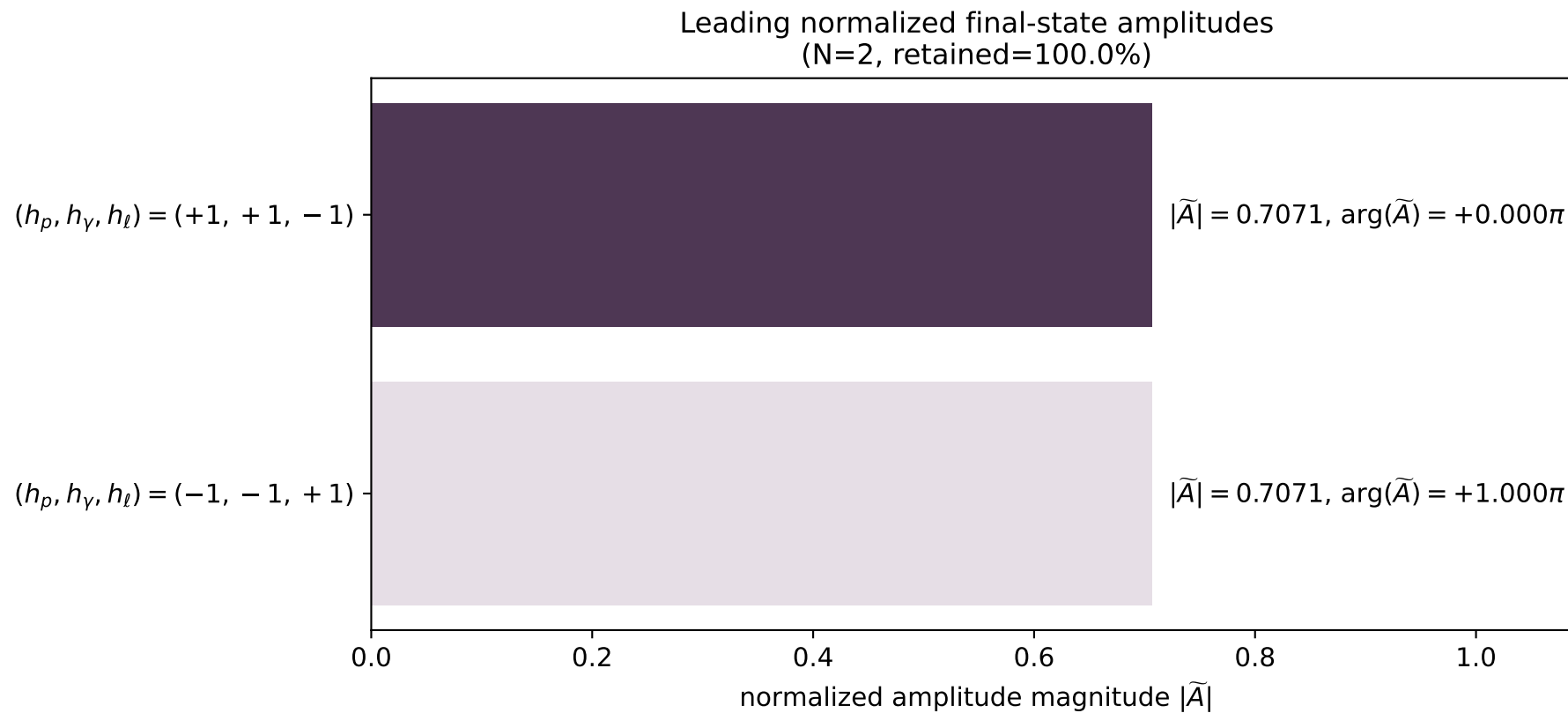
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_153
 $d_{\text{GHZ}}=9.35668\text{e-}08$
 region: polarization_cluster_04_local_minimum_153
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0153
 lepton mixing angle: $\alpha_e=2.6674747$ rad
 incoming lepton: $-0.889696|+\rangle +0.456554|-\rangle$
 proton mixing angle: $\alpha_p=2.0449223$ rad
 incoming proton: $-0.456561|+\rangle +0.889692|-\rangle$

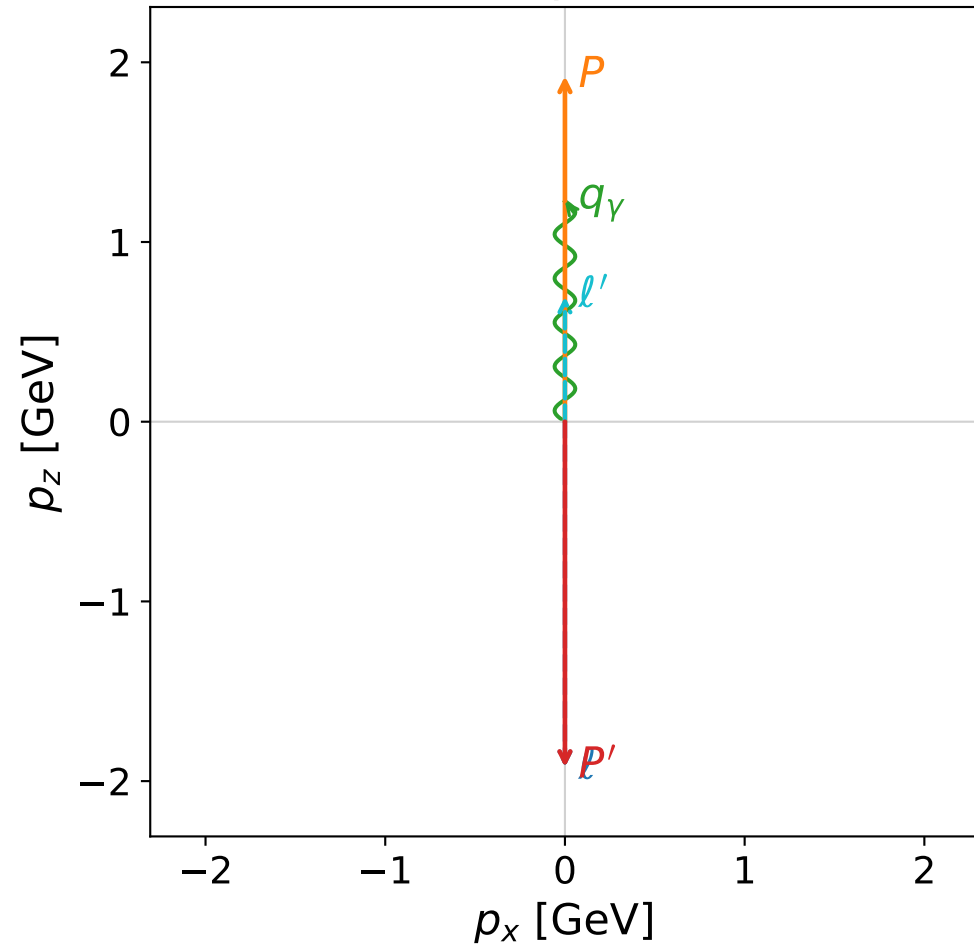
$s=21.6182$, $\sqrt{s}=4.64954$, $|\vec{P}|=2.23015$, $|\vec{P}'|=2.23015$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.53792$, $\phi_\gamma=3.63958$
 $E_\gamma=1.45997$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.67951$
 $Q^2=6.8705$, $x_B=0.336017$, $t=-19.8944$, $W^2=14.4563$, $y=0.985945$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2302$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 P $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2302$
 ℓ' $E= 0.7702$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.7702$
 P' $E= 2.4194$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2302$
 q_γ $E= 1.4600$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4600$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

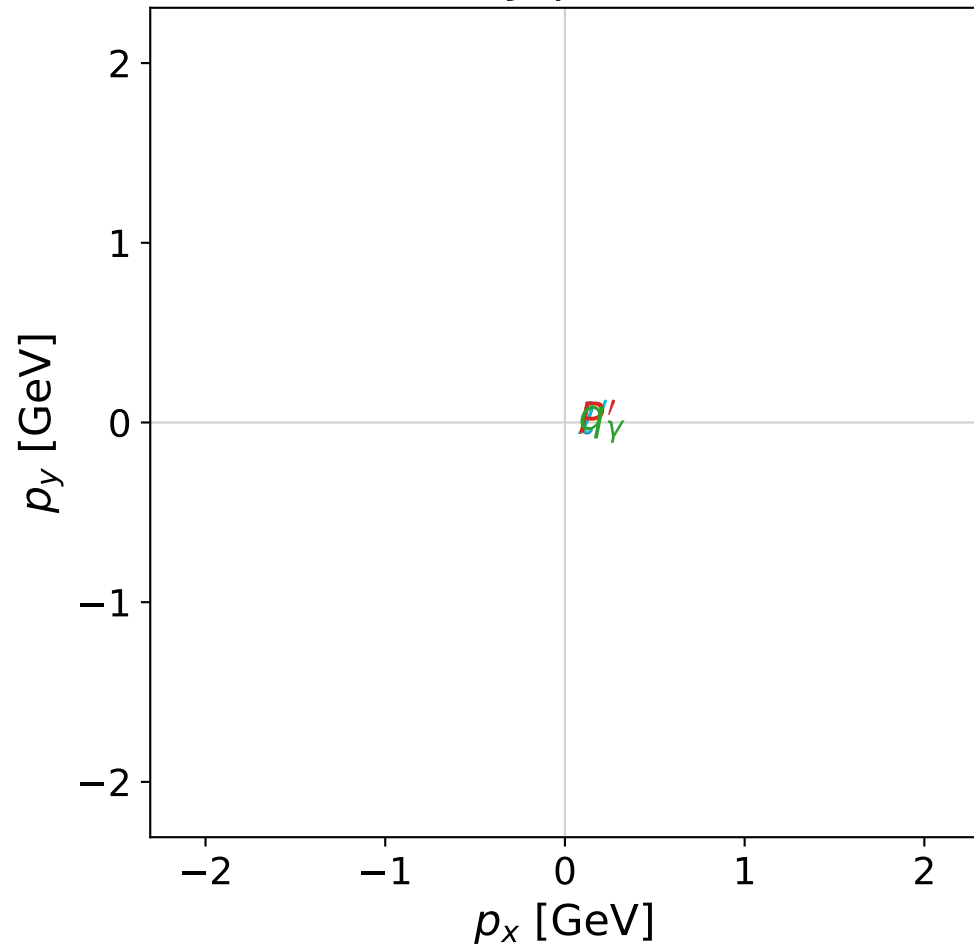


dghz_mixing_angles_polarization_04_local_minimum_155
 $d_{\text{GHZ}}=9.76885\text{e-}08$
 region: polarization_cluster_04_local_minimum_155
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0155
 lepton mixing angle: $\alpha_e=2.379922$ rad
 incoming lepton: $-0.723684|+\rangle +0.690131|-\rangle$
 proton mixing angle: $\alpha_p=2.3324913$ rad
 incoming proton: $-0.690149|+\rangle +0.723667|-\rangle$

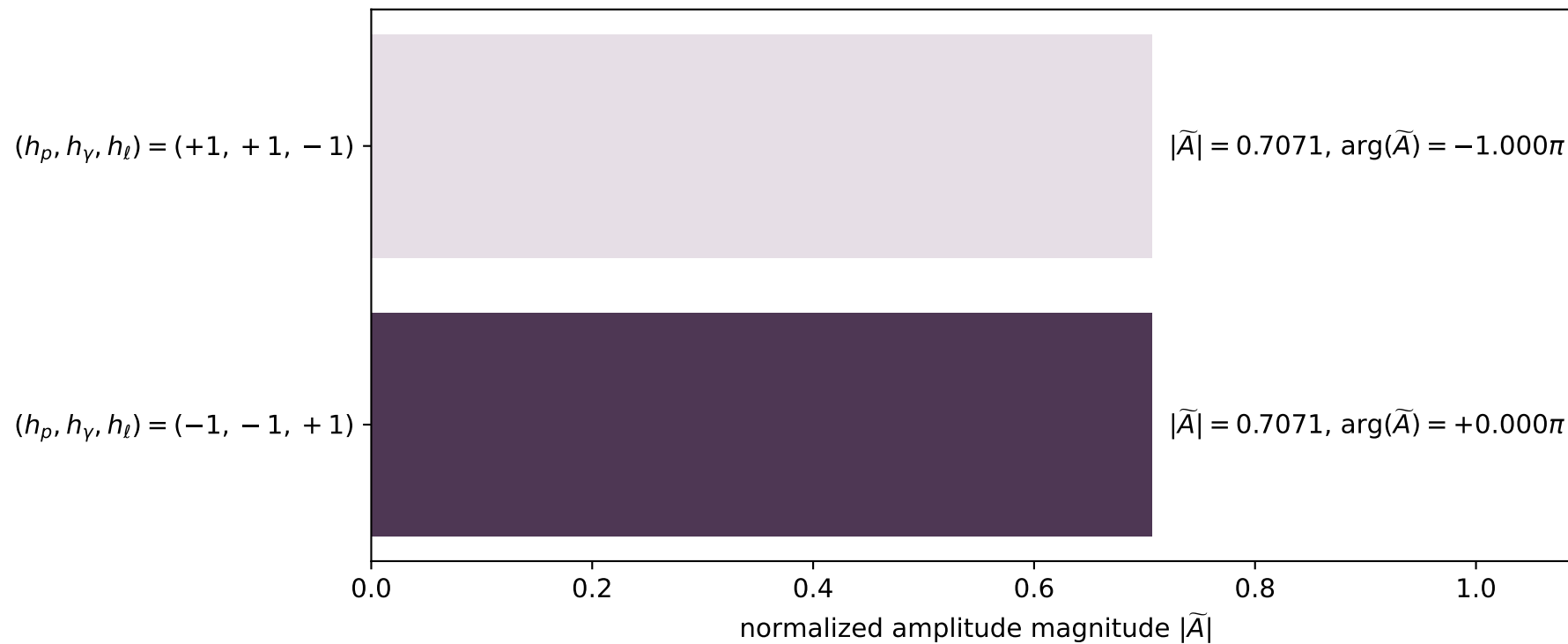
$s=16.5062$, $\sqrt{s}=4.06278$, $|\vec{P}|=1.92311$, $|\vec{P}'|=1.92311$
 $\theta_{P'}=3.14159$, $\theta_\gamma=0$, $\phi_{P'}=4.19286$, $\phi_\gamma=0.182475$
 $E_\gamma=1.22715$, $\theta_{\ell'}=0$, $\phi_{\ell'}=1.05127$
 $Q^2=5.35358$, $x_B=0.34934$, $t=-14.7934$, $W^2=10.8511$, $y=0.98071$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9231$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9231$
 P $E= 2.1397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9231$
 ℓ' $E= 0.6960$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6960$
 P' $E= 2.1397$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.9231$
 q_γ $E= 1.2272$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2272$

x--y plane

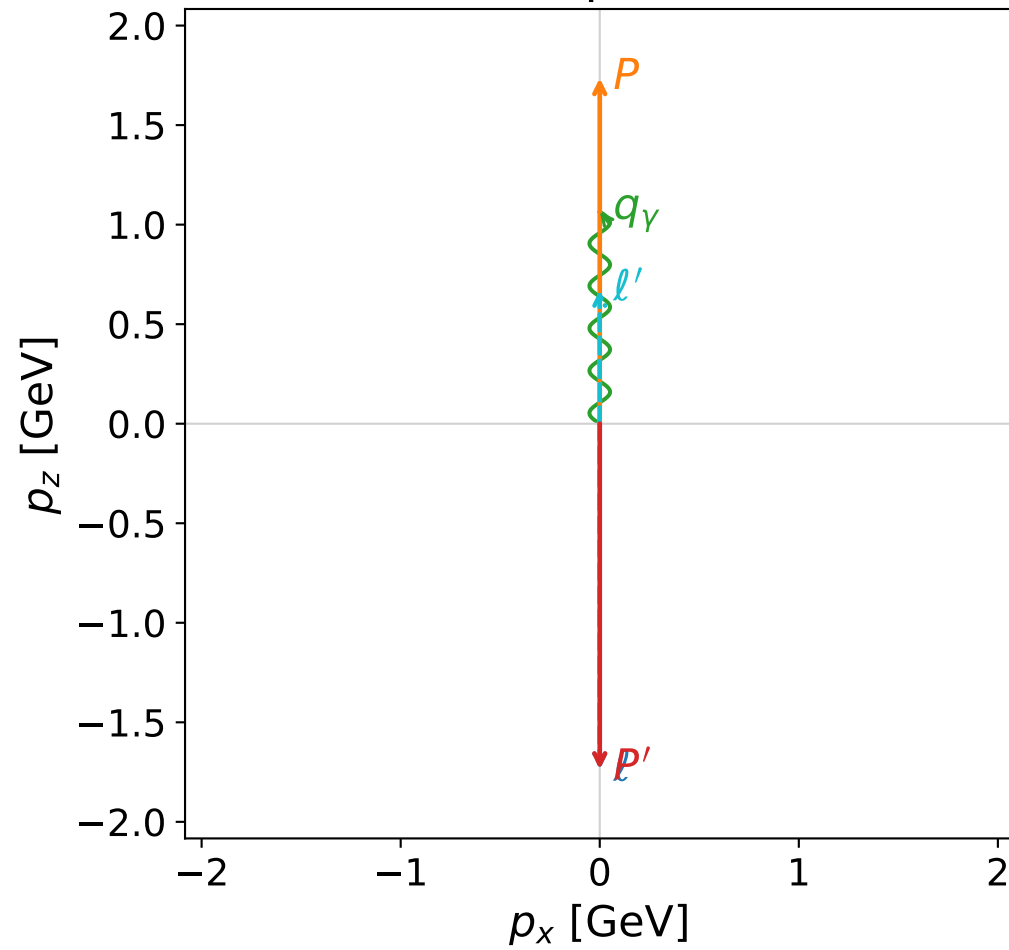


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

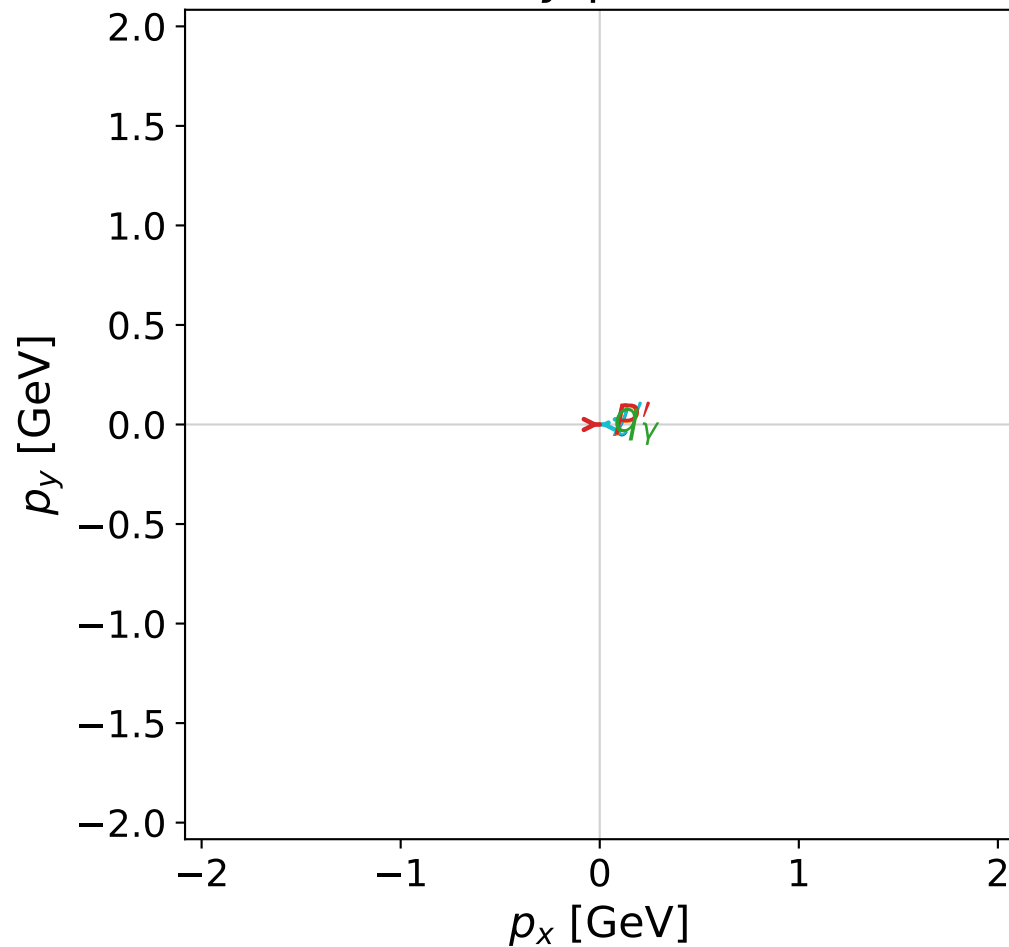


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



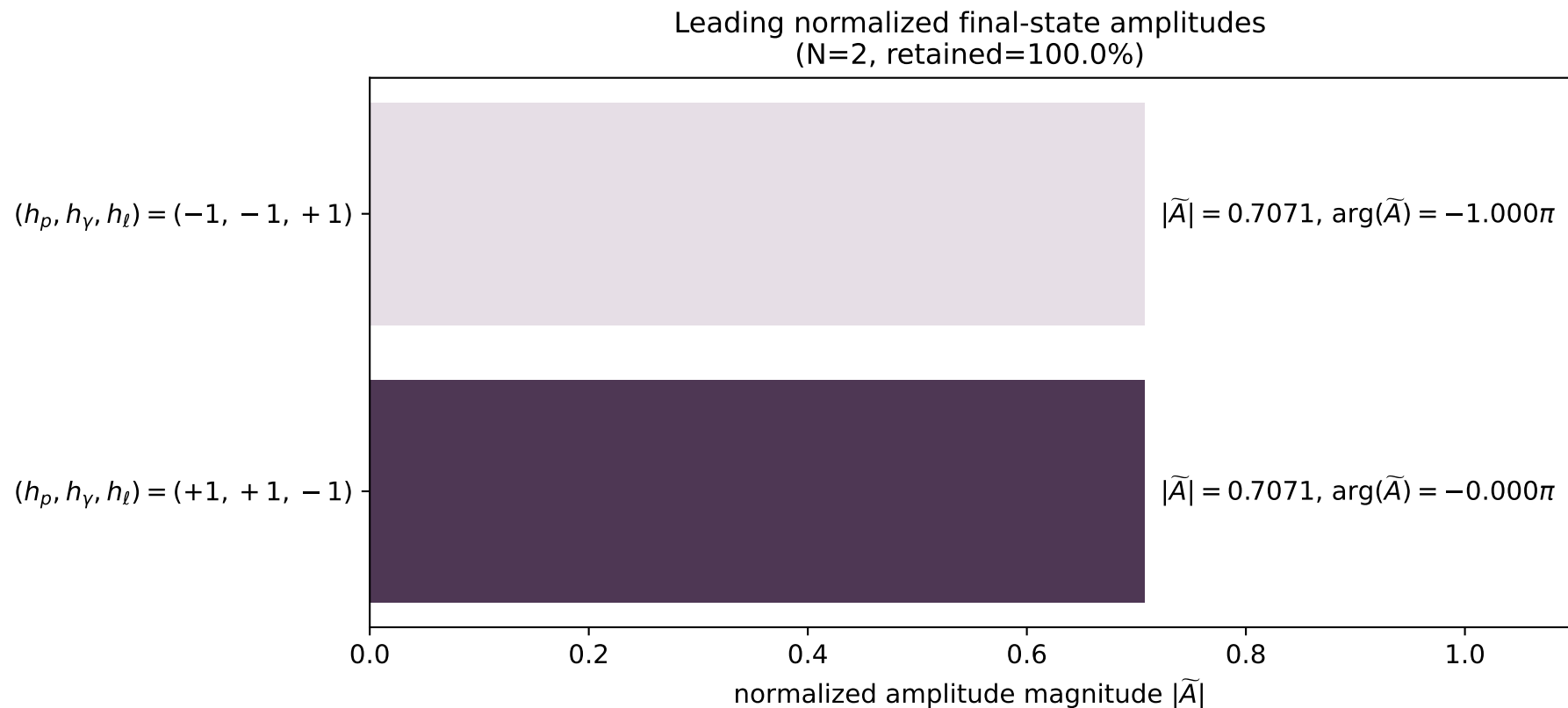
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_162
 $d_{\{\rm GHZ\}}=1.14455\text{e-}07$
 region: polarization_cluster_04_local_minimum_162
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0162
 lepton mixing angle: $\alpha_e=2.254464$ rad
 incoming lepton: $-0.631641|+\rangle +0.775261|-\rangle$
 proton mixing angle: $\alpha_p=2.4579197$ rad
 incoming proton: $-0.775258|+\rangle +0.631645|-\rangle$

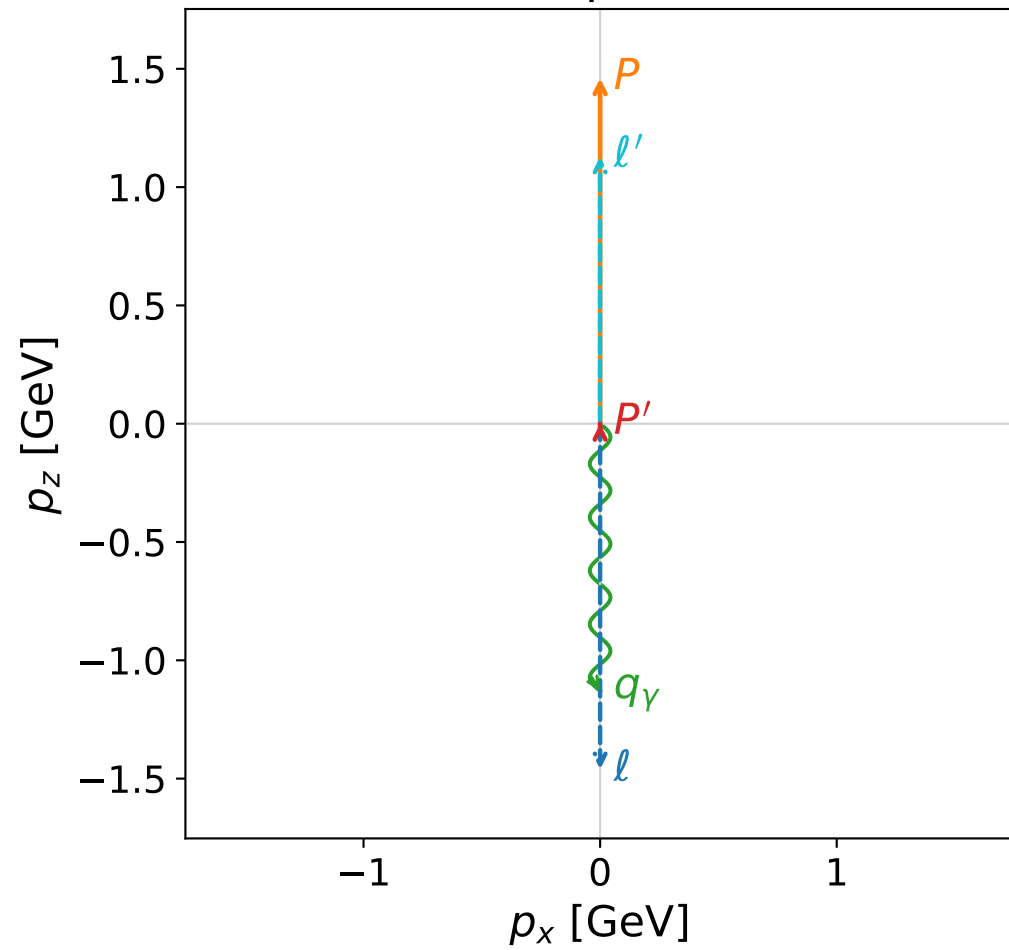
$s=13.7588$, $\sqrt{s}=3.70928$, $|\vec{P}|=1.73604$, $|\vec{P}'|=1.73604$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.143001$, $\phi_\gamma=2.52234$
 $E_\gamma=1.06513$, $\theta_{l'}=0$, $\phi_{l'}=3.28459$
 $Q^2=4.65894$, $x_B=0.370915$, $t=-12.0554$, $W^2=8.78157$, $y=0.975287$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7360$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 P $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7360$
 ℓ' $E= 0.6709$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.6709$
 P' $E= 1.9732$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7360$
 q_γ $E= 1.0651$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.0651$

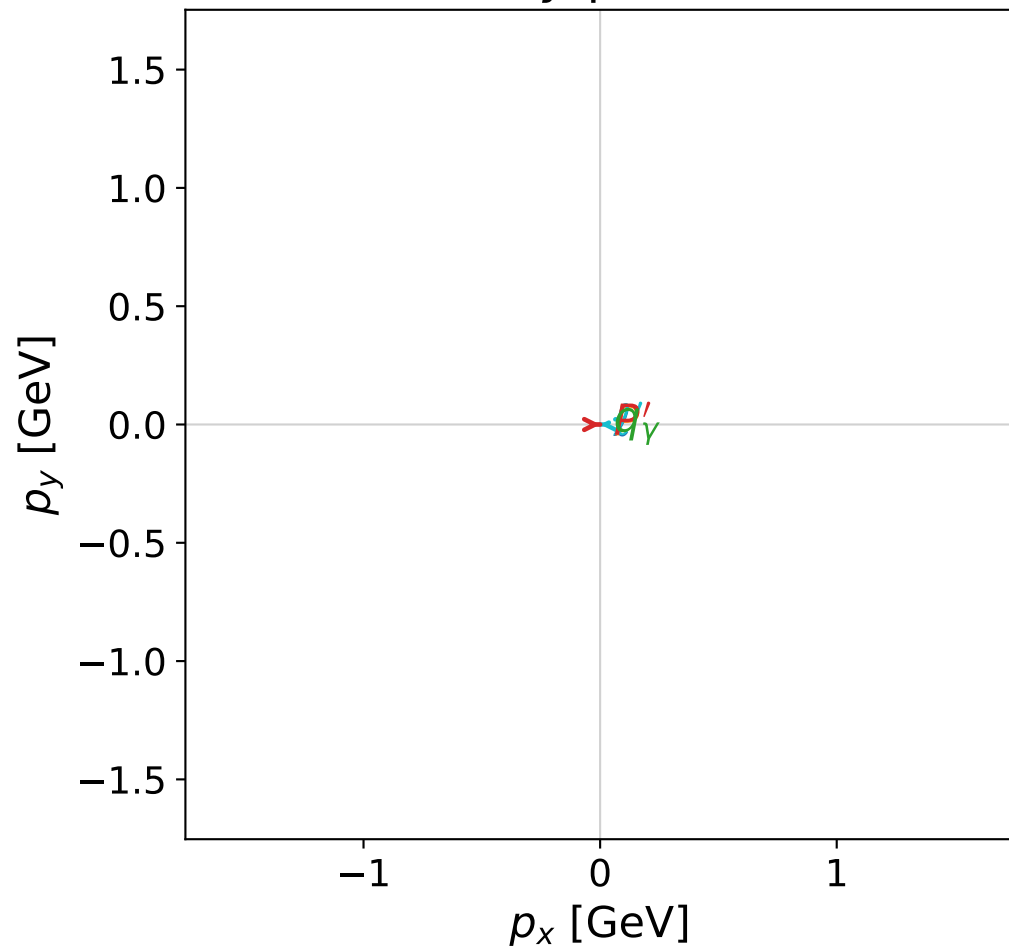


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

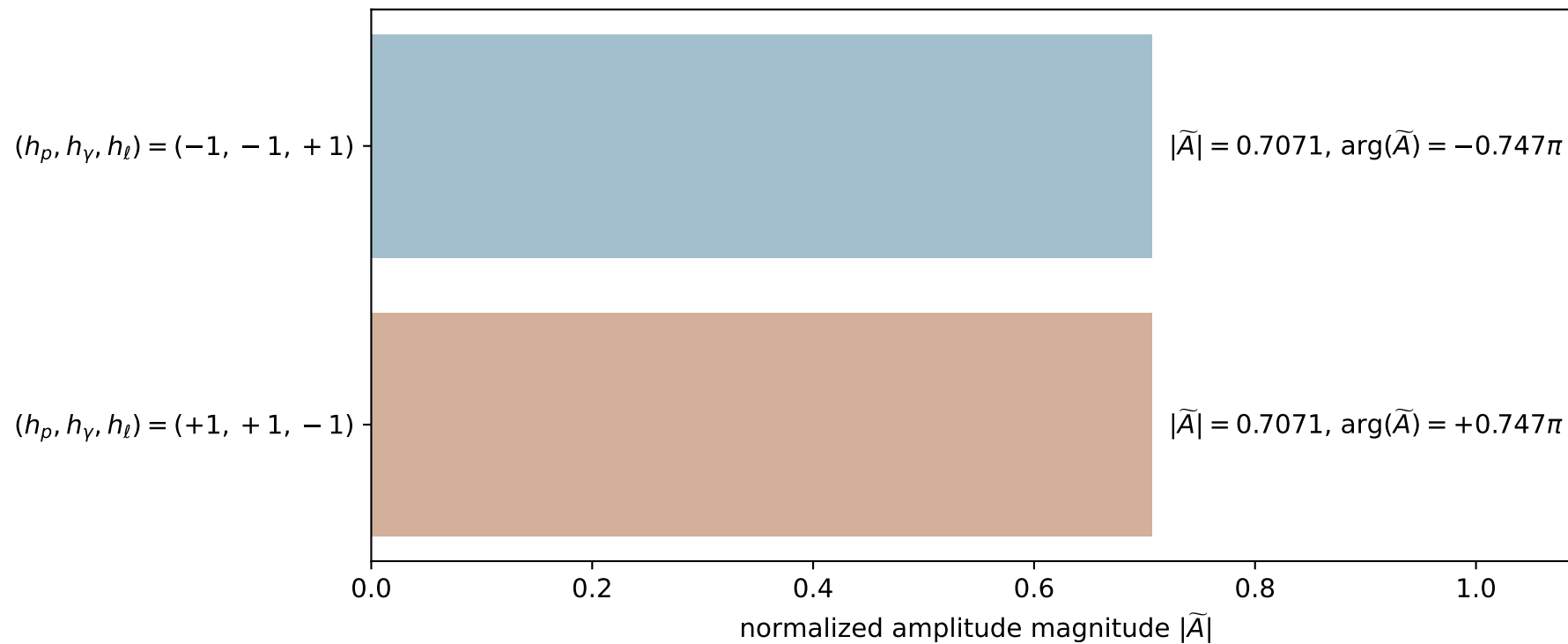


dghz_mixing_angles_polarization_04_local_minimum_166
 $d_{\text{GHZ}}=1.3556\text{e-}07$
 region: polarization_cluster_04_local_minimum_166
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0166
 lepton mixing angle: $\alpha_e=1.8819764$ rad
 incoming lepton: $-0.306182|+\rangle + 0.951973|-\rangle$
 proton mixing angle: $\alpha_p=1.8819874$ rad
 incoming proton: $-0.306193|+\rangle + 0.95197|-\rangle$

$s=10.2184$, $\sqrt{s}=3.19662$, $|\vec{P}|=1.46069$, $|\vec{P}'|=0.000294288$
 $\theta_{P'}=0.000613329$, $\theta_Y=3.14159$, $\phi_{P'}=0.00132073$, $\phi_Y=3.93698$
 $E_Y=1.12946$, $\theta_{\ell'}=1.59797\text{e-}07$, $\phi_{\ell'}=3.14291$
 $Q^2=6.59744$, $x_B=0.75685$, $t=-1.49606$, $W^2=2.99937$, $y=0.933439$

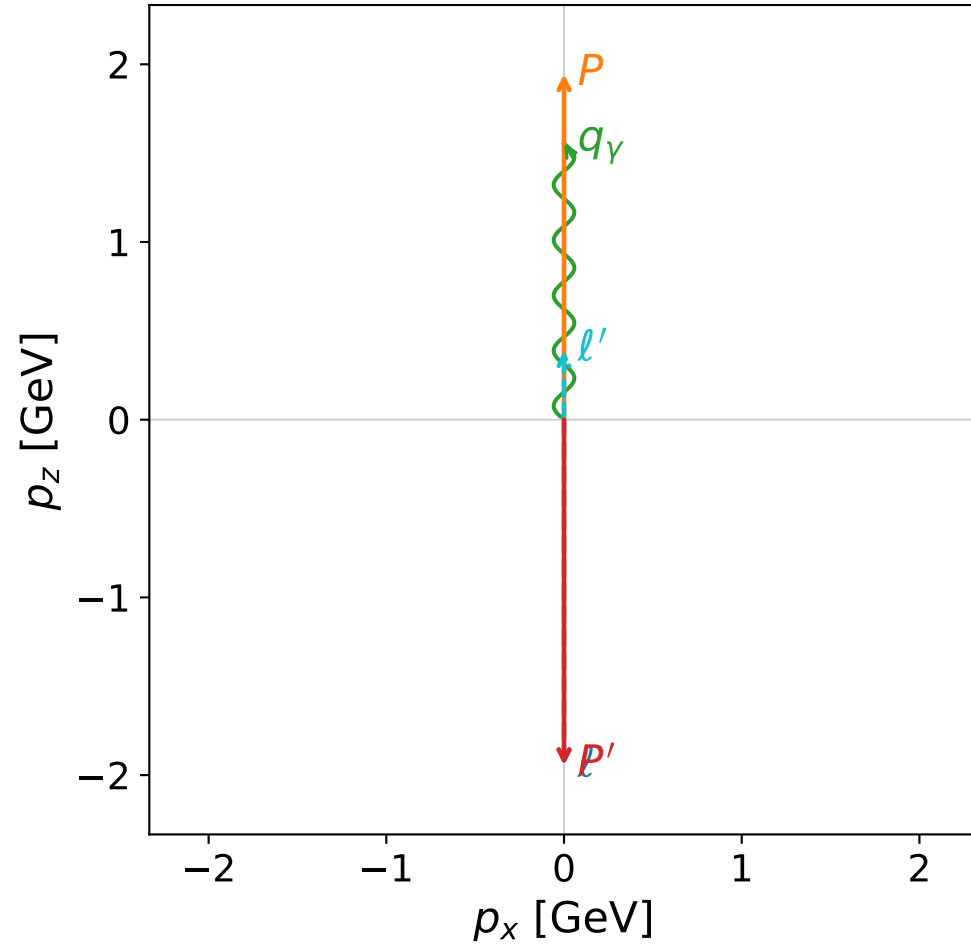
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.4607$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4607$
 P $E= 1.7359$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4607$
 ℓ' $E= 1.1292$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.1292$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0003$
 q_Y $E= 1.1295$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.1295$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

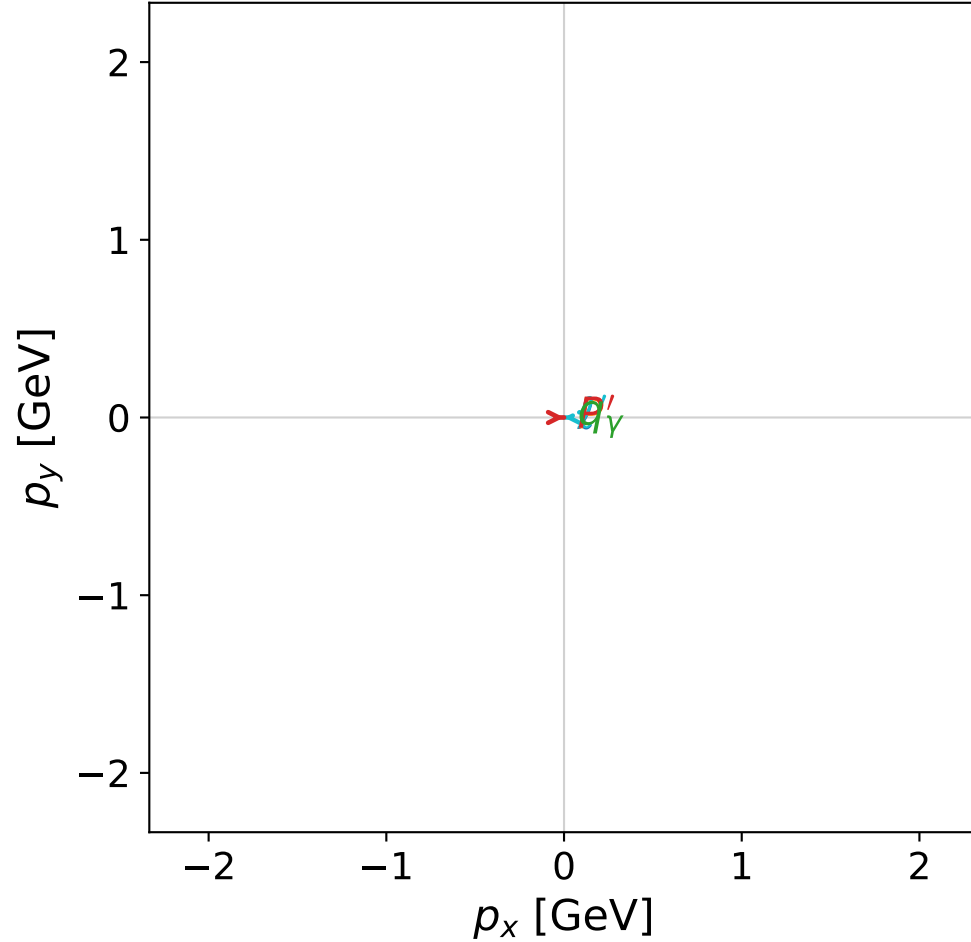


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



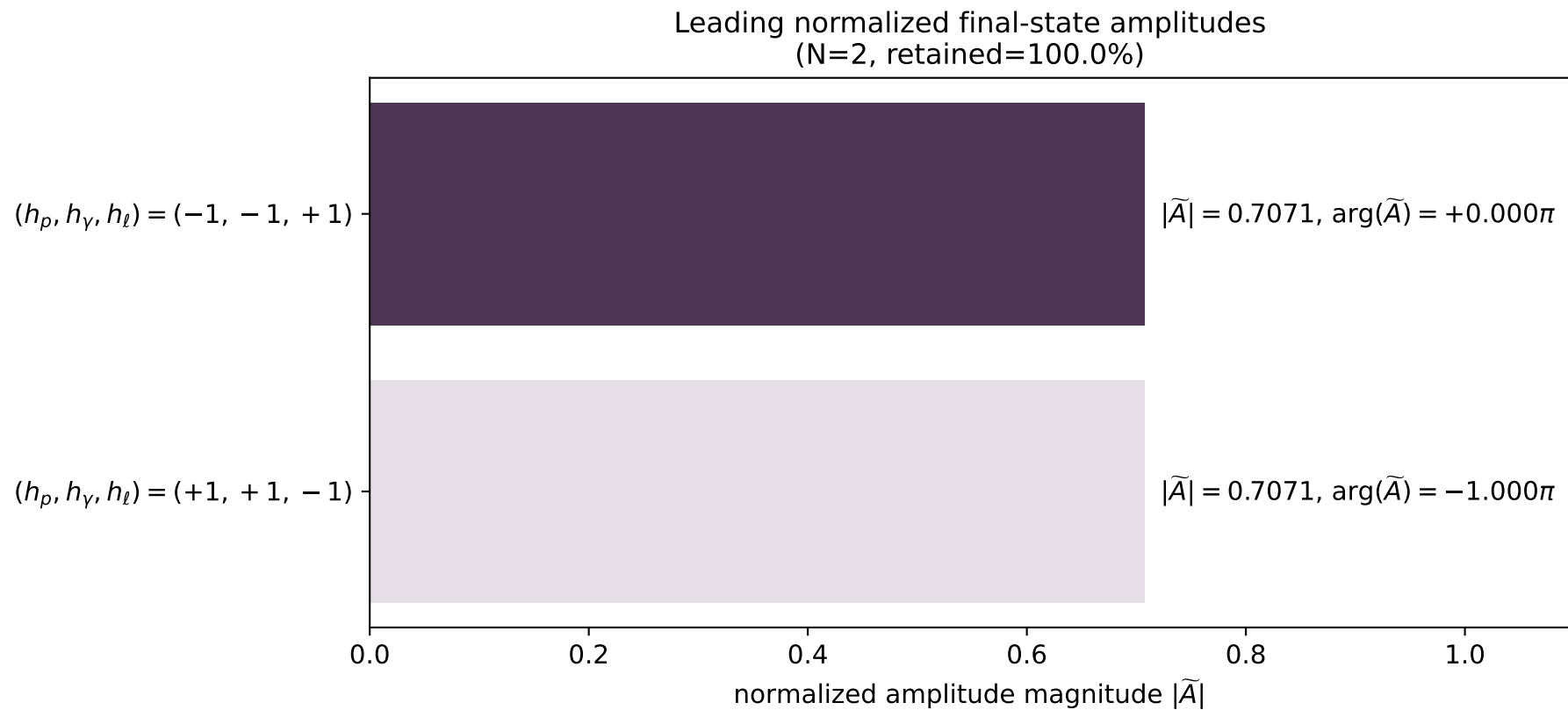
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_169
 $d_{\{\text{rm GHZ}\}}=1.74025\text{e-}07$
 region: polarization_cluster_04_local_minimum_169
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0169
 lepton mixing angle: $\alpha_e=2.2827776$ rad
 incoming lepton: $-0.653335|+\rangle +0.757069|-\rangle$
 proton mixing angle: $\alpha_p=2.4296097$ rad
 incoming proton: $-0.757068|+\rangle +0.653336|-\rangle$

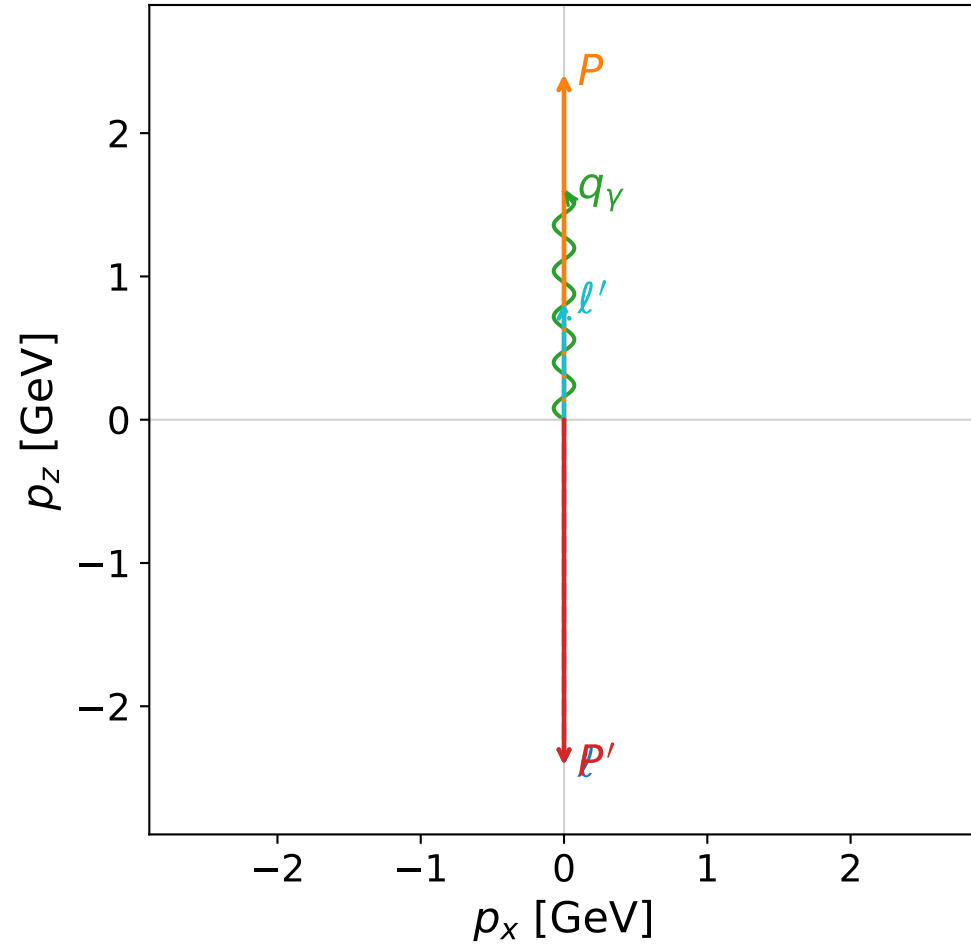
$s=16.843$, $\sqrt{s}=4.10403$, $|\vec{P}|=1.94482$, $|\vec{P}'|=1.94482$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=6.13063$, $\phi_\gamma=0.266108$
 $E_\gamma=1.55486$, $\theta_{\ell'}=0$, $\phi_{\ell'}=2.98904$
 $Q^2=3.03359$, $x_B=0.192048$, $t=-15.1293$, $W^2=13.6422$, $y=0.989526$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9448$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9448$
 P $E= 2.1592$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9448$
 ℓ' $E= 0.3900$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.3900$
 P' $E= 2.1592$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9448$
 q_γ $E= 1.5549$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5549$

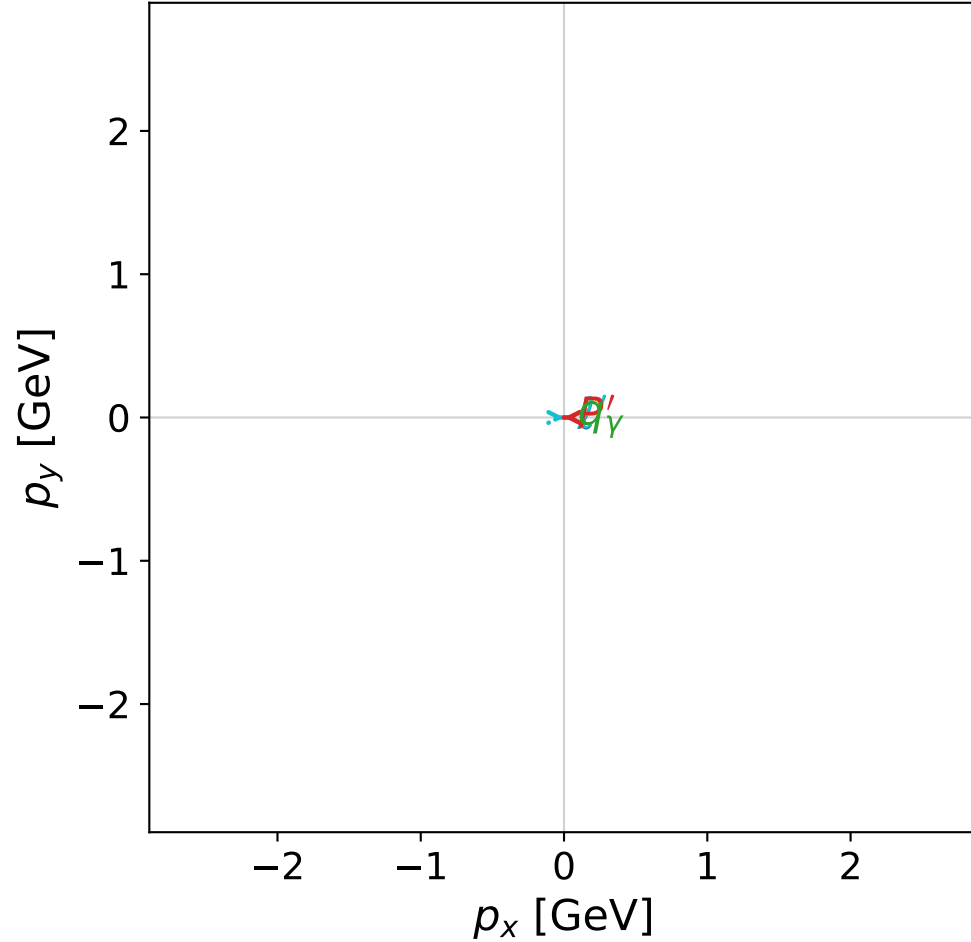


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



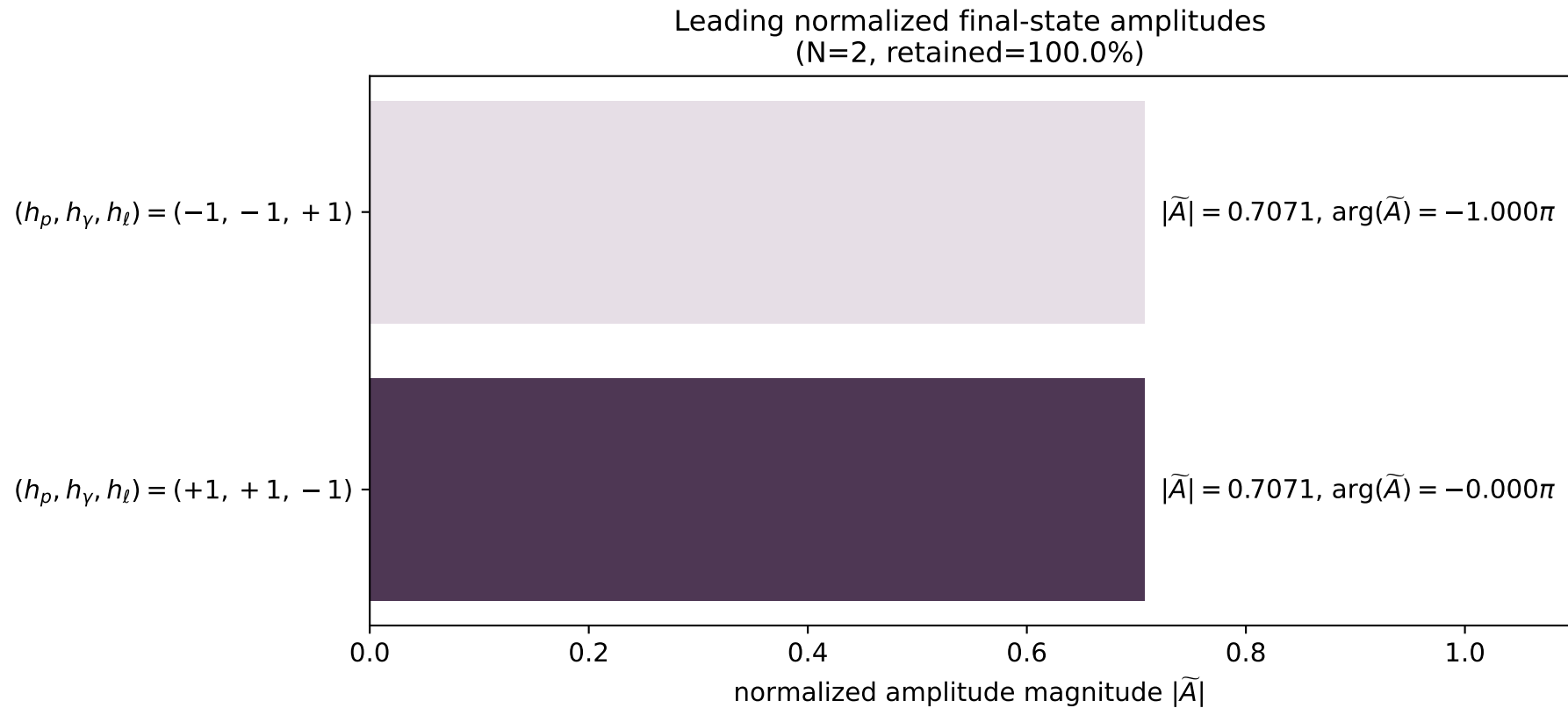
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_176
 $d_{\{\rm GHZ\}}=3.49411\text{e-}07$
 region: polarization_cluster_04_local_minimum_176
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0176
 lepton mixing angle: $\alpha_e=3.0459519$ rad
 incoming lepton: $-0.99543|+\rangle +0.095495|-\rangle$
 proton mixing angle: $\alpha_p=1.6664366$ rad
 incoming proton: $-0.0954945|+\rangle +0.99543|-\rangle$

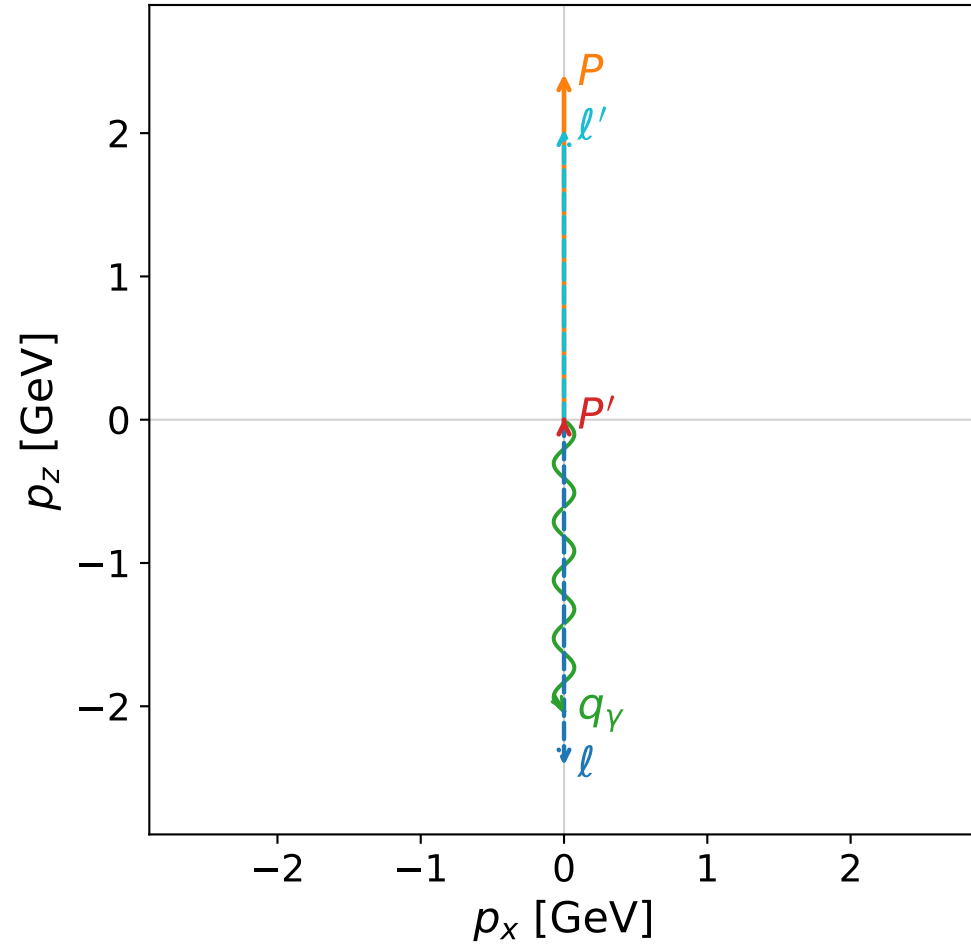
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.3165$, $\phi_\gamma=2.32512$
 $E_\gamma=1.59717$, $\theta_{l'}=0$, $\phi_{l'}=0.17491$
 $Q^2=7.86165$, $x_B=0.329859$, $t=-23.2713$, $W^2=16.8516$, $y=0.988111$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.8148$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.8148$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 1.5972$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5972$

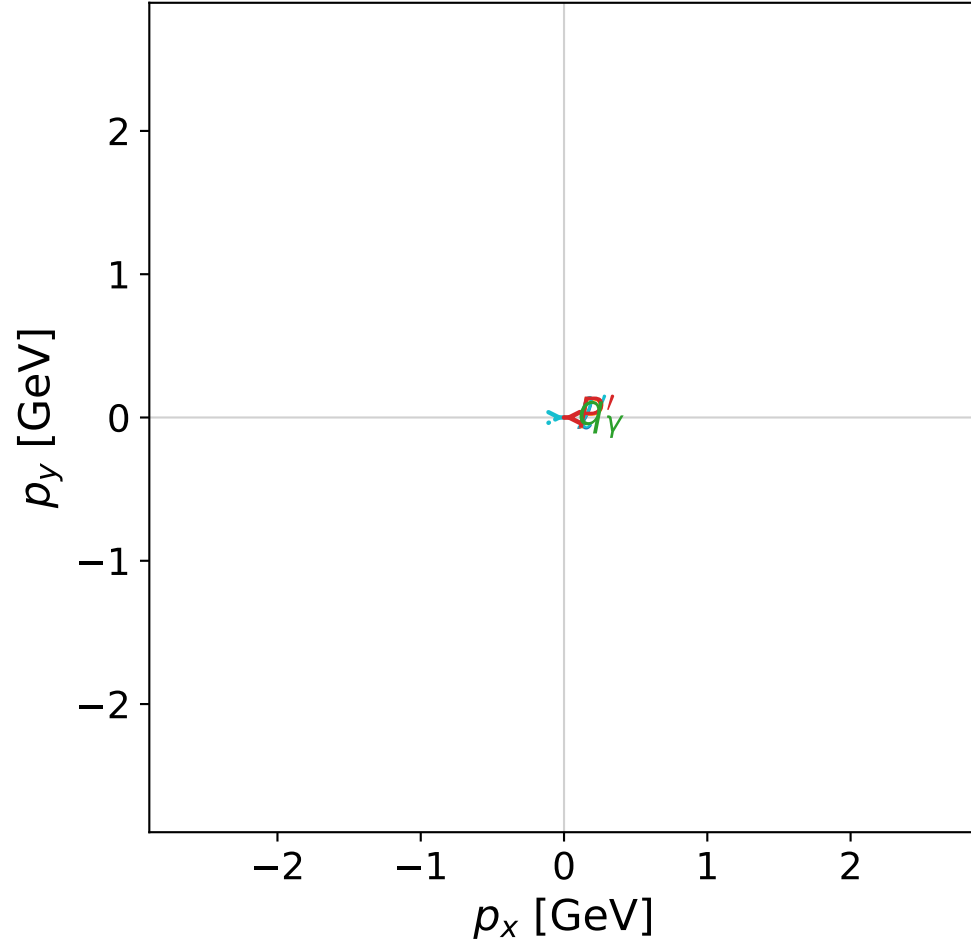


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



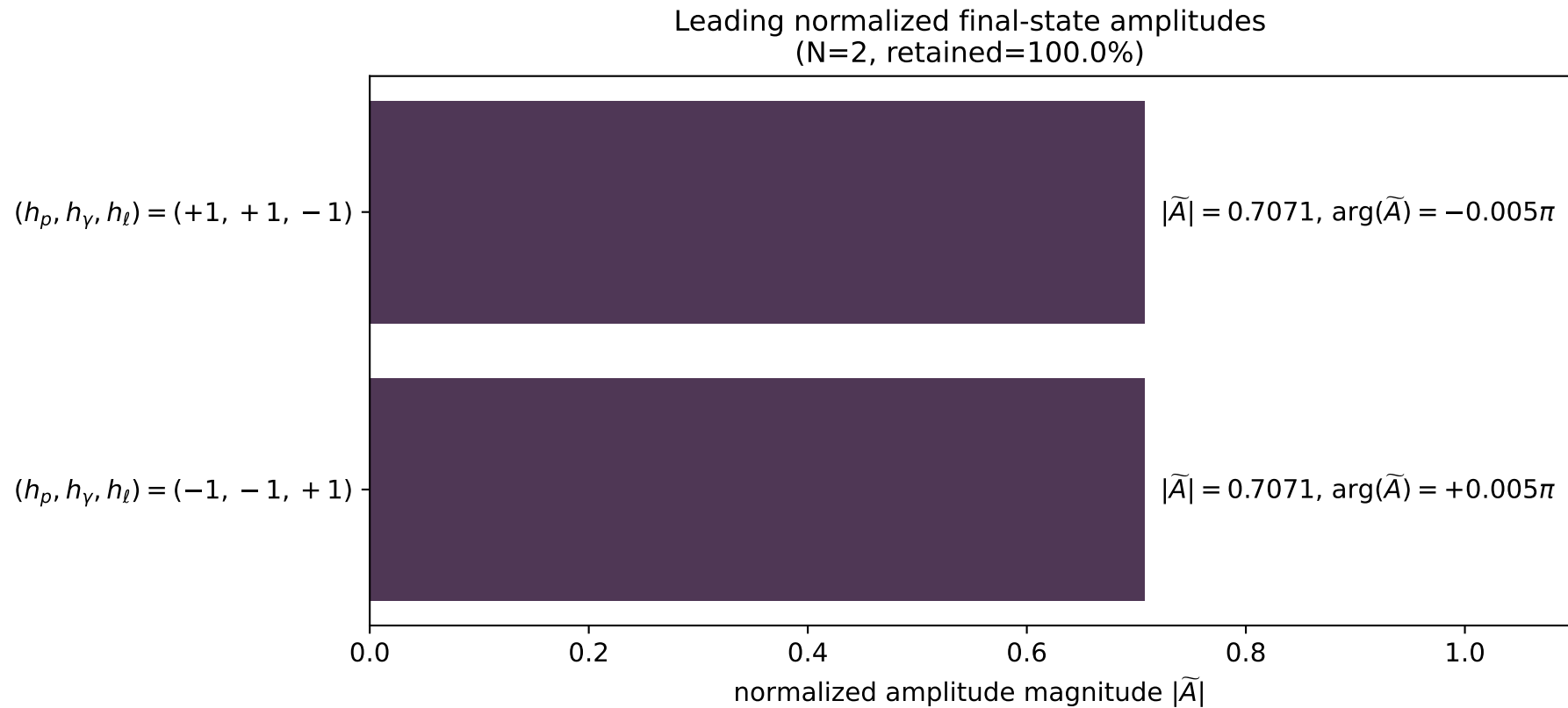
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_177
 $d_{\{\text{rm GHZ}\}}=4.14871\text{e-}07$
 region: polarization_cluster_04_local_minimum_177
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0177
 lepton mixing angle: $\alpha_e=3.1093625$ rad
 incoming lepton: $-0.999481|+\rangle +0.0322245|-\rangle$
 proton mixing angle: $\alpha_p=3.1093621$ rad
 incoming proton: $-0.999481|+\rangle +0.0322249|-\rangle$

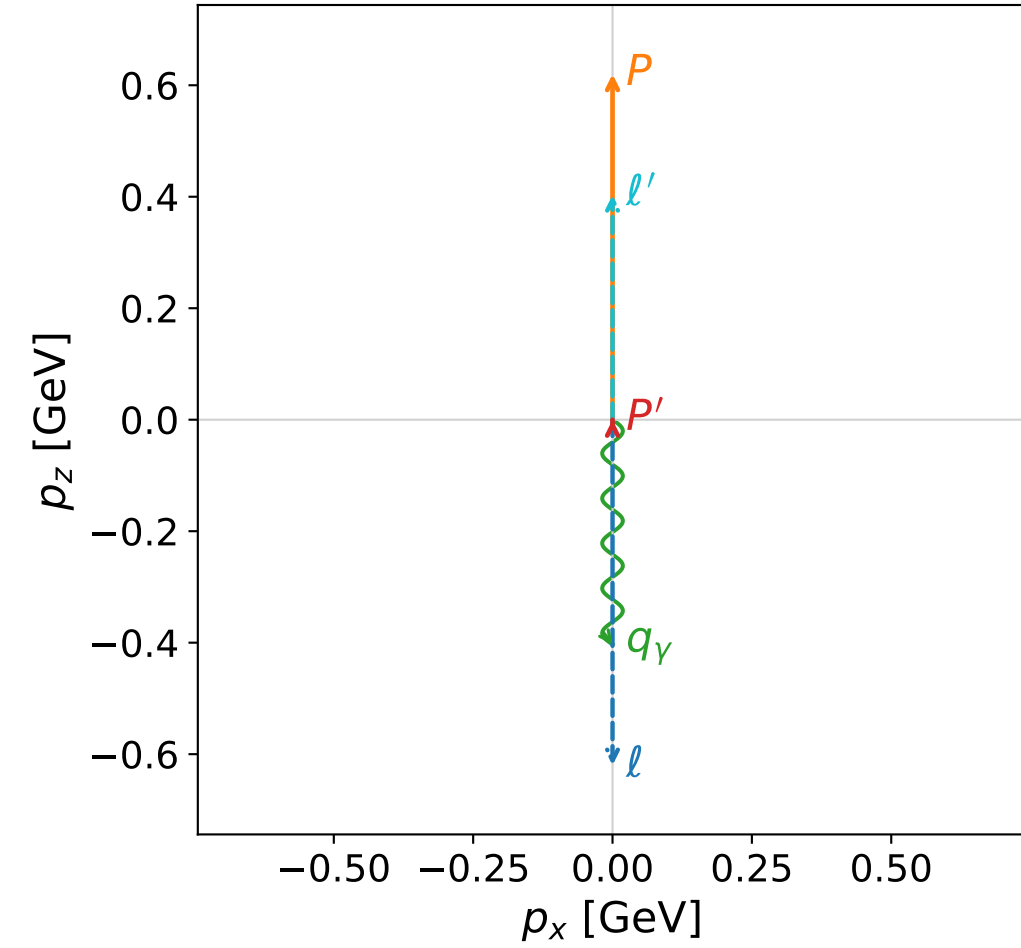
$s=24.9987$, $\sqrt{s}=4.99987$, $|\vec{P}|=2.41195$, $|\vec{P}'|=0.00931426$
 $\theta_{P'}=0.00233638$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14277$, $\phi_\gamma=0.0168325$
 $E_\gamma=2.03557$, $\theta_{\ell'}=1.07398\text{e-}05$, $\phi_{\ell'}=0.00117805$
 $Q^2=19.5489$, $x_B=0.835218$, $t=-3.05056$, $W^2=4.73668$, $y=0.970433$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0263$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0263$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0093$
 q_γ $E= 2.0356$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0356$

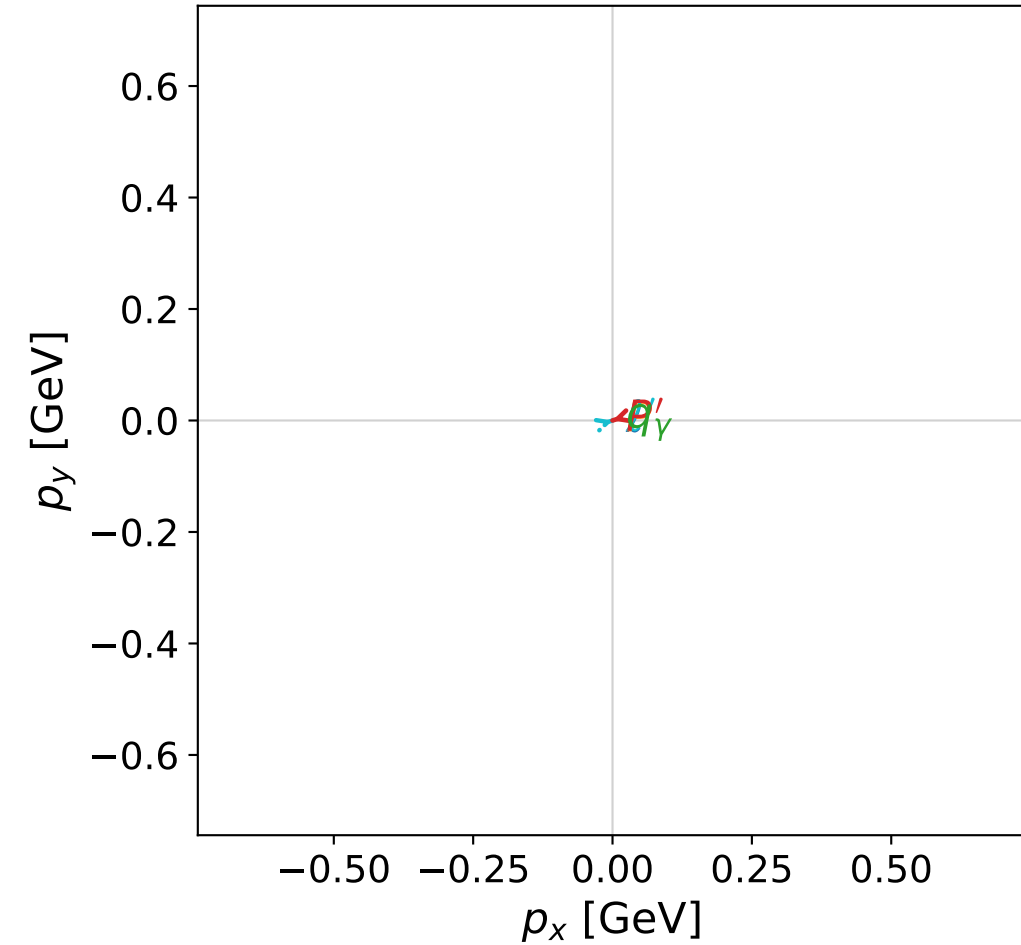


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

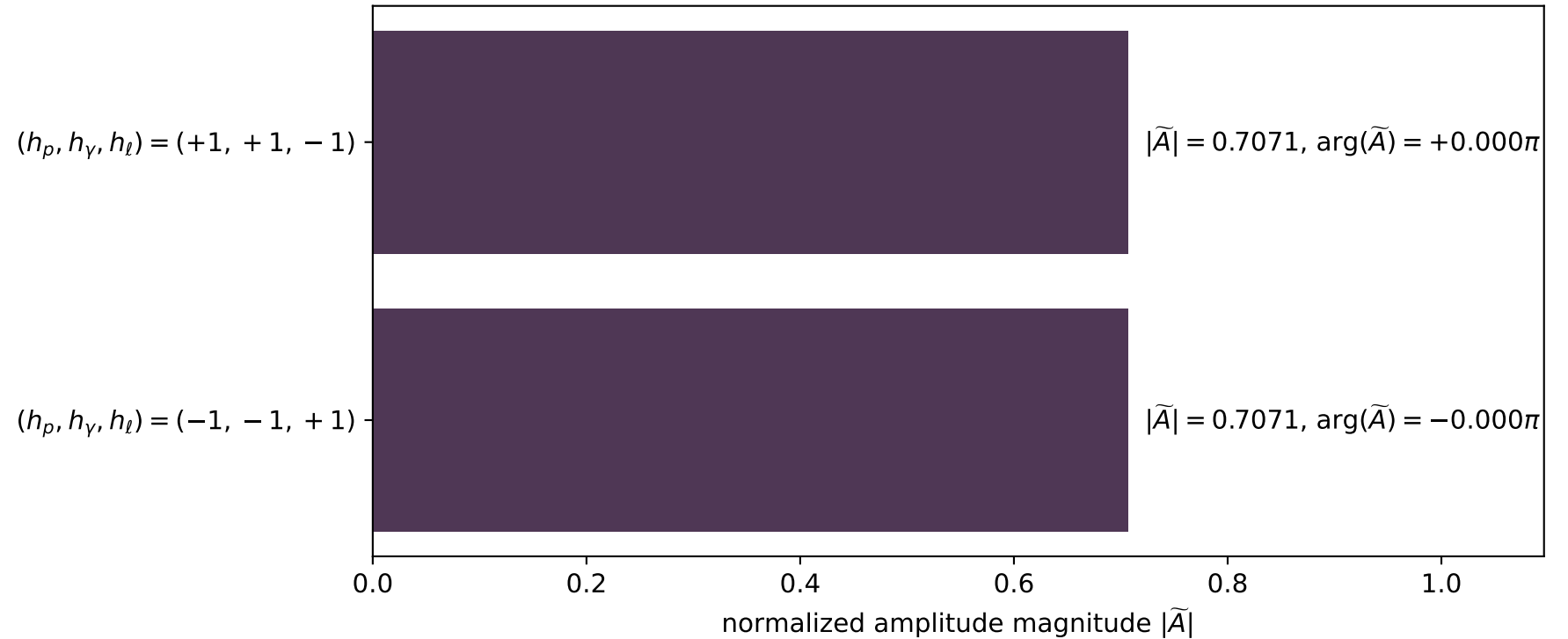


dghz_mixing_angles_polarization_04_local_minimum_183
 $d_{\text{GHZ}}=1.00228\text{e-}06$
 region: polarization_cluster_04_local_minimum_183
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0183
 lepton mixing angle: $\alpha_e=1.8447186$ rad
 incoming lepton: $-0.27051|+\rangle +0.962717|-\rangle$
 proton mixing angle: $\alpha_p=1.8447094$ rad
 incoming proton: $-0.270501|+\rangle +0.96272|-\rangle$

$s=3.04272$, $\sqrt{s}=1.74434$, $|\vec{P}|=0.61997$, $|\vec{P}'|=1.00646\text{e-}06$
 $\theta_{P'}=0.00076699$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.44967$, $\phi_\gamma=6.28171$
 $E_\gamma=0.40317$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.308074$
 $Q^2=0.999811$, $x_B=0.569317$, $t=-0.349628$, $W^2=1.63619$, $y=0.811955$

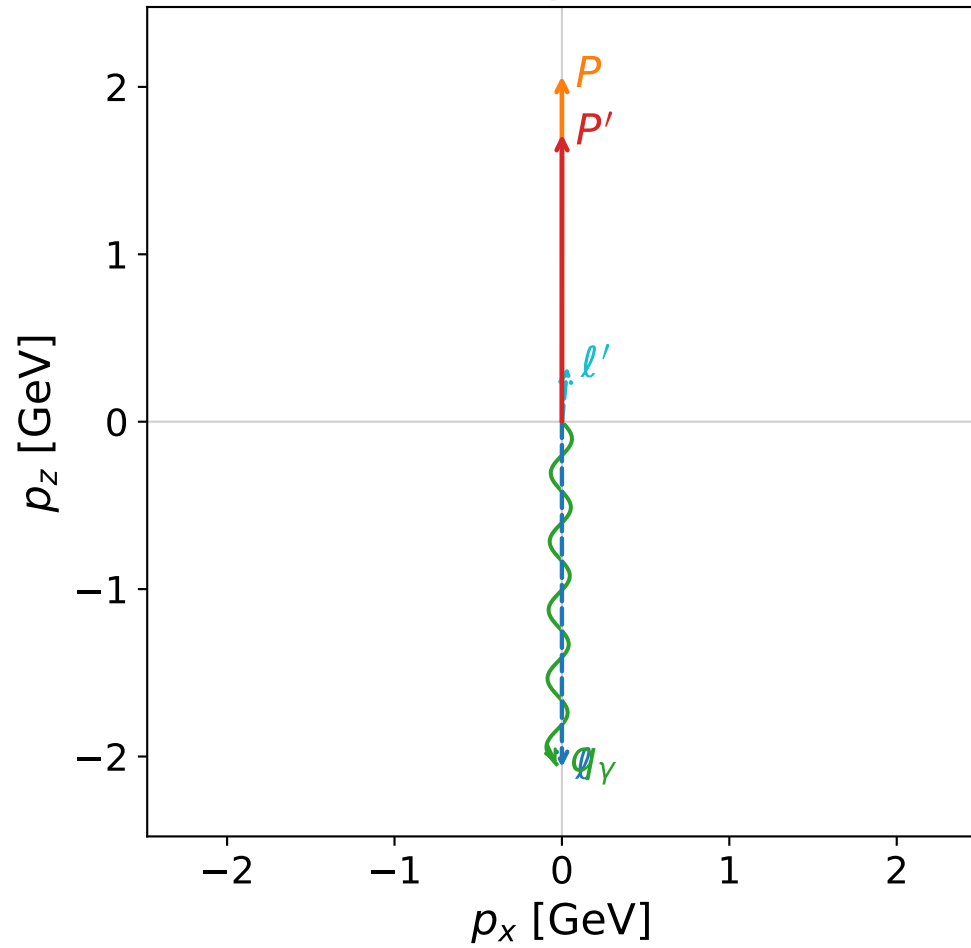
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6200$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6200$
 P $E= 1.1244$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6200$
 ℓ' $E= 0.4032$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4032$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.4032$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.4032$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

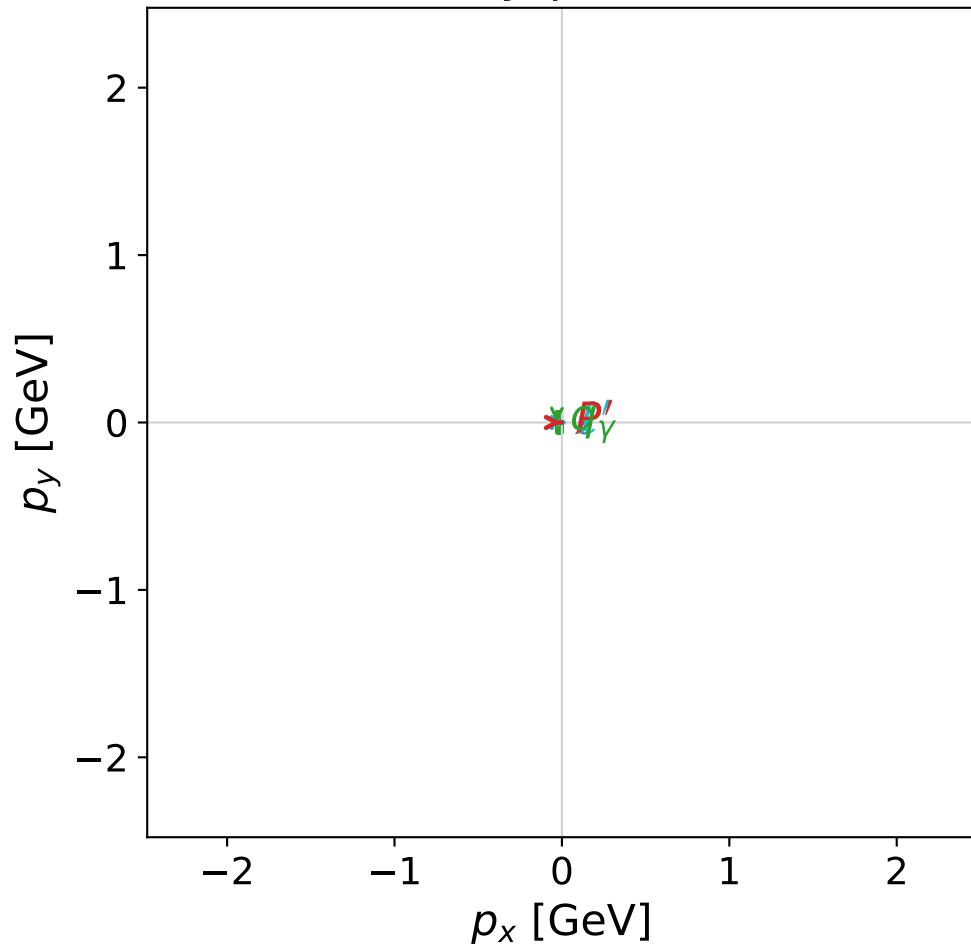


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



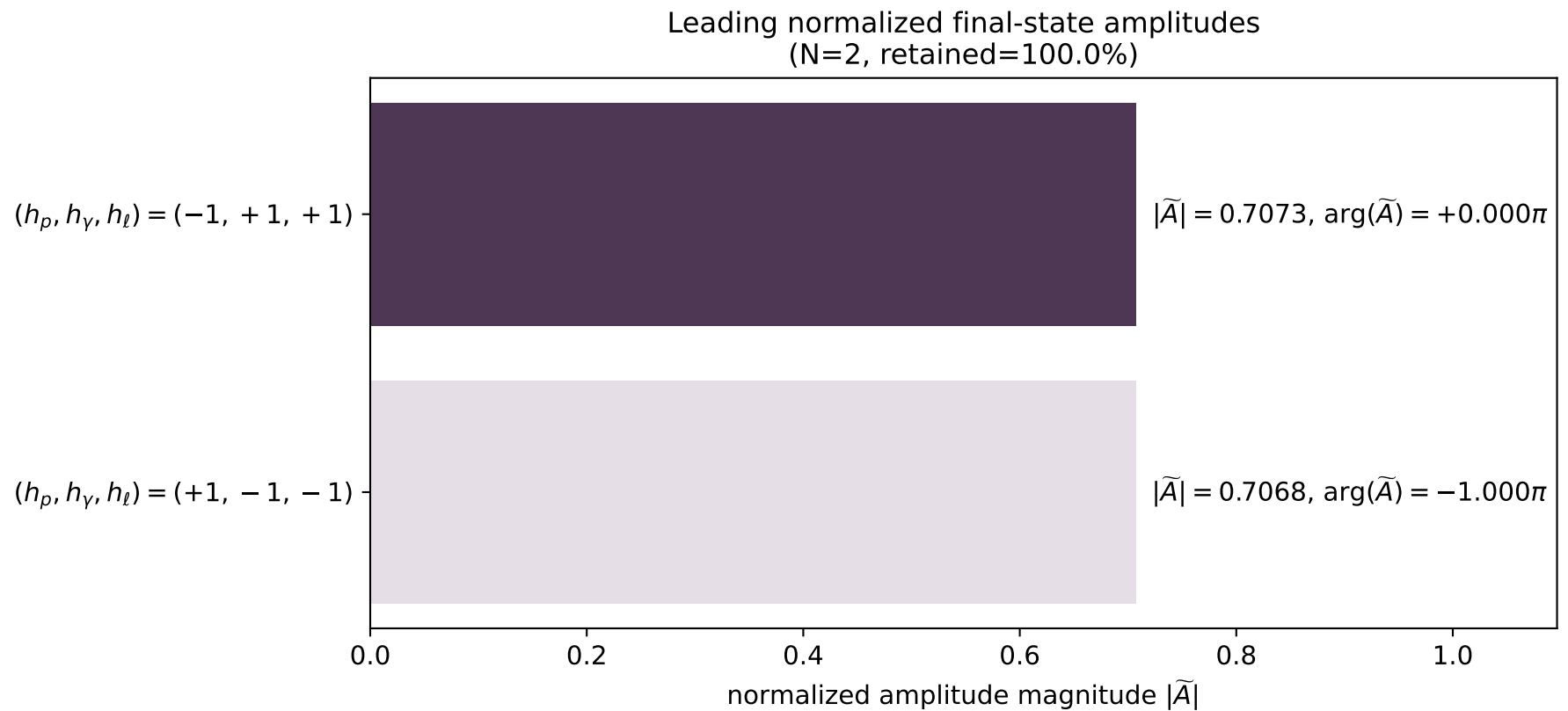
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_184
 $d_{\text{GHZ}}=2.62168\text{e-}06$
 region: polarization_cluster_04_local_minimum_184
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0184
 lepton mixing angle: $\alpha_e=2.1744032$ rad
 incoming lepton: $-0.567616|+\rangle +0.823294|-\rangle$
 proton mixing angle: $\alpha_p=2.5444445$ rad
 incoming proton: $-0.826943|+\rangle +0.562286|-\rangle$

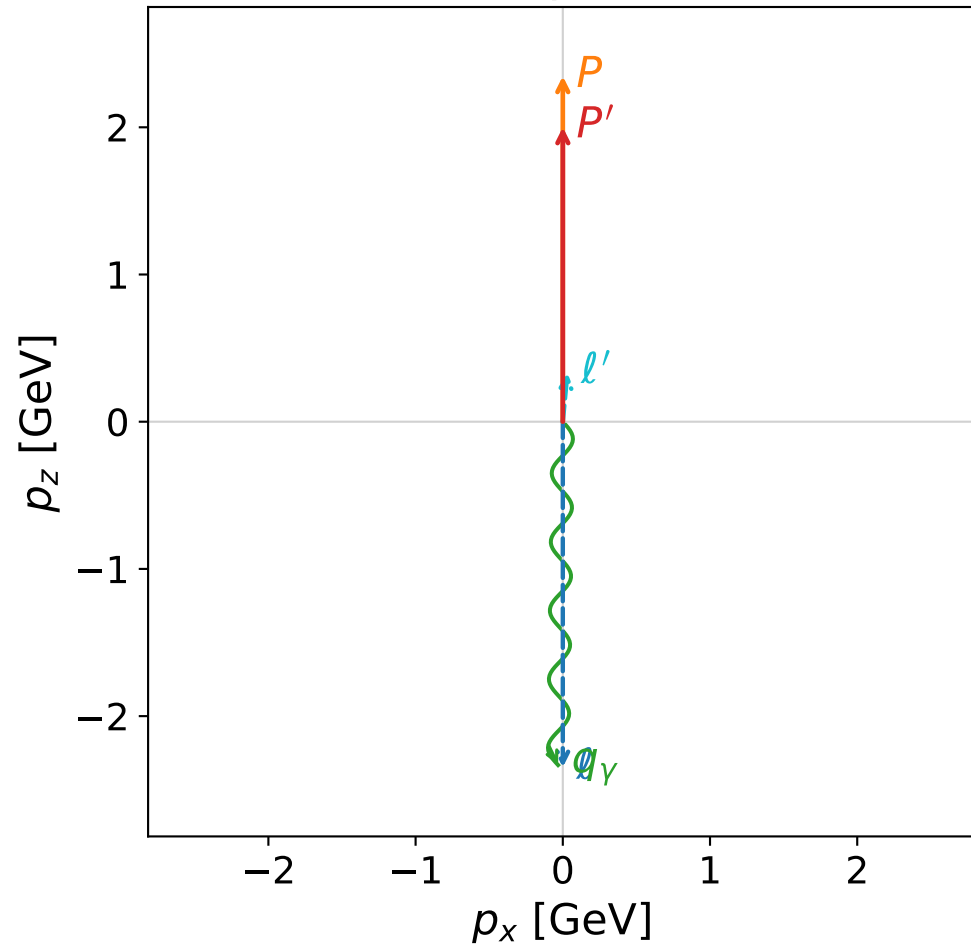
$s=18.765$, $\sqrt{s}=4.33186$, $|\vec{P}|=2.06438$, $|\vec{P}'|=1.71704$
 $\theta_{P'}=0.000299242$, $\theta_\gamma=3.12573$, $\phi_{P'}=8.42046\text{e-}05$, $\phi_\gamma=3.14159$
 $E_\gamma=2.04553$, $\theta_{l'}=0.0969549$, $\phi_{l'}=6.28318$
 $Q^2=2.71682$, $x_B=0.153105$, $t=-0.023959$, $W^2=15.9079$, $y=0.992152$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0644$
 P $E= 2.2675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0644$
 l' $E= 0.3298$ $p_x= 0.0319$ $p_y= -0.0000$ $p_z= 0.3282$
 P' $E= 1.9565$ $p_x= 0.0005$ $p_y= 0.0000$ $p_z= 1.7170$
 q_γ $E= 2.0455$ $p_x= -0.0324$ $p_y= 0.0000$ $p_z= -2.0453$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_186

$d_{\{\rm GHZ\}}=7.14153\text{e-}06$

region: polarization_cluster_04_local_minimum_186

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0186

lepton mixing angle: $\alpha_e=2.2229182$ rad

incoming lepton: $-0.606874|+\rangle +0.794798|-\rangle$

proton mixing angle: $\alpha_p=2.4957371$ rad

incoming proton: $-0.798585|+\rangle +0.601882|-\rangle$

$s=23.7724$, $\sqrt{s}=4.8757$, $|\vec{P}|=2.34762$, $|\vec{P}'|=2.00059$

$\theta_{P'}=0.000172458$, $\theta_\gamma=3.12791$, $\phi_{P'}=0.000436125$, $\phi_\gamma=3.14159$

$E_\gamma=2.33272$, $\theta_{l'}=0.0948293$, $\phi_{l'}=6.28318$

$Q^2=3.12386$, $x_B=0.137221$, $t=-0.0189857$, $W^2=20.5212$, $y=0.994436$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.3476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3476$

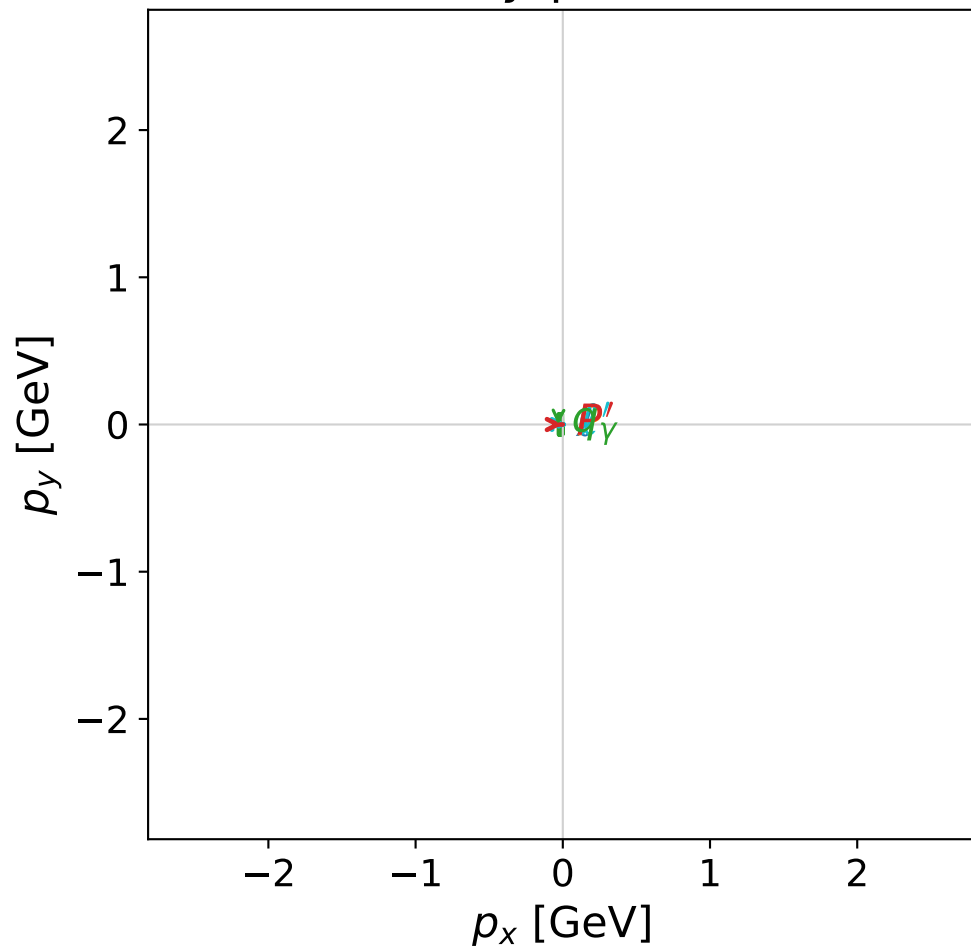
P $E= 2.5281$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3476$

l' $E= 0.3334$ $p_x= 0.0316$ $p_y= -0.0000$ $p_z= 0.3319$

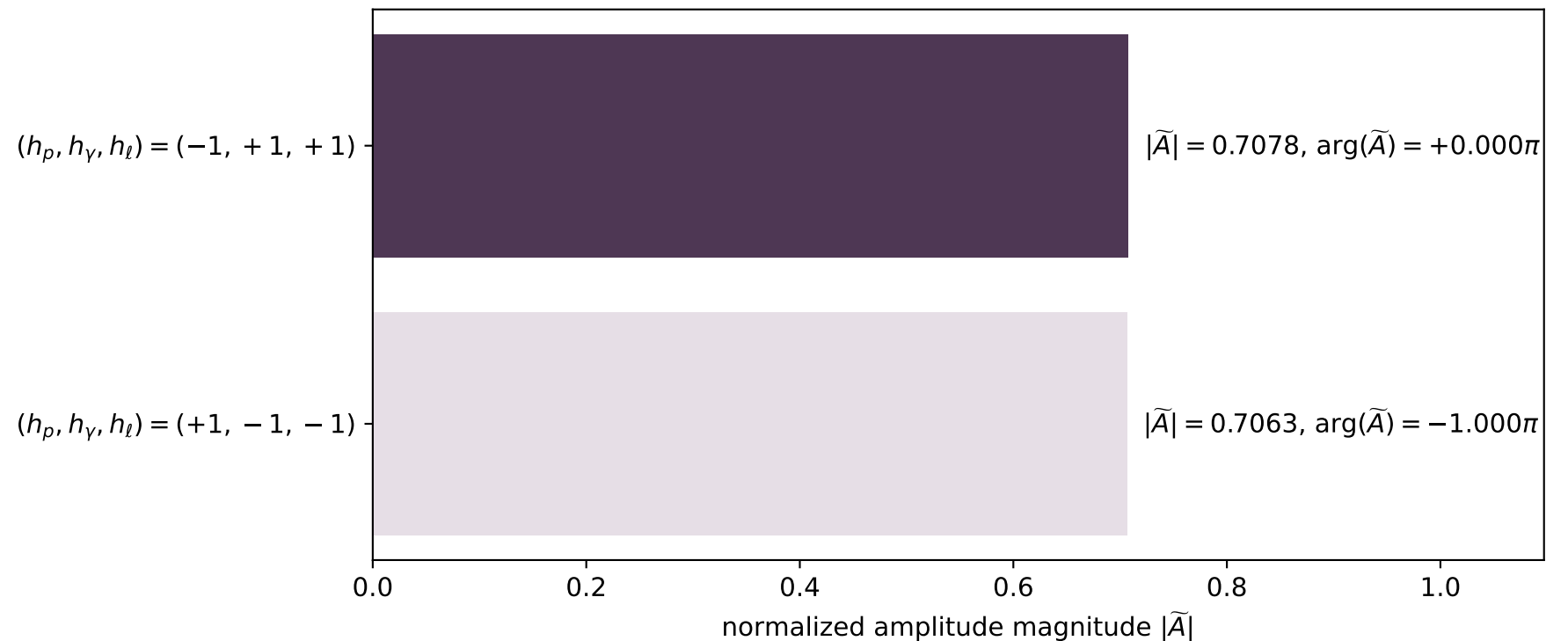
P' $E= 2.2096$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 2.0006$

q_γ $E= 2.3327$ $p_x= -0.0319$ $p_y= 0.0000$ $p_z= -2.3325$

x--y plane

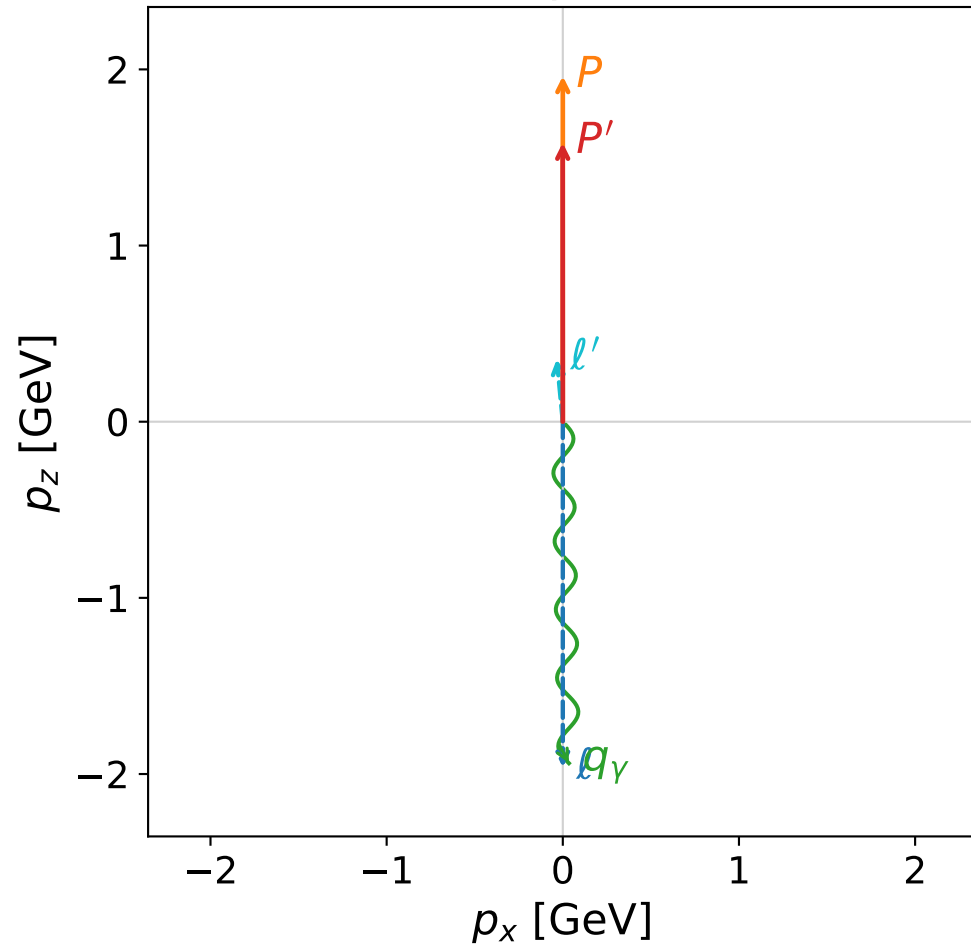


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

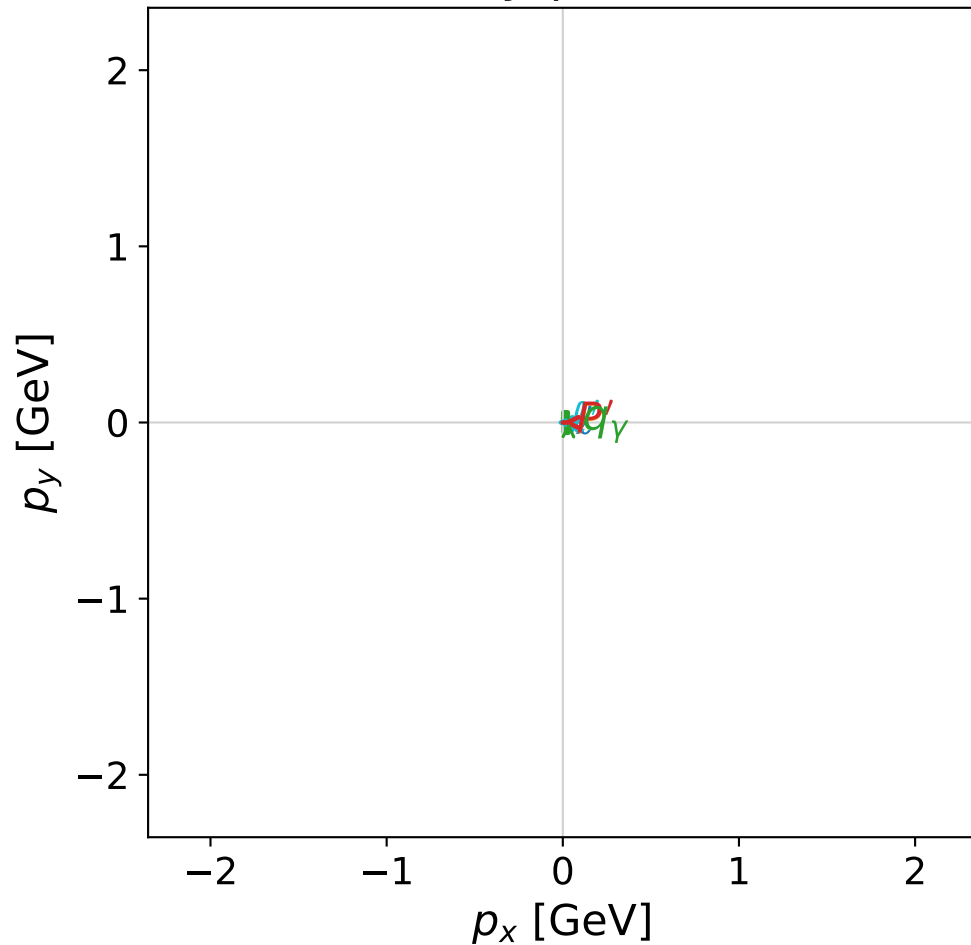


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_205

$d_{\{\rm GHZ\}}=2.5832\text{e-}05$

region: polarization_cluster_04_local_minimum_205

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0205

lepton mixing angle: $\alpha_e=2.5877528$ rad

incoming lepton: $-0.850511|+\rangle +0.525957|-\rangle$

proton mixing angle: $\alpha_p=2.116265$ rad

incoming proton: $-0.518819|+\rangle +0.854884|-\rangle$

$s=17.1102$, $\sqrt{s}=4.13645$, $|\vec{P}|=1.96187$, $|\vec{P}'|=1.58344$

$\theta_{p'}=0.000451651$, $\theta_Y=3.1237$, $\phi_{p'}=3.14162$, $\phi_Y=6.28317$

$E_Y=1.93908$, $\theta_{\ell'}=0.0953256$, $\phi_{\ell'}=3.14157$

$Q^2=2.79482$, $x_B=0.173893$, $t=-0.0315471$, $W^2=14.1571$, $y=0.990252$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.9619$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9619$

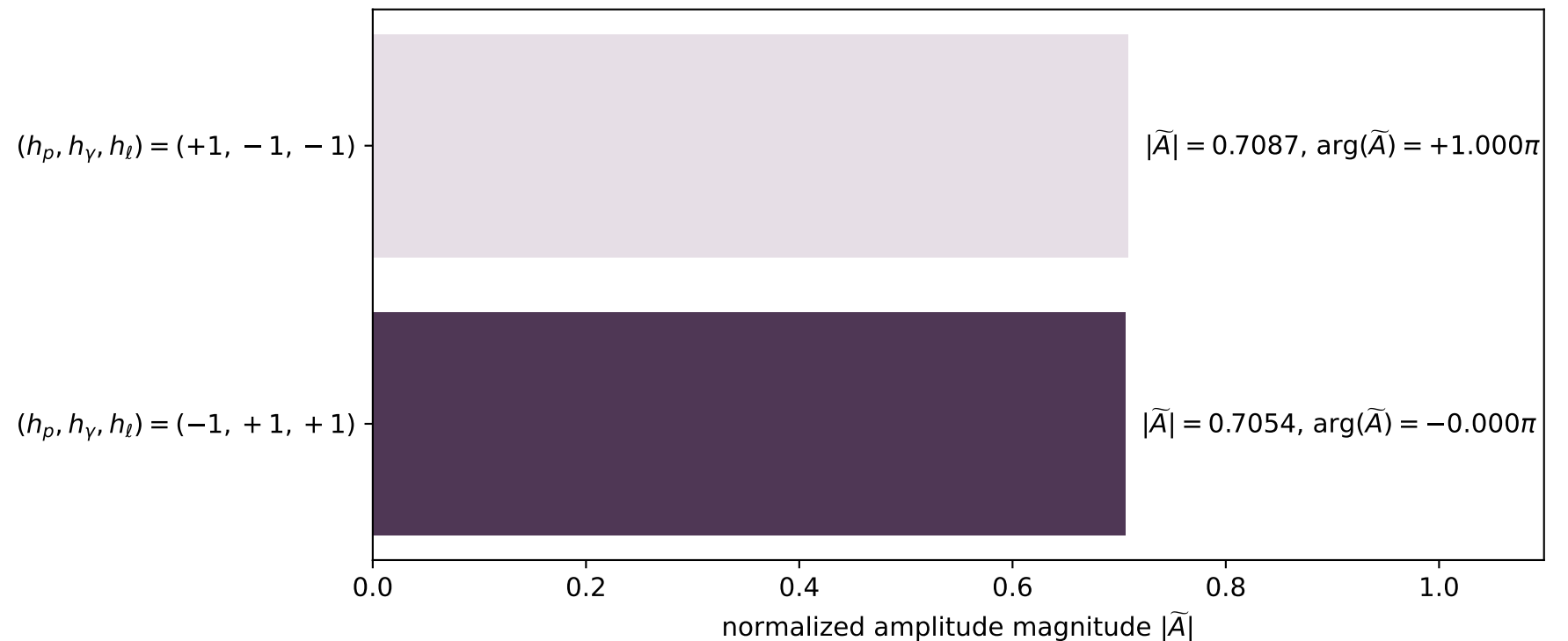
P $E= 2.1746$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9619$

ℓ' $E= 0.3570$ $p_x= -0.0340$ $p_y= 0.0000$ $p_z= 0.3553$

P' $E= 1.8404$ $p_x= -0.0007$ $p_y= -0.0000$ $p_z= 1.5834$

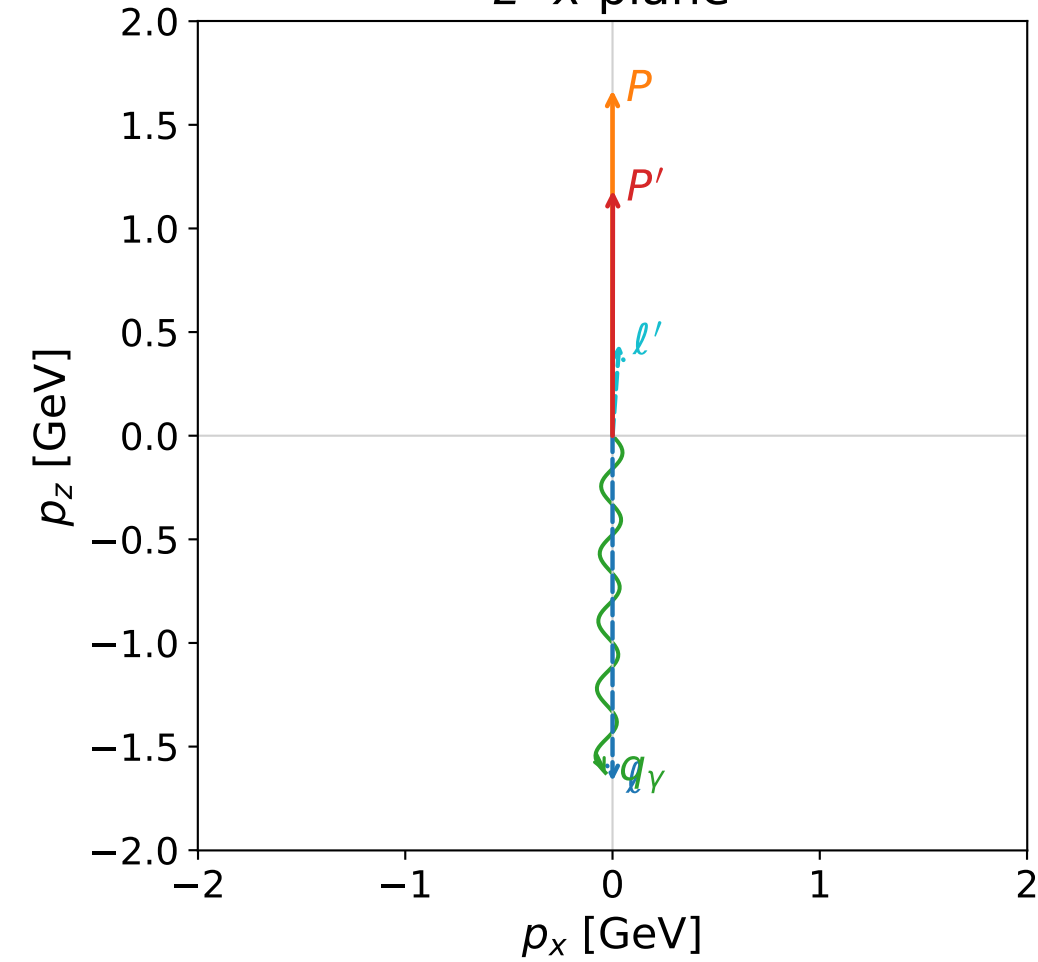
q_Y $E= 1.9391$ $p_x= 0.0347$ $p_y= -0.0000$ $p_z= -1.9388$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

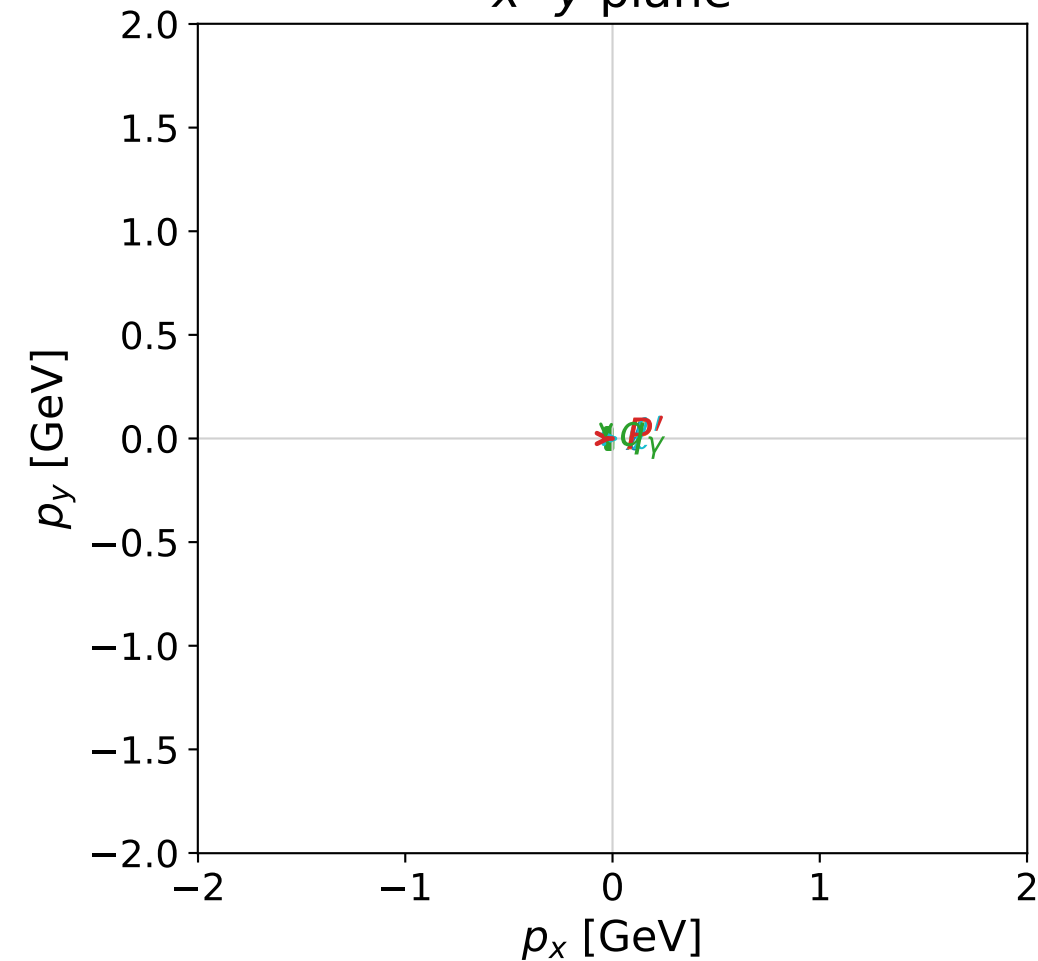


dghz_mixing_angles_polarization_04_local_minimum_210
 $d_{\{\rm GHZ\}}=2.82327\text{e-}05$
 region: polarization_cluster_04_local_minimum_210
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0210
 lepton mixing angle: $\alpha_e=1.938937$ rad
 incoming lepton: $-0.359881|+\rangle +0.932998|-\rangle$
 proton mixing angle: $\alpha_p=2.7798744$ rad
 incoming proton: $-0.93529|+\rangle +0.353882|-\rangle$

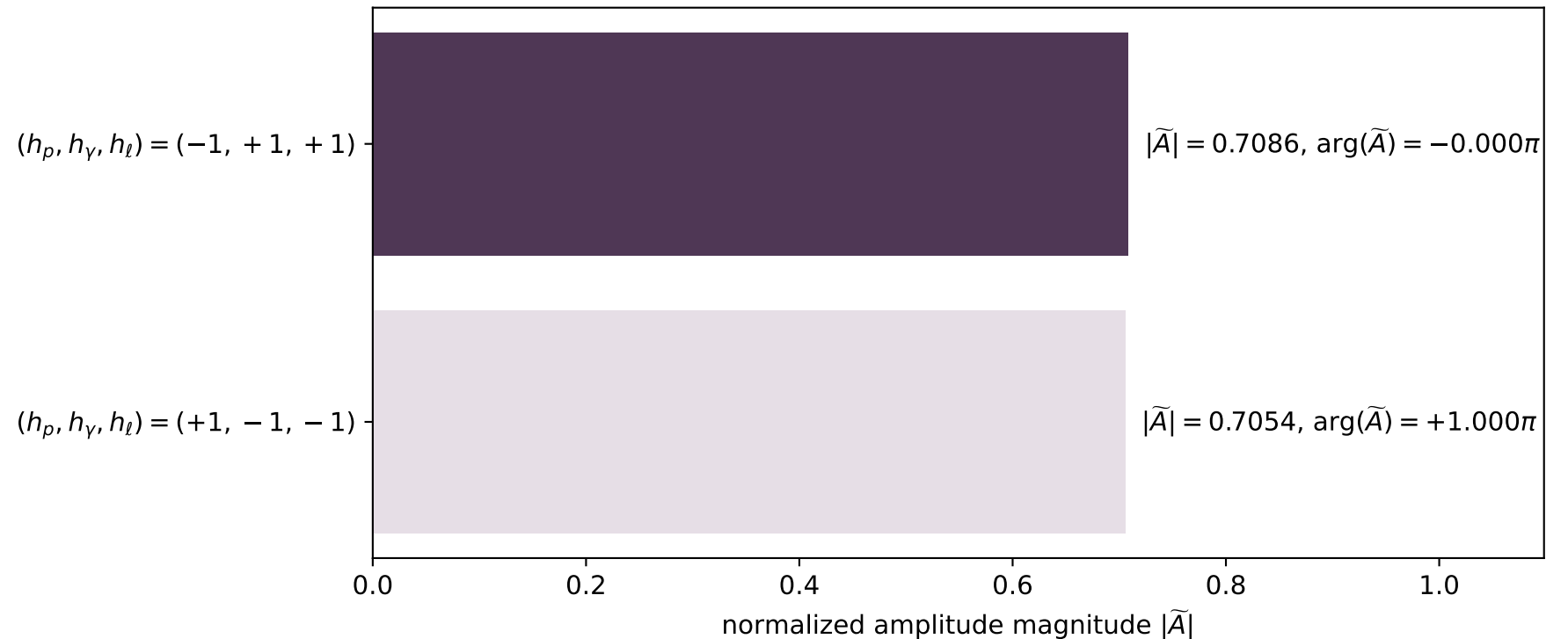
$s=12.8213$, $\sqrt{s}=3.58068$, $|\vec{P}|=1.66748$, $|\vec{P}'|=1.1839$
 $\theta_{p'}=0.00120427$, $\theta_\gamma=3.12072$, $\phi_{p'}=6.28227$, $\phi_\gamma=3.14164$
 $E_\gamma=1.62664$, $\theta_{l'}=0.0733953$, $\phi_{l'}=8.56722\text{e-}05$
 $Q^2=2.95466$, $x_B=0.252116$, $t=-0.0716446$, $W^2=9.64462$, $y=0.981411$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.6675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6675$
 P $E= 1.9132$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6675$
 ℓ' $E= 0.4436$ $p_x= 0.0325$ $p_y= 0.0000$ $p_z= 0.4424$
 P' $E= 1.5105$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1839$
 q_γ $E= 1.6266$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= -1.6263$

x--y plane

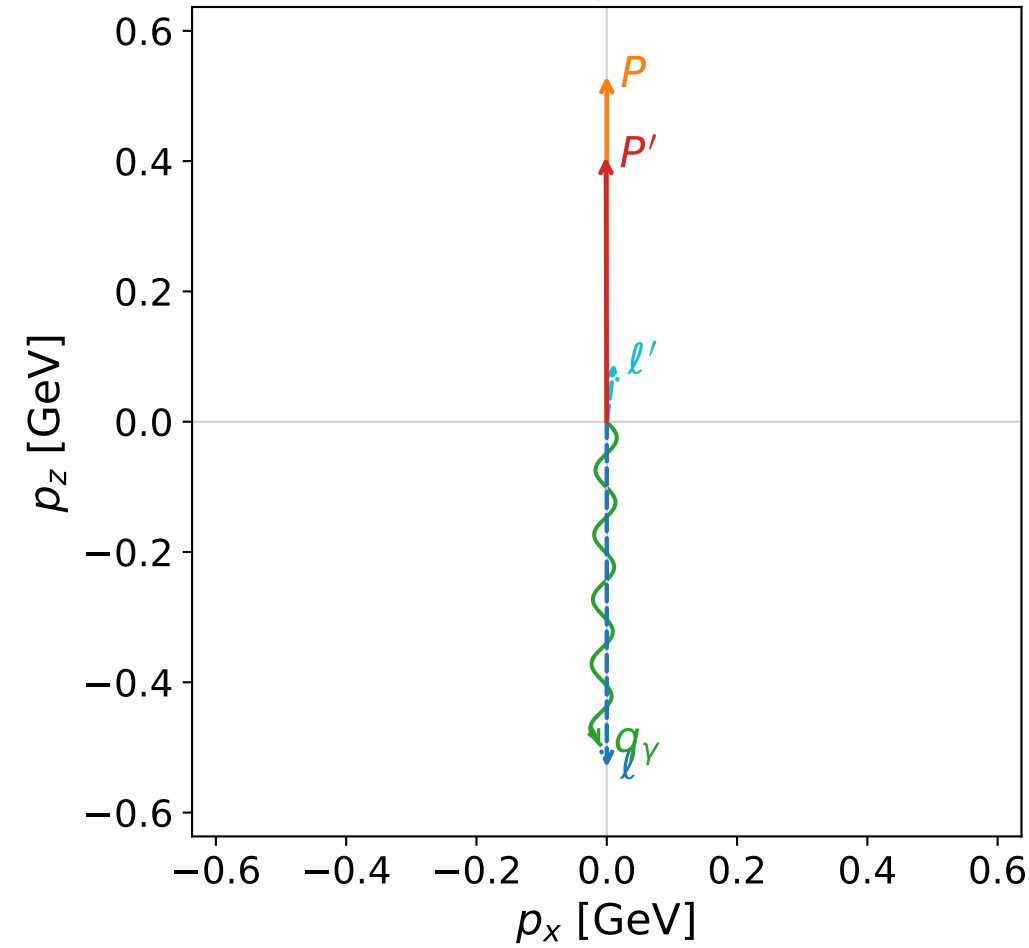


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

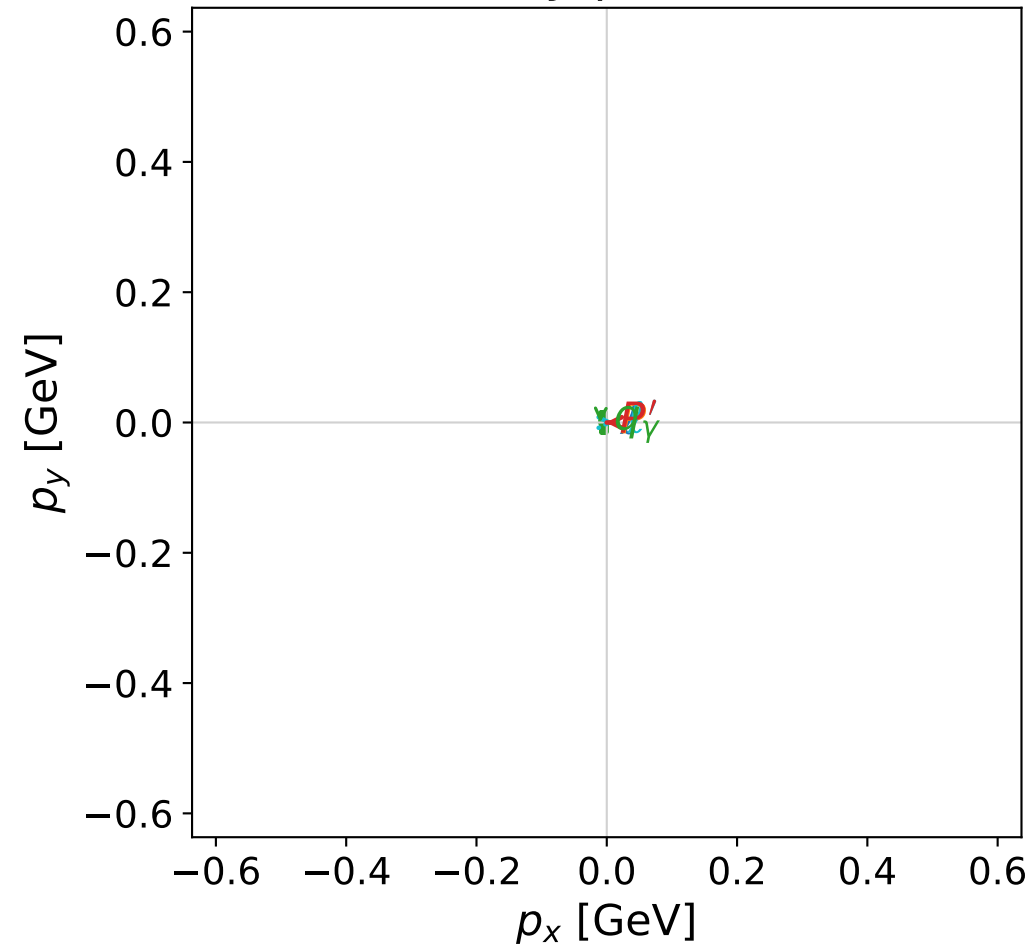


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

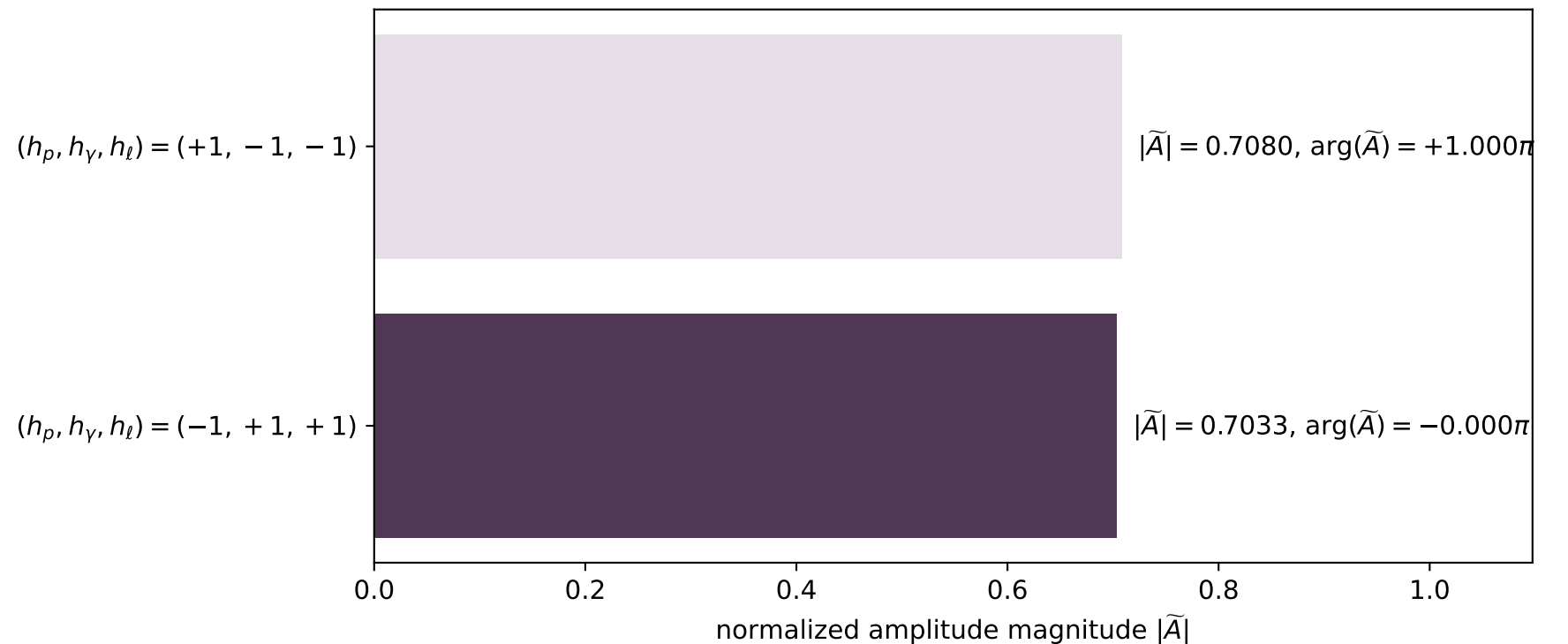


dghz_mixing_angles_polarization_04_local_minimum_221
 $d_{\text{GHZ}}=4.54196\text{e-}05$
 region: polarization_cluster_04_local_minimum_221
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0221
 lepton mixing angle: $\alpha_e=1.989916$ rad
 incoming lepton: $-0.406956|+\rangle + 0.913448|-\rangle$
 proton mixing angle: $\alpha_p=2.6885858$ rad
 incoming proton: $-0.899135|+\rangle + 0.437671|-\rangle$

$s=2.58602$, $\sqrt{s}=1.60811$, $|\vec{P}|=0.53049$, $|\vec{P}'|=0.406256$
 $\theta_{P'}=0.00318369$, $\theta_\gamma=3.12141$, $\phi_{P'}=3.14161$, $\phi_\gamma=3.14159$
 $E_\gamma=0.495777$, $\theta_{l'}=0.125661$, $\phi_{l'}=6.28318$
 $Q^2=0.190506$, $x_B=0.118564$, $t=-0.0123647$, $W^2=2.29612$, $y=0.941749$

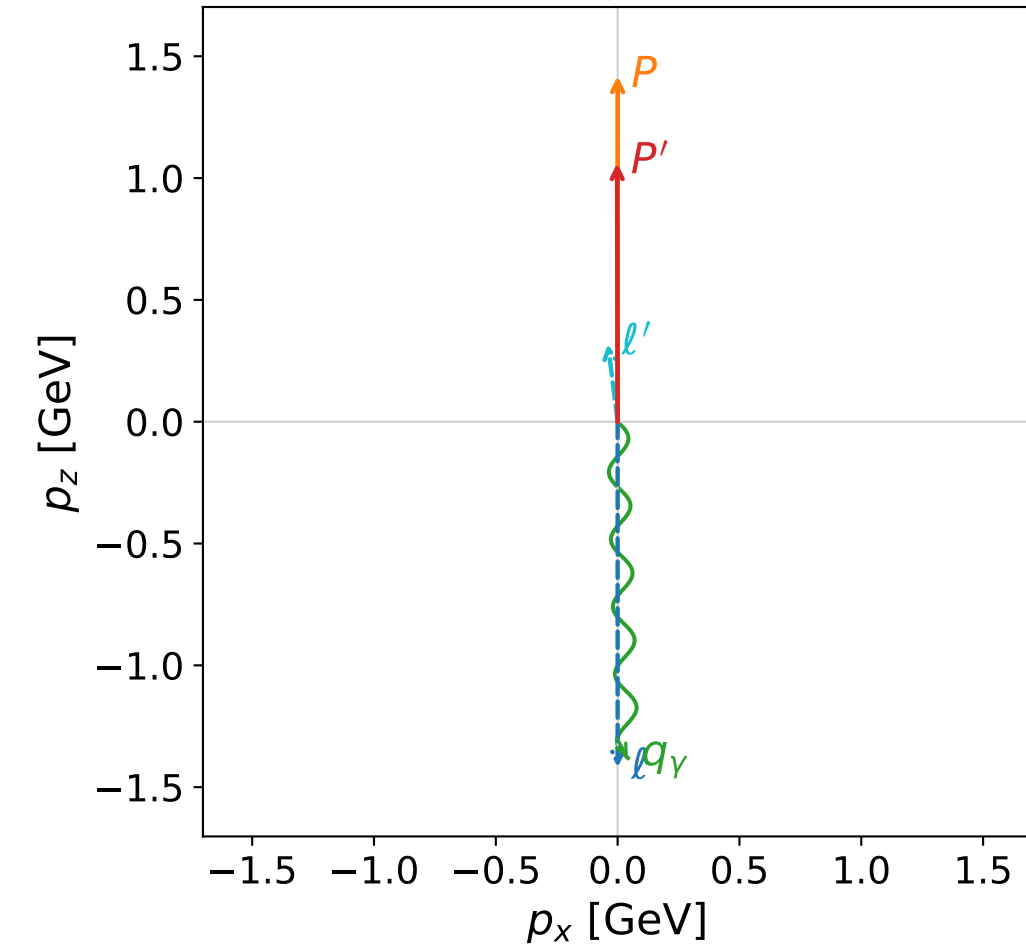
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5305$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5305$
 P $E= 1.0776$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5305$
 l' $E= 0.0901$ $p_x= 0.0113$ $p_y= -0.0000$ $p_z= 0.0894$
 P' $E= 1.0222$ $p_x= -0.0013$ $p_y= -0.0000$ $p_z= 0.4063$
 q_γ $E= 0.4958$ $p_x= -0.0100$ $p_y= 0.0000$ $p_z= -0.4957$

Leading normalized final-state amplitudes
(N=2, retained=99.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

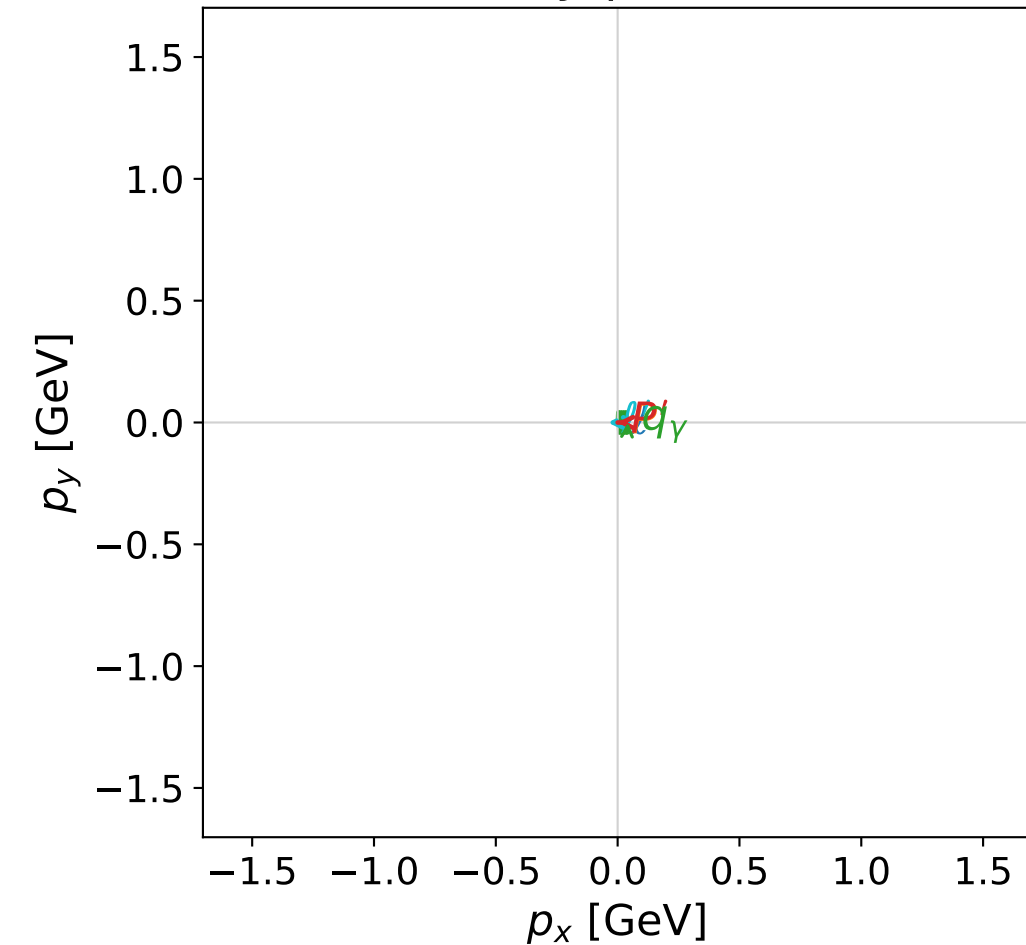


dghz_mixing_angles_polarization_04_local_minimum_228
 $d_{\text{GHZ}}=5.43438\text{e-}05$
 region: polarization_cluster_04_local_minimum_228
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0228
 lepton mixing angle: $\alpha_e=2.7339948$ rad
 incoming lepton: $-0.918076|+\rangle + 0.396405|-\rangle$
 proton mixing angle: $\alpha_p=1.9650585$ rad
 incoming proton: $-0.384127|+\rangle + 0.92328|-\rangle$

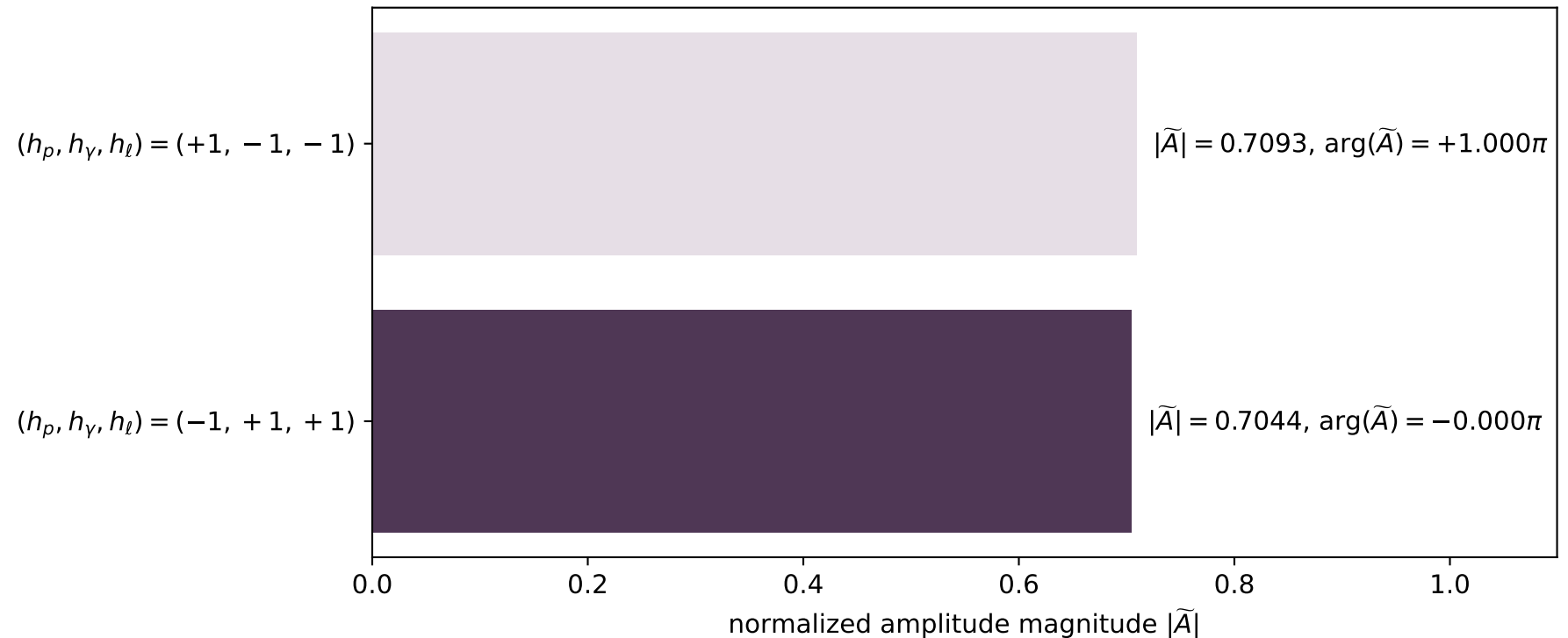
$s=9.72826$, $\sqrt{s}=3.11902$, $|\vec{P}|=1.41846$, $|\vec{P}'|=1.06154$
 $\theta_{P'}=0.00179591$, $\theta_\gamma=3.11125$, $\phi_{P'}=3.1417$, $\phi_\gamma=2.99541\text{e-}05$
 $E_\gamma=1.38105$, $\theta_{l'}=0.124754$, $\phi_{l'}=3.14162$
 $Q^2=1.81636$, $x_B=0.209741$, $t=-0.0467601$, $W^2=7.7235$, $y=0.978708$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.4185$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4185$
 P $E= 1.7006$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4185$
 l' $E= 0.3214$ $p_x= -0.0400$ $p_y= -0.0000$ $p_z= 0.3189$
 P' $E= 1.4166$ $p_x= -0.0019$ $p_y= -0.0000$ $p_z= 1.0615$
 q_γ $E= 1.3811$ $p_x= 0.0419$ $p_y= 0.0000$ $p_z= -1.3804$

x--y plane

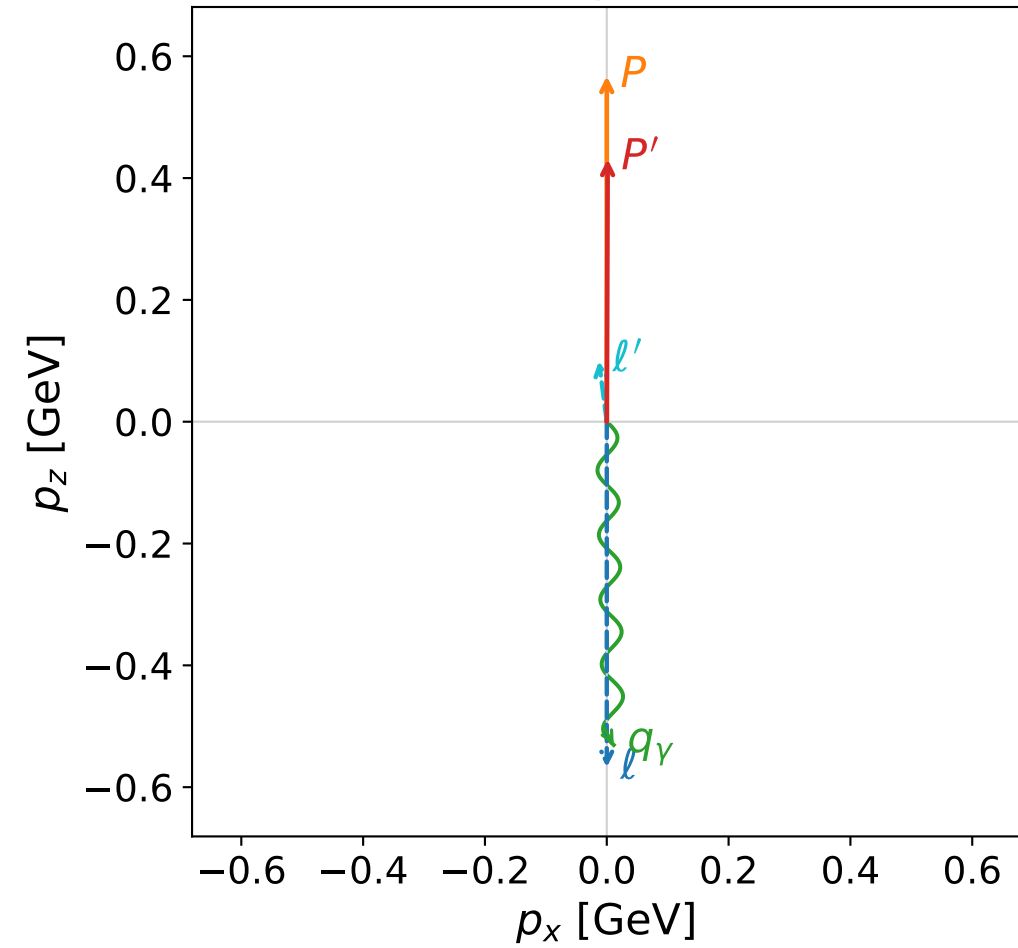


Leading normalized final-state amplitudes
 (N=2, retained=99.9%)

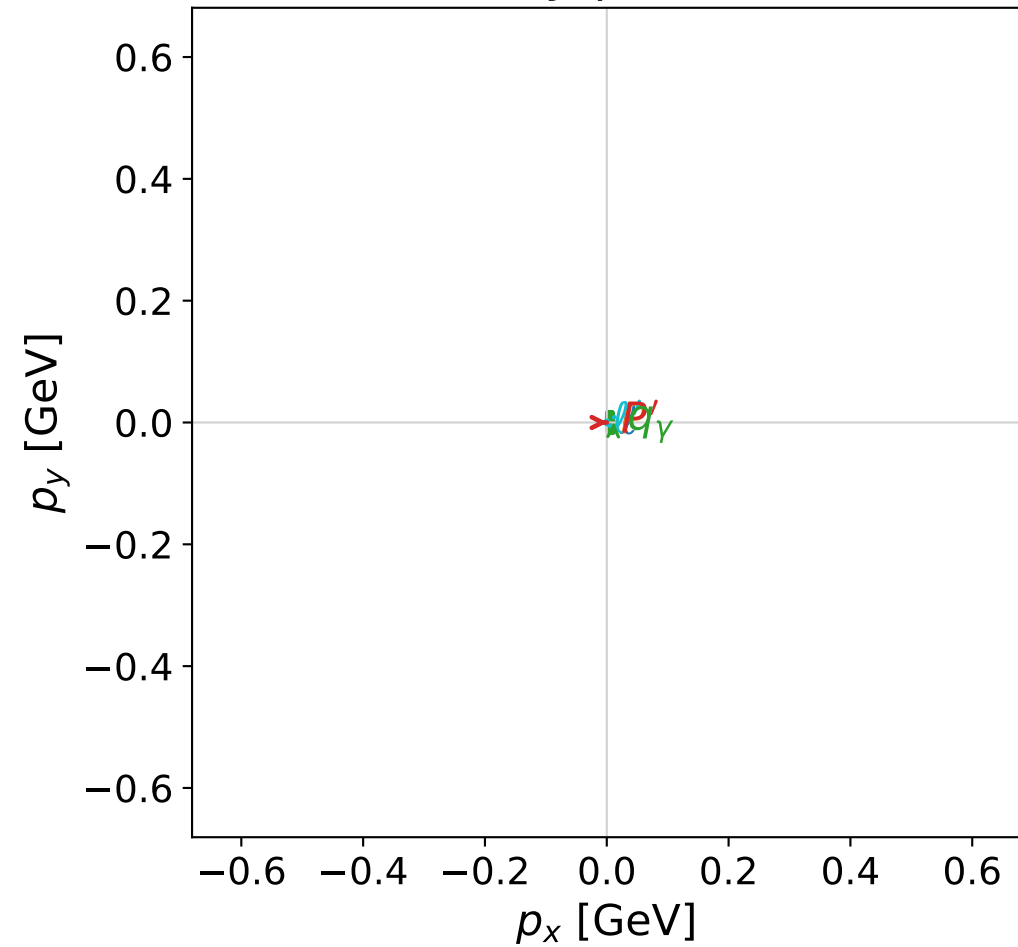


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

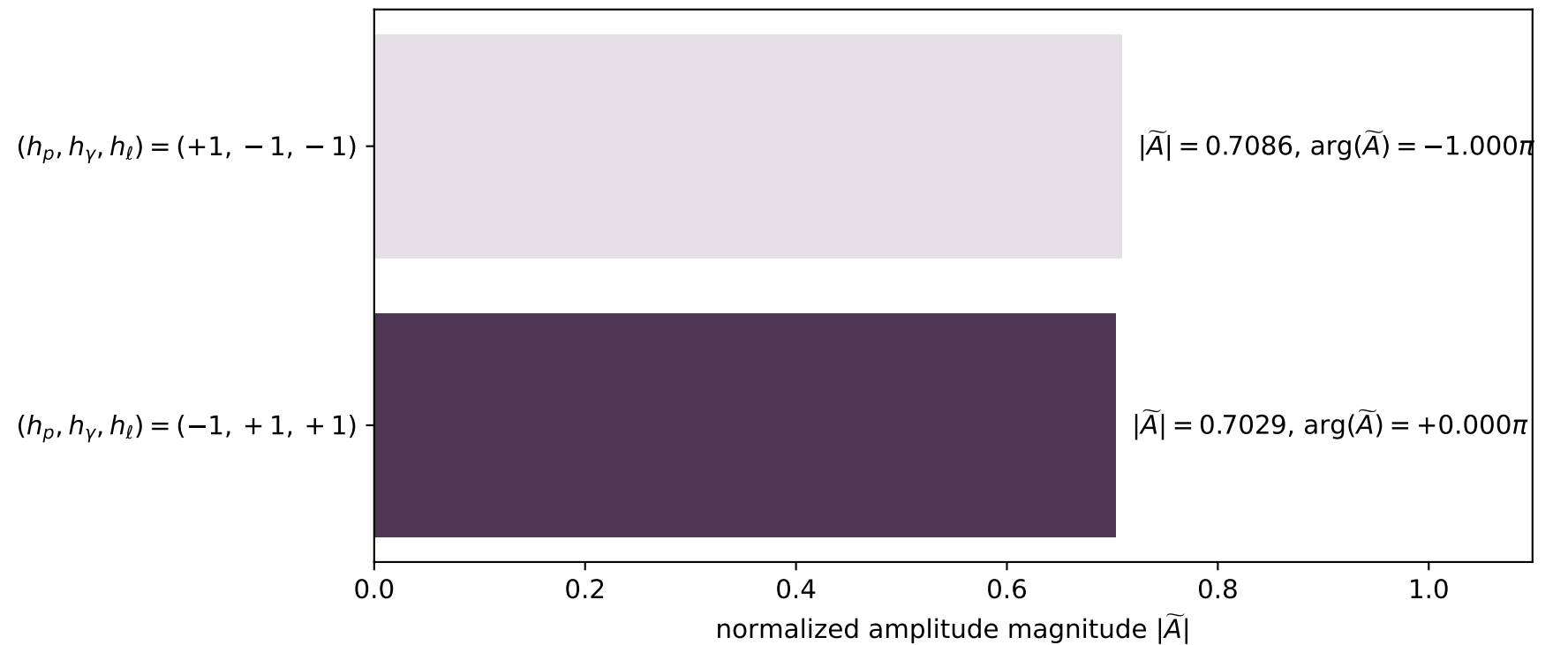


dghz_mixing_angles_polarization_04_local_minimum_235
 $d_{\text{GHZ}}=6.62947\text{e-}05$
 region: polarization_cluster_04_local_minimum_235
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0235
 lepton mixing angle: $\alpha_e=2.7359514$ rad
 incoming lepton: $-0.91885|+\rangle +0.394608|-\rangle$
 proton mixing angle: $\alpha_p=2.0031027$ rad
 incoming proton: $-0.418966|+\rangle +0.908002|-\rangle$

$s=2.76767$, $\sqrt{s}=1.66363$, $|\vec{P}|=0.567381$, $|\vec{P}'|=0.431117$
 $\theta_{p'}=0.00334598$, $\theta_Y=3.12$, $\phi_{p'}=4.16802\text{e-}06$, $\phi_Y=6.28316$
 $E_Y=0.530853$, $\theta_{\ell'}=0.128826$, $\phi_{\ell'}=3.14157$
 $Q^2=0.227024$, $x_B=0.127496$, $t=-0.0144849$, $W^2=2.43345$, $y=0.943218$

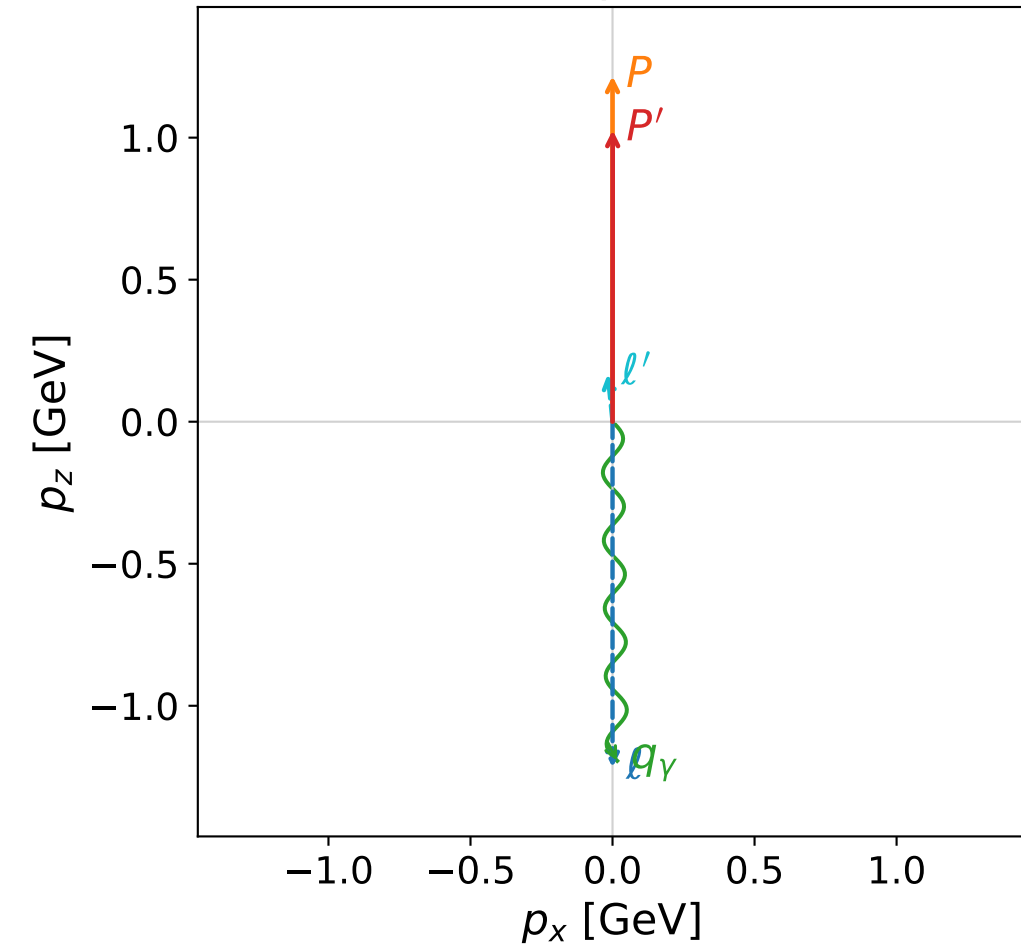
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5674$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5674$
 P $E= 1.0963$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5674$
 ℓ' $E= 0.1004$ $p_x= -0.0129$ $p_y= 0.0000$ $p_z= 0.0996$
 P' $E= 1.0323$ $p_x= 0.0014$ $p_y= 0.0000$ $p_z= 0.4311$
 q_Y $E= 0.5309$ $p_x= 0.0115$ $p_y= -0.0000$ $p_z= -0.5307$

Leading normalized final-state amplitudes
 (N=2, retained=99.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



dghz_mixing_angles_polarization_04_local_minimum_238

$d_{\{\rm GHZ\}}=7.24626\text{e-}05$

region: polarization_cluster_04_local_minimum_238

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0238

lepton mixing angle: $\alpha_e=2.5683975$ rad

incoming lepton: $-0.840172|+\rangle +0.542319|-\rangle$

proton mixing angle: $\alpha_p=2.1590186$ rad

incoming proton: $-0.554883|+\rangle +0.831928|-\rangle$

$s=7.57815$, $\sqrt{s}=2.75284$, $|\vec{P}|=1.21662$, $|\vec{P}'|=1.02635$

$\theta_{p'}=0.000568325$, $\theta_Y=3.12781$, $\phi_{p'}=0.000173411$, $\phi_Y=8.59629\text{e-}06$

$E_Y=1.19402$, $\theta_{l'}=0.101329$, $\phi_{l'}=3.14161$

$Q^2=0.817507$, $x_B=0.12408$, $t=-0.0149387$, $W^2=6.65088$, $y=0.983613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.2166$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2166$

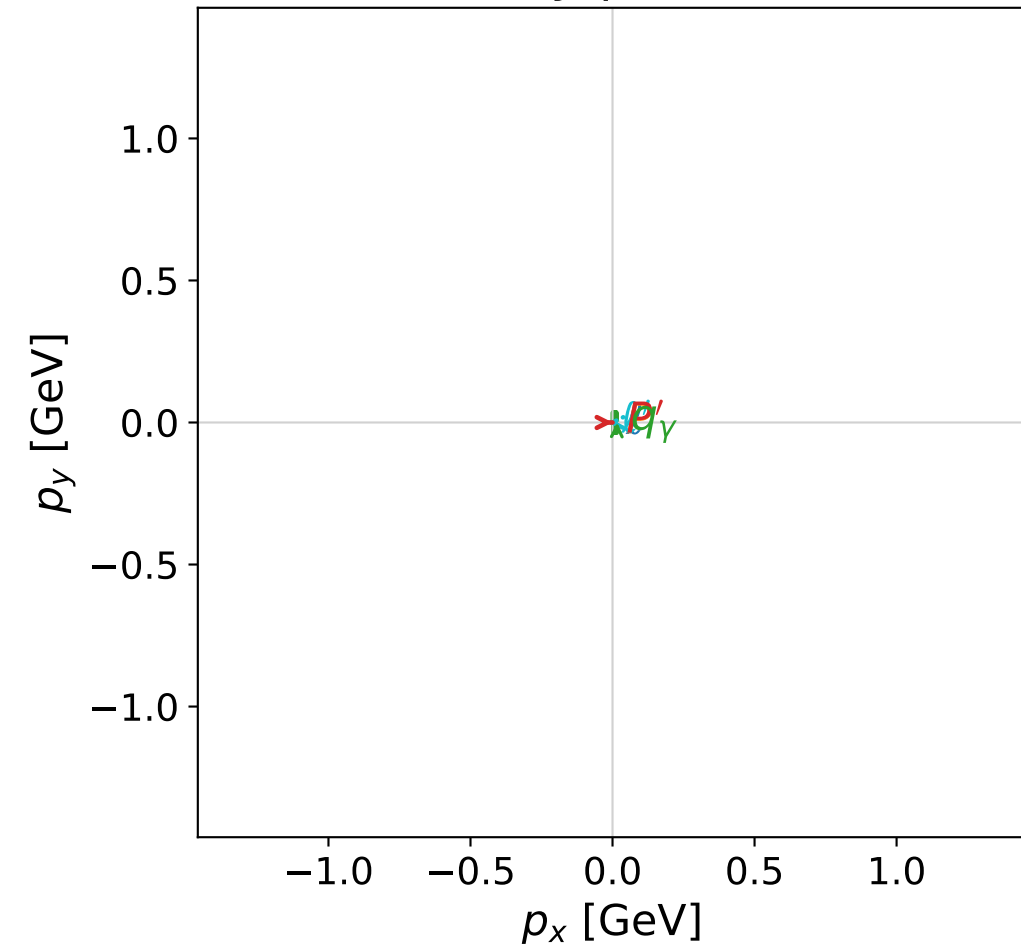
P $E= 1.5362$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2166$

ℓ' $E= 0.1684$ $p_x= -0.0170$ $p_y= -0.0000$ $p_z= 0.1676$

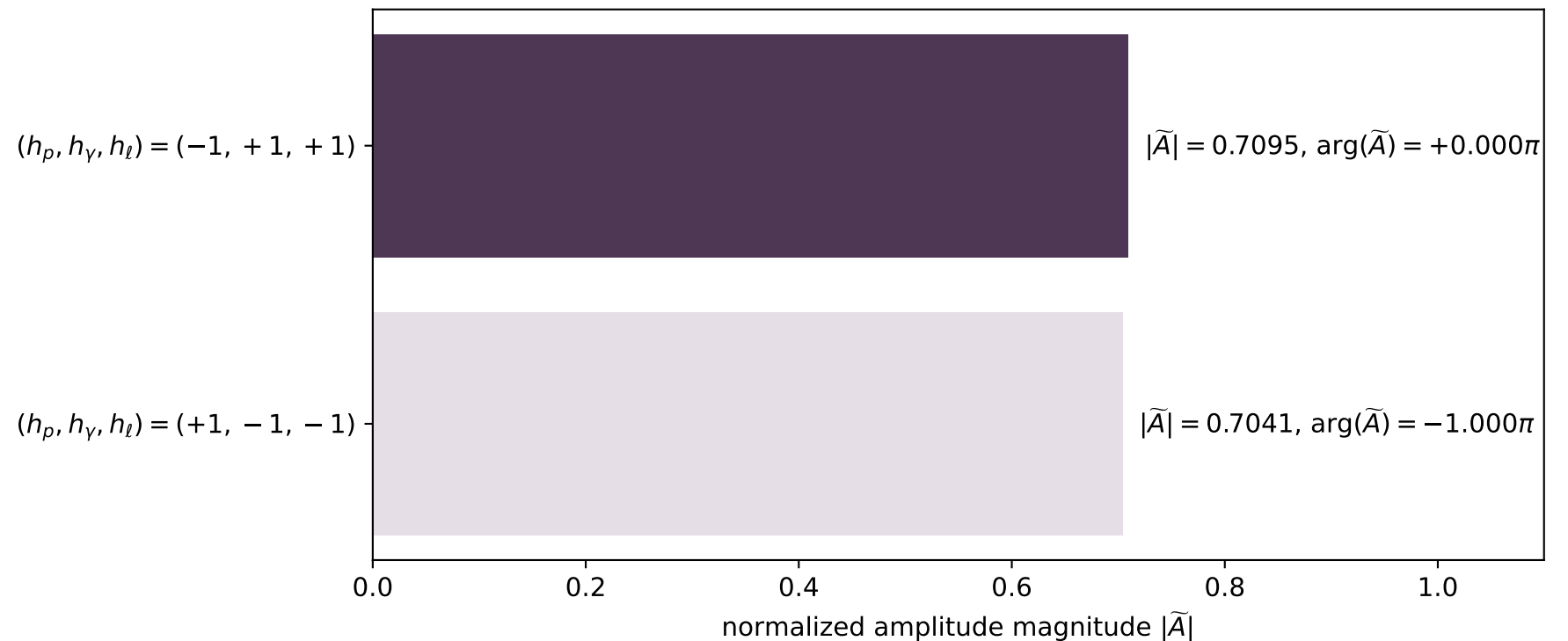
P' $E= 1.3904$ $p_x= 0.0006$ $p_y= 0.0000$ $p_z= 1.0263$

q_Y $E= 1.1940$ $p_x= 0.0165$ $p_y= 0.0000$ $p_z= -1.1939$

x--y plane

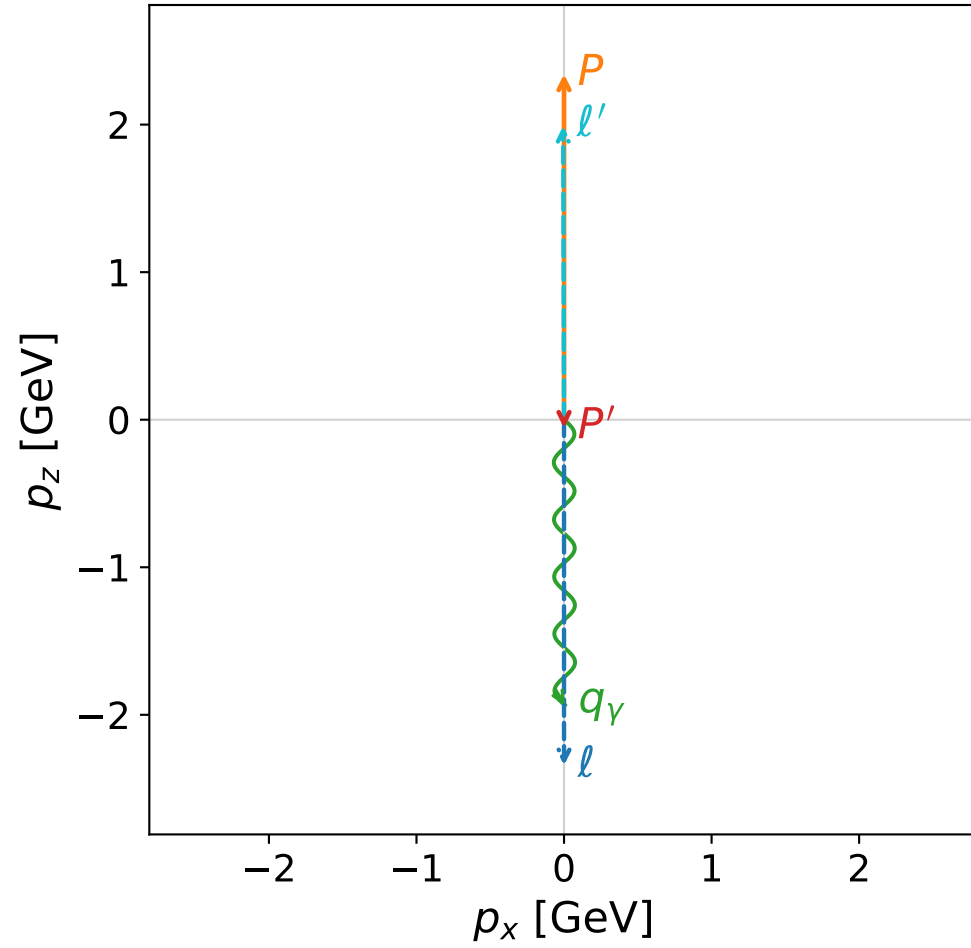


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

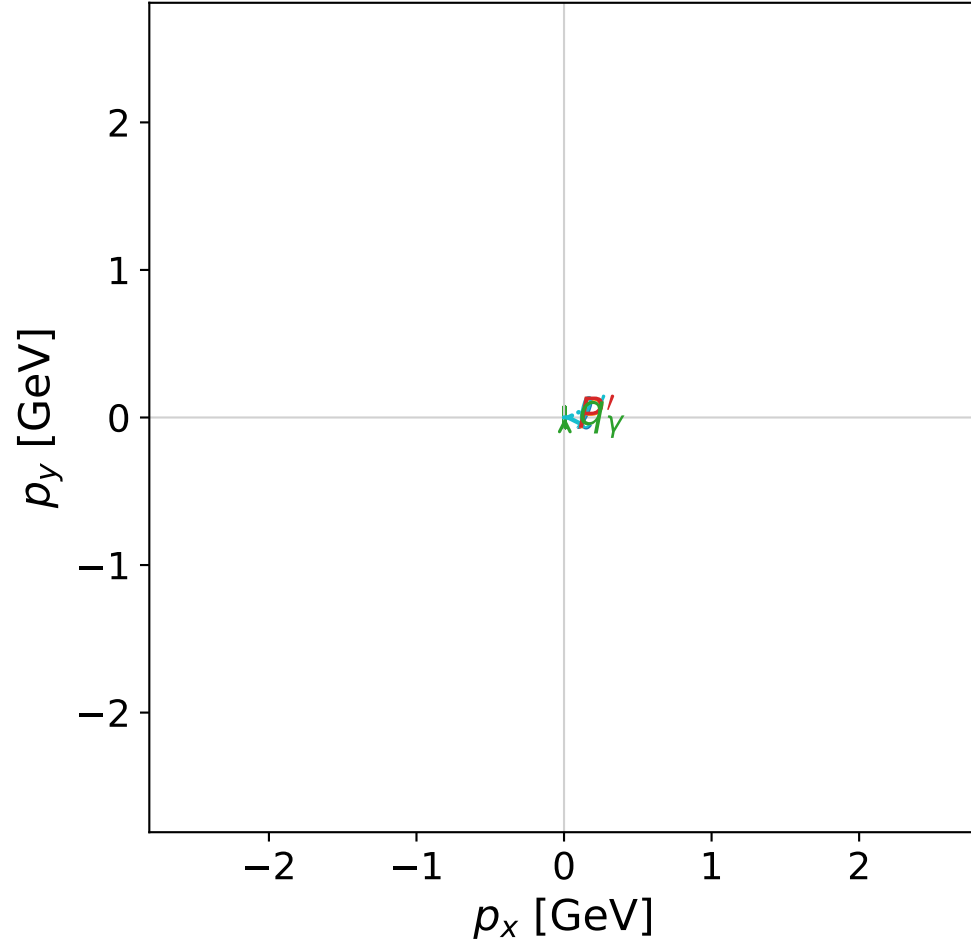


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



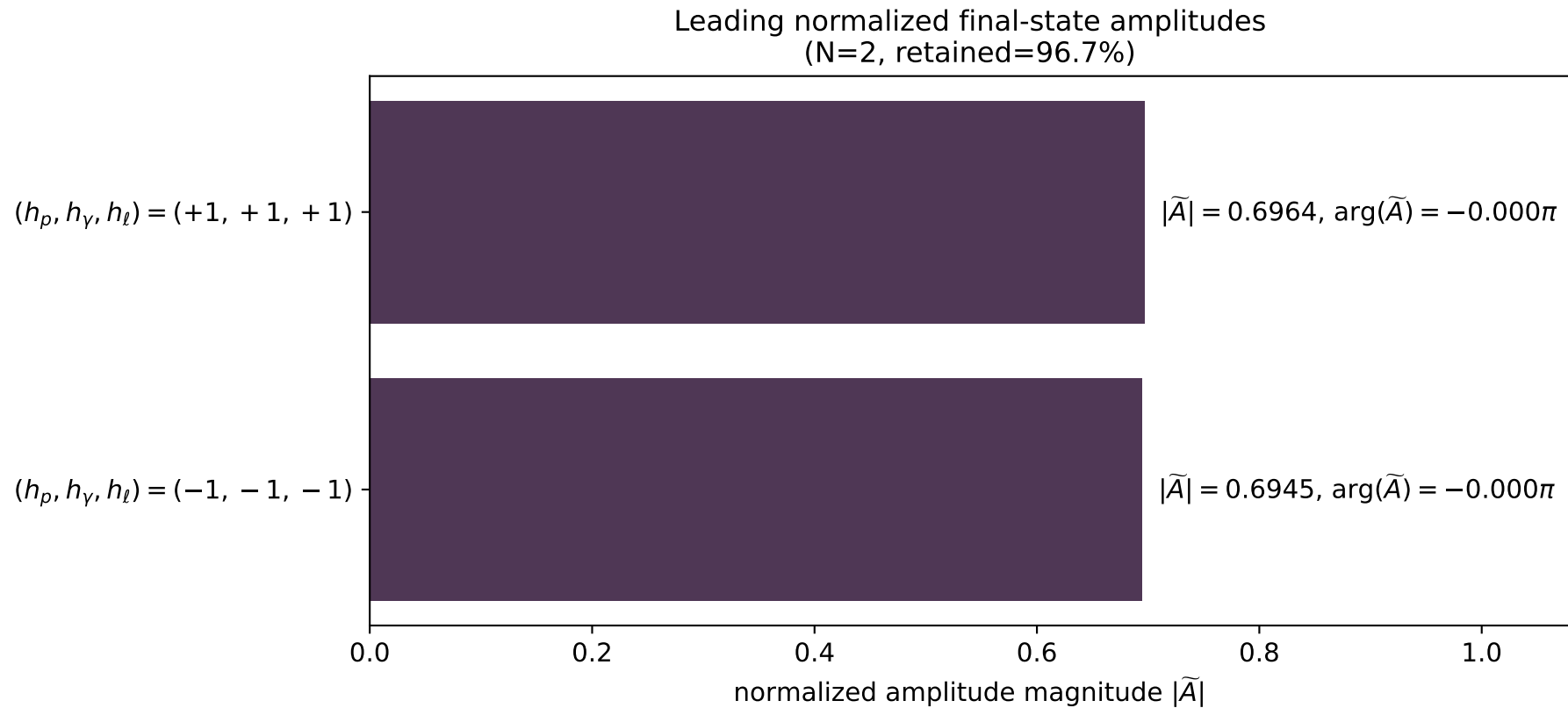
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_245
 $d_{\{\text{rm GHZ}\}}=0.000119118$
 region: polarization_cluster_04_local_minimum_245
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0245
 lepton mixing angle: $\alpha_e=2.9733431$ rad
 incoming lepton: $-0.985879|+\rangle +0.167457|-\rangle$
 proton mixing angle: $\alpha_p=1.7395034$ rad
 incoming proton: $-0.167908|+\rangle +0.985803|-\rangle$

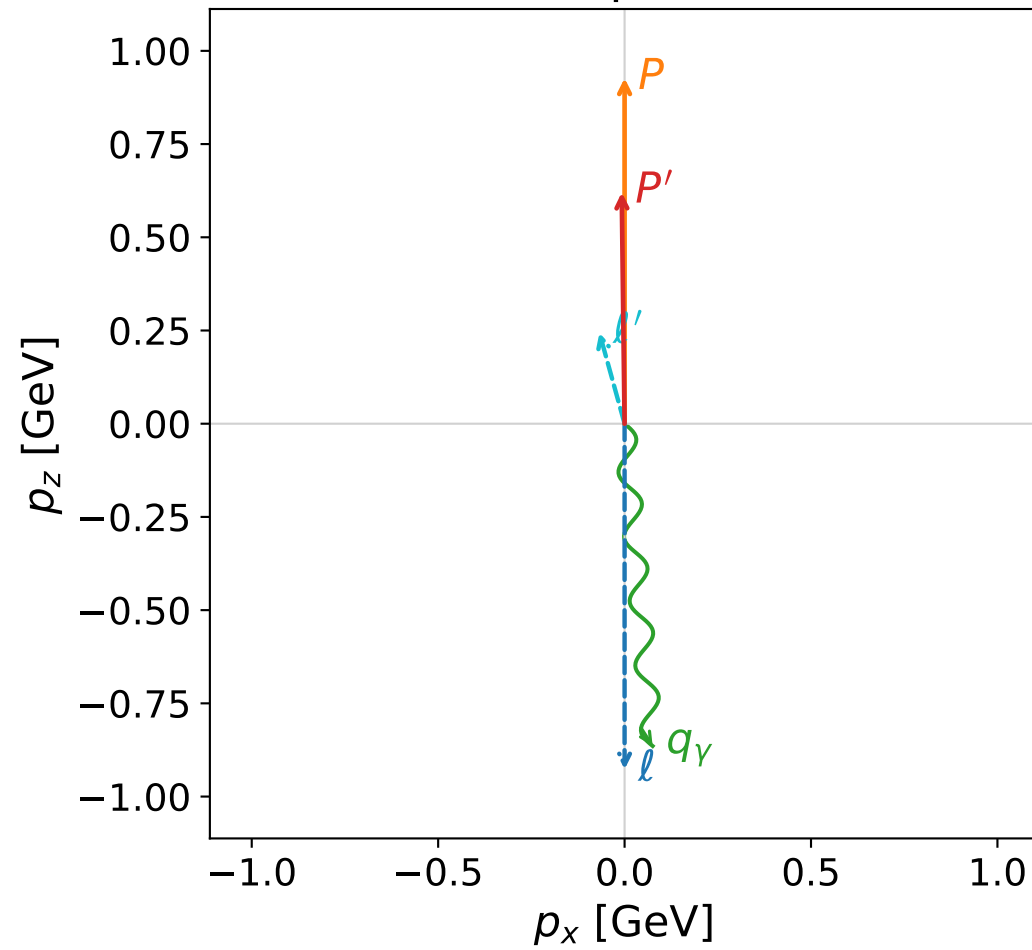
$s=23.6705$, $\sqrt{s}=4.86523$, $|\vec{P}|=2.3422$, $|\vec{P}'|=0.0567288$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.13863$, $\phi_{p'}=2.24178$, $\phi_\gamma=4.57737\text{e-}07$
 $E_\gamma=1.9344$, $\theta_{\ell'}=0.00287975$, $\phi_{\ell'}=3.14159$
 $Q^2=18.6544$, $x_B=0.845219$, $t=-3.24792$, $W^2=4.29593$, $y=0.968399$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3422$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3422$
 P $E= 2.5230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3422$
 ℓ' $E= 1.9911$ $p_x= -0.0057$ $p_y= -0.0000$ $p_z= 1.9911$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0567$
 q_γ $E= 1.9344$ $p_x= 0.0057$ $p_y= 0.0000$ $p_z= -1.9344$

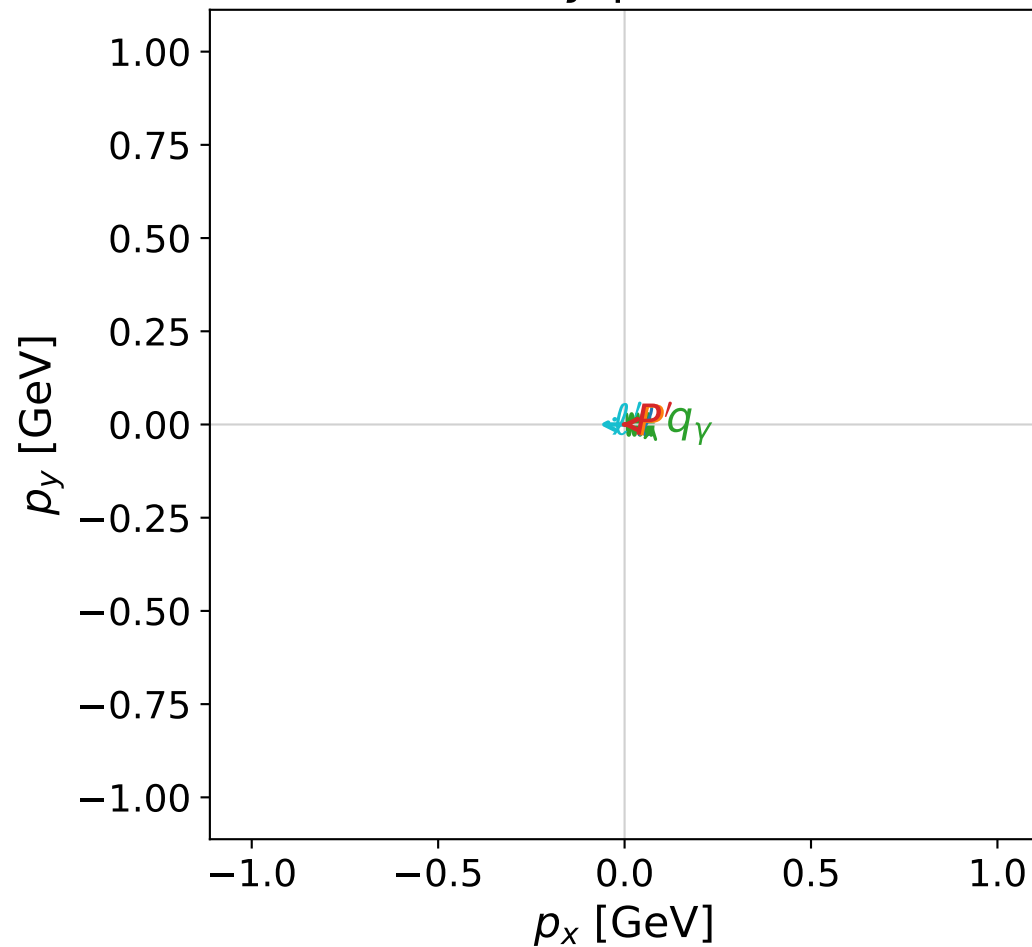


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

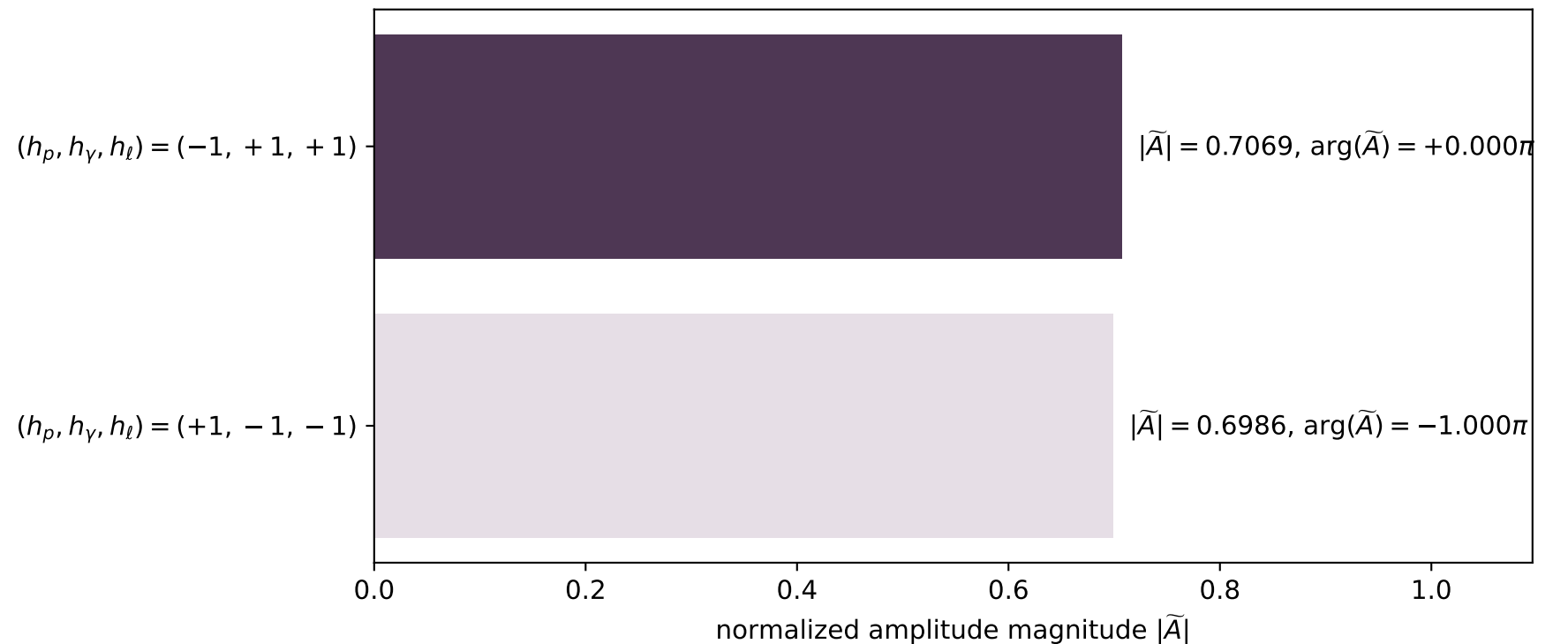


dghz_mixing_angles_polarization_04_local_minimum_251
 $d_{\text{GHZ}}=0.000133889$
 region: polarization_cluster_04_local_minimum_251
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0251
 lepton mixing angle: $\alpha_e=2.7320295$ rad
 incoming lepton: $-0.917295|+\rangle +0.398209|-\rangle$
 proton mixing angle: $\alpha_p=1.949413$ rad
 incoming proton: $-0.369635|+\rangle +0.929177|-\rangle$

$s=5.04265$, $\sqrt{s}=2.24559$, $|\vec{P}|=0.926887$, $|\vec{P}'|=0.620376$
 $\theta_{p'}=0.0123959$, $\theta_\gamma=3.05471$, $\phi_{p'}=3.14155$, $\phi_\gamma=6.28317$
 $E_\gamma=0.867702$, $\theta_{l'}=0.270165$, $\phi_{l'}=3.14158$
 $Q^2=0.922051$, $x_B=0.233591$, $t=-0.0563611$, $W^2=3.90509$, $y=0.948228$

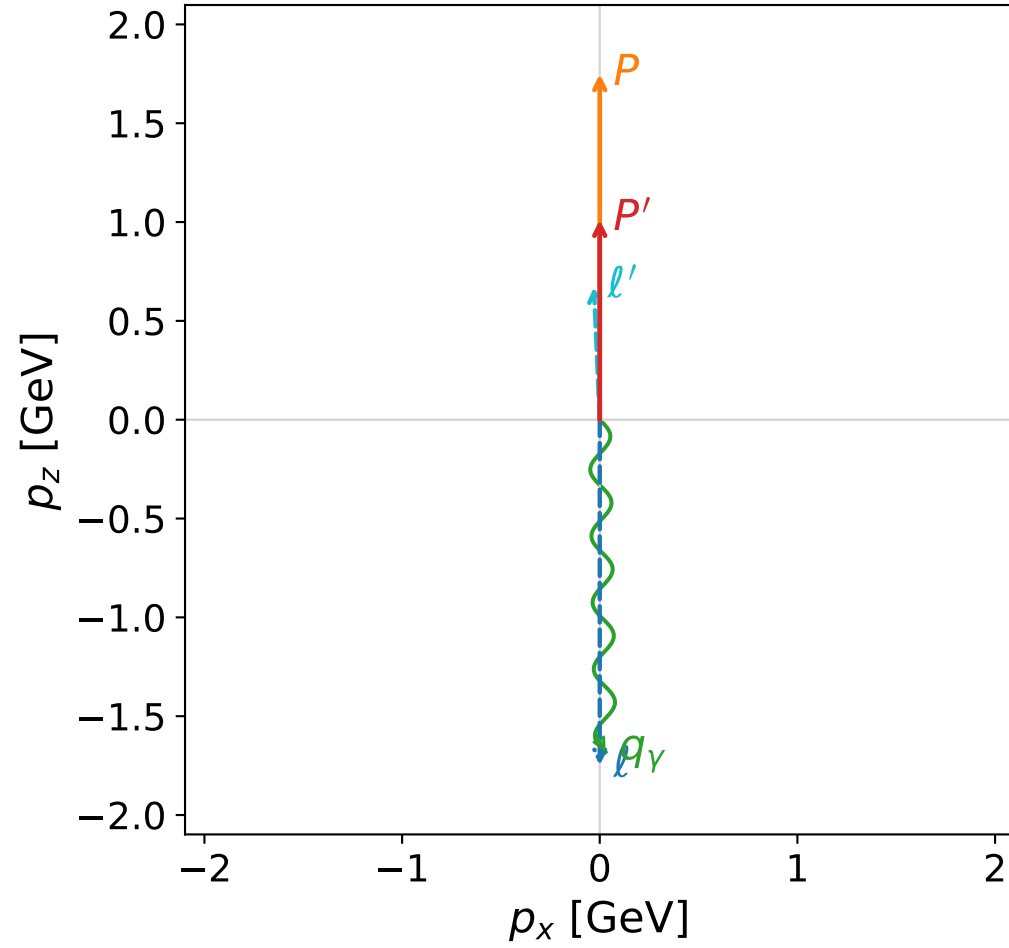
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9269$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9269$
 l' $E= 0.2533$ $p_x= -0.0676$ $p_y= 0.0000$ $p_z= 0.2441$
 P' $E= 1.1246$ $p_x= -0.0077$ $p_y= 0.0000$ $p_z= 0.6203$
 q_γ $E= 0.8677$ $p_x= 0.0753$ $p_y= -0.0000$ $p_z= -0.8644$

Leading normalized final-state amplitudes
(N=2, retained=98.8%)

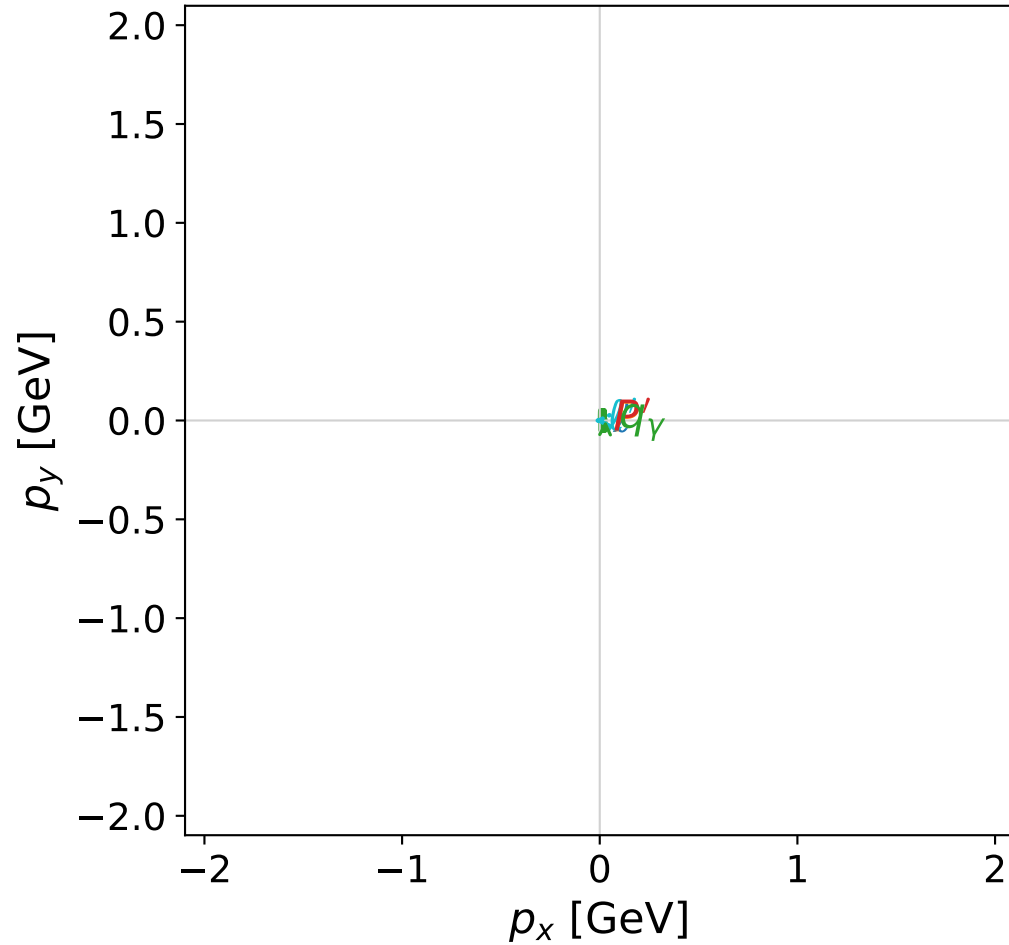


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



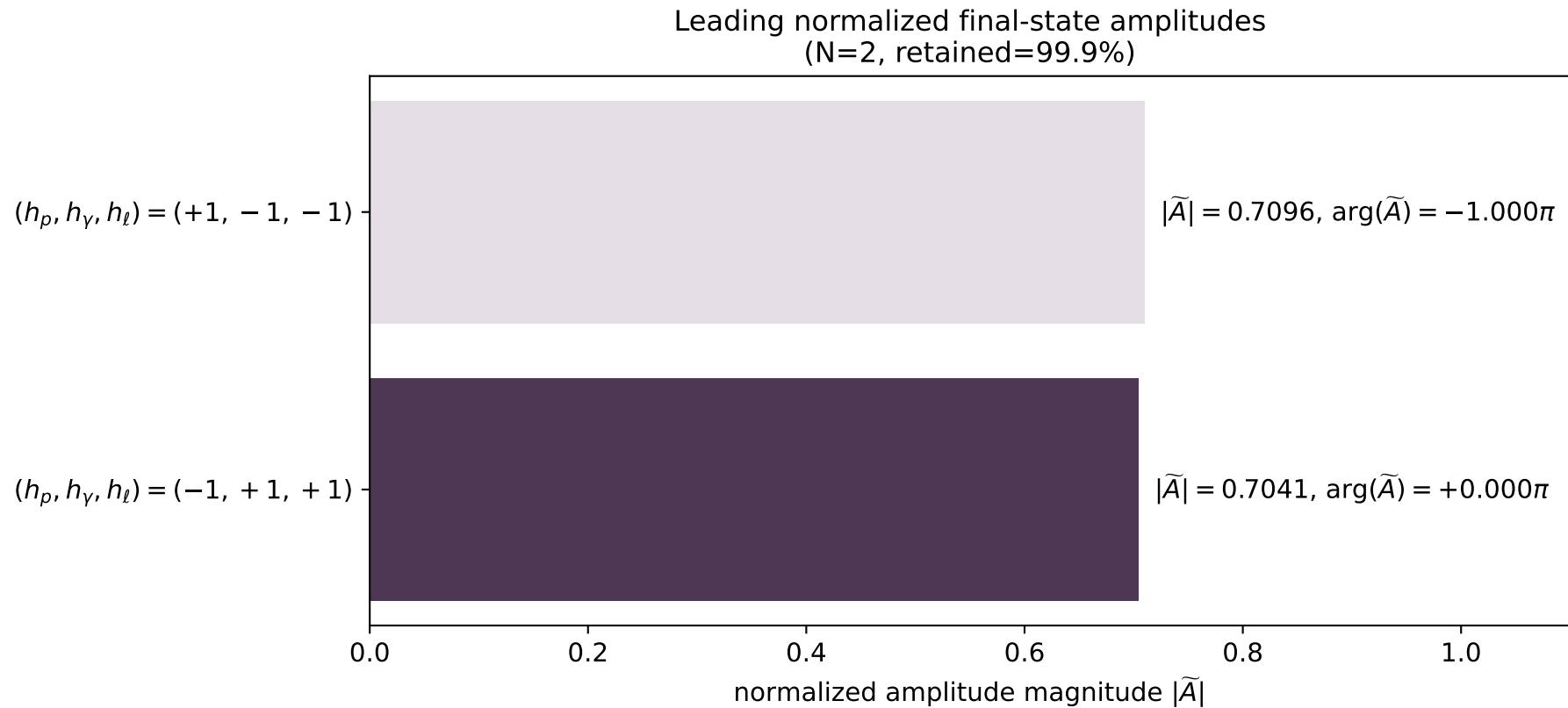
x--y plane



dghz_mixing_angles_polarization_04_local_minimum_252
 $d_{\text{GHZ}}=0.000137569$
 region: polarization_cluster_04_local_minimum_252
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0252
 lepton mixing angle: $\alpha_e=2.8965137$ rad
 incoming lepton: $-0.970118|+\rangle +0.242633|-\rangle$
 proton mixing angle: $\alpha_p=1.8140654$ rad
 incoming proton: $-0.240877|+\rangle +0.970556|-\rangle$

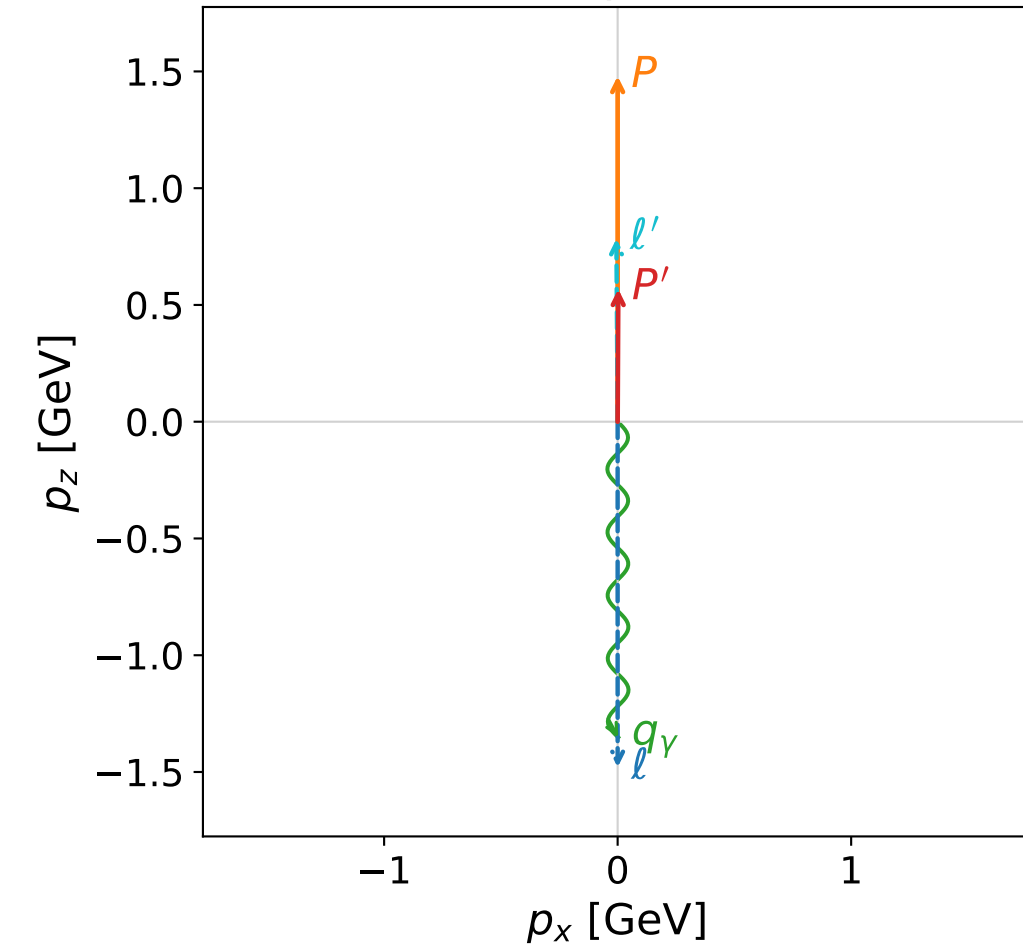
$s=13.9325$, $\sqrt{s}=3.73262$, $|\vec{P}|=1.74845$, $|\vec{P}'|=1.01096$
 $\theta_{P'}=0$, $\theta_\gamma=3.12412$, $\phi_{P'}=5.03716$, $\phi_\gamma=6.28317$
 $E_\gamma=1.68205$, $\theta_{l'}=0.0437779$, $\phi_{l'}=3.14158$
 $Q^2=4.69394$, $x_B=0.368621$, $t=-0.177771$, $W^2=8.9197$, $y=0.975575$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7485$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7485$
 P $E= 1.9842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7485$
 l' $E= 0.6715$ $p_x= -0.0294$ $p_y= 0.0000$ $p_z= 0.6708$
 P' $E= 1.3791$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0110$
 q_γ $E= 1.6821$ $p_x= 0.0294$ $p_y= -0.0000$ $p_z= -1.6818$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

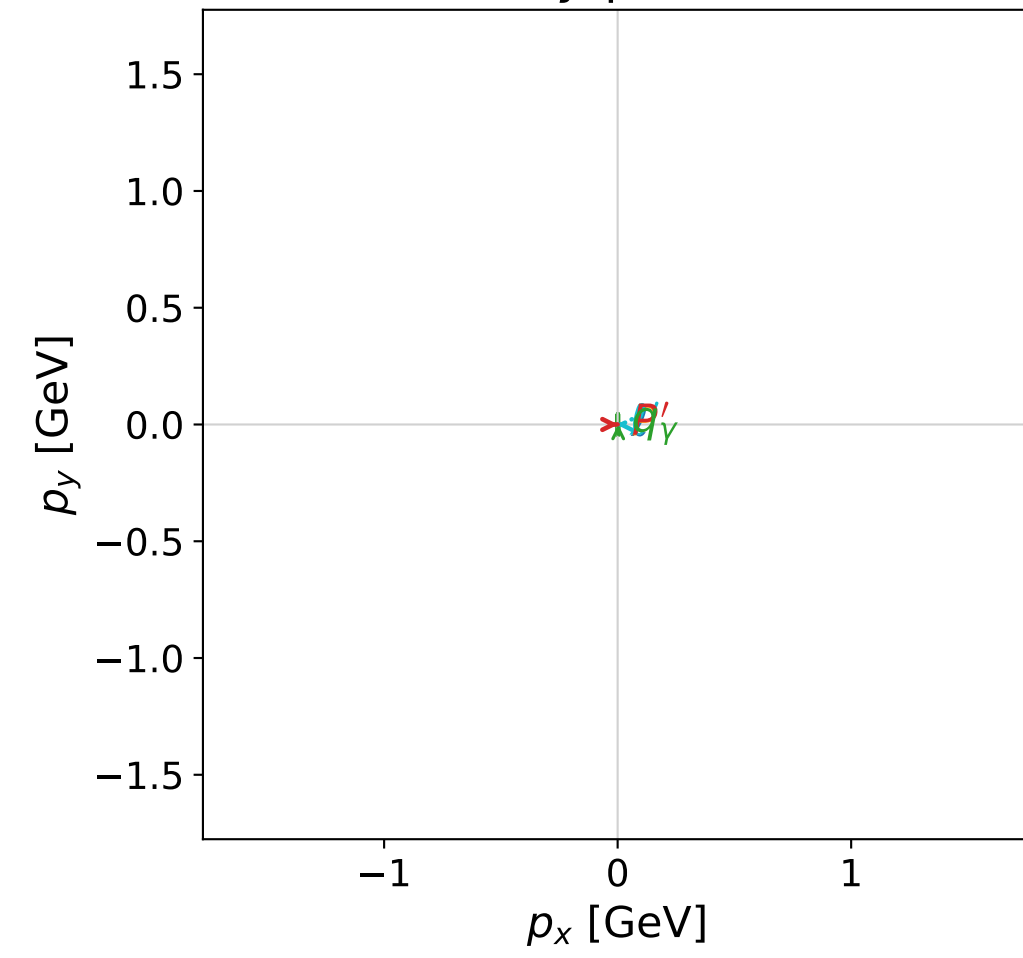


dghz_mixing_angles_polarization_04_local_minimum_293
 $d_{\text{GHZ}}=0.000295565$
 region: polarization_cluster_04_local_minimum_293
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0293
 lepton mixing angle: $\alpha_e=2.6023939$ rad
 incoming lepton: $-0.85812|+\rangle +0.513449|-\rangle$
 proton mixing angle: $\alpha_p=2.1130396$ rad
 incoming proton: $-0.516059|+\rangle +0.856553|-\rangle$

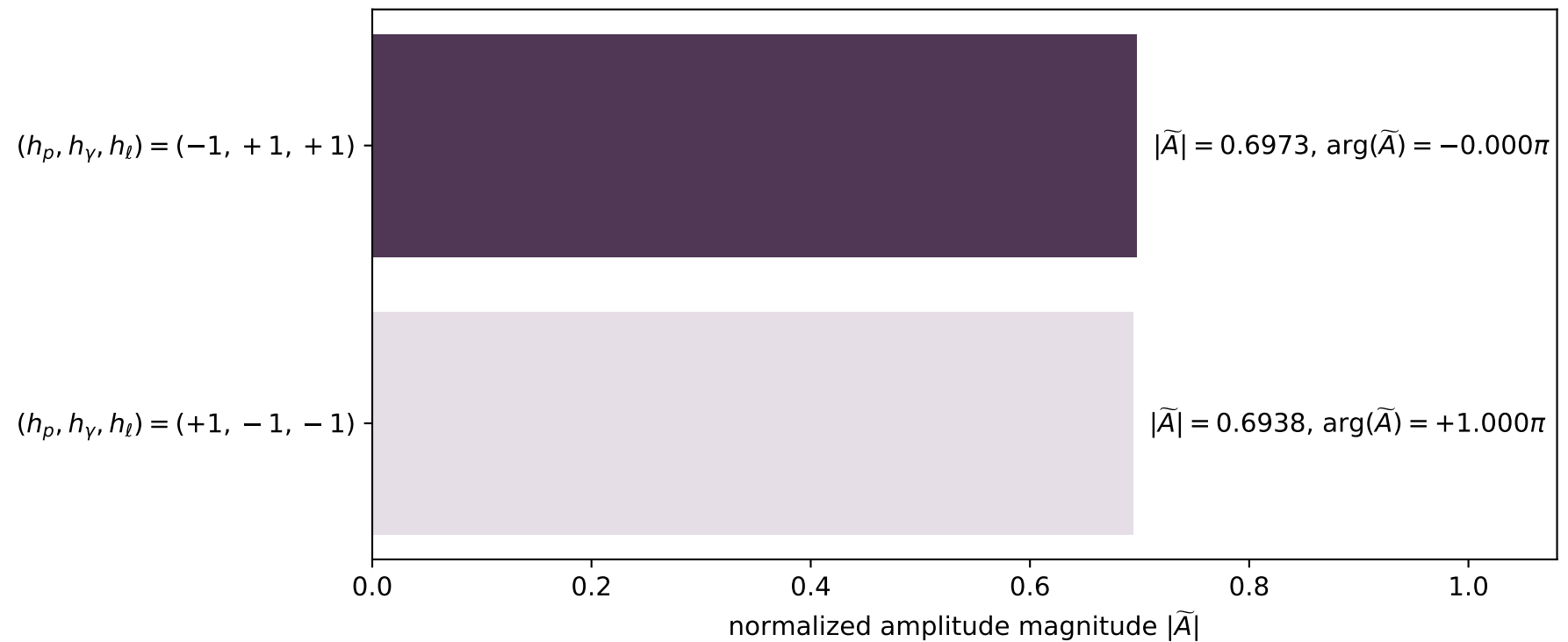
$s=10.4463$, $\sqrt{s}=3.23207$, $|\vec{P}|=1.47992$, $|\vec{P}'|=0.568077$
 $\theta_{P'}=0.00483204$, $\theta_\gamma=3.13964$, $\phi_{P'}=6.28121$, $\phi_\gamma=0$
 $E_\gamma=1.35176$, $\theta_{l'}=0.00687461$, $\phi_{l'}=3.14059$
 $Q^2=4.63923$, $x_B=0.507591$, $t=-0.401758$, $W^2=5.38032$, $y=0.955392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.4799$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4799$
 P $E= 1.7521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4799$
 l' $E= 0.7837$ $p_x= -0.0054$ $p_y= 0.0000$ $p_z= 0.7837$
 P' $E= 1.0966$ $p_x= 0.0027$ $p_y= -0.0000$ $p_z= 0.5681$
 q_γ $E= 1.3518$ $p_x= 0.0026$ $p_y= 0.0000$ $p_z= -1.3518$

x--y plane

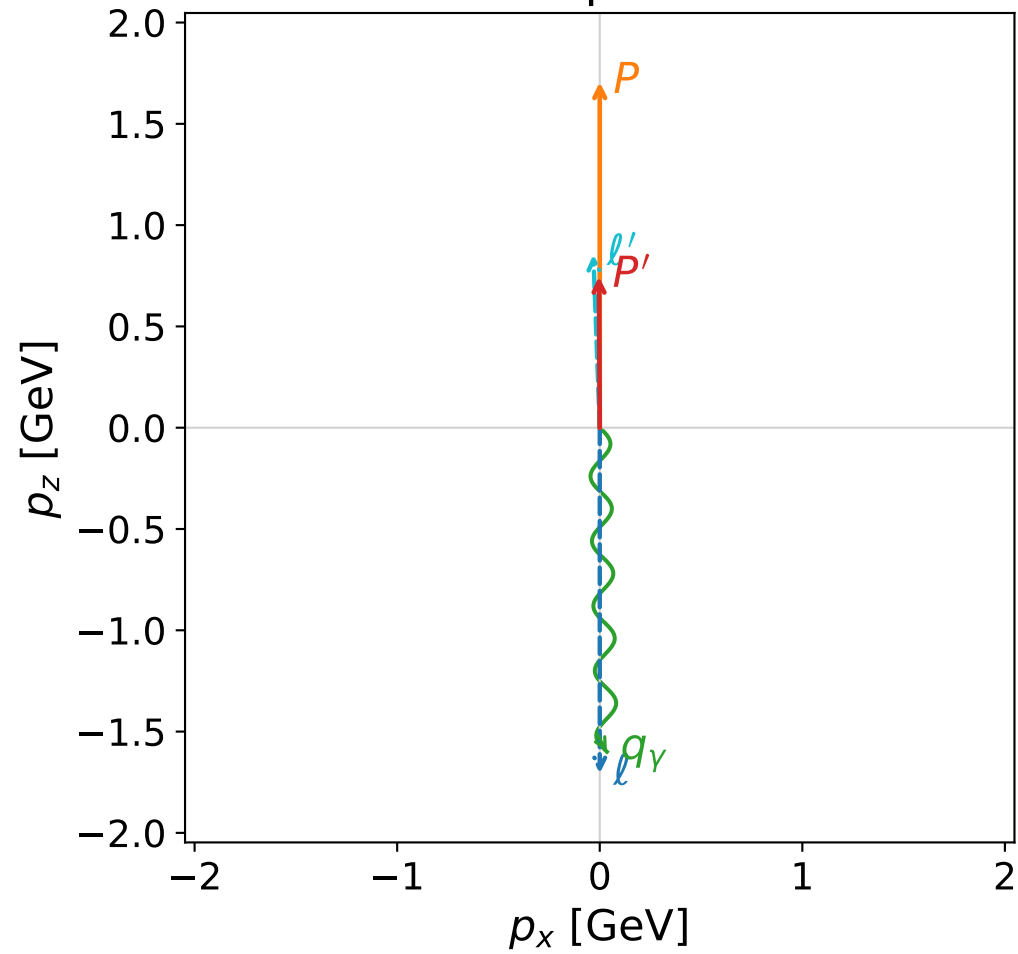


Leading normalized final-state amplitudes
 (N=2, retained=96.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

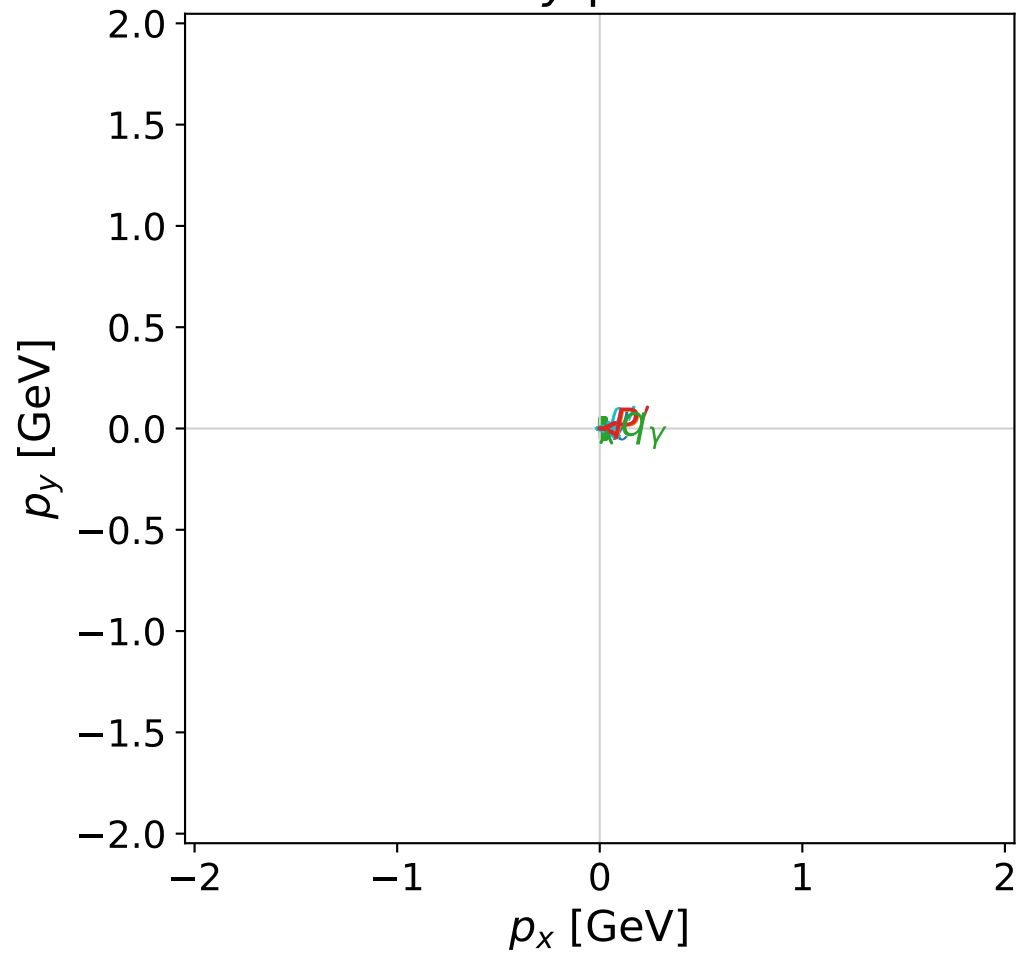


dghz_mixing_angles_polarization_04_local_minimum_297
 $d_{\{\rm GHZ\}}=0.000306617$
 region: polarization_cluster_04_local_minimum_297
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0297
 lepton mixing angle: $\alpha_e=3.0107596$ rad
 incoming lepton: $-0.991454|+\rangle +0.13046|-\rangle$
 proton mixing angle: $\alpha_p=1.6989407$ rad
 incoming proton: $-0.127794|+\rangle +0.991801|-\rangle$

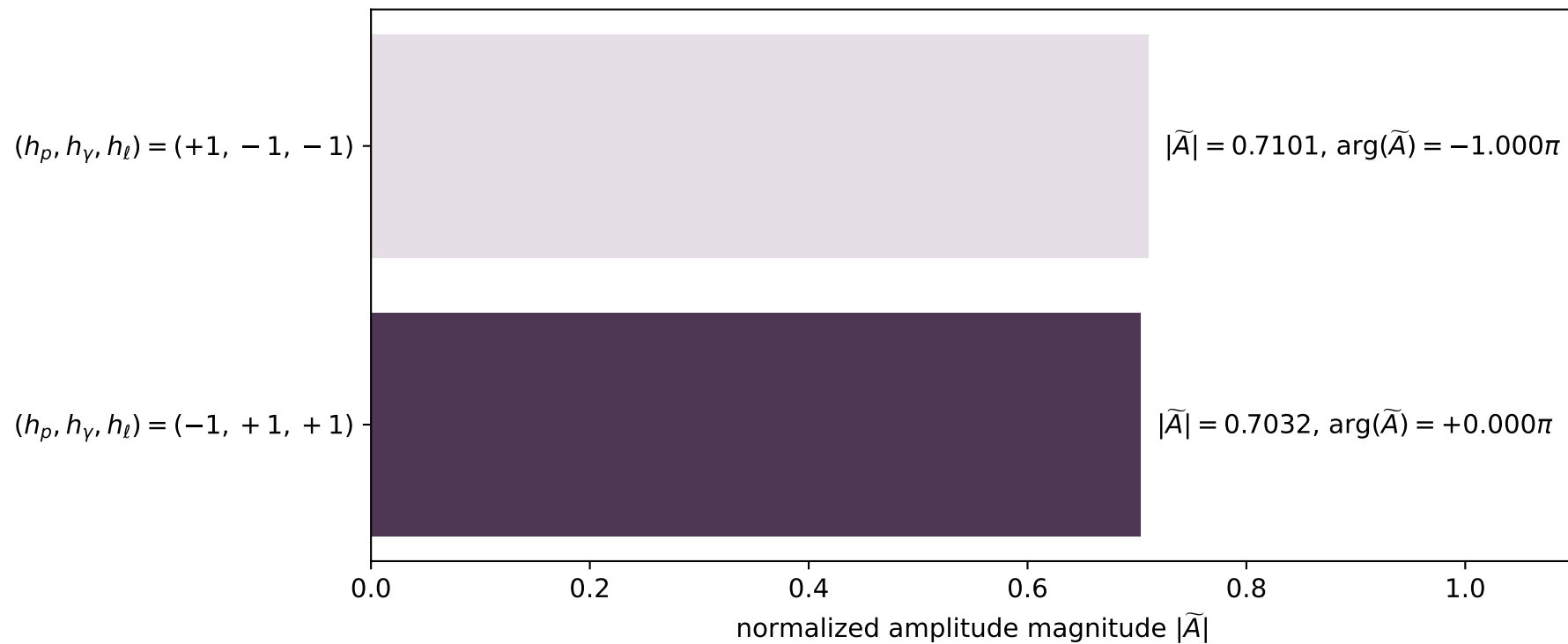
$s=13.3426$, $\sqrt{s}=3.65276$, $|\vec{P}|=1.70594$, $|\vec{P}'|=0.745662$
 $\theta_{p'}=0.00520095$, $\theta_\gamma=3.11943$, $\phi_{p'}=3.14107$, $\phi_\gamma=0$
 $E_\gamma=1.59997$, $\theta_{l'}=0.0369583$, $\phi_{l'}=3.14166$
 $Q^2=5.82901$, $x_B=0.483769$, $t=-0.361859$, $W^2=7.09999$, $y=0.96681$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7059$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7059$
 P $E= 1.9468$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7059$
 l' $E= 0.8545$ $p_x= -0.0316$ $p_y= -0.0000$ $p_z= 0.8539$
 P' $E= 1.1983$ $p_x= -0.0039$ $p_y= 0.0000$ $p_z= 0.7457$
 q_γ $E= 1.6000$ $p_x= 0.0355$ $p_y= 0.0000$ $p_z= -1.5996$

x--y plane

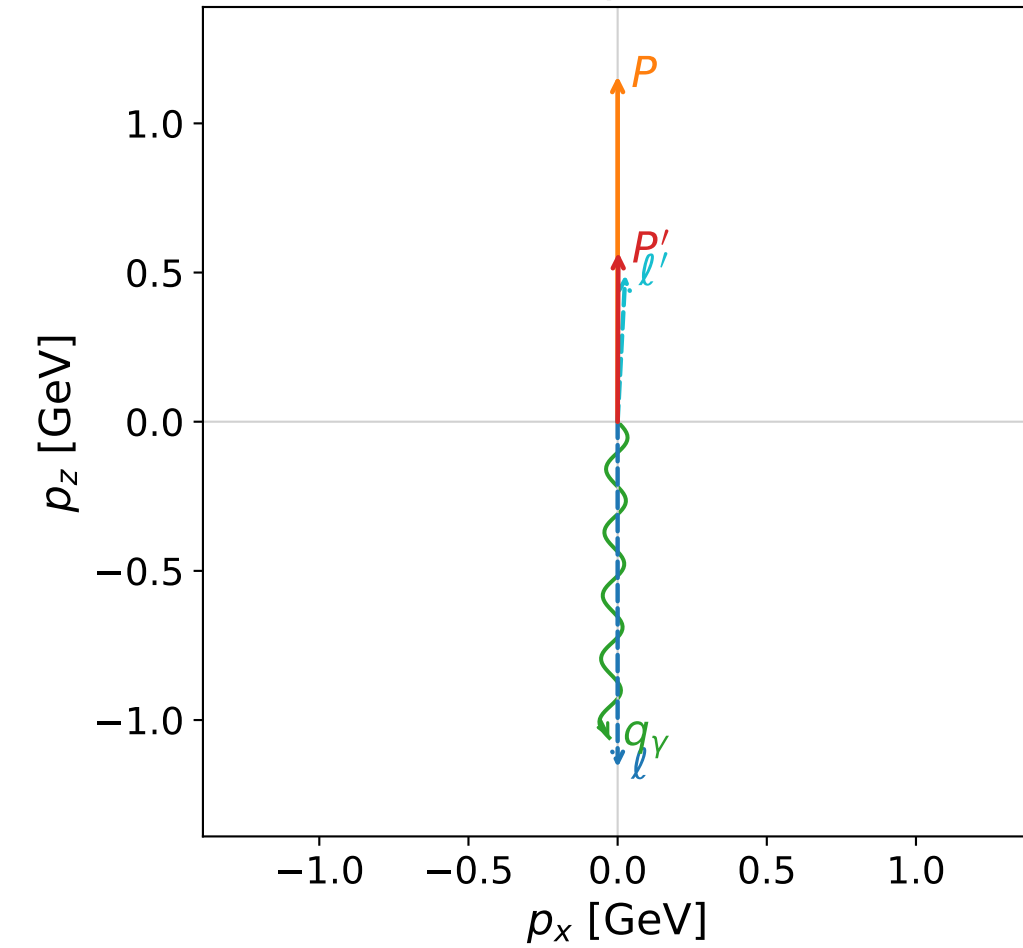


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

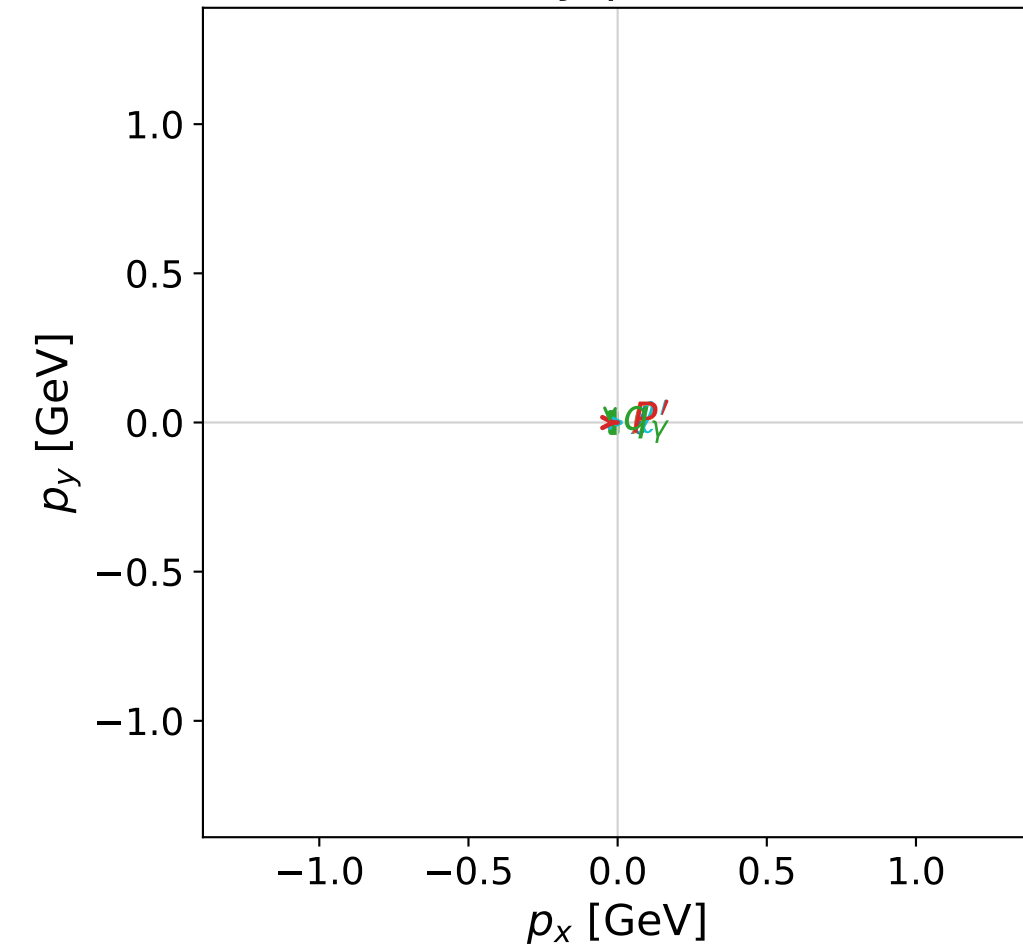


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

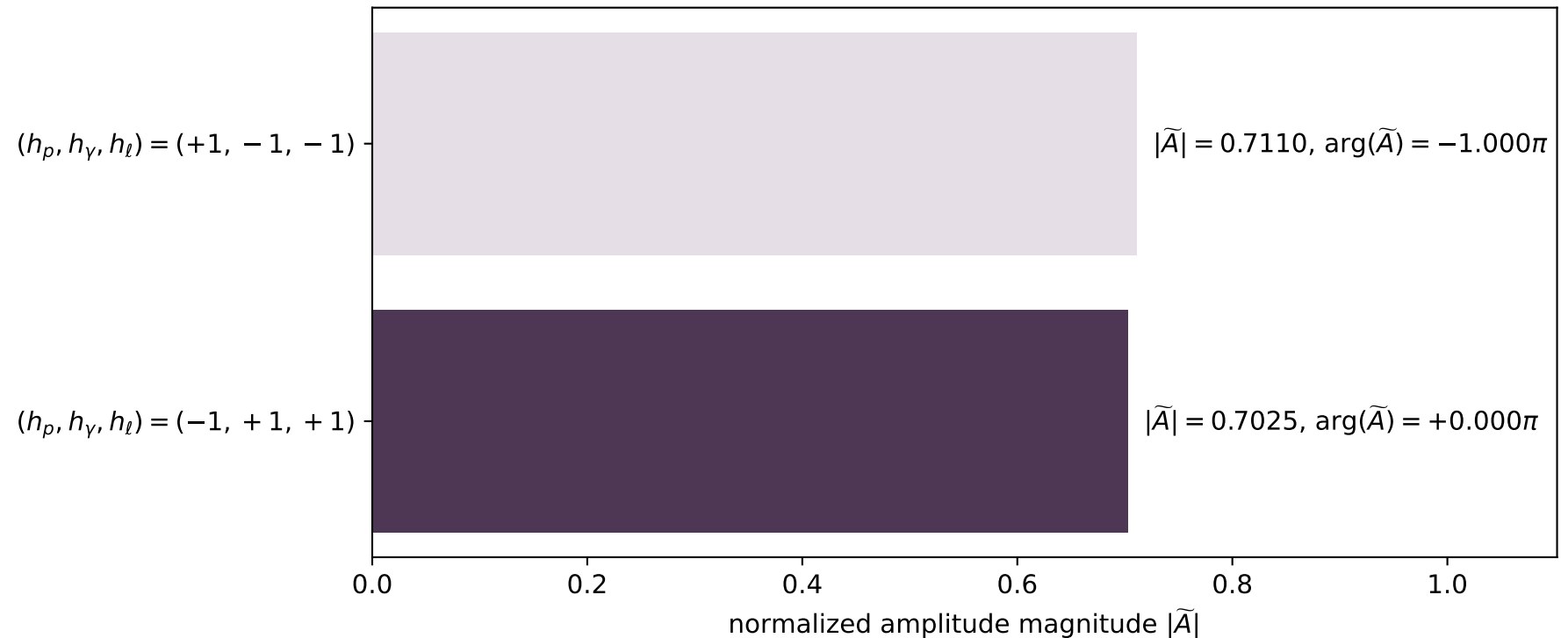


dghz_mixing_angles_polarization_04_local_minimum_319
 $d_{\text{GHZ}}=0.000411475$
 region: polarization_cluster_04_local_minimum_319
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0319
 lepton mixing angle: $\alpha_e=1.704901$ rad
 incoming lepton: $-0.133703|+\rangle +0.991021|-\rangle$
 proton mixing angle: $\alpha_p=3.0080191$ rad
 incoming proton: $-0.991092|+\rangle +0.133177|-\rangle$

$s=7.01694$, $\sqrt{s}=2.64895$, $|\vec{P}|=1.1584$, $|\vec{P}'|=0.567777$
 $\theta_{p'}=0.00378308$, $\theta_Y=3.11539$, $\phi_{p'}=0.00339324$, $\phi_Y=3.14154$
 $E_Y=1.05998$, $\theta_{l'}=0.0520539$, $\phi_{l'}=6.28284$
 $Q^2=2.28056$, $x_B=0.392633$, $t=-0.193536$, $W^2=4.40765$, $y=0.946437$

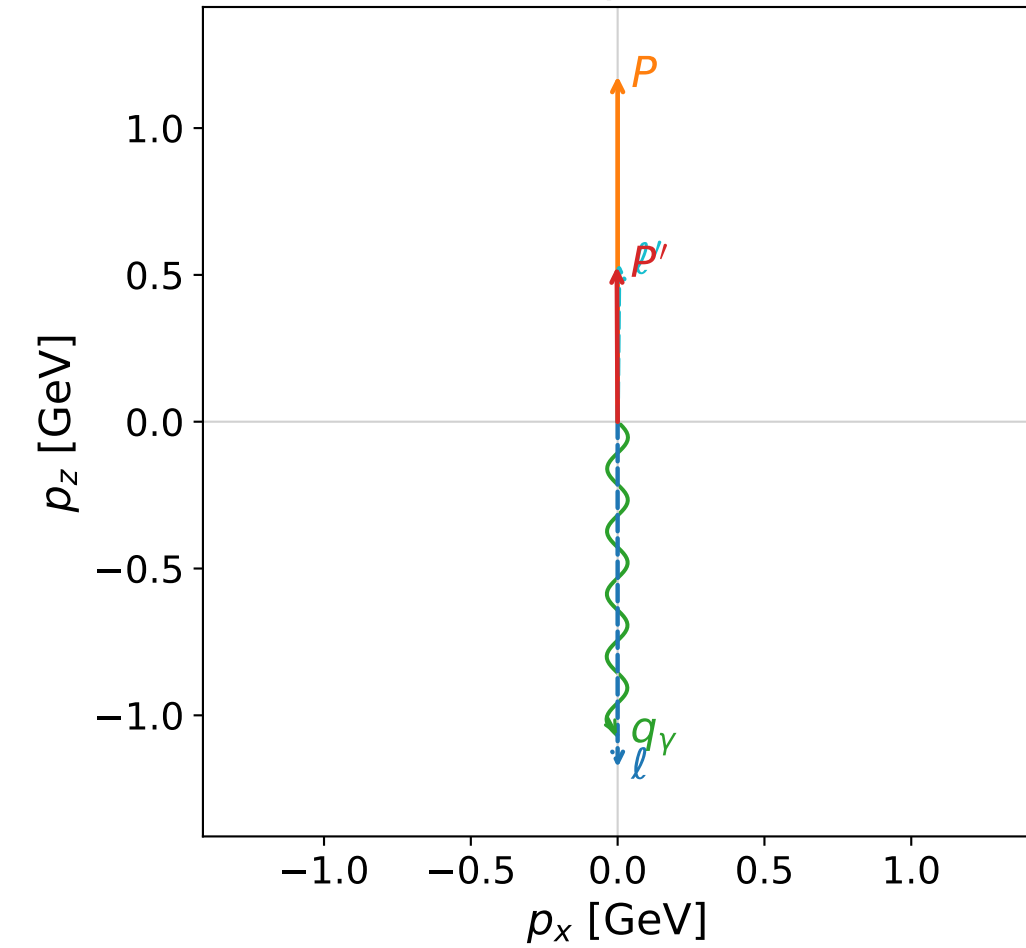
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.1584$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1584$
 P $E= 1.4905$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1584$
 ℓ' $E= 0.4925$ $p_x= 0.0256$ $p_y= -0.0000$ $p_z= 0.4918$
 P' $E= 1.0965$ $p_x= 0.0021$ $p_y= 0.0000$ $p_z= 0.5678$
 q_Y $E= 1.0600$ $p_x= -0.0278$ $p_y= 0.0000$ $p_z= -1.0596$

Leading normalized final-state amplitudes
 (N=2, retained=99.9%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

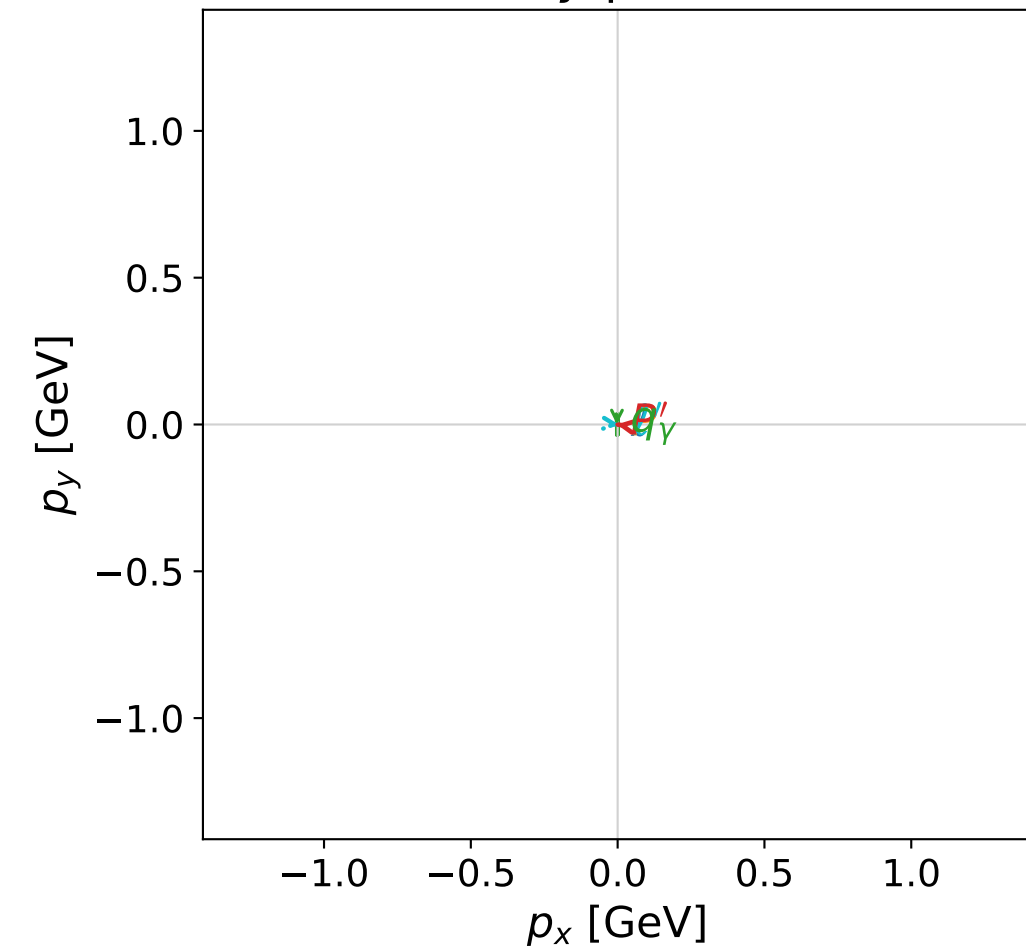


dghz_mixing_angles_polarization_04_local_minimum_336
 $d_{\text{GHZ}}=0.000558565$
 region: polarization_cluster_04_local_minimum_336
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0336
 lepton mixing angle: $\alpha_e=2.0949563$ rad
 incoming lepton: $-0.500486|+\rangle +0.865745|-\rangle$
 proton mixing angle: $\alpha_p=2.6121434$ rad
 incoming proton: $-0.863085|+\rangle +0.505058|-\rangle$

$s=7.19362$, $\sqrt{s}=2.68209$, $|\vec{P}|=1.17702$, $|\vec{P}'|=0.528084$
 $\theta_{P'}=0.00510742$, $\theta_\gamma=3.13917$, $\phi_{P'}=2.97399$, $\phi_\gamma=3.14159$
 $E_\gamma=1.06685$, $\theta_{l'}=0.00977172$, $\phi_{l'}=6.19762$
 $Q^2=2.53666$, $x_B=0.425599$, $t=-0.237414$, $W^2=4.30339$, $y=0.944002$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.1770$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1770$
 P $E= 1.5051$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1770$
 l' $E= 0.5388$ $p_x= 0.0052$ $p_y= -0.0004$ $p_z= 0.5388$
 P' $E= 1.0764$ $p_x= -0.0027$ $p_y= 0.0004$ $p_z= 0.5281$
 q_γ $E= 1.0669$ $p_x= -0.0026$ $p_y= -0.0000$ $p_z= -1.0669$

x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=96.6%)

